机械工程基础

几何体

2024.3.12

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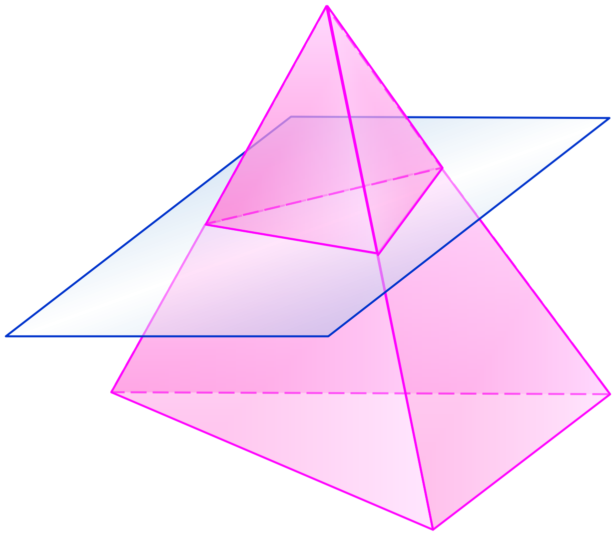
- 交线
- 组合体
- 读图

■交线

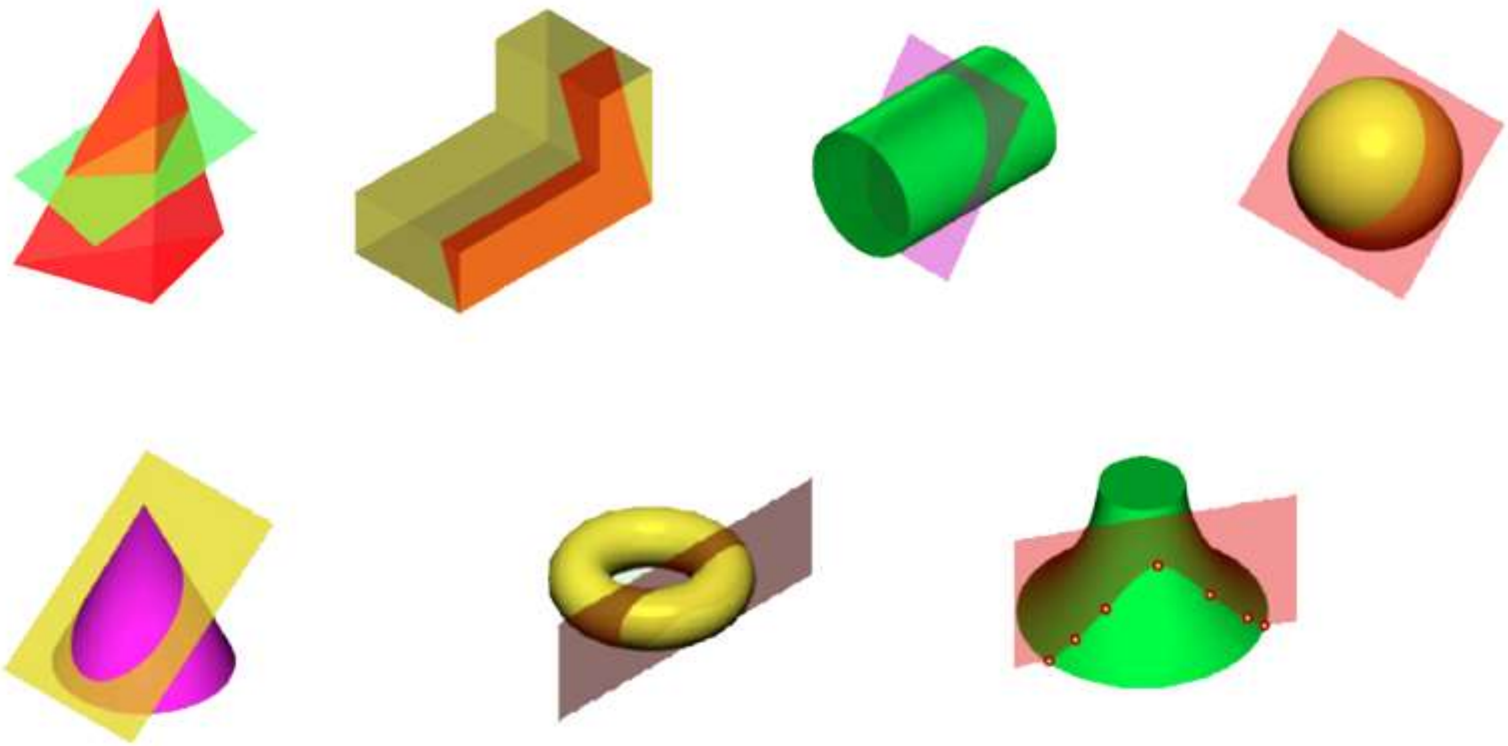
截交 用平面与立体相交

- **截平面** 用以截切立体的平面
- **截交线** 截平面与立体表面的交线
- **截断面** 因截平面的截切，在立体上形成的平面

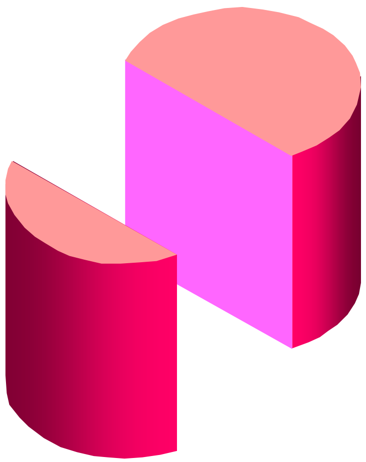


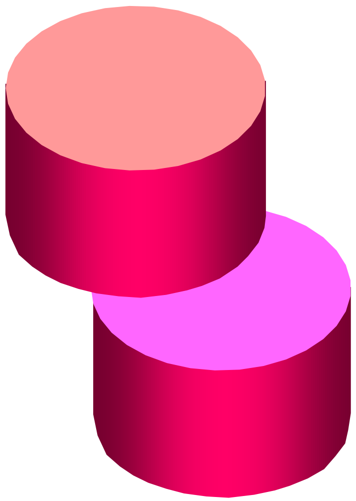
截交

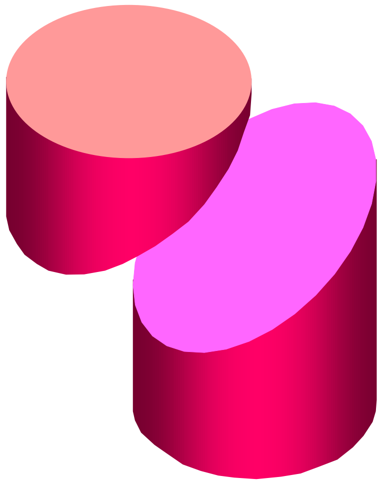
- ✓ 截断面是一封闭的平面多边形
- ✓ 截交线的形状取决于被截立体的形状及截平面与立体的相对位置
- ✓ 截交线是截平面与立体表面的共有线



截交

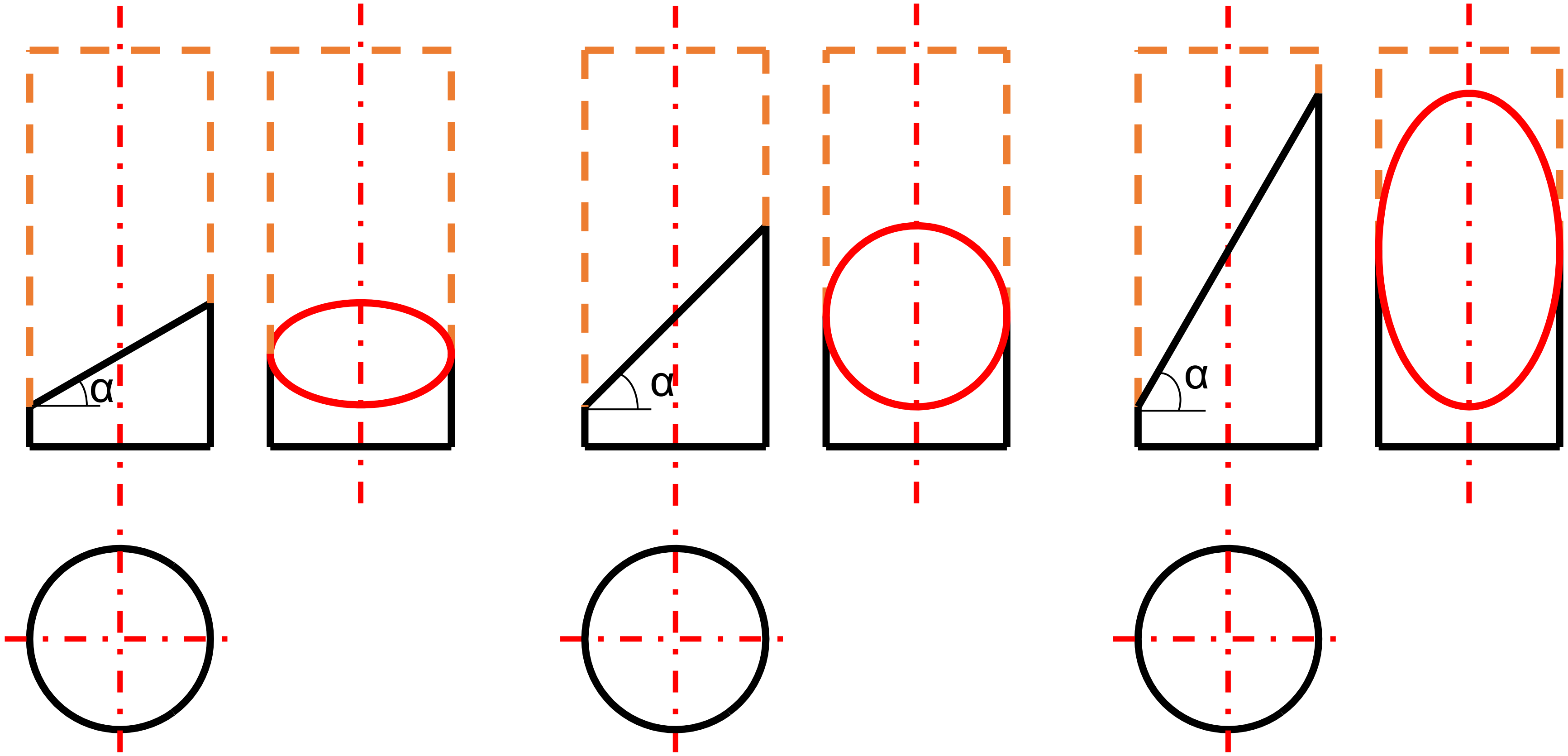






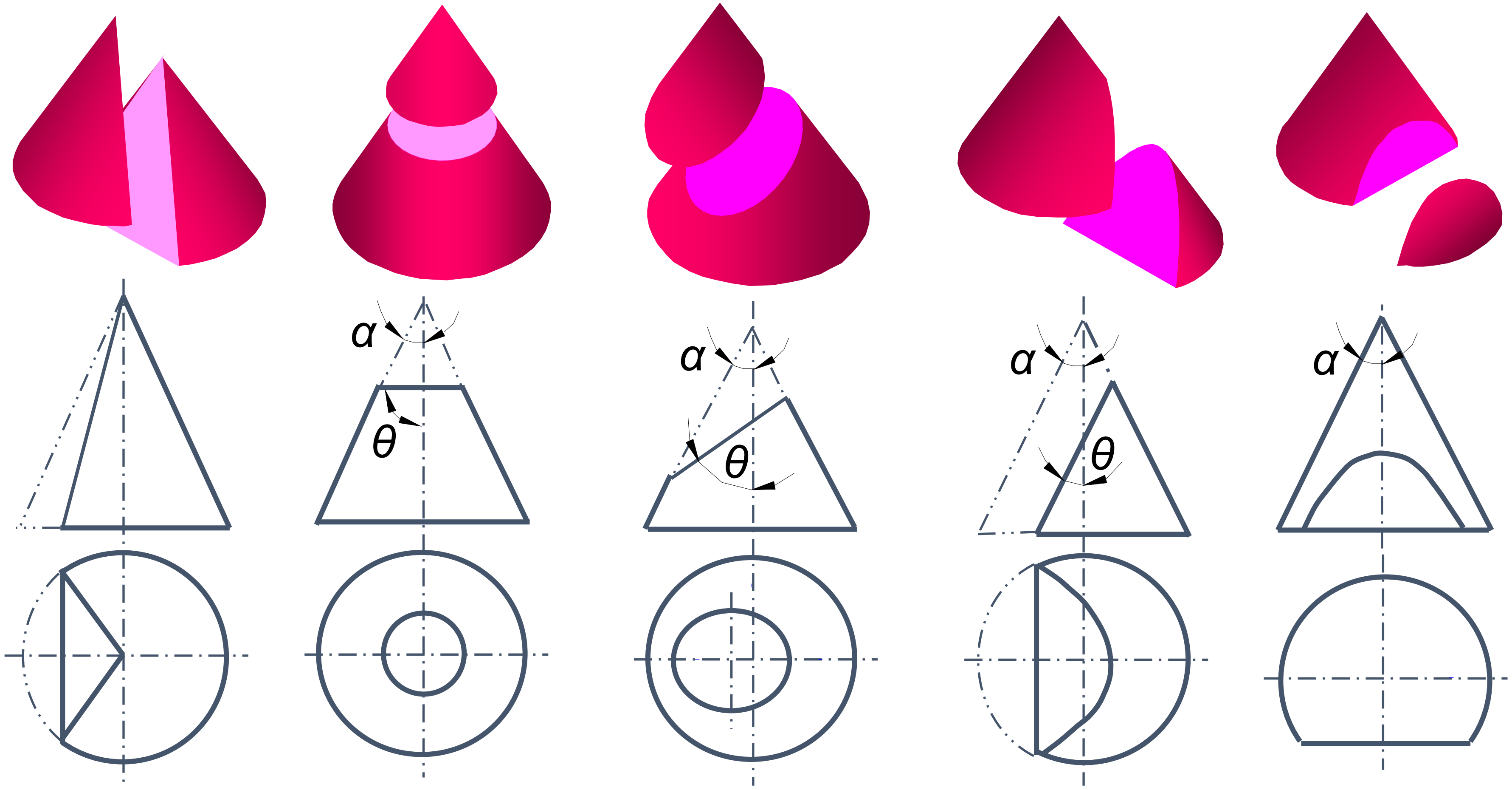
平行	垂直	倾斜
两平行直线	圆	椭圆

截交



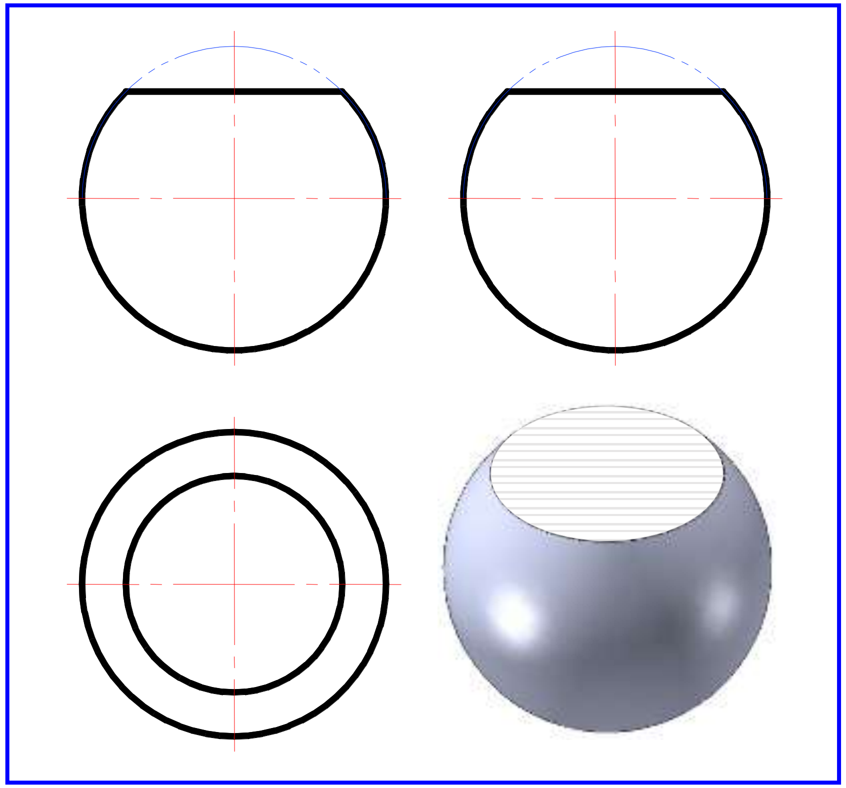
（ α < 45° ）　　（ α = 45° ）　　（ α > 45° ）

截交



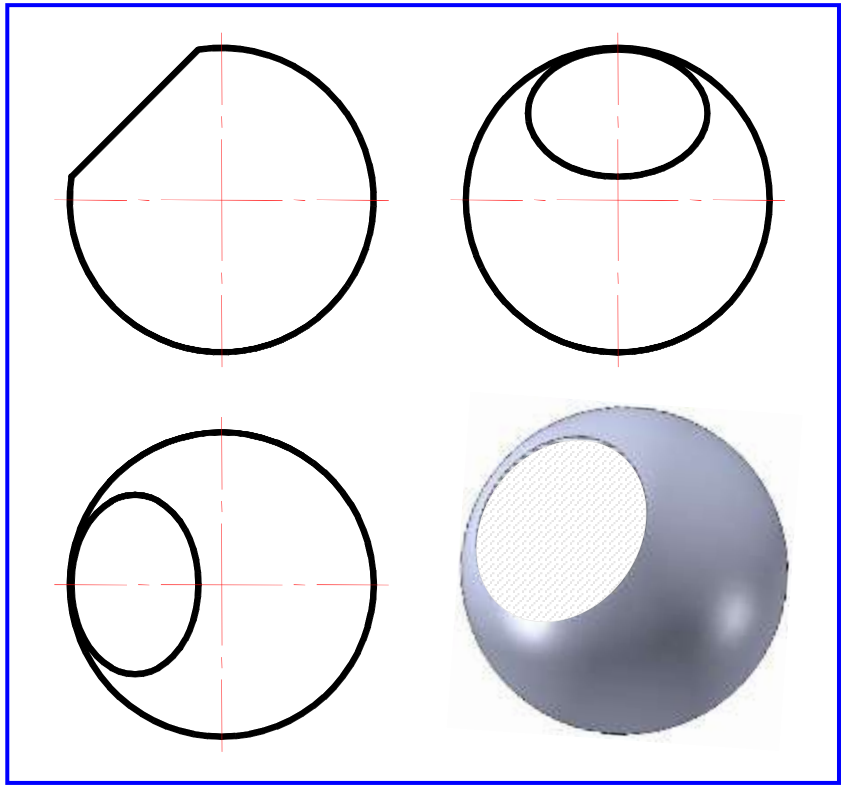
过锥顶	$\theta = 90^\circ$	$90^\circ > \theta > \alpha$	$\theta = \alpha$	$0^\circ \le \theta < \alpha$
两相交直线	圆	椭圆	抛物线	双曲线

截交



截平面 // 投影面

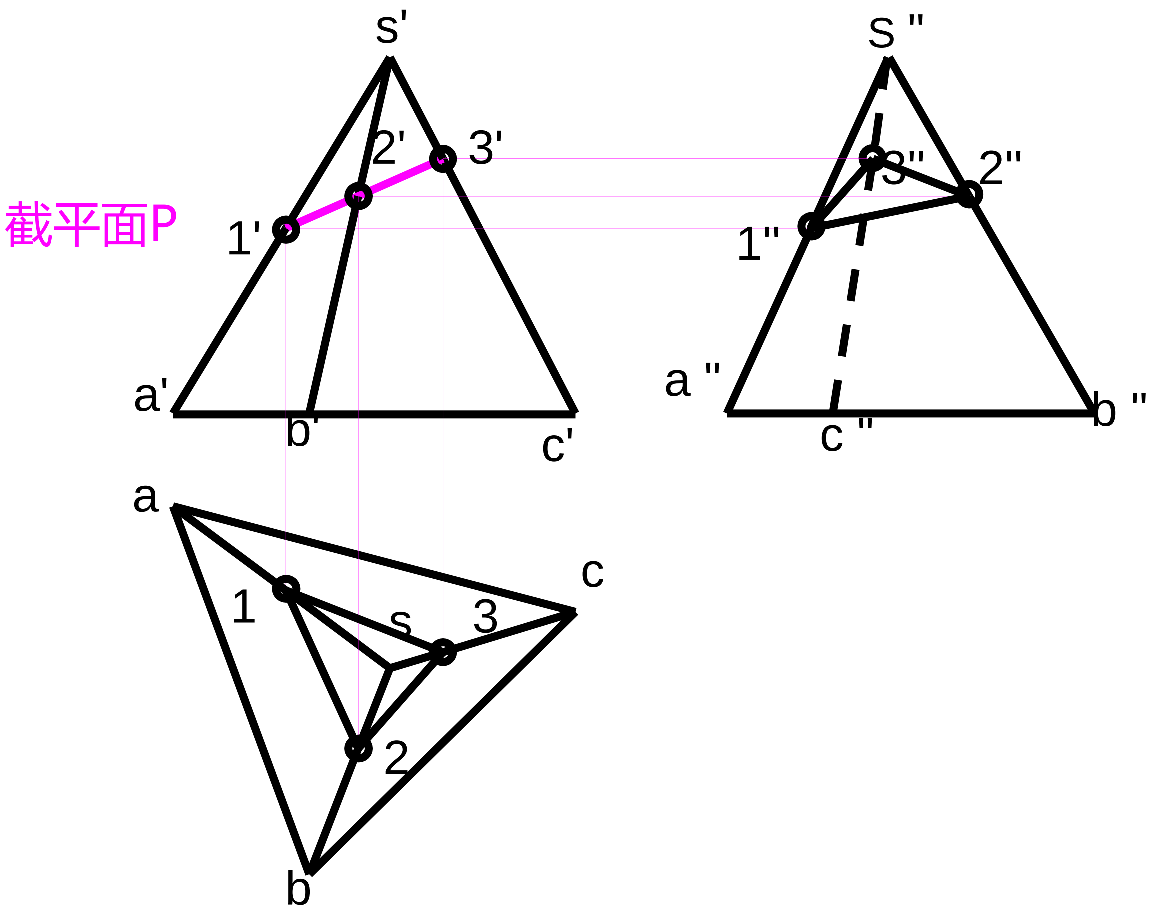
截交线的投影是圆、直线

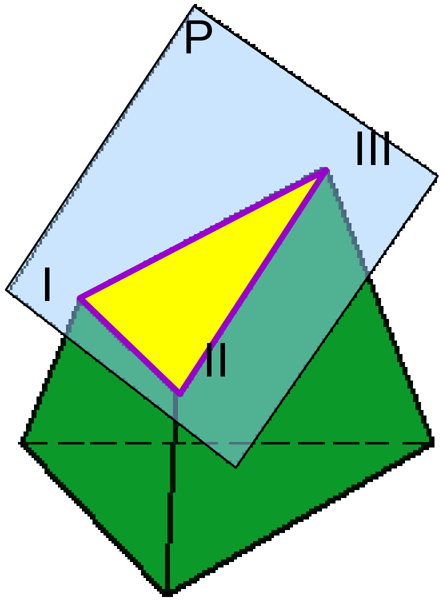


截平面 ∠ 投影面

截交线的投影是椭圆、直线

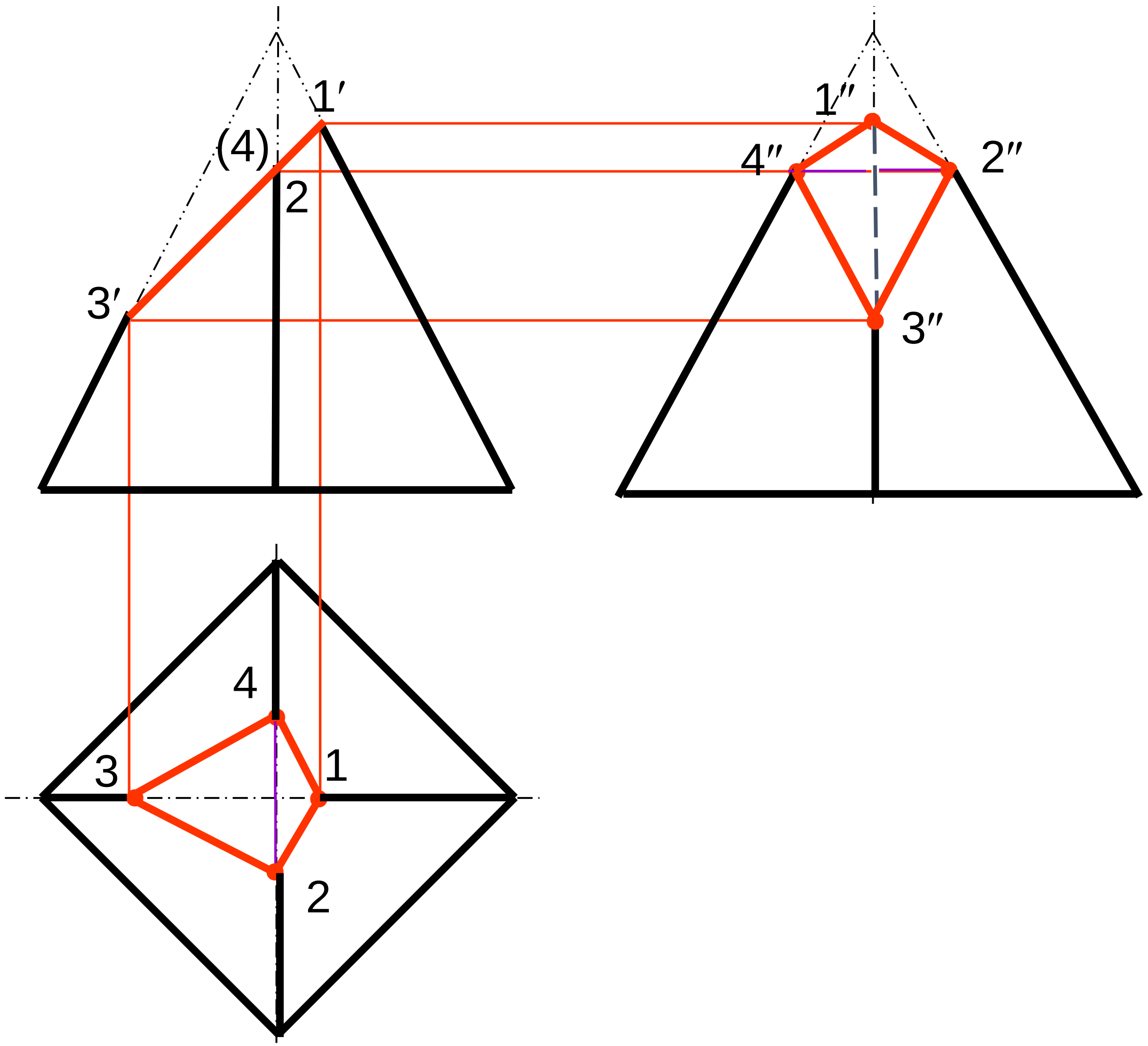
求被正垂面截切的三棱锥的另外两面投影。

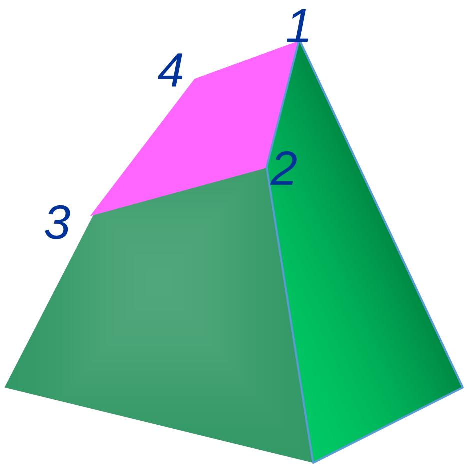




【例】

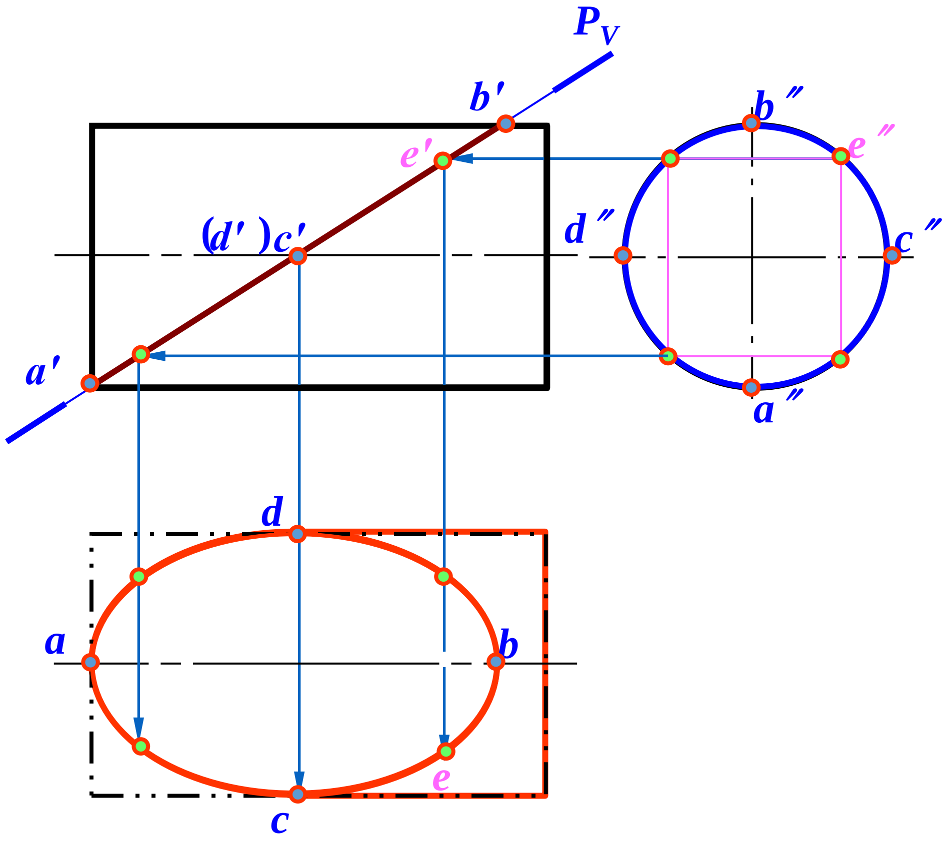
求四棱锥被截切后的俯视图和左视图。

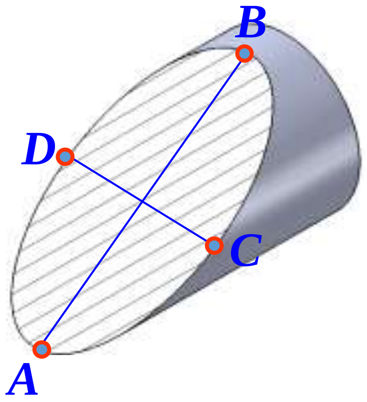




【例】

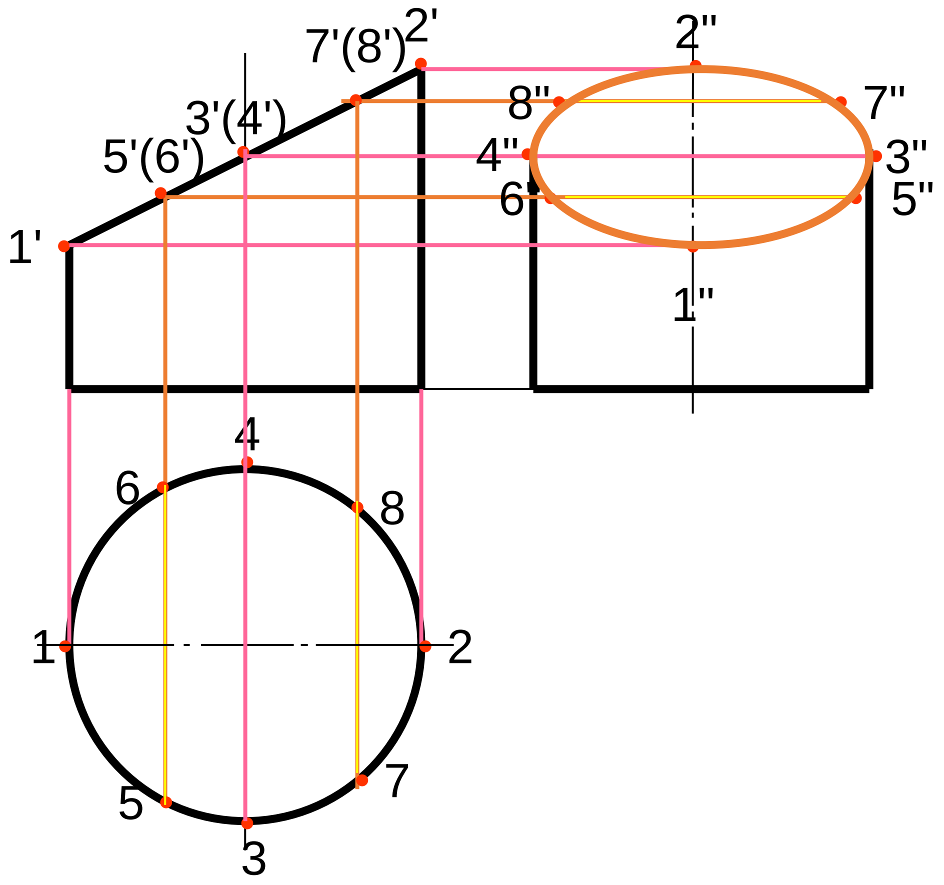
作俯视图。

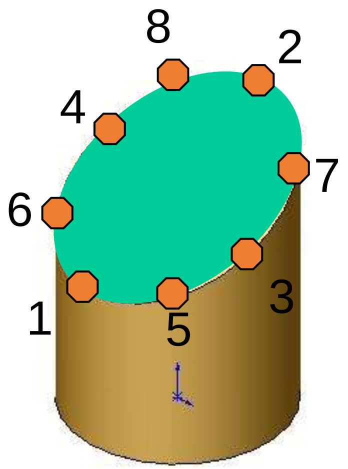




【例】

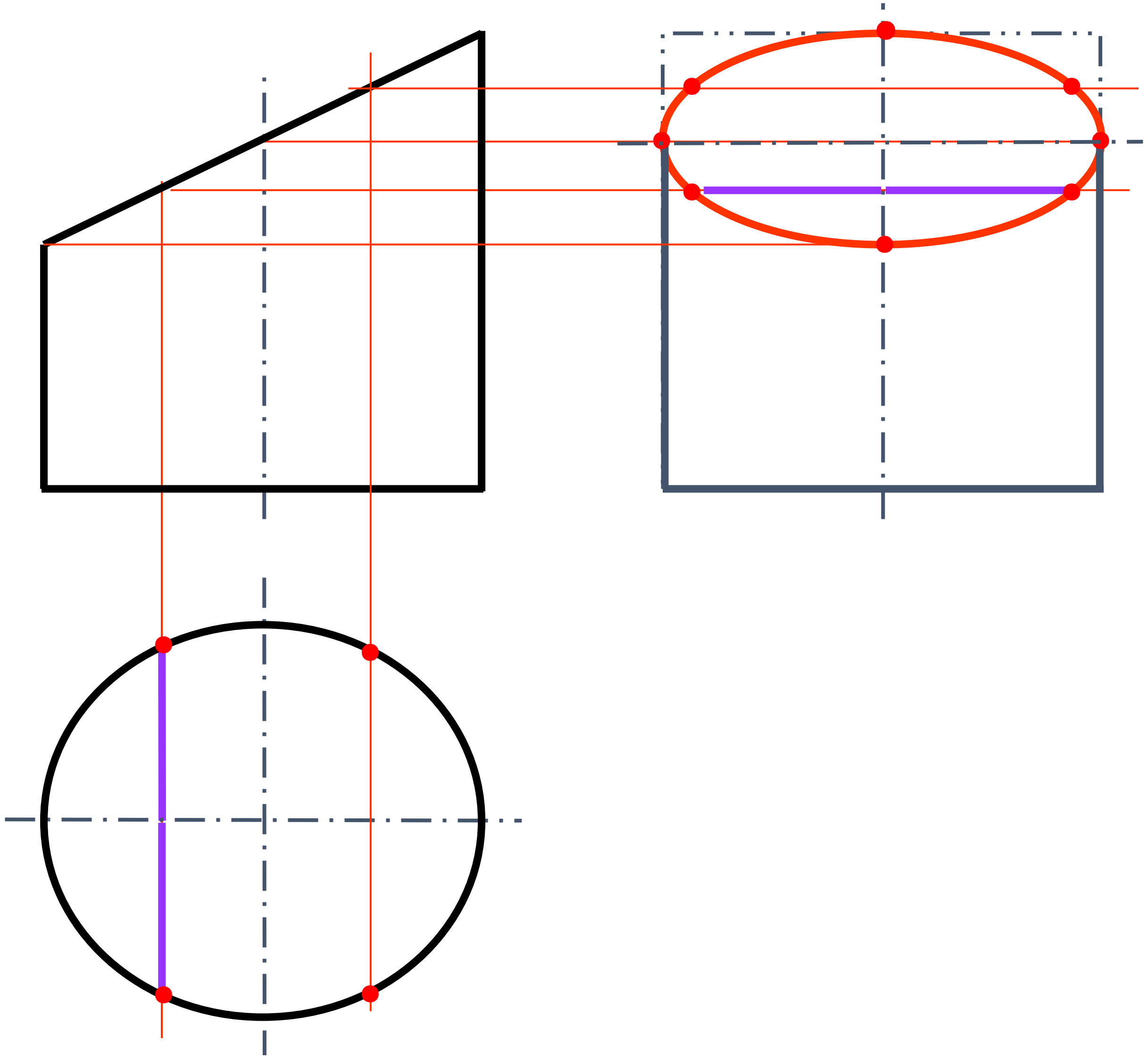
作左视图。





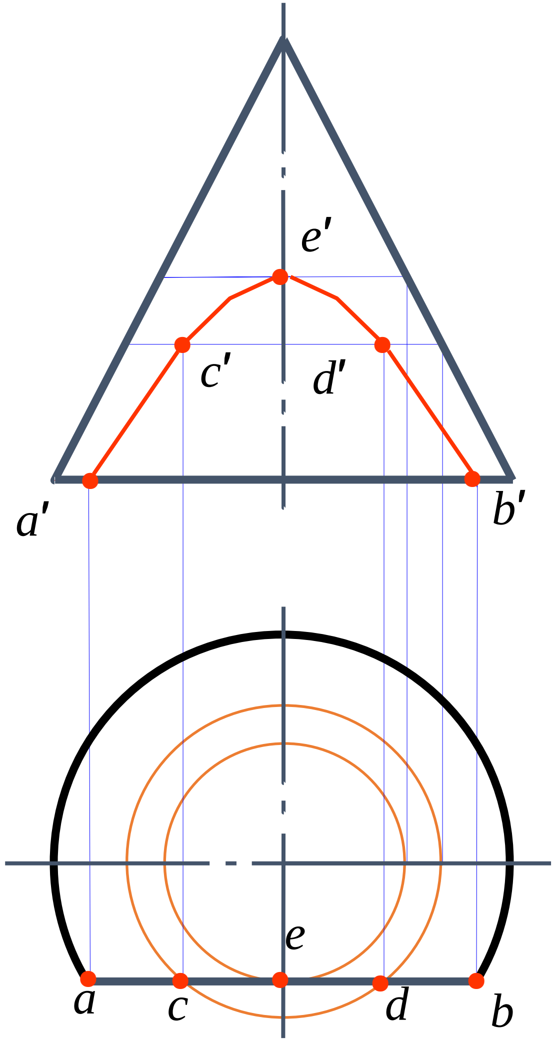
【例】

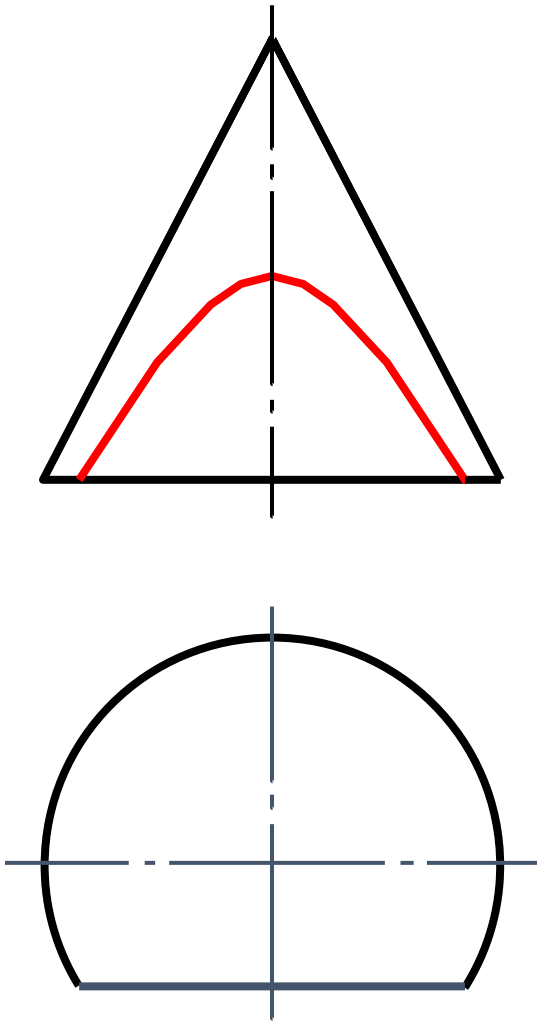
作左视图。

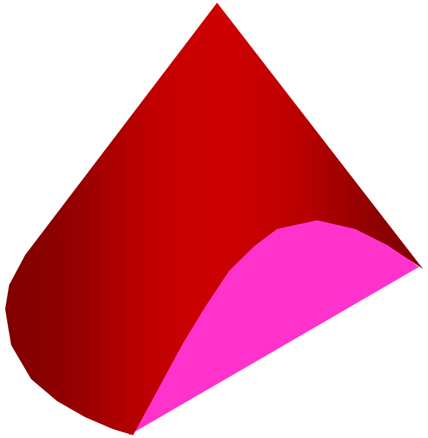


【例】

圆锥被正平面截切，补全主视图。

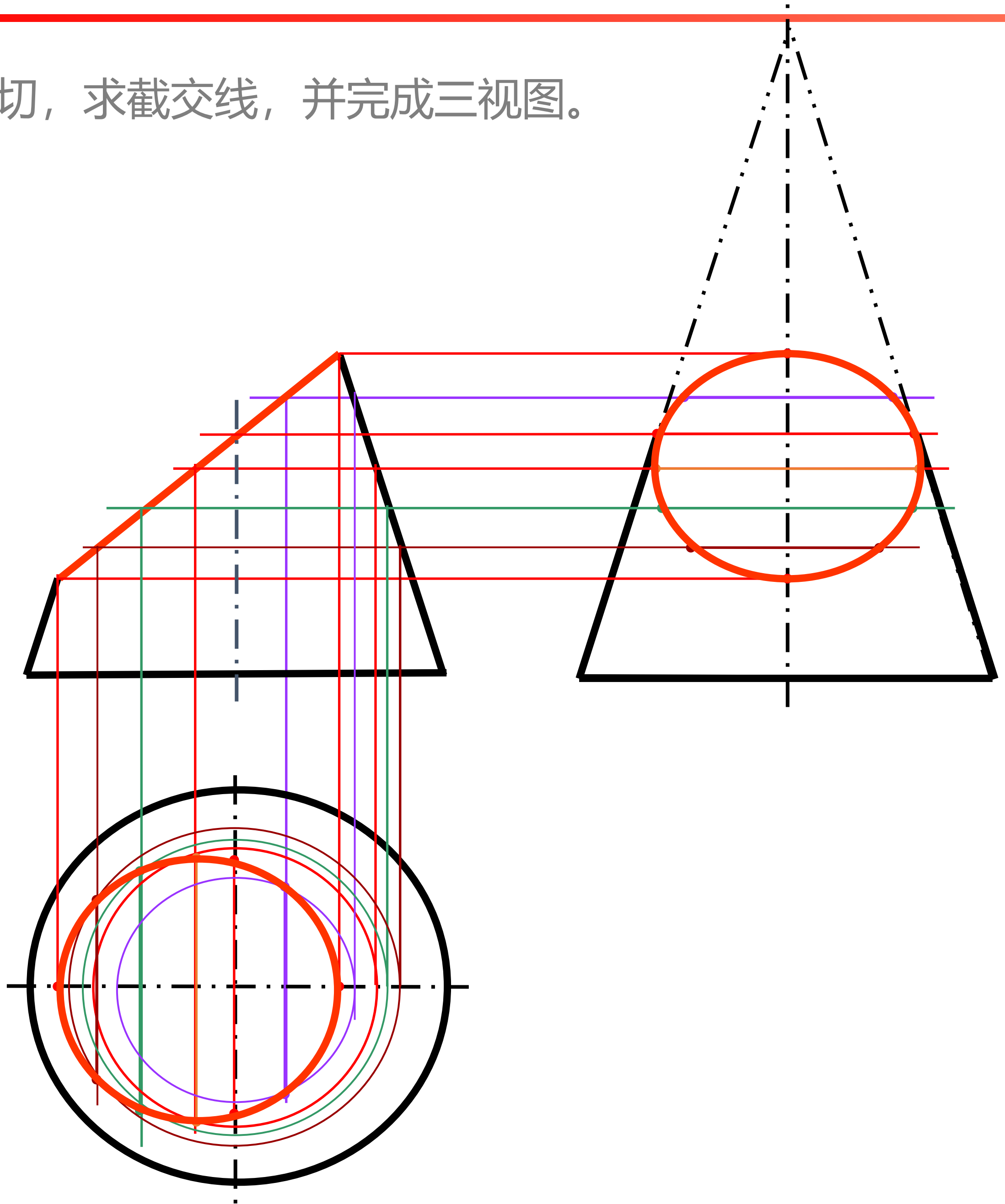


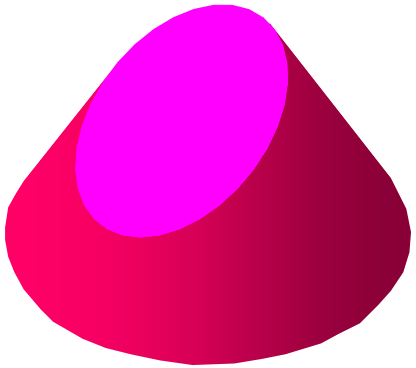




【例】

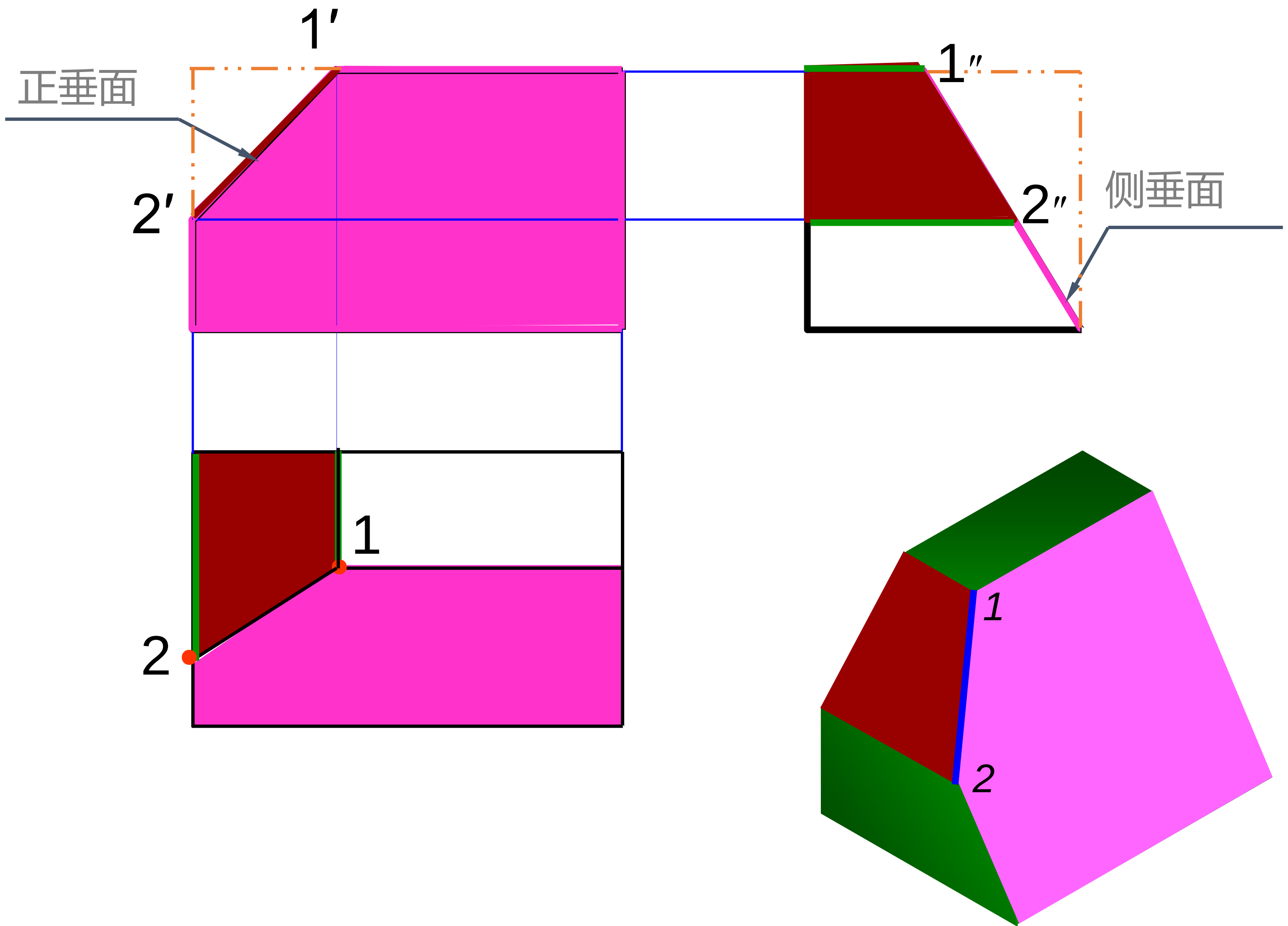
圆锥被正垂面截切，求截交线，并完成三视图。



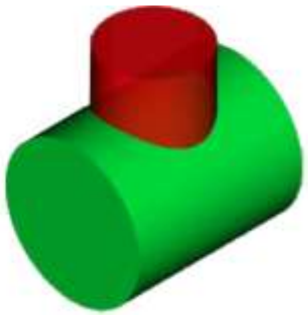


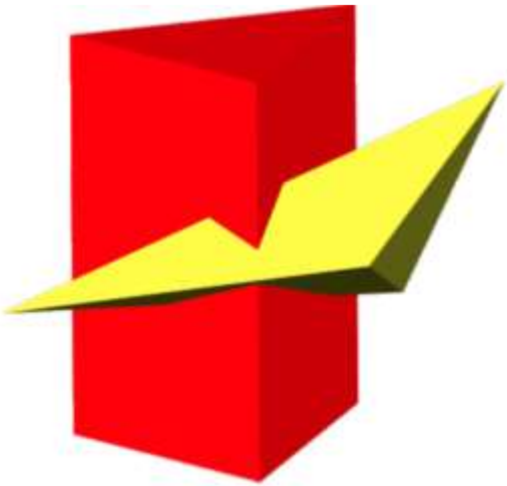
【例】

作俯视图。



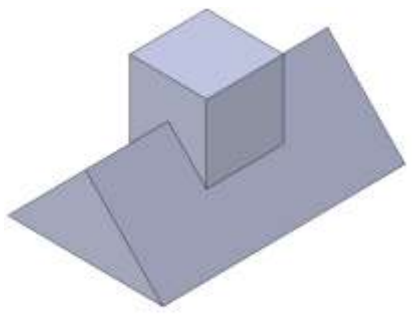
- **相贯** 立体与立体相交
- **相贯线** 立体表面的交线



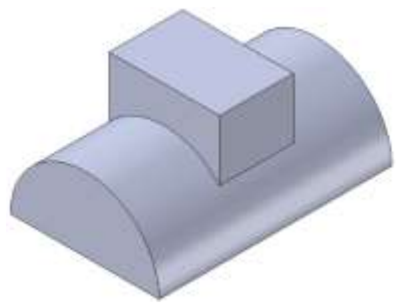


相贯

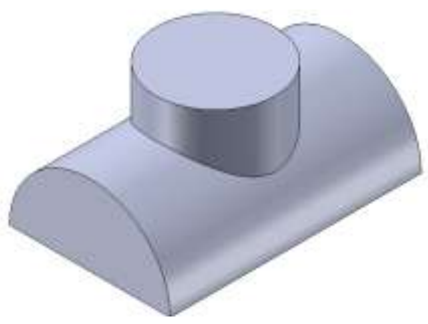
- **表面性** 相贯线位于两立体的表面上
- **封闭性** 相贯线一般是封闭的空间折线或空间曲线
- **共有性** 相贯线是两立体表面的共有线



两平面立体相交



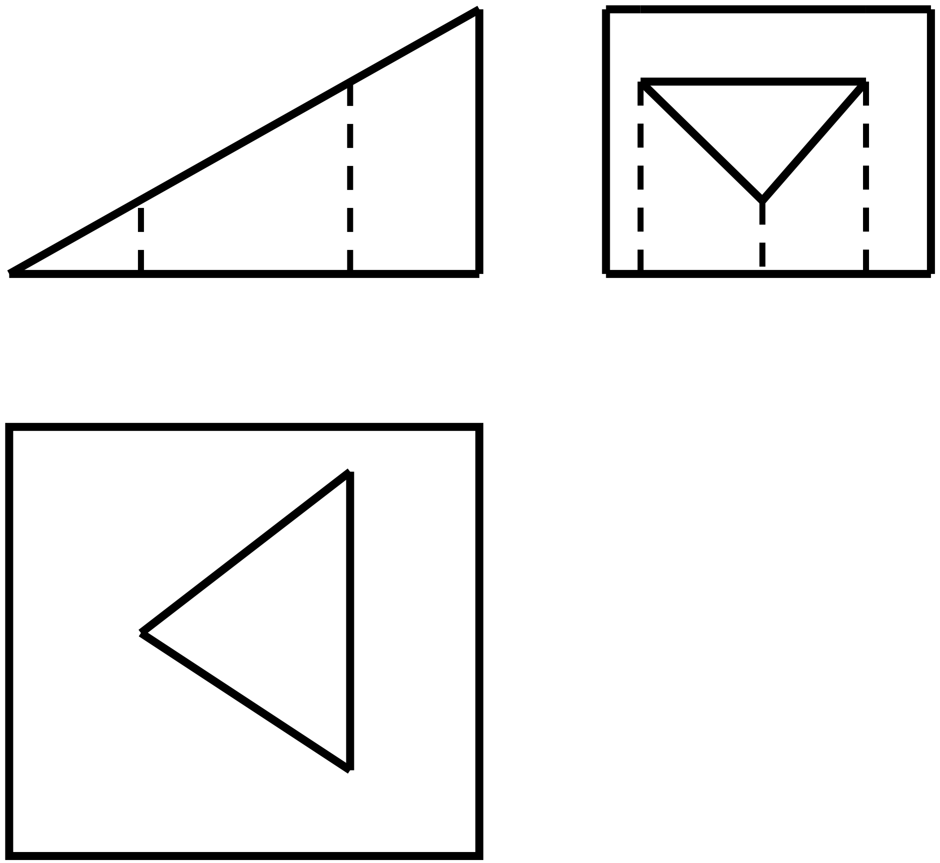
平面立体与曲面立体相交

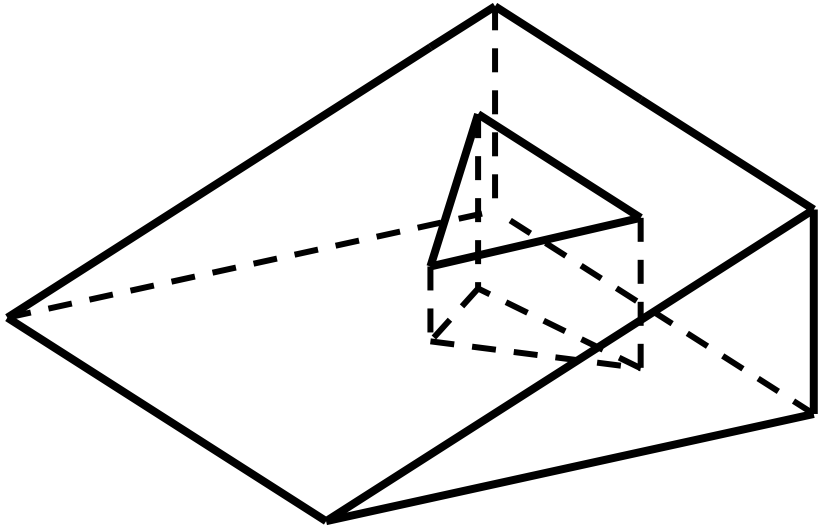


两曲面立体相交

【例】

补全视图。





#【例】

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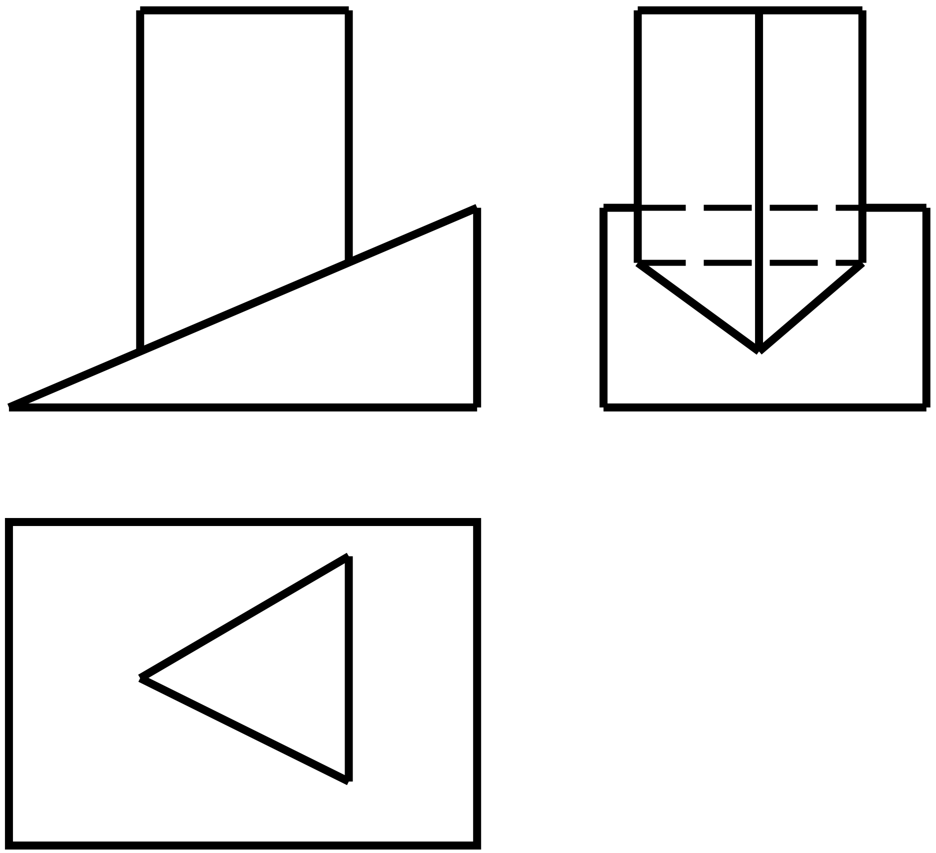
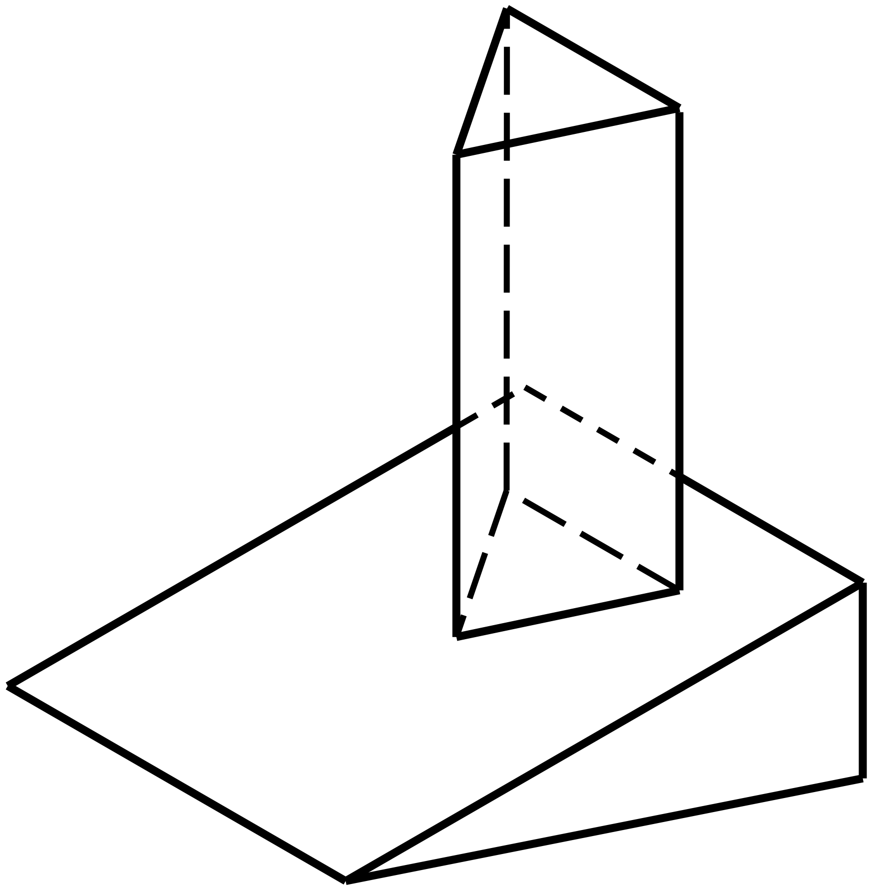
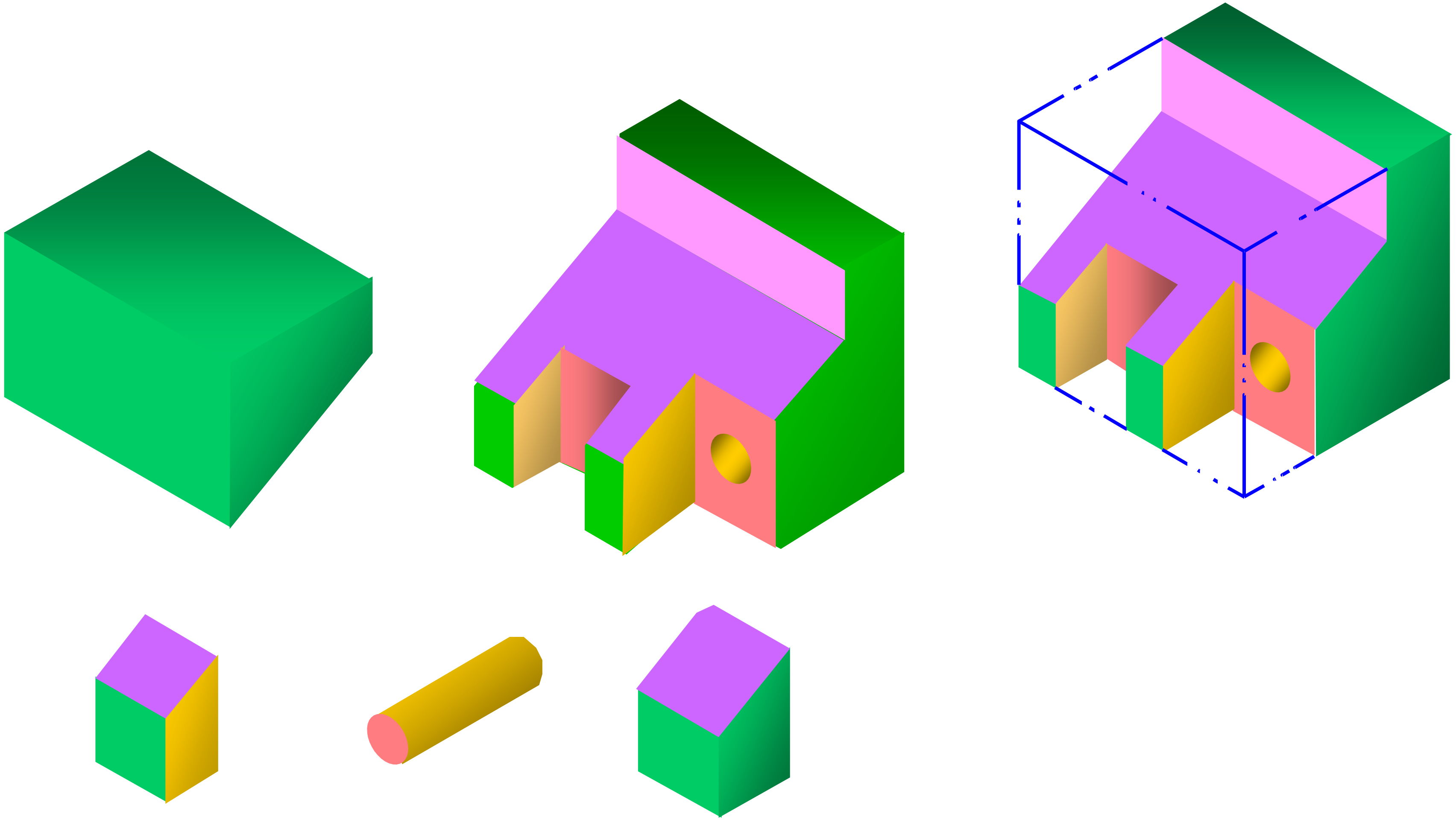
补全视图。





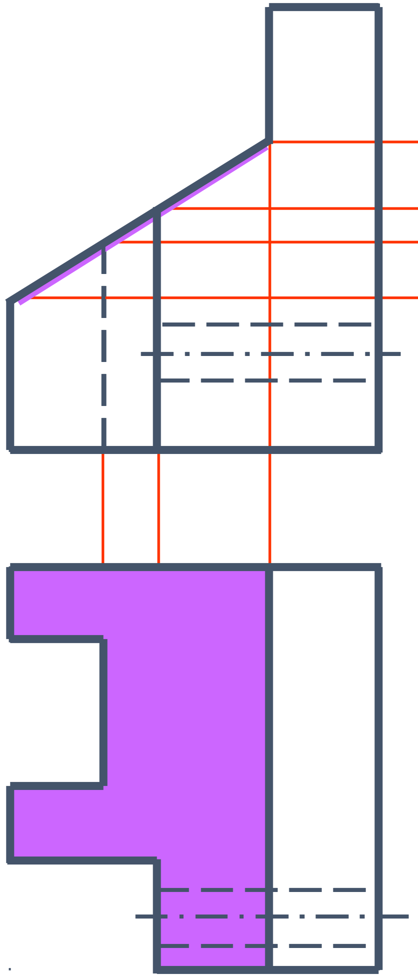
【例】

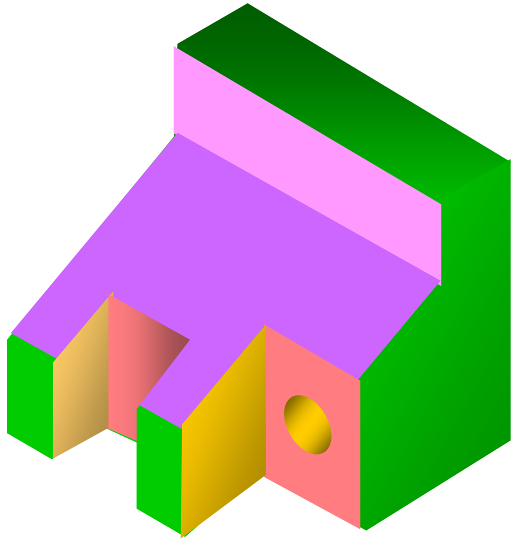
导向块的三视图画法。



【例】

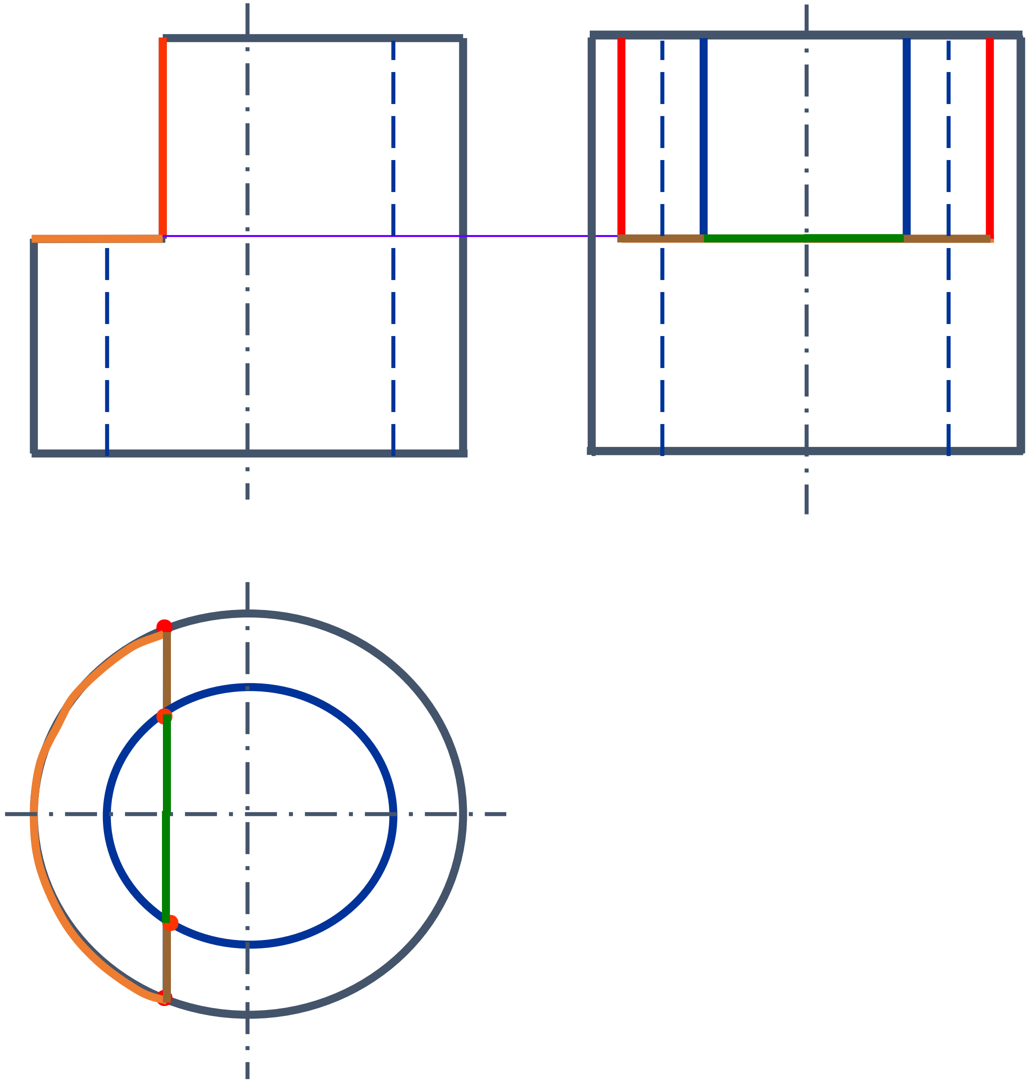
导向块的三视图画法。

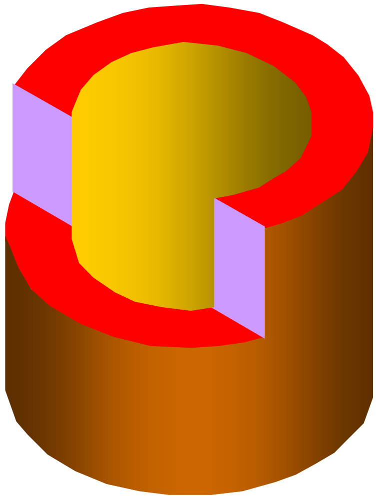


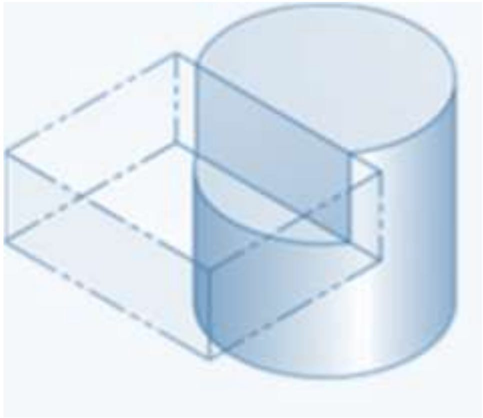


#【例】

作左视图。

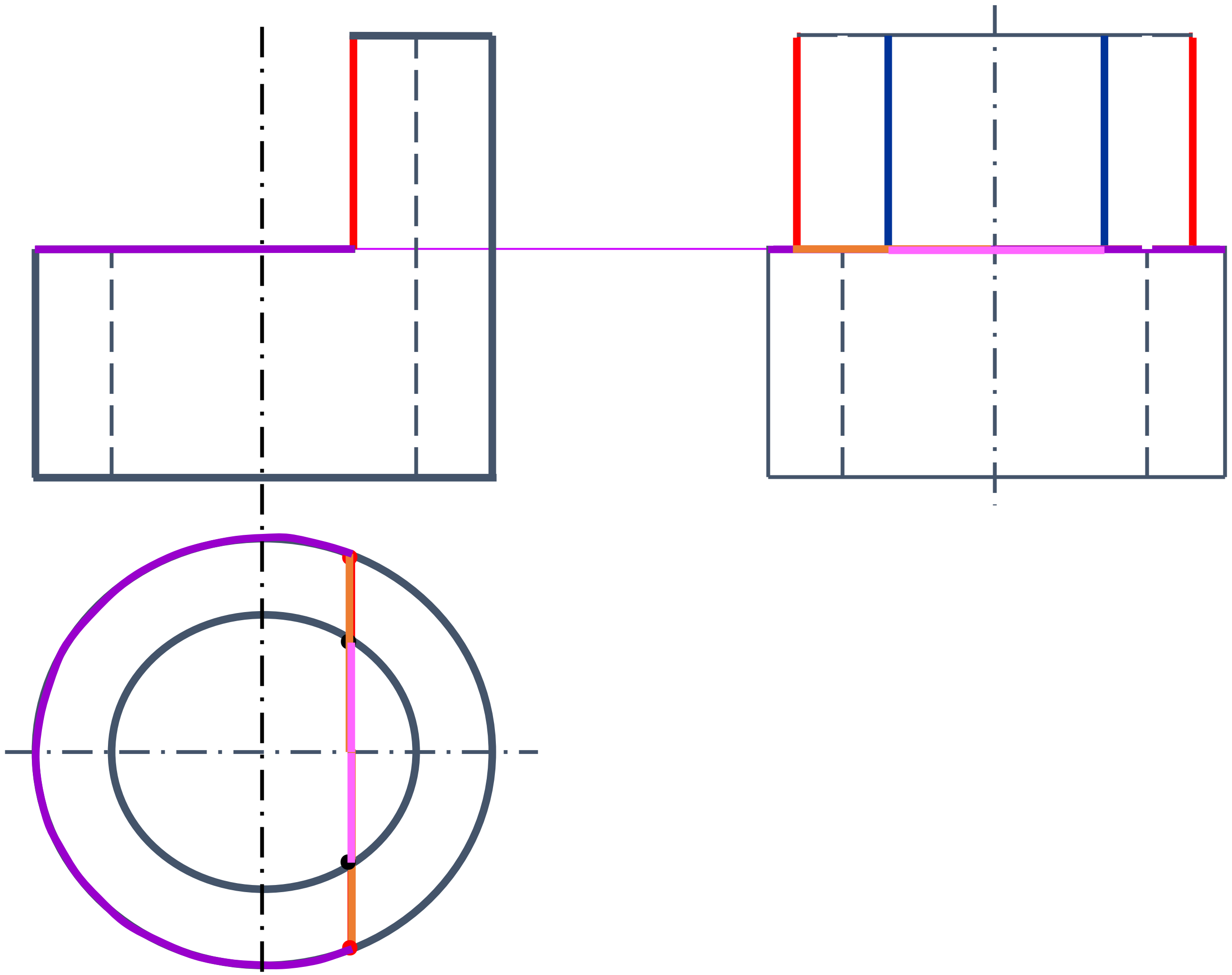


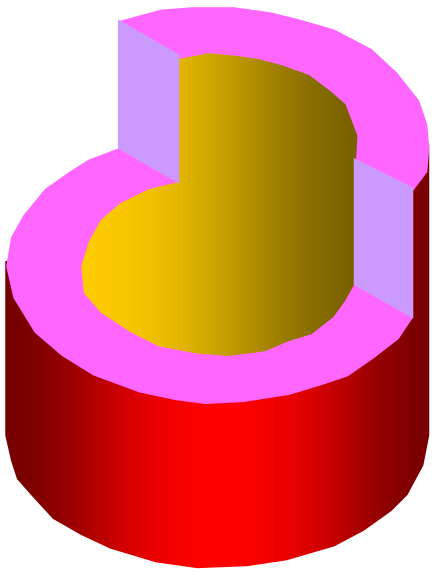


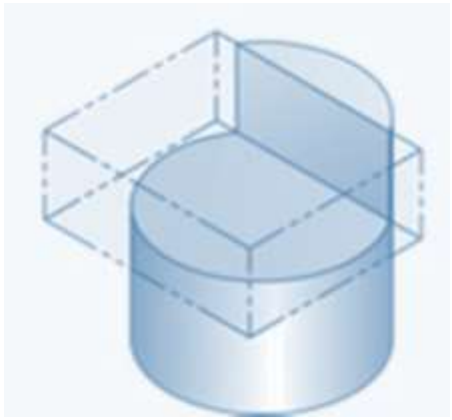


【例】

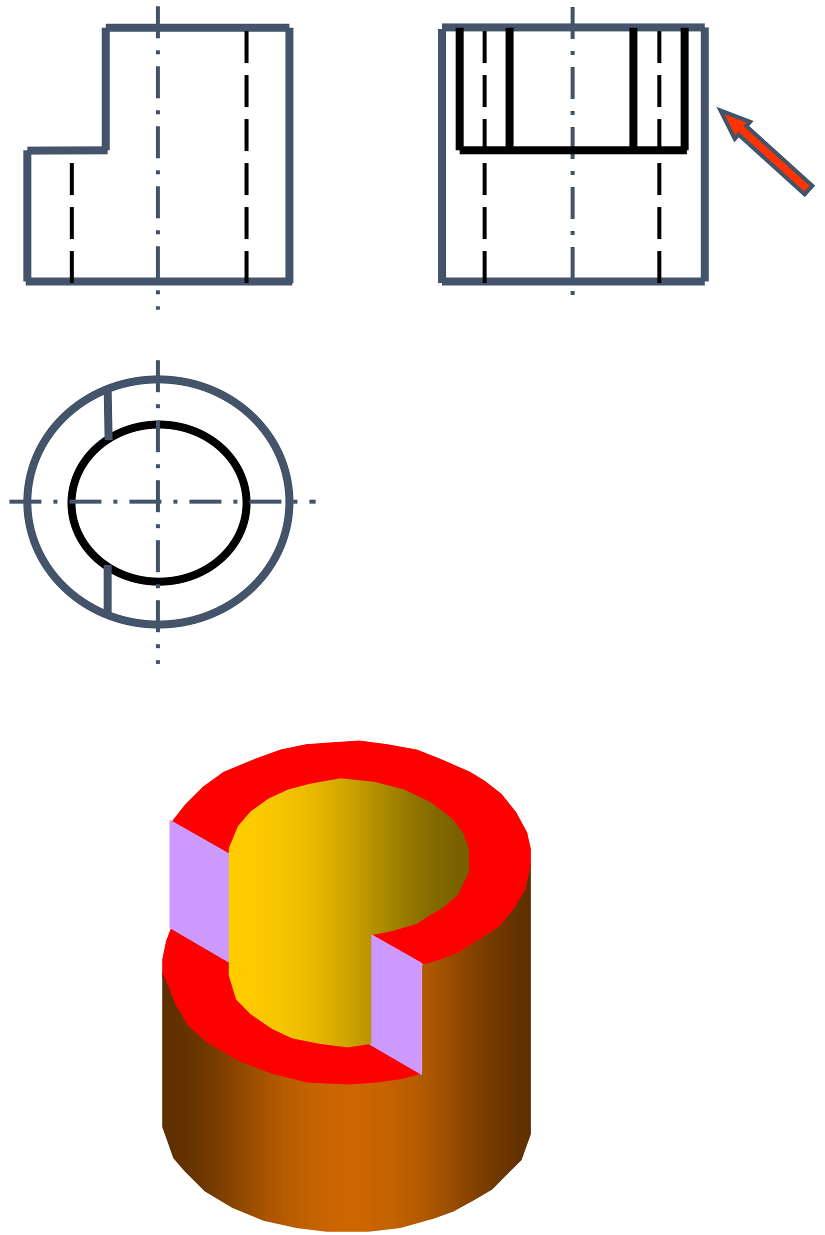
作左视图。

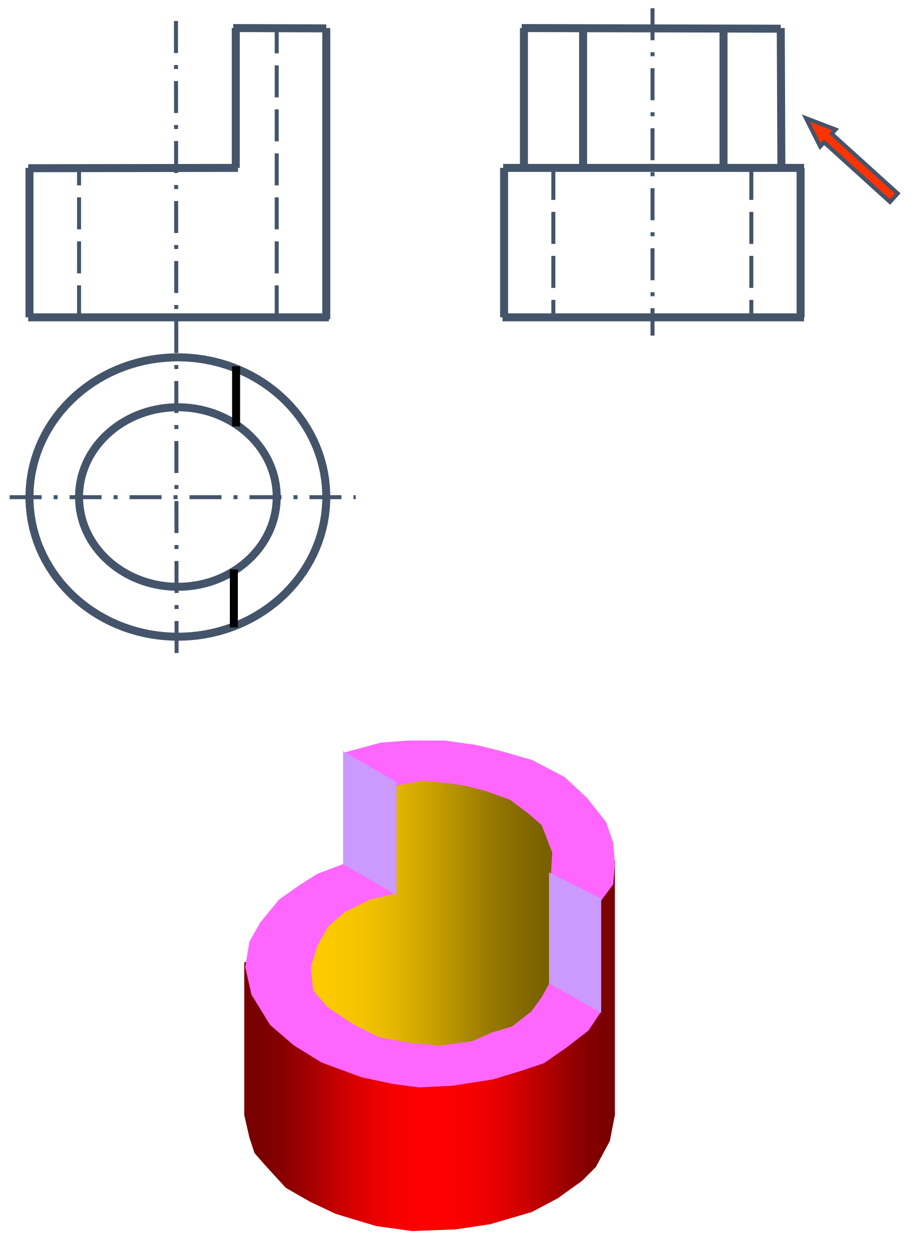






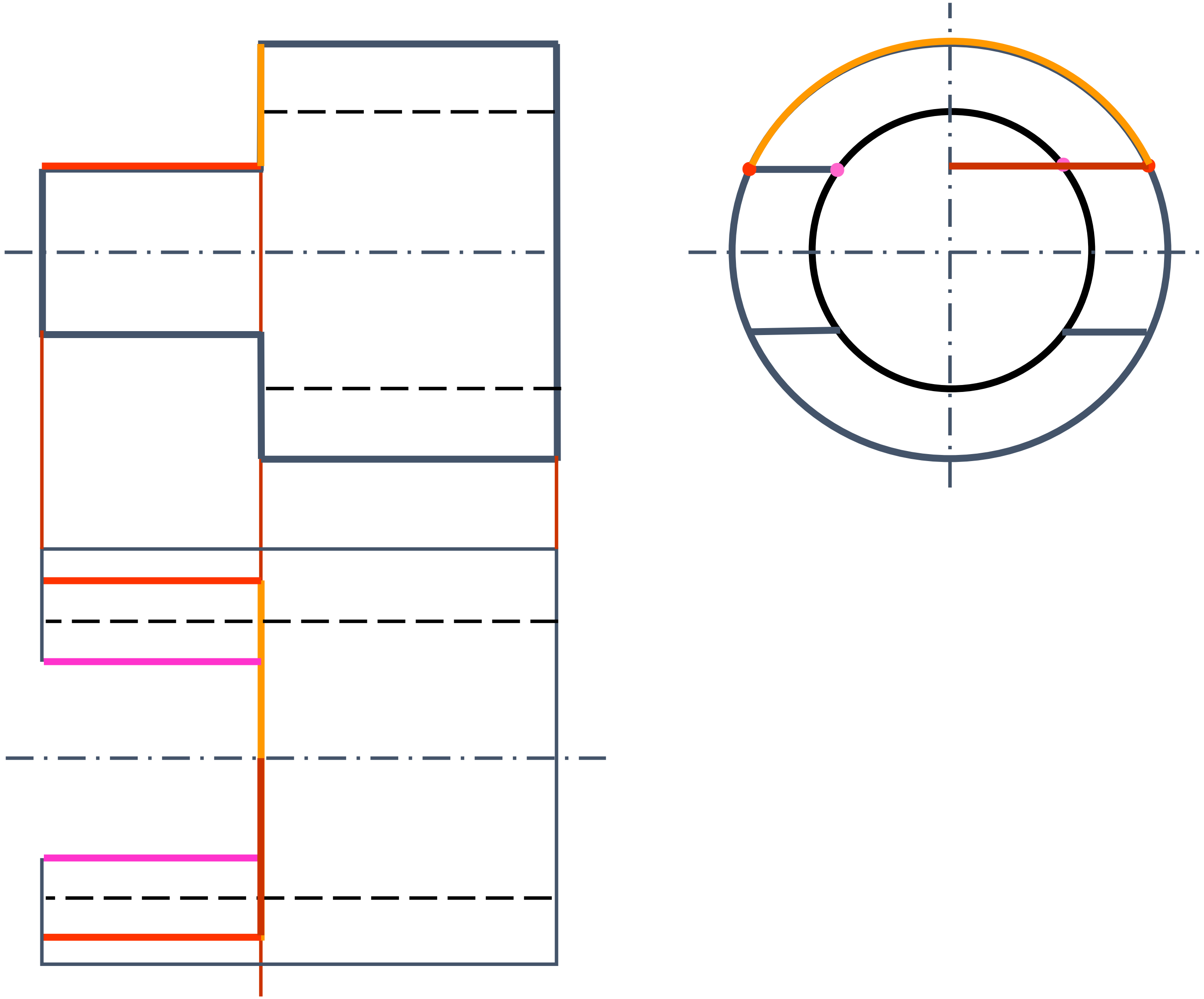
【例】

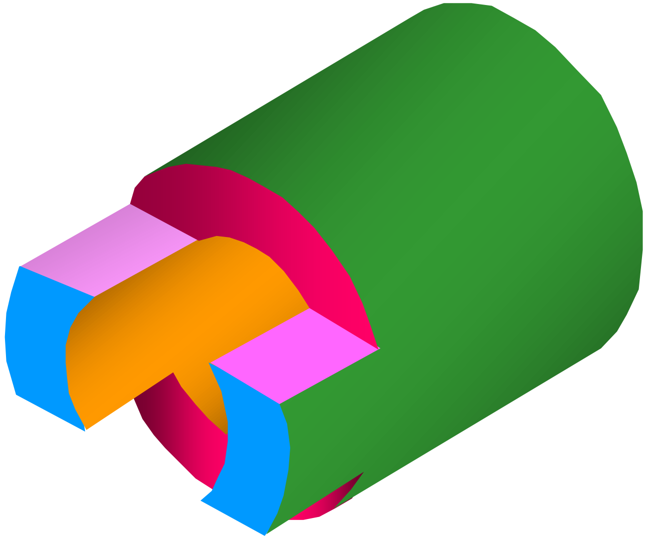


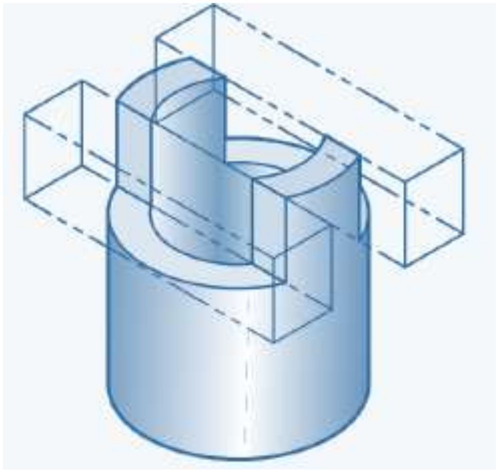


【例】

作俯视图。

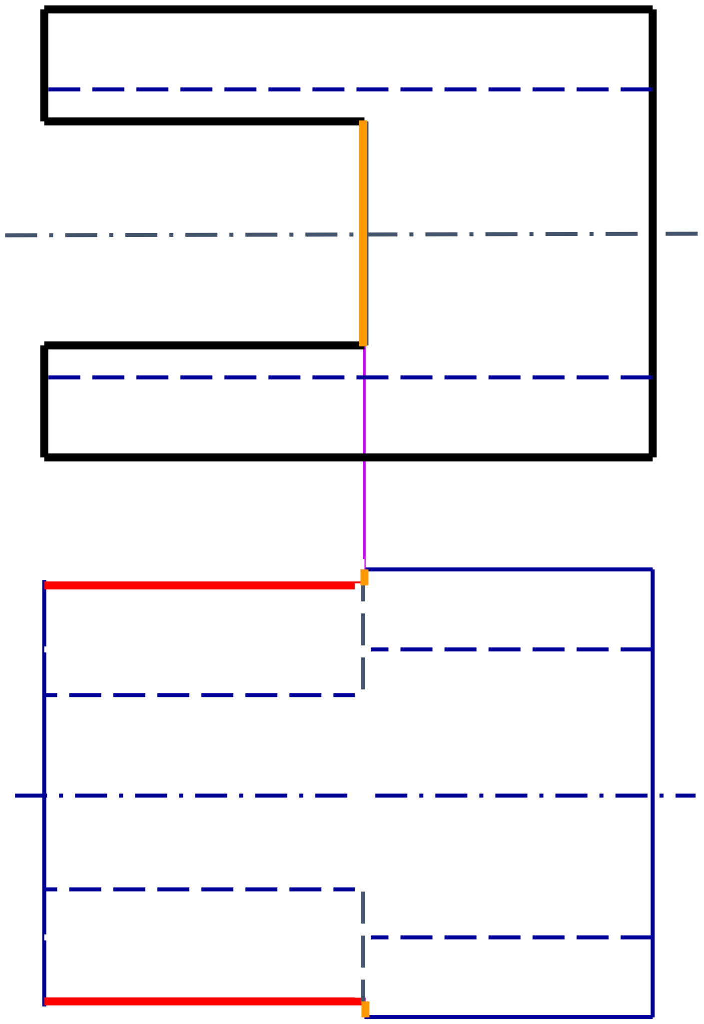


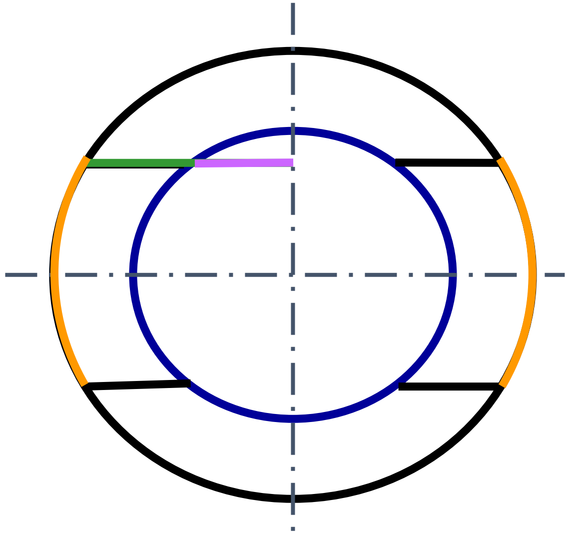


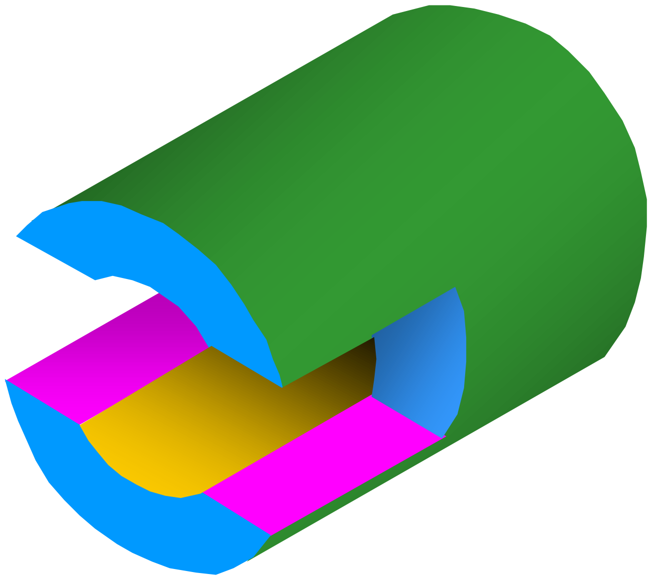


【例】

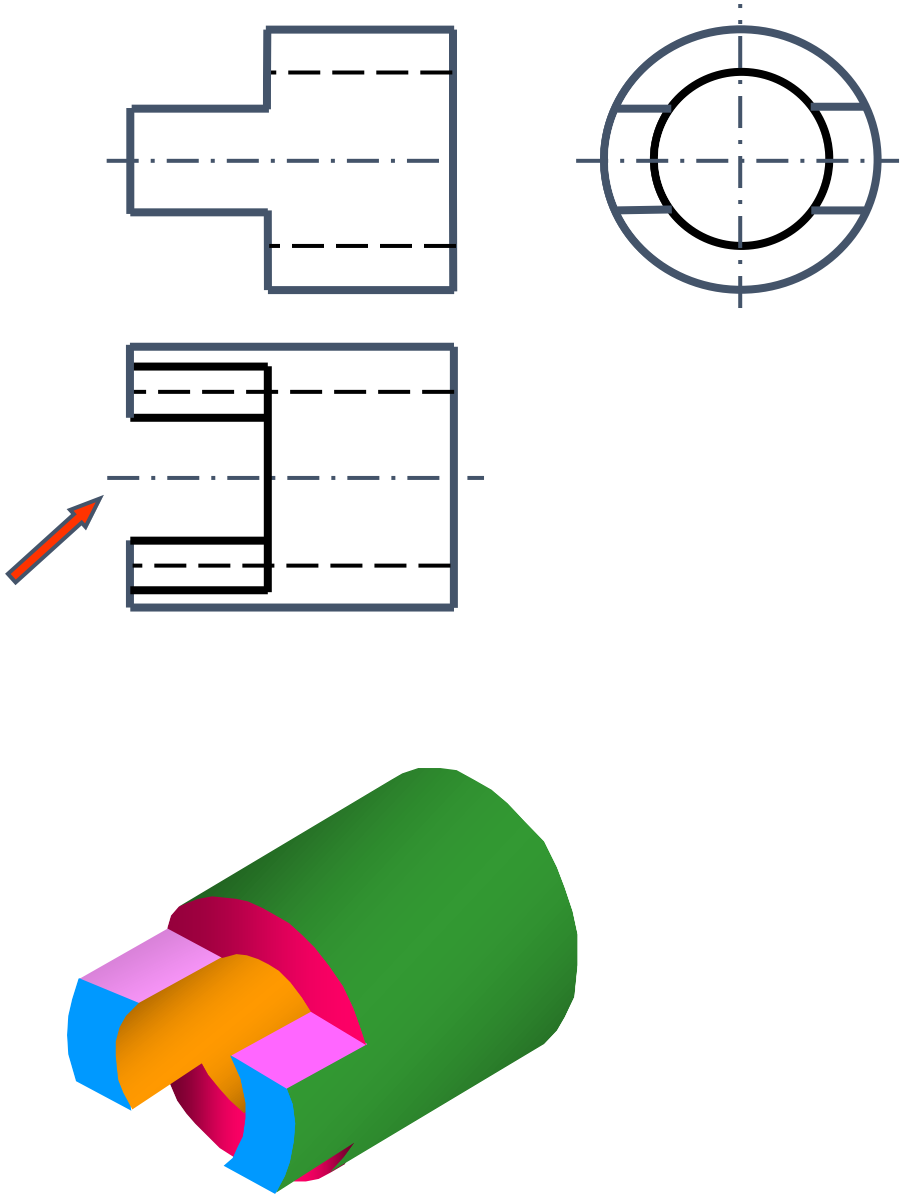
作俯视图。

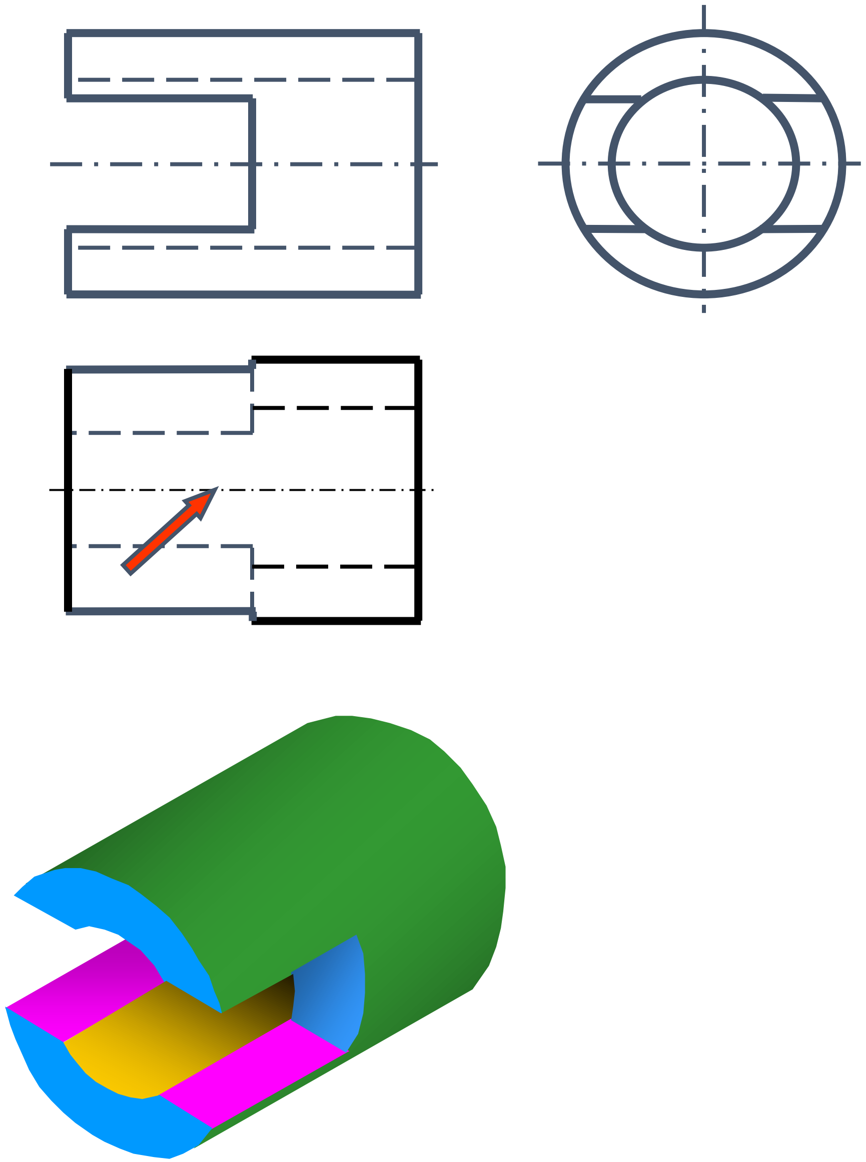






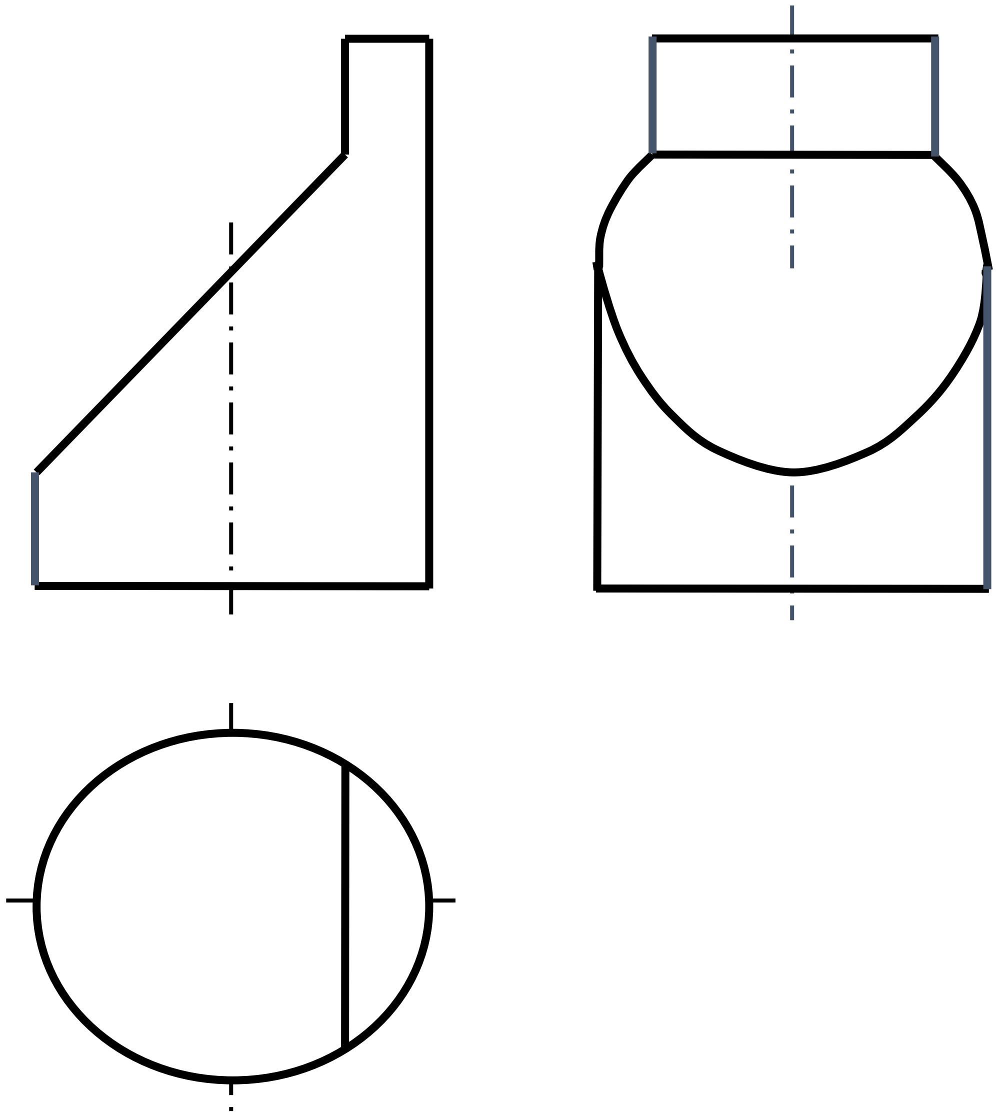
【例】

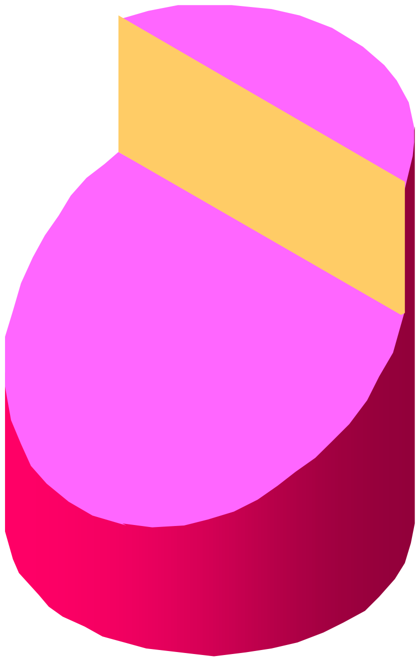




【例】

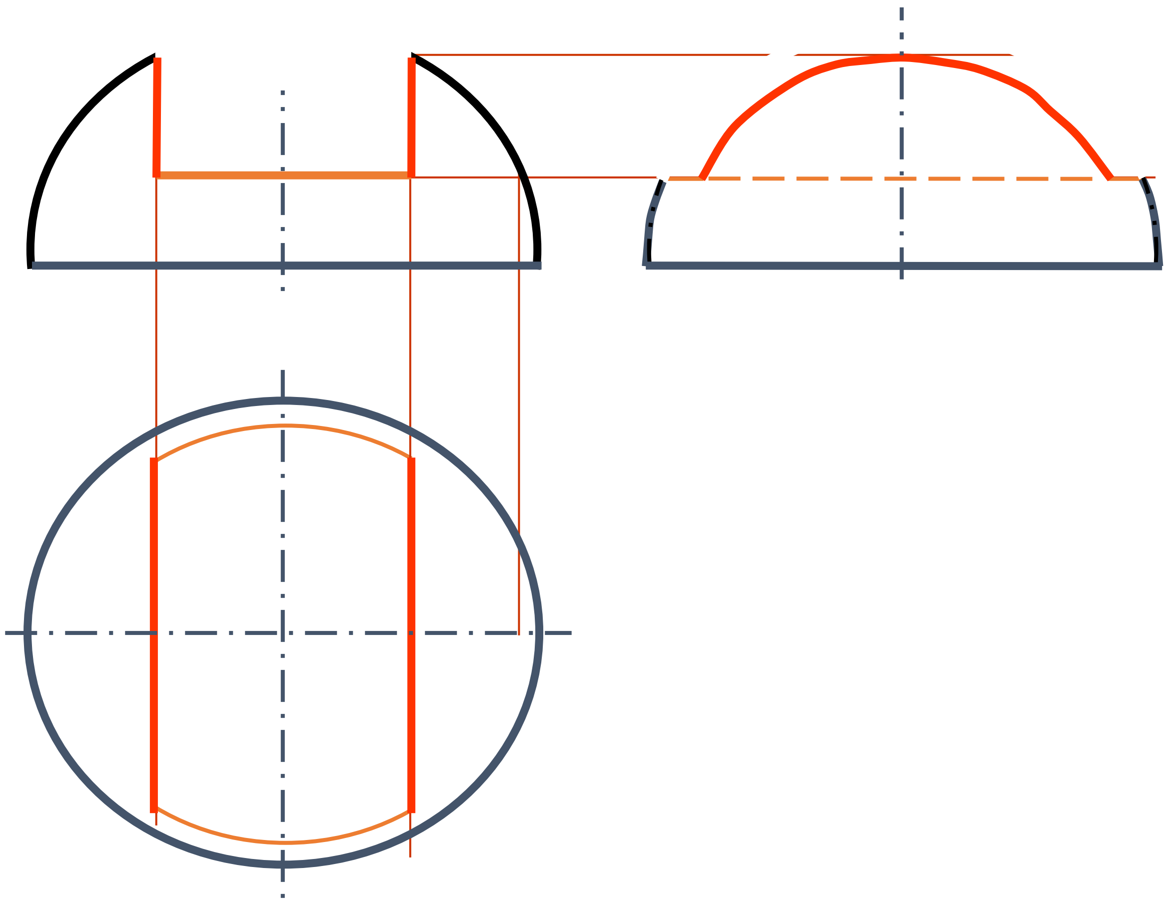
作左视图。

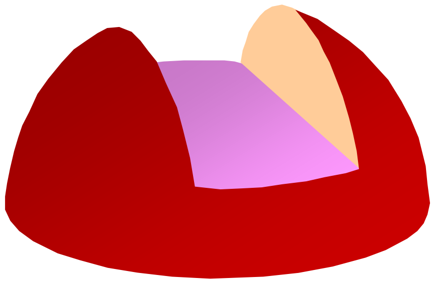


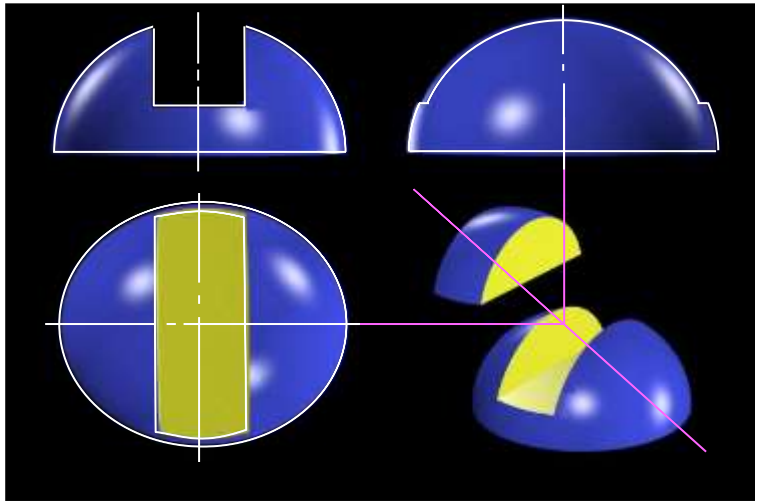


【例】

作半球体截切后的俯视图和左视图。







【例】

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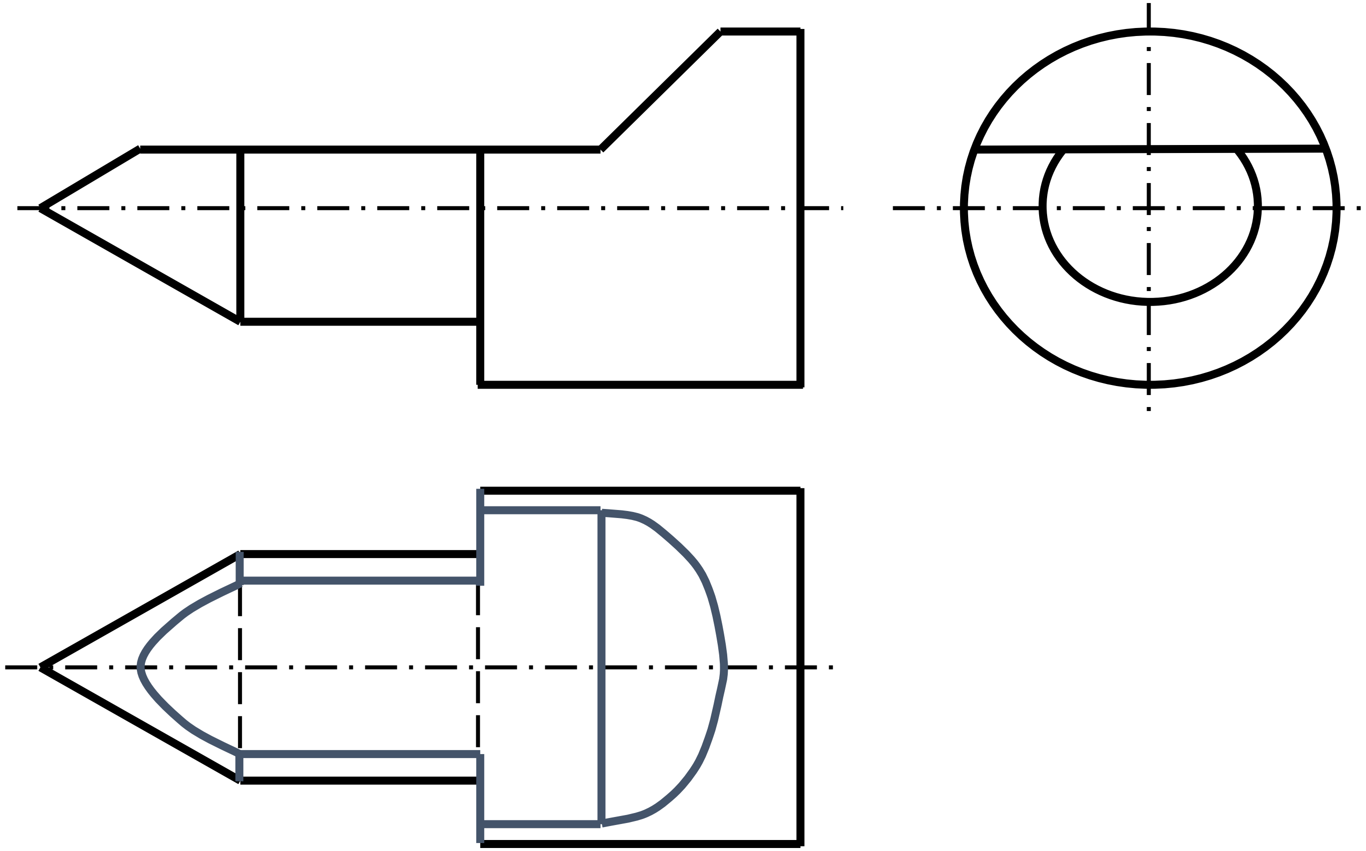
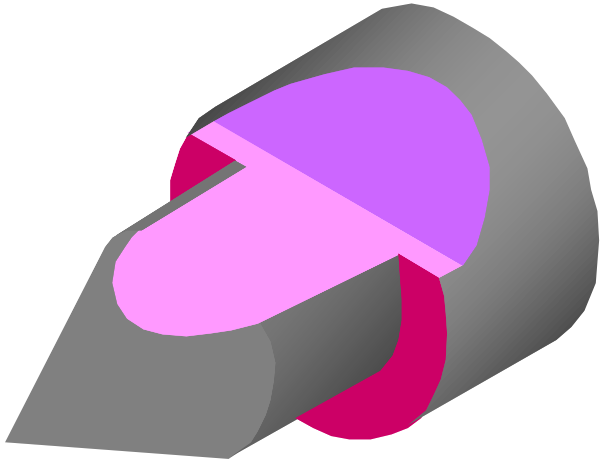
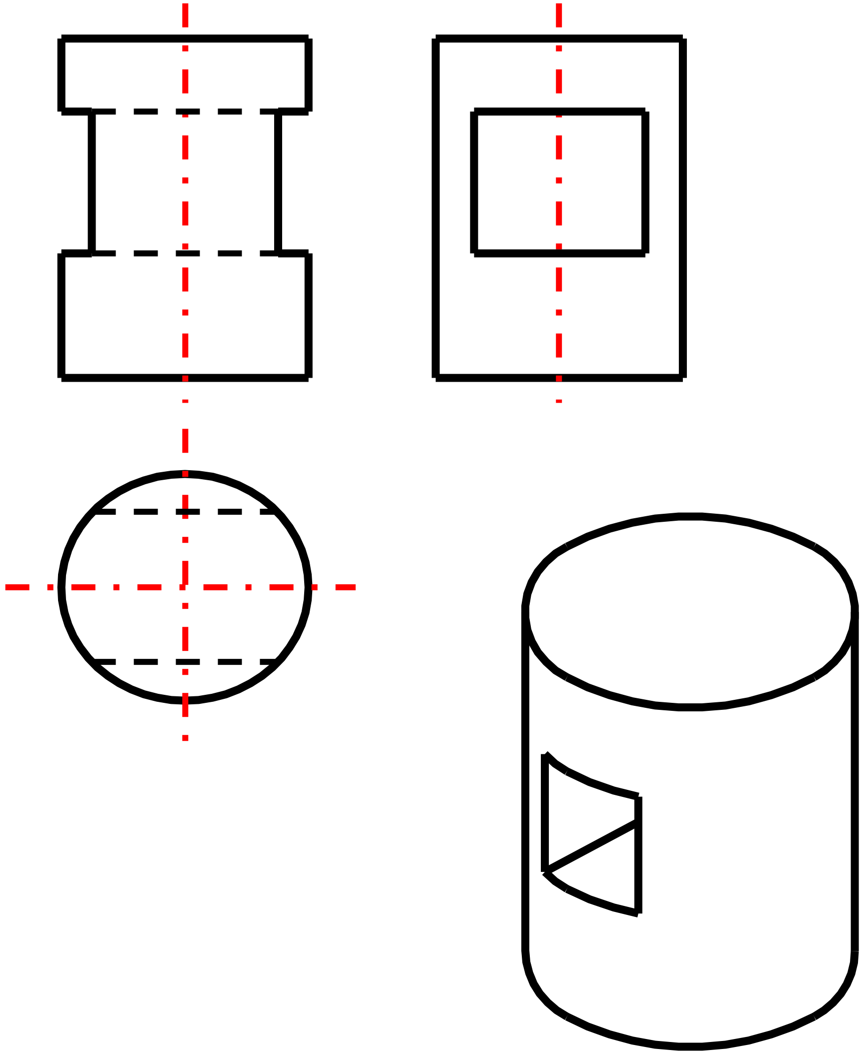
作顶尖的俯视图。

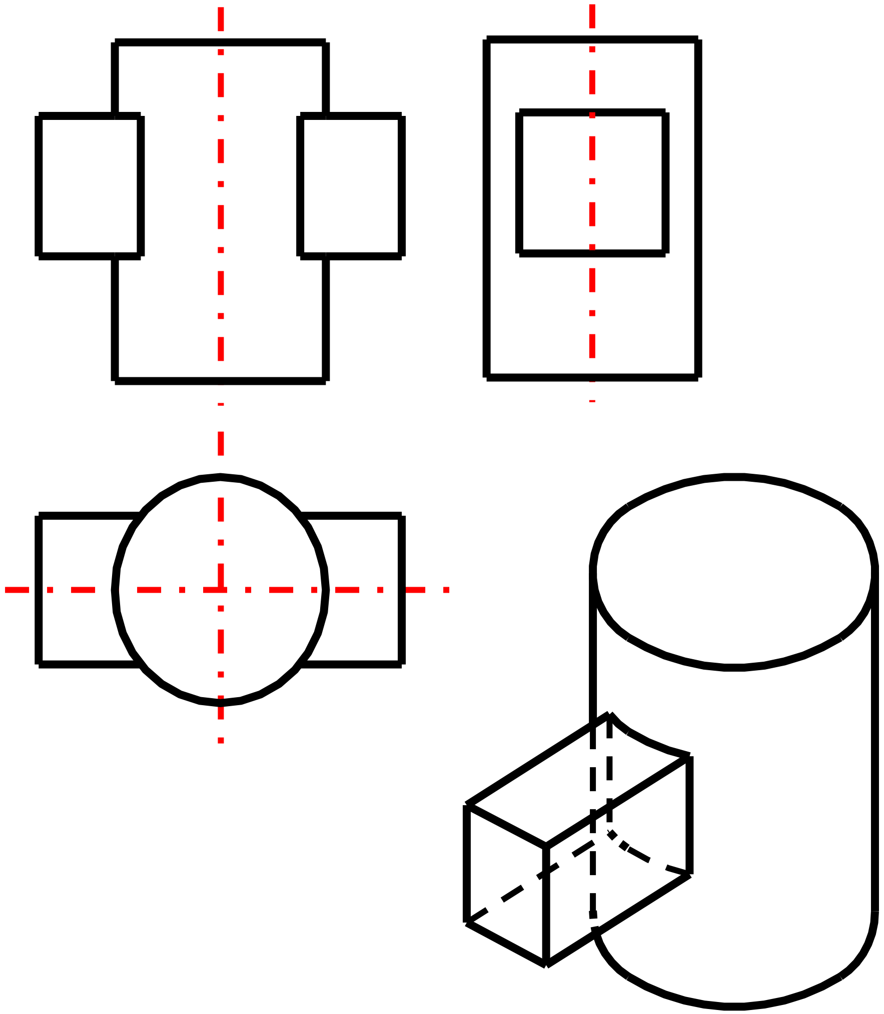




【例】



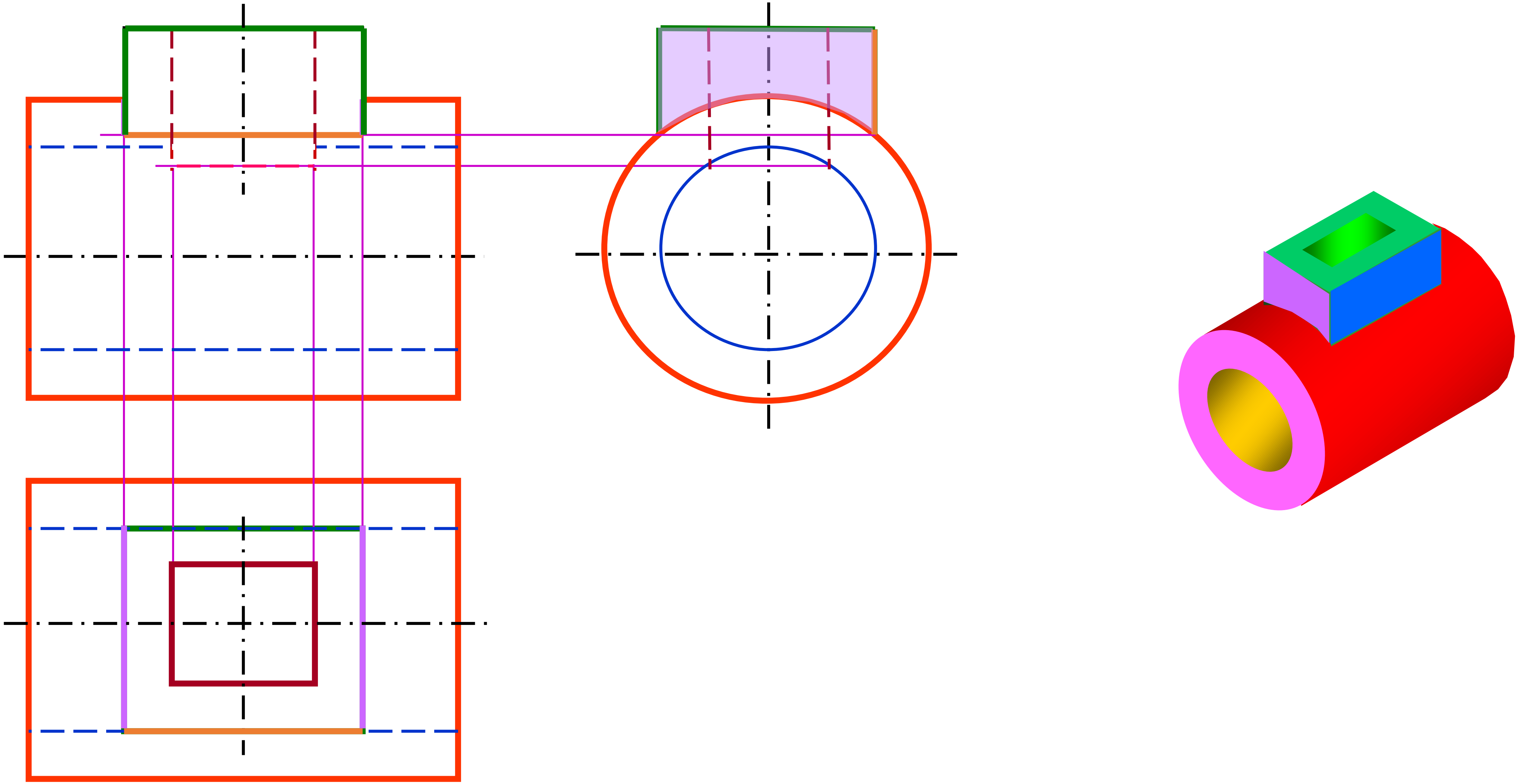
(切割,差)



(叠加，并)

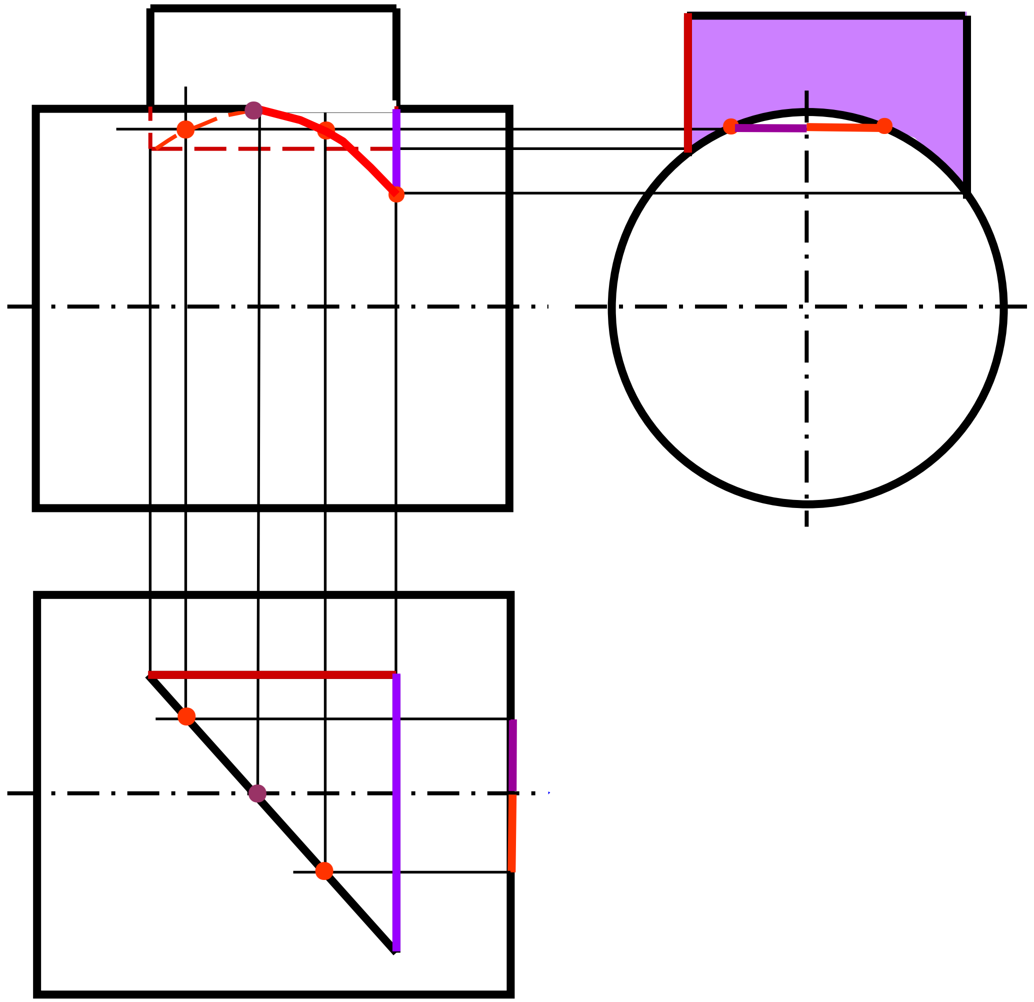
【例】

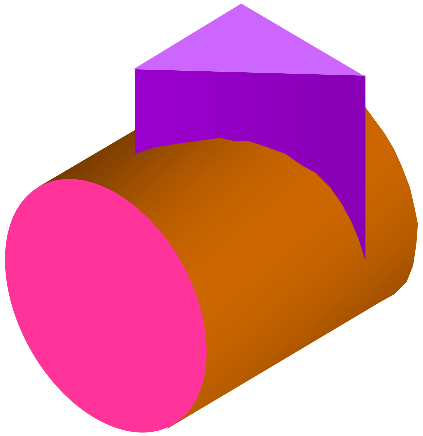
补全主视图。



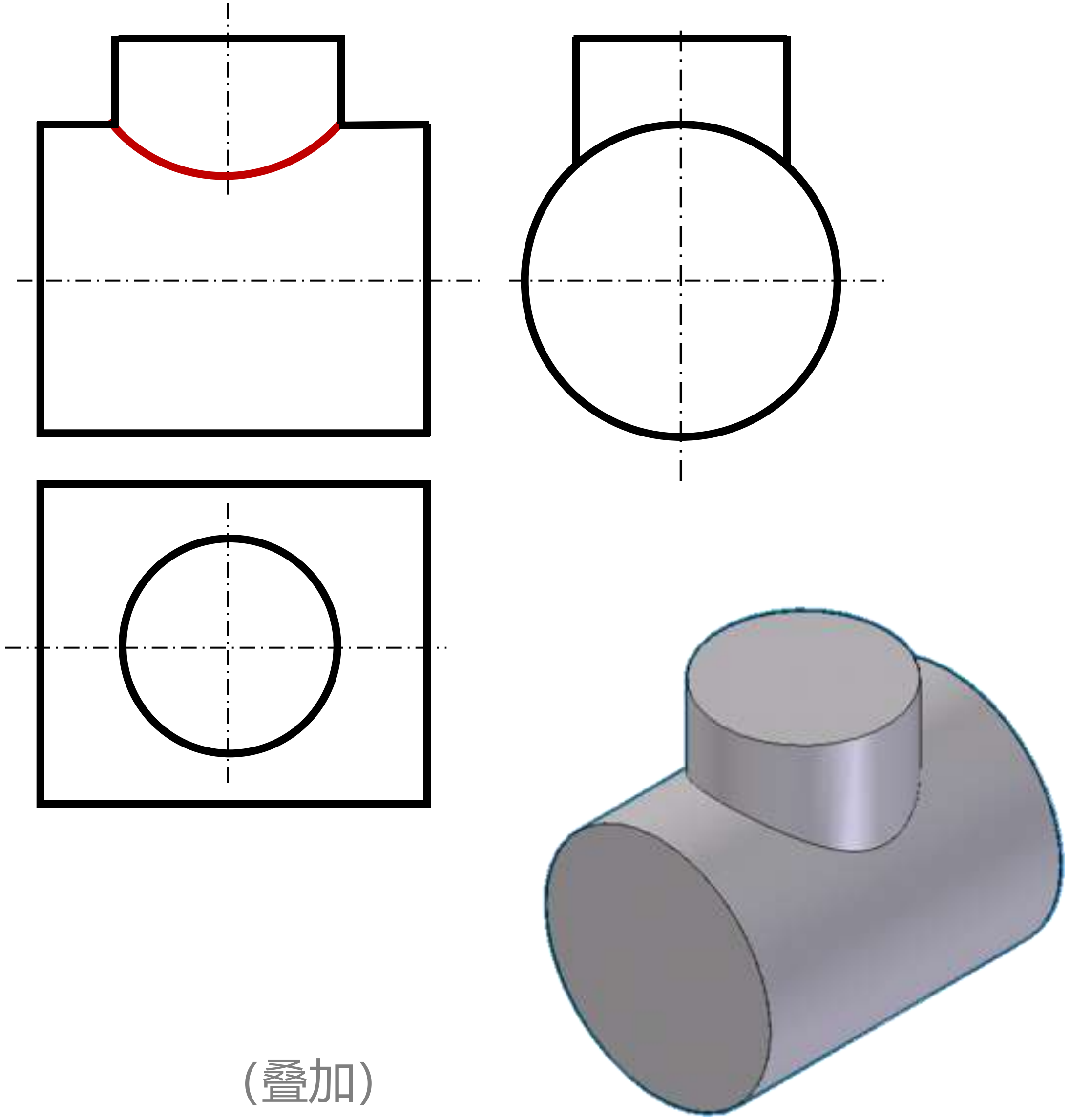
【例】

补全主视图。

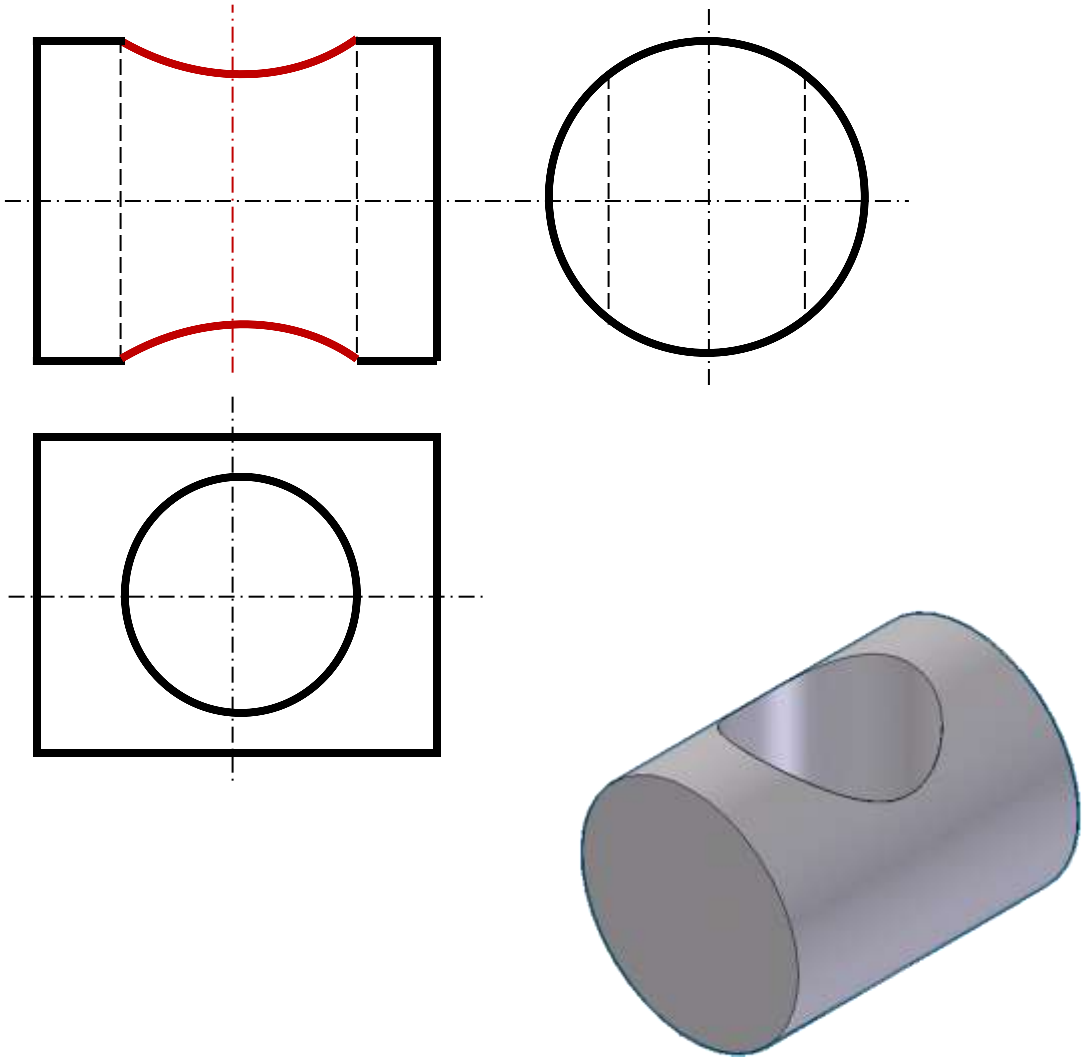




相贯



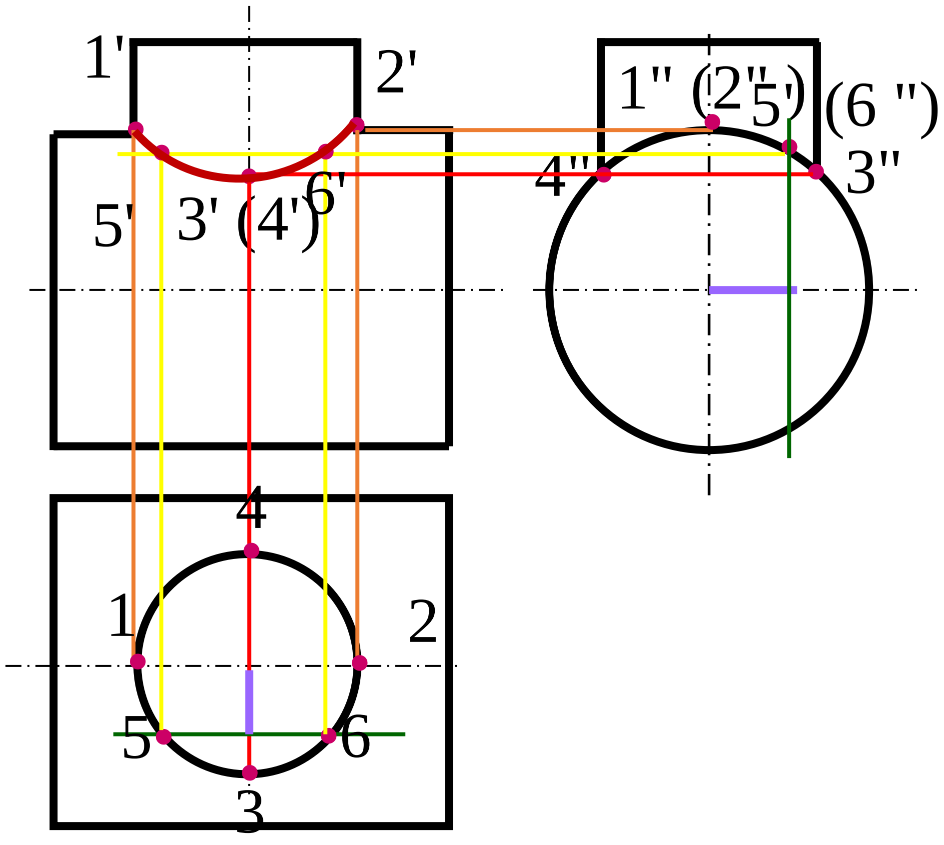
(叠加)

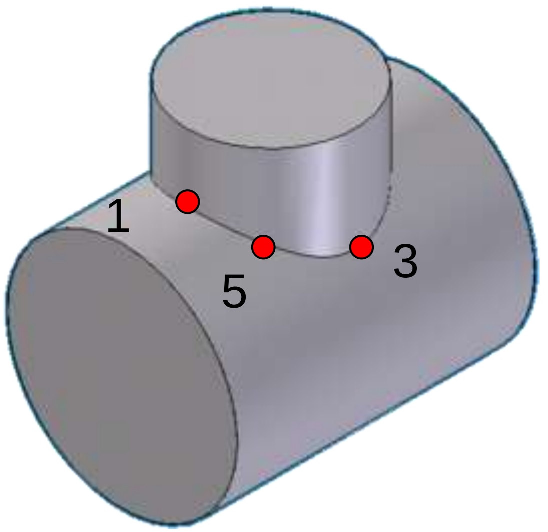


(切割)

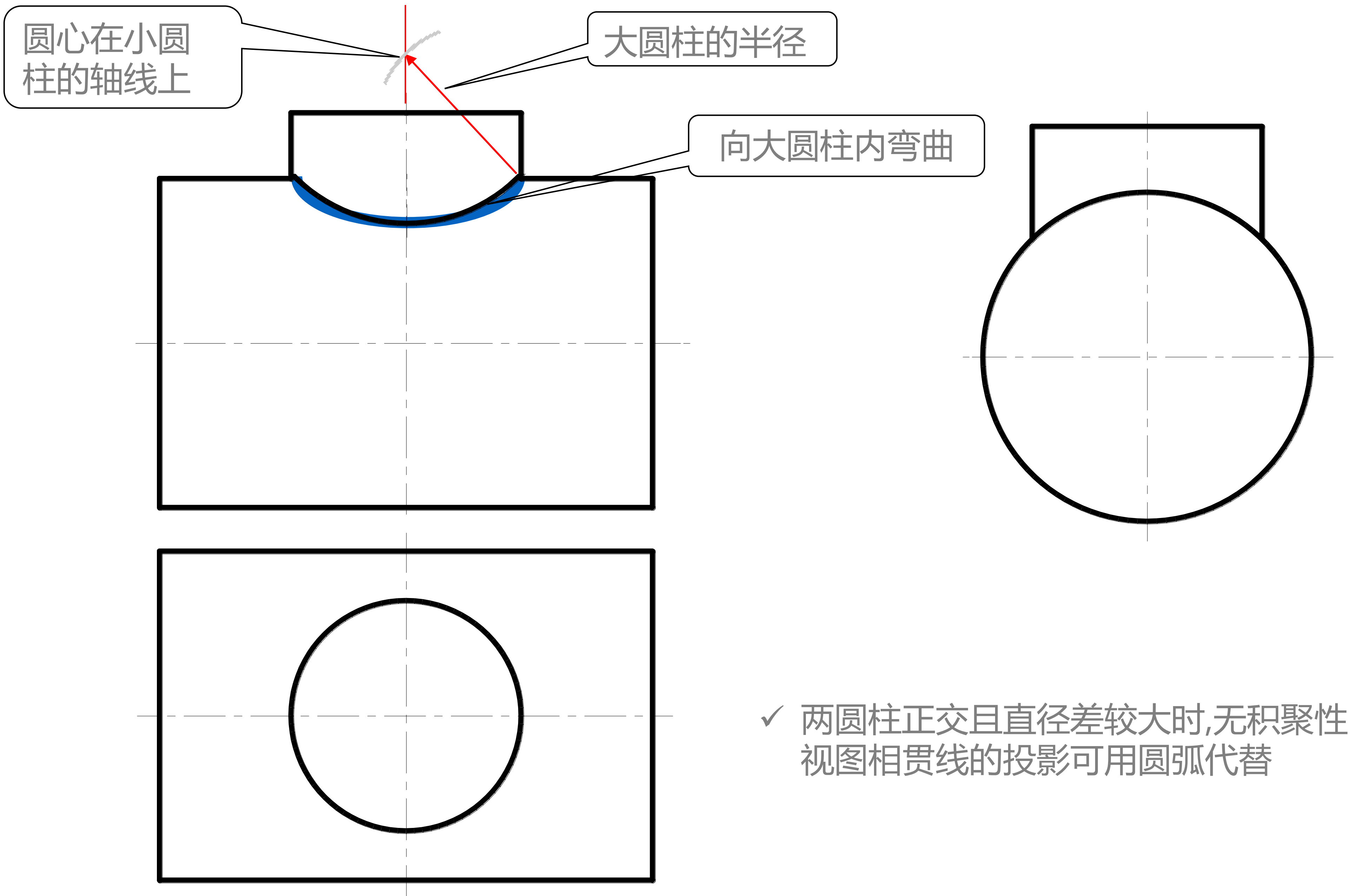
【例】

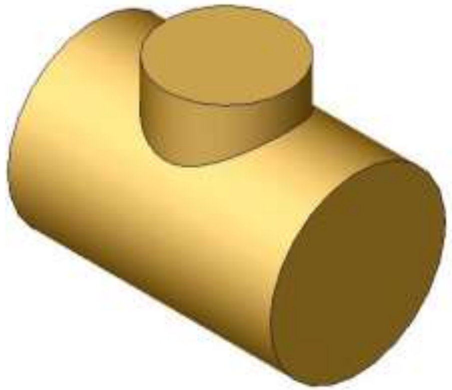
作两圆柱正交。





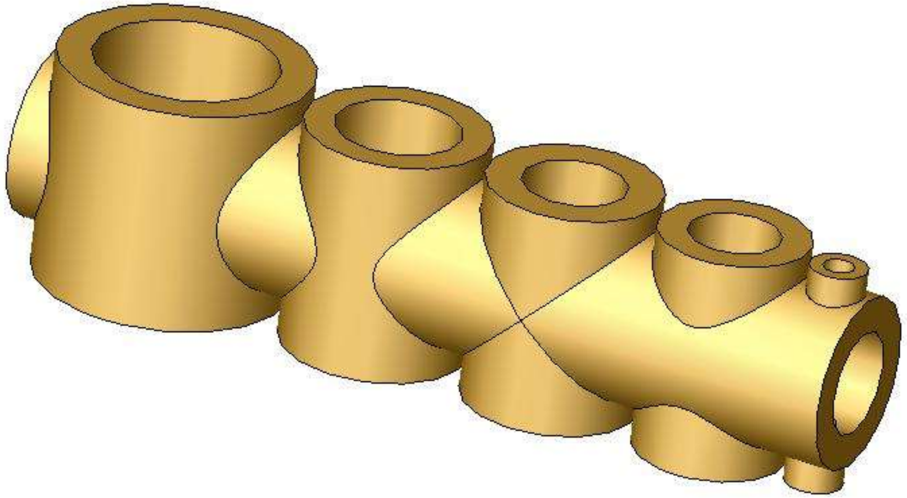
相贯线简化画法

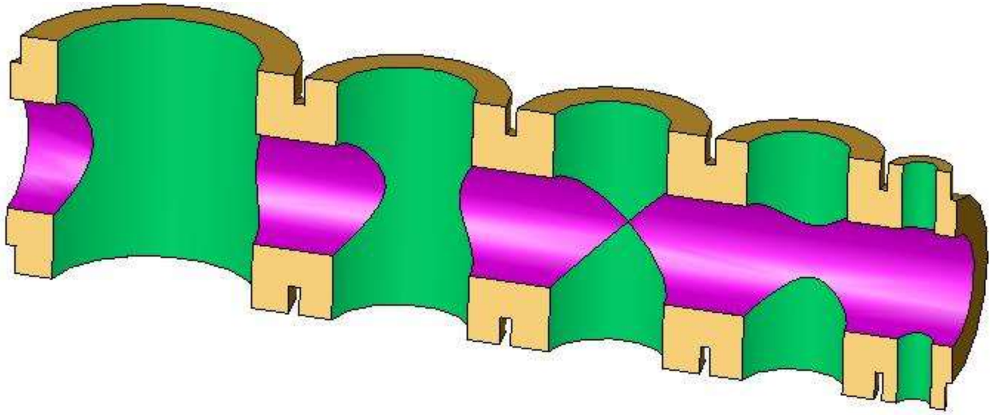




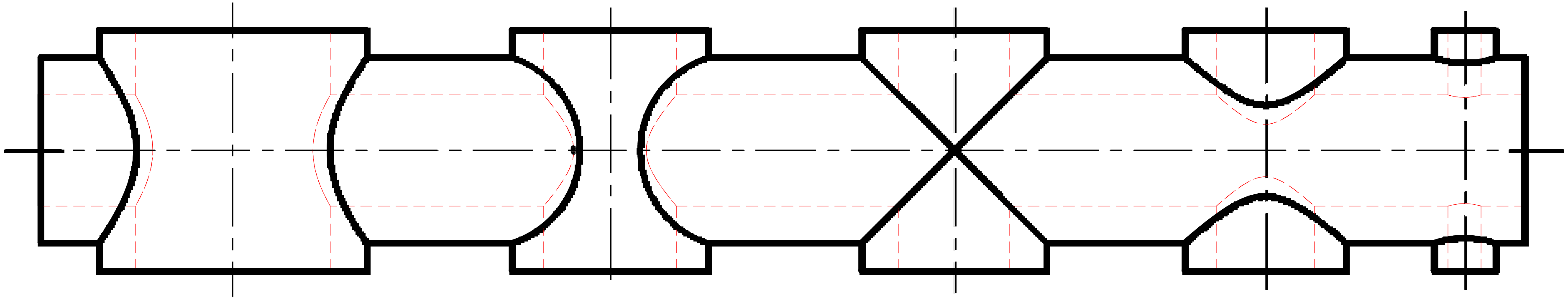
✓ 两圆柱正交且直径差较大时,无积聚性视图相贯线的投影可用圆弧代替

两圆柱正交相贯线

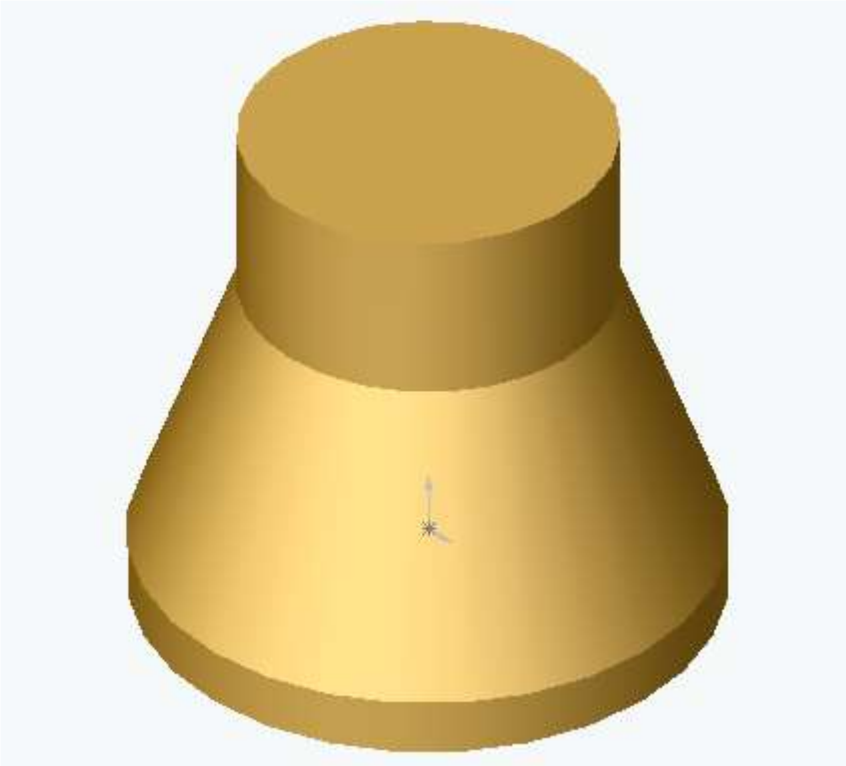


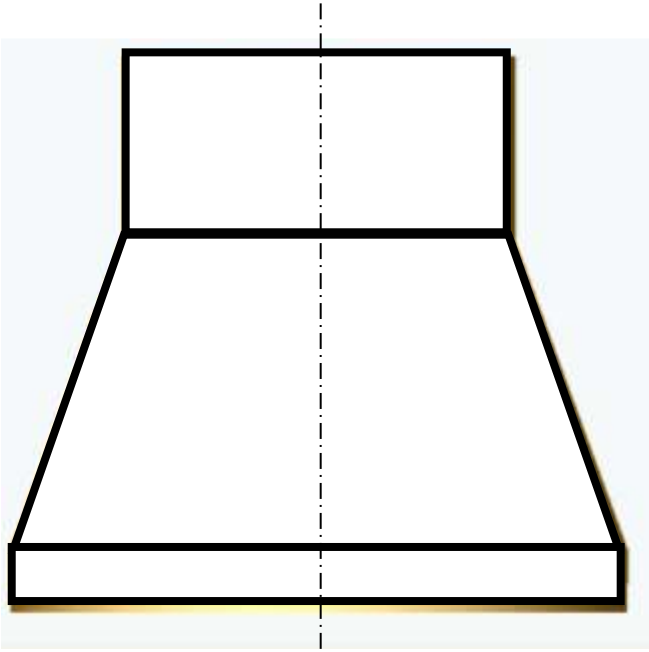


半剖



相贯线的特殊情况

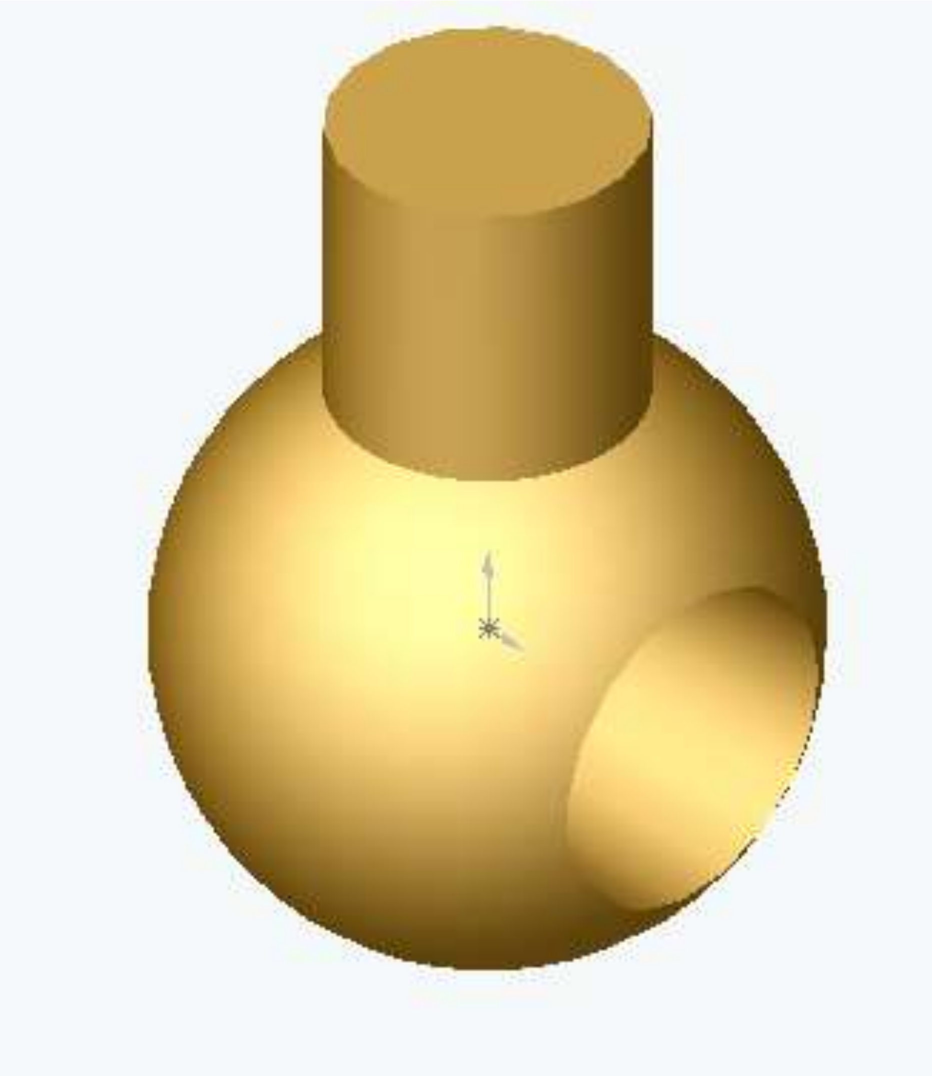


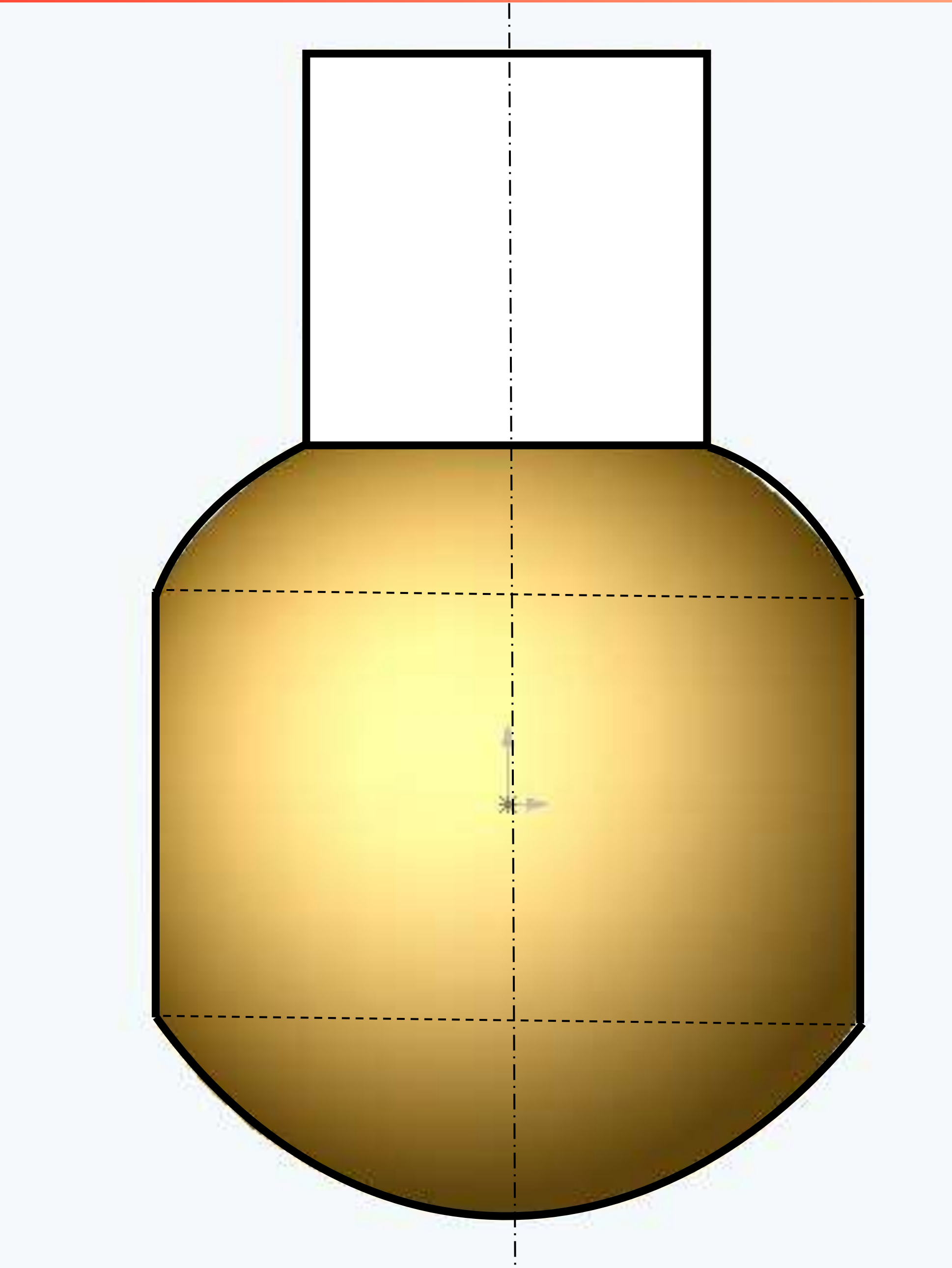


✓ 圆柱与圆锥同轴线相交，交线退化为圆，在该圆所垂直的投影面内积聚为直线。

相贯线的特殊情况

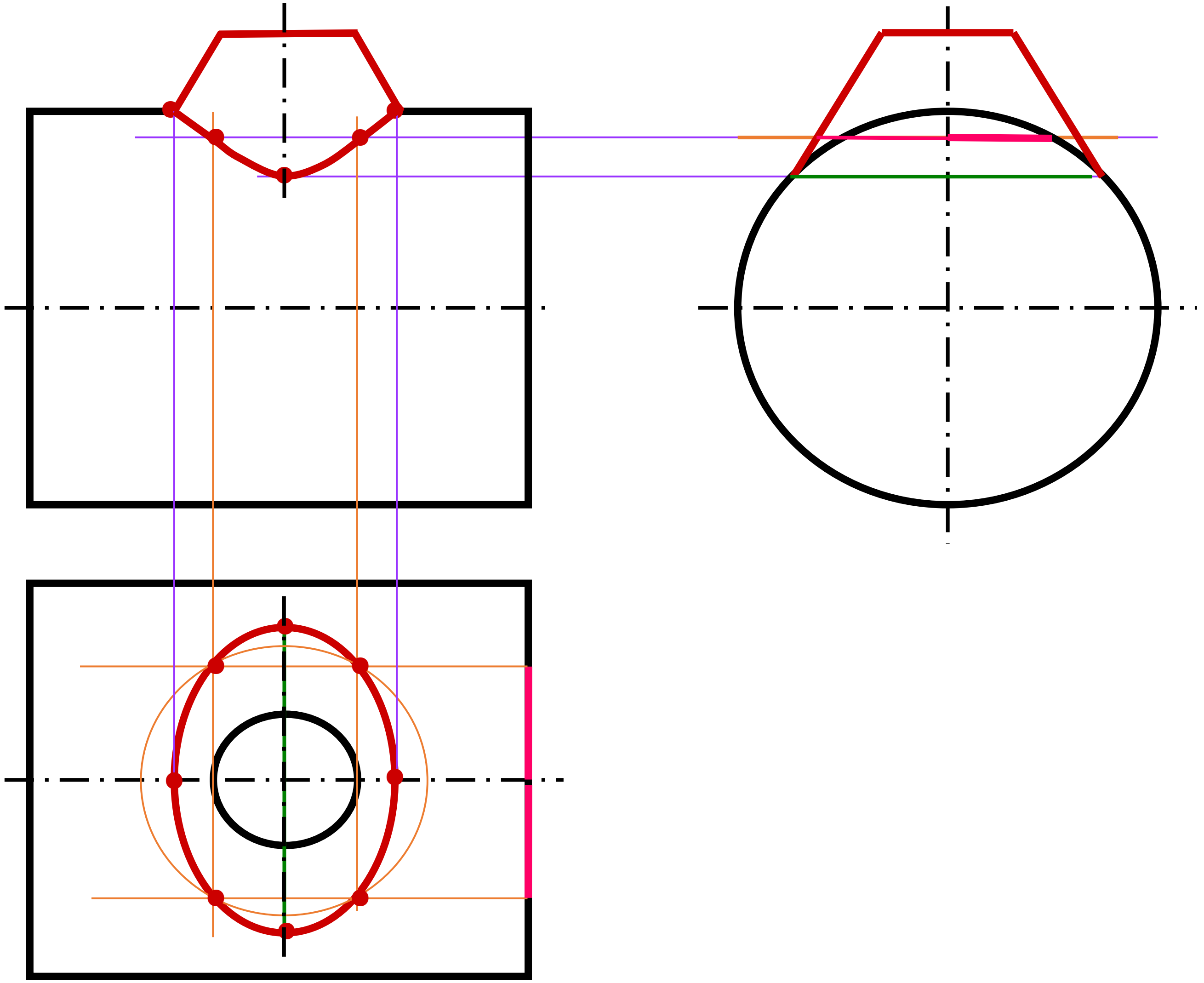
✓ 圆柱轴线通过球心

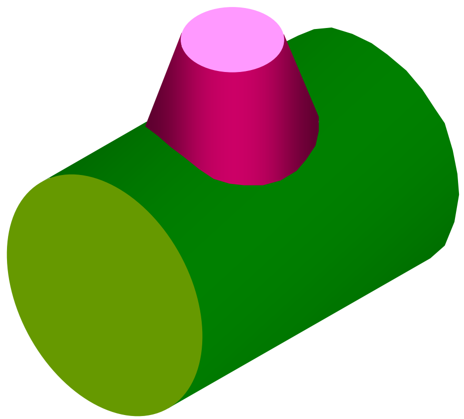




【例】

圆柱与圆锥相贯，求其相贯线。

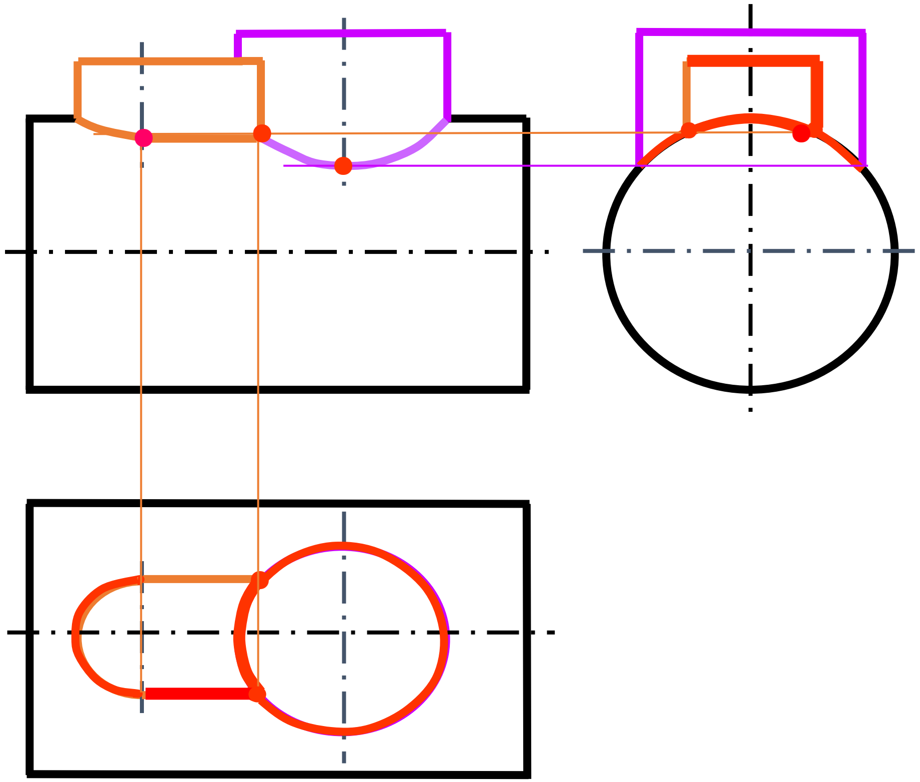


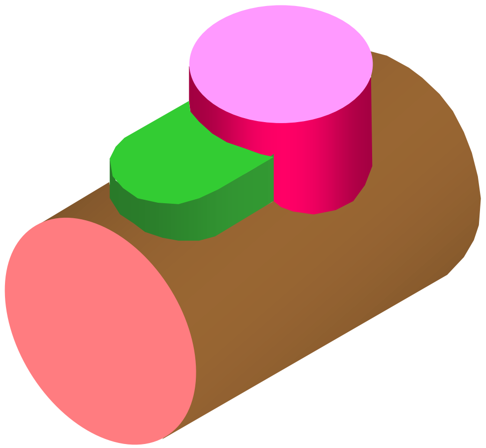


#【例】

北京理工大學 BEIJING INSTITUTE OF TECHNOLOGY

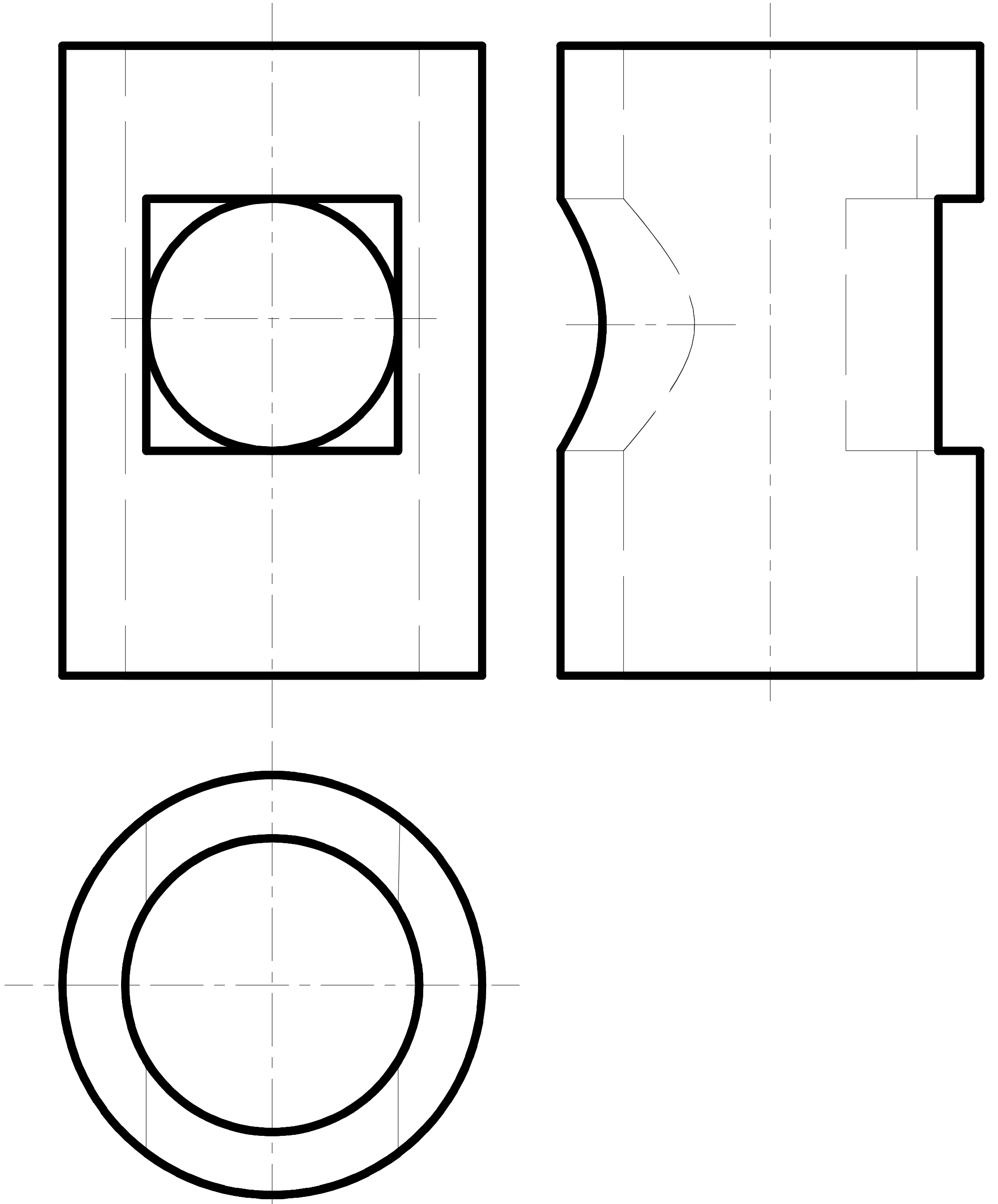
补全主视图。

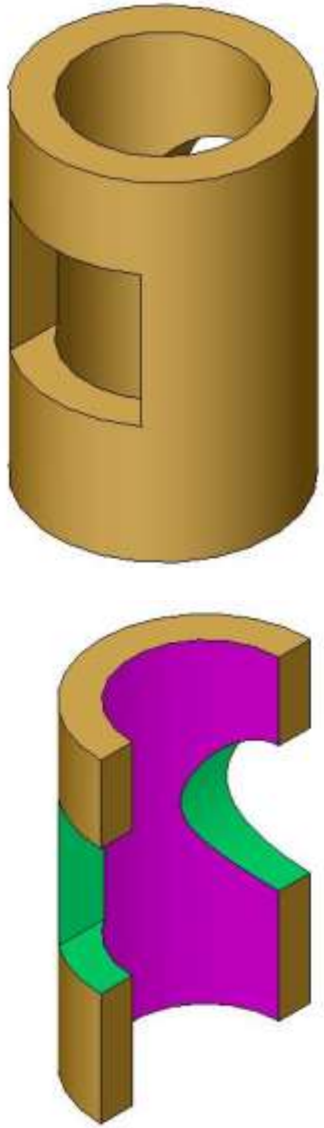




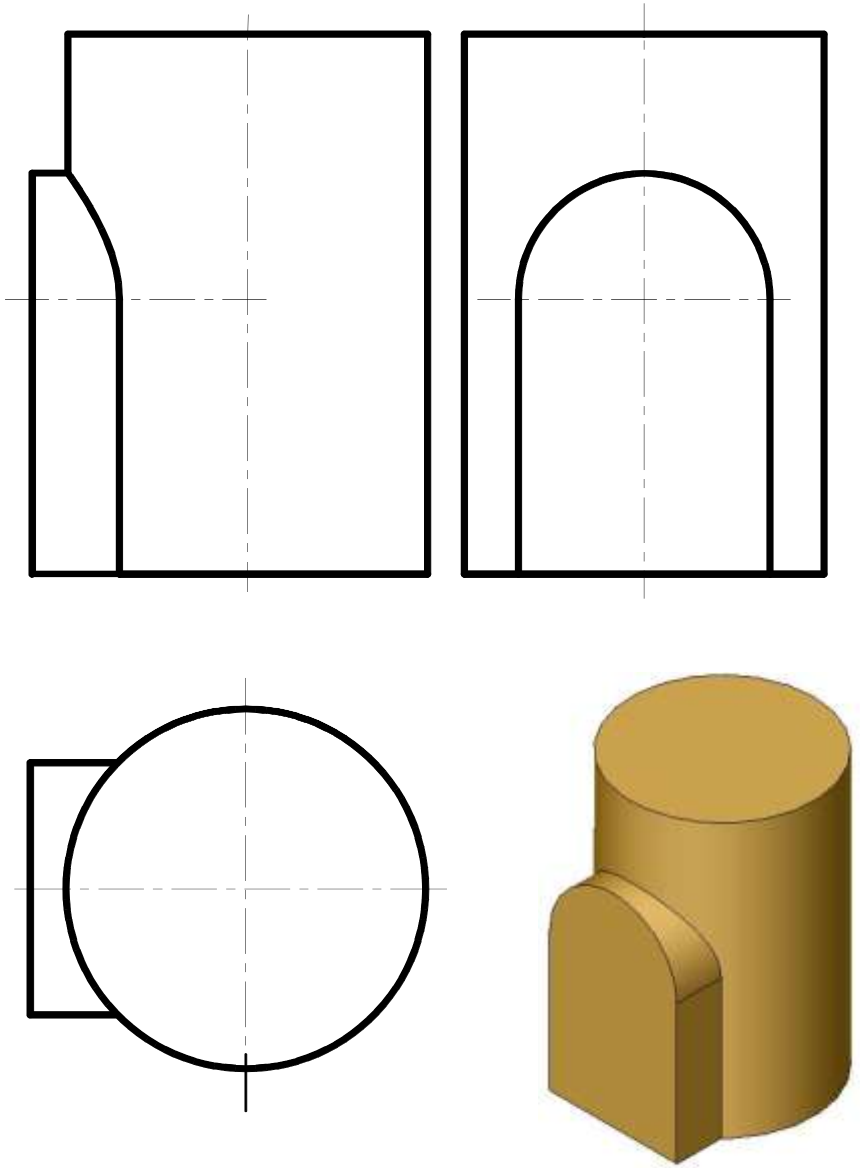
【例】

作左视图。

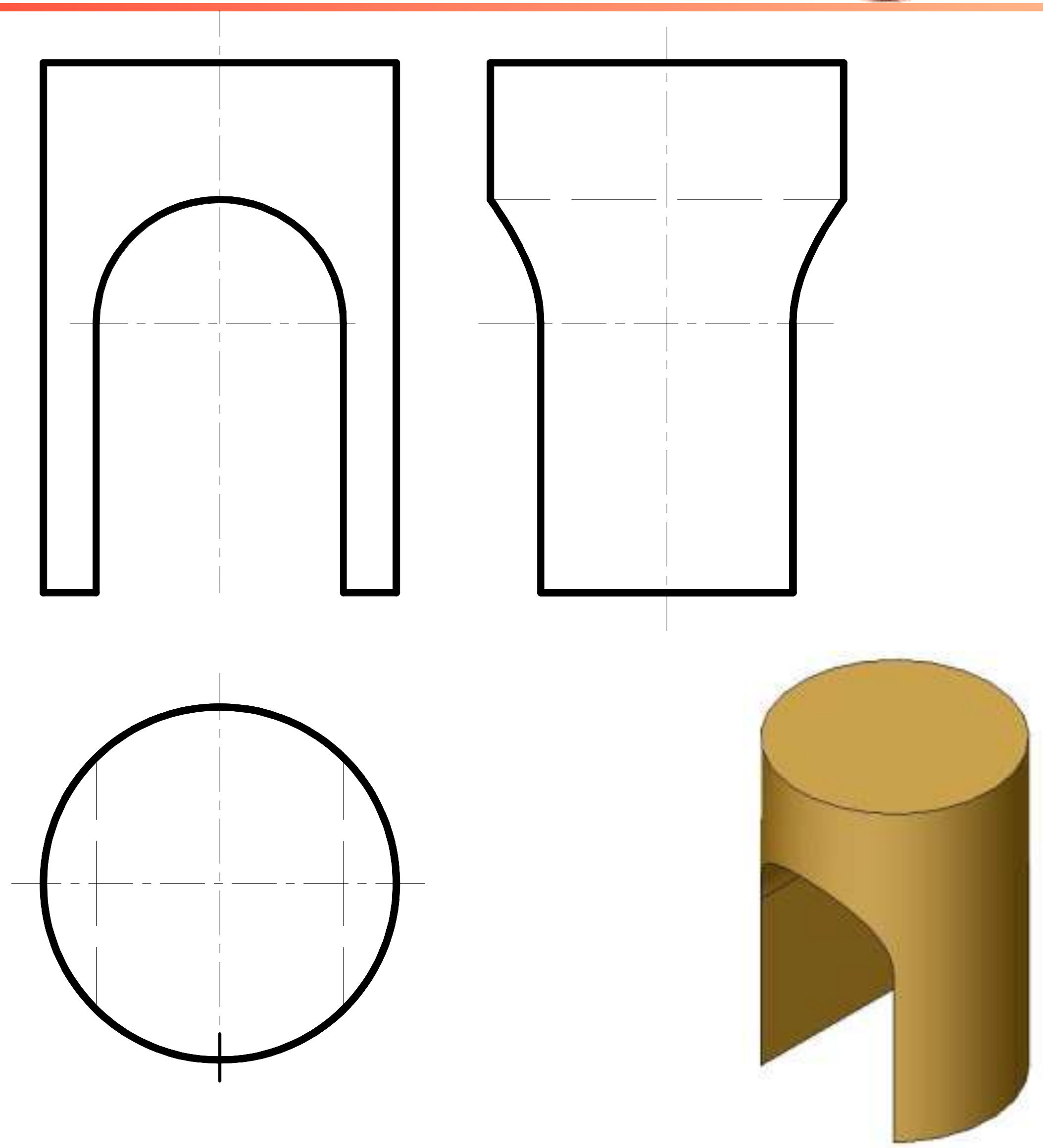




【例】

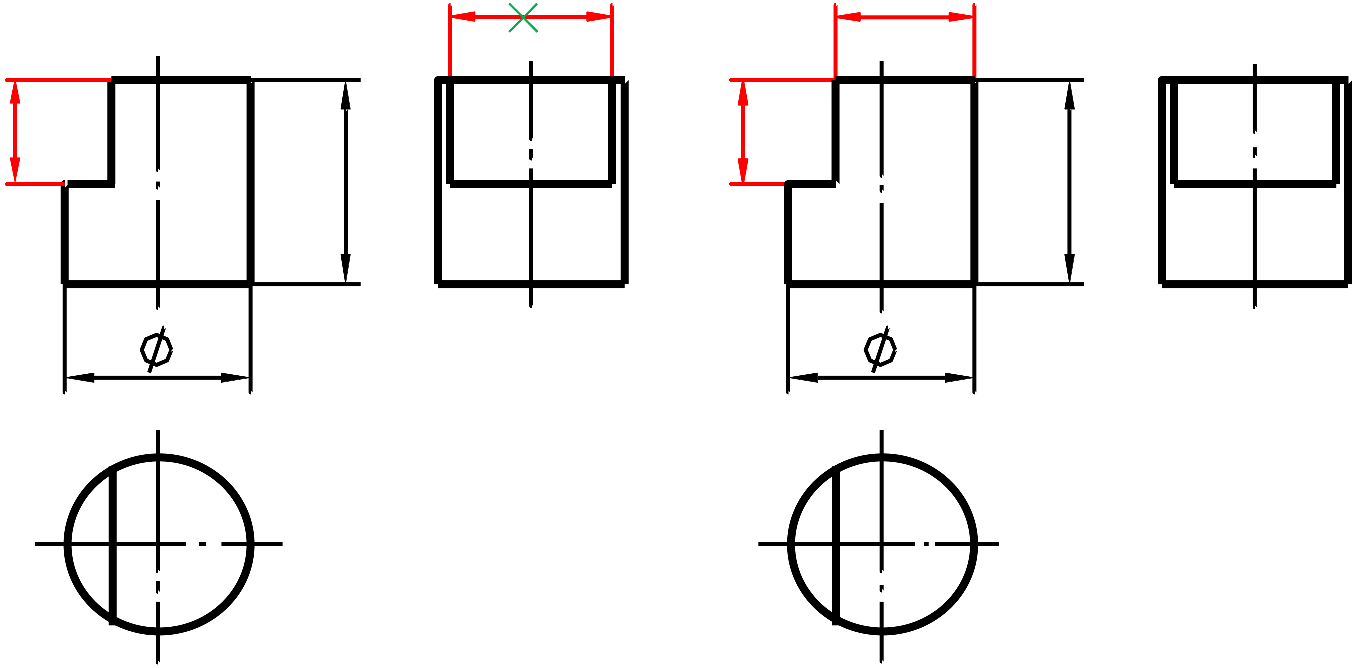


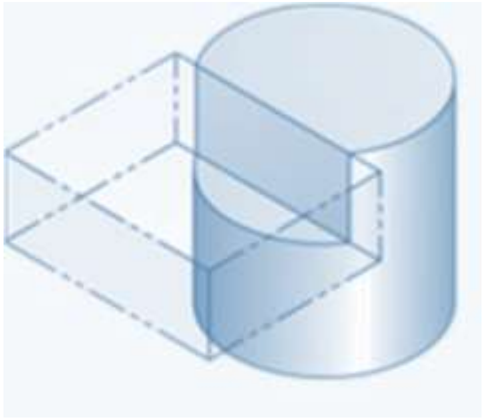
(叠加)



(切割)

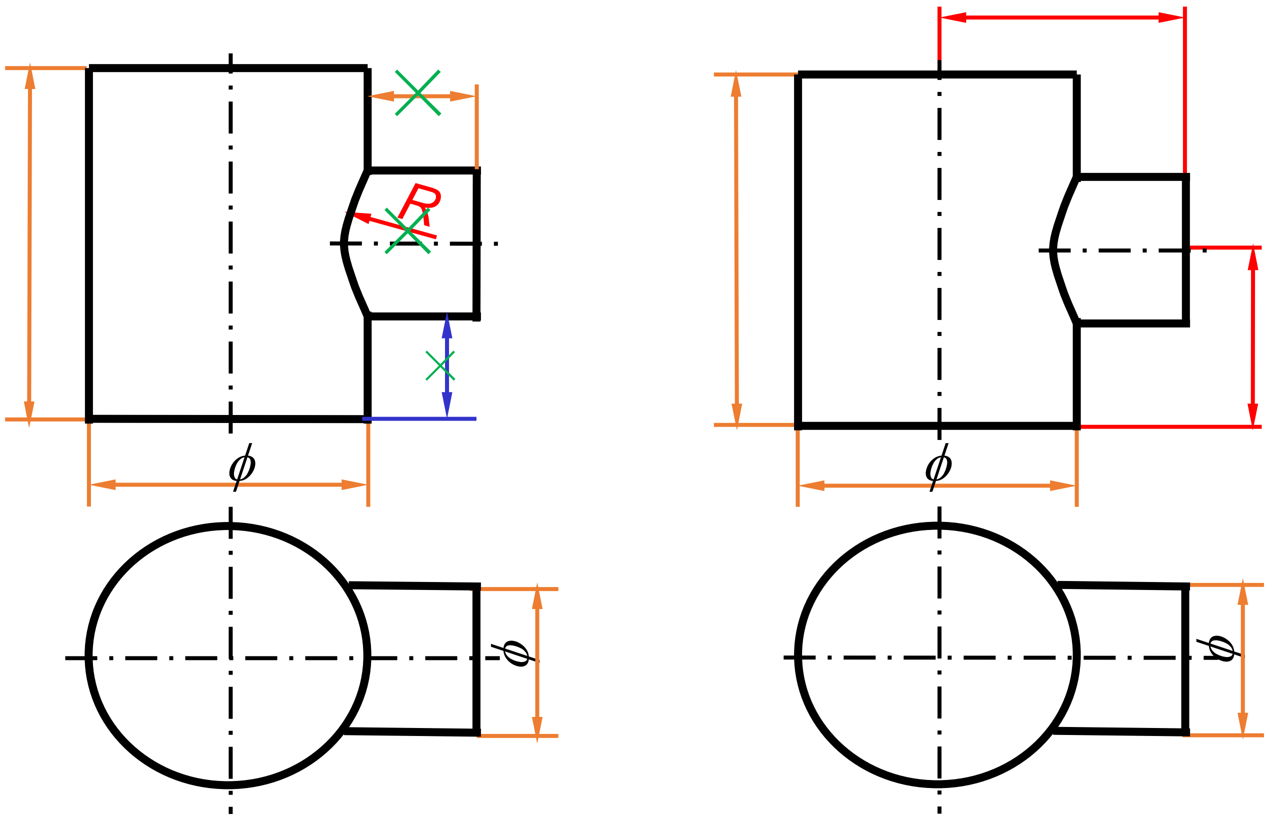
交线尺寸

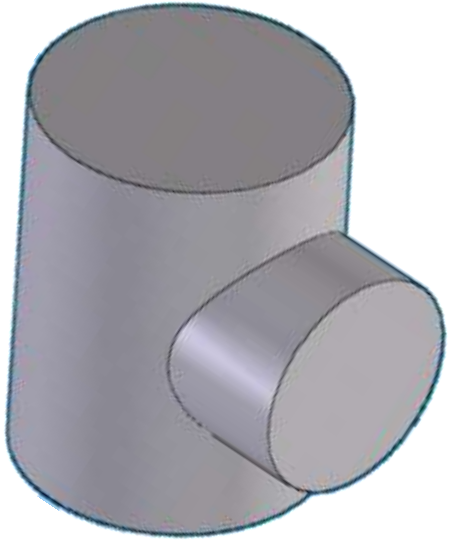




- ✓ 两相交体各自的尺寸
- ✓ 彼此间的相对位置尺寸
- ✓ 由于交线是自然形成的，所以在截交线上不标注尺寸

交线尺寸





- ✓ 两相交体各自的尺寸
- ✓ 彼此间的相对位置尺寸
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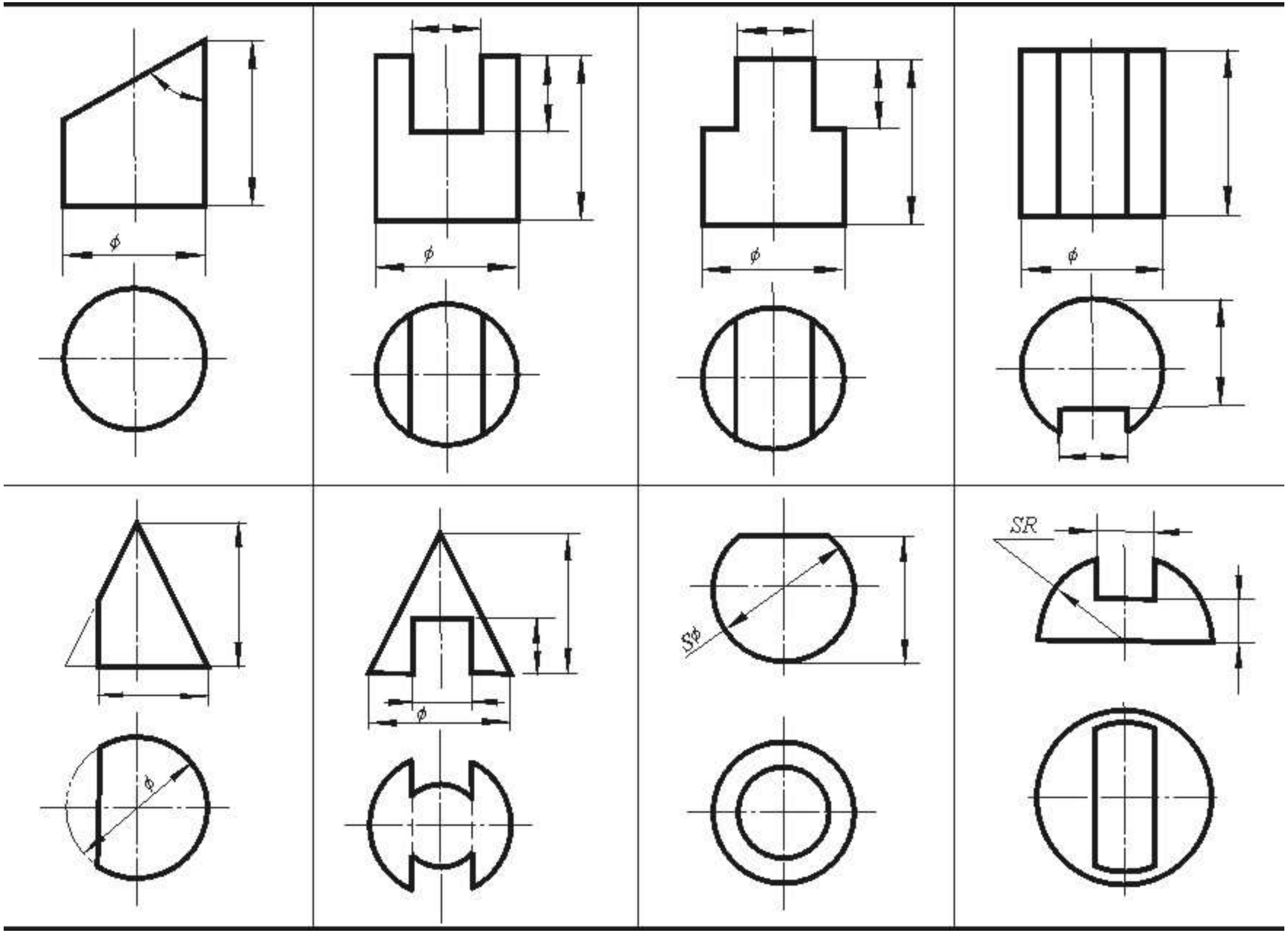
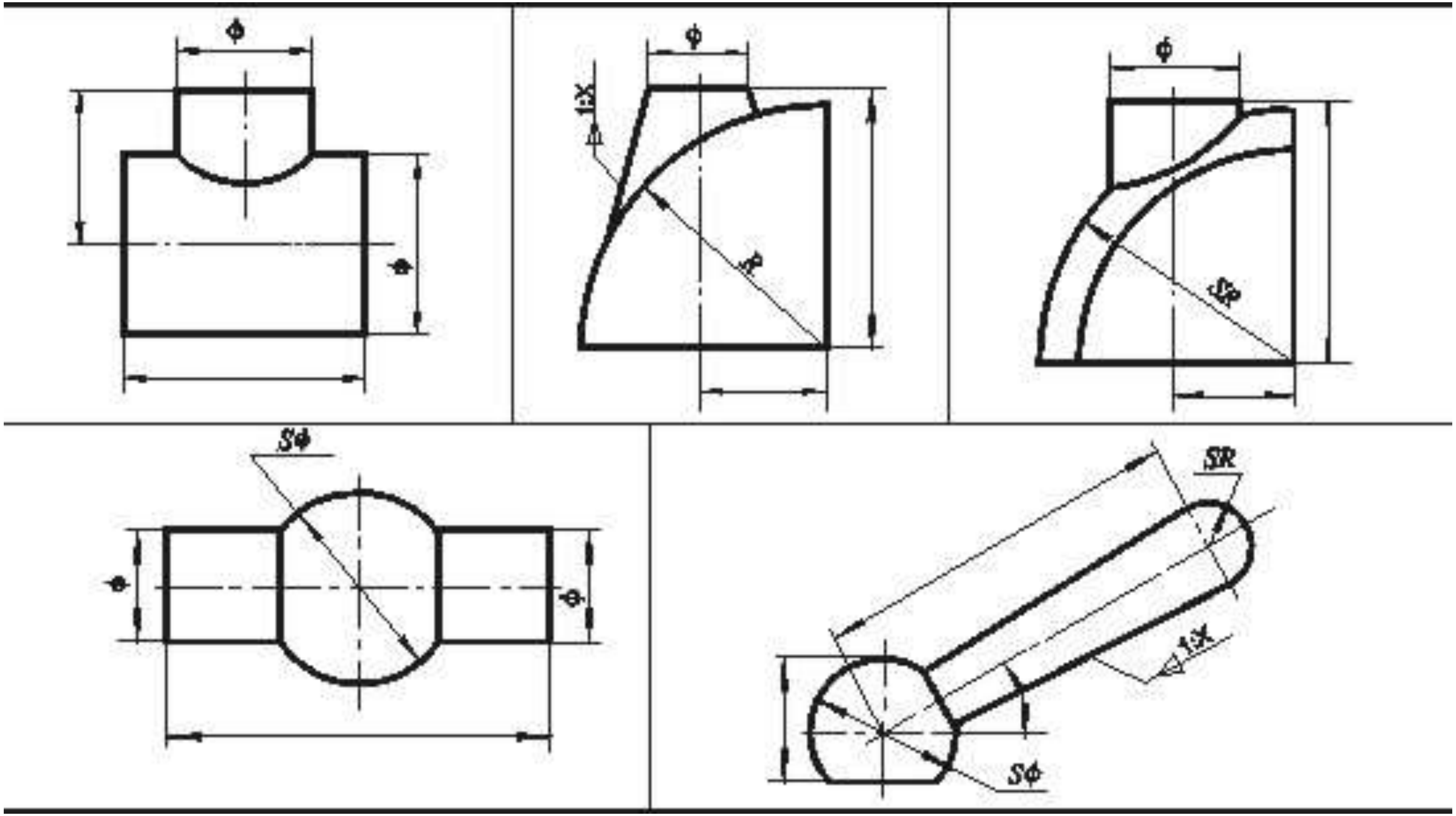
交线尺寸

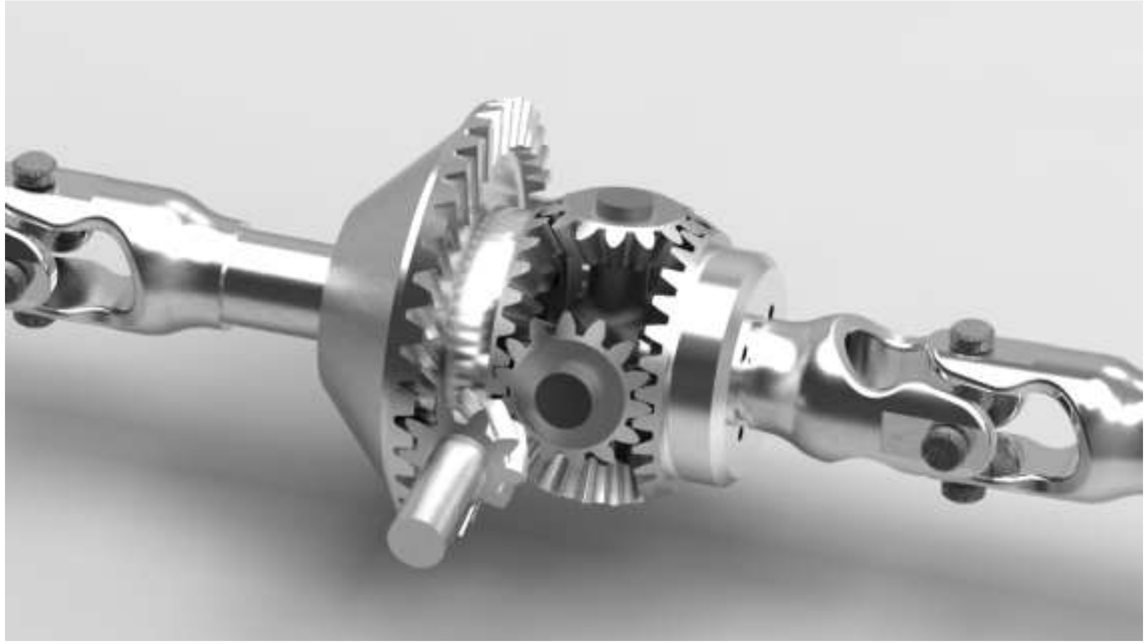


交线尺寸

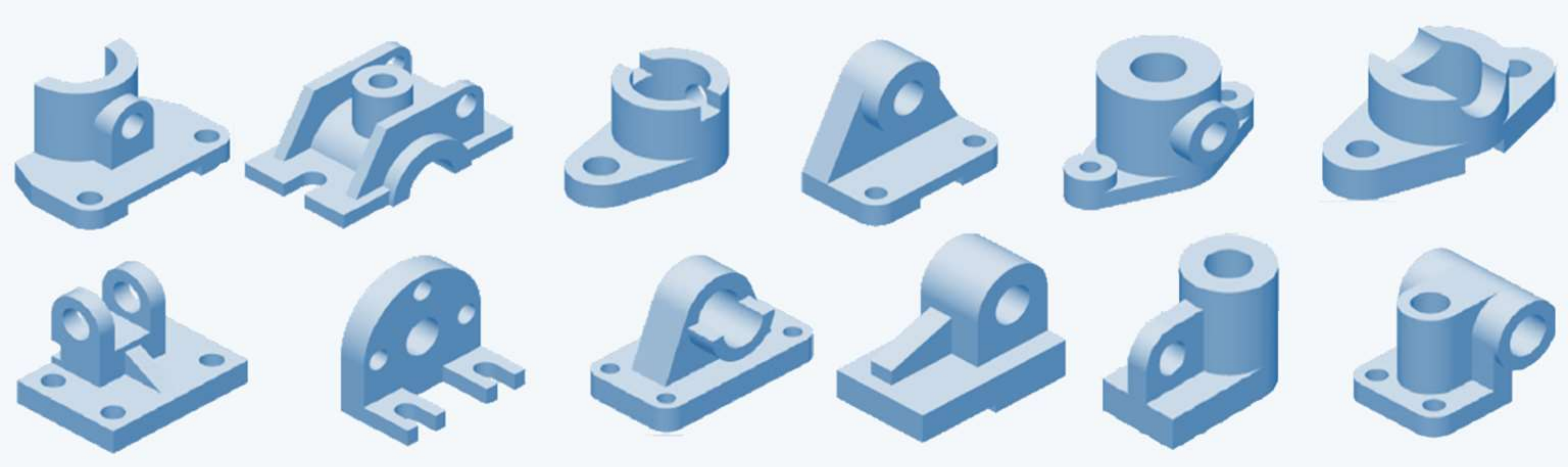
北京理工大學
BEIJING INSTITUTE OF TECHNOLOGY



组合体



组合体



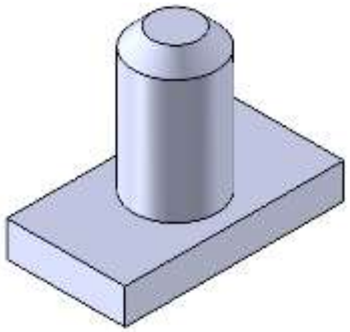
✓ 组合体由多个基本几何体所构成

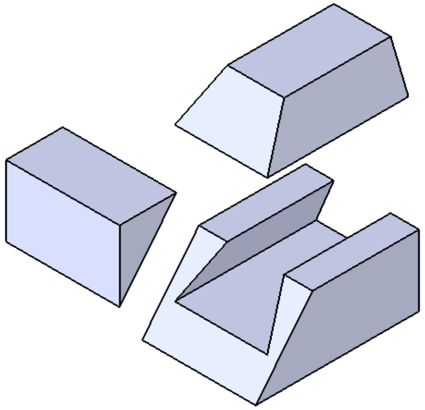
组合体

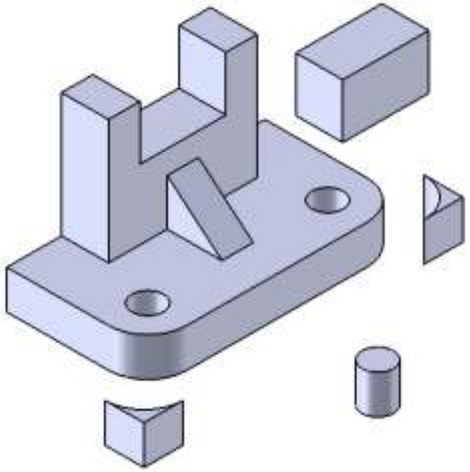
- ✓ 组合体由多个基本几何体所构成

组合体的构成

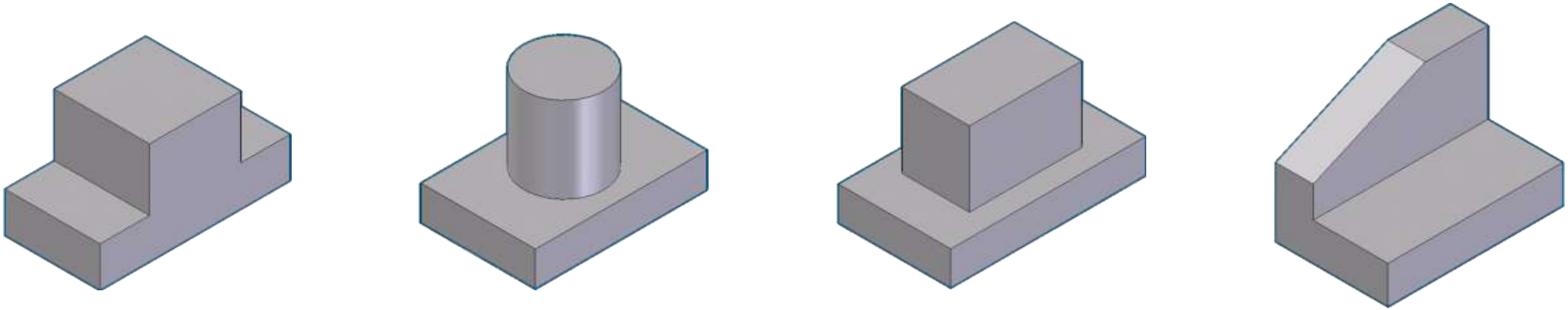
- 叠加
- 切割
- 综合



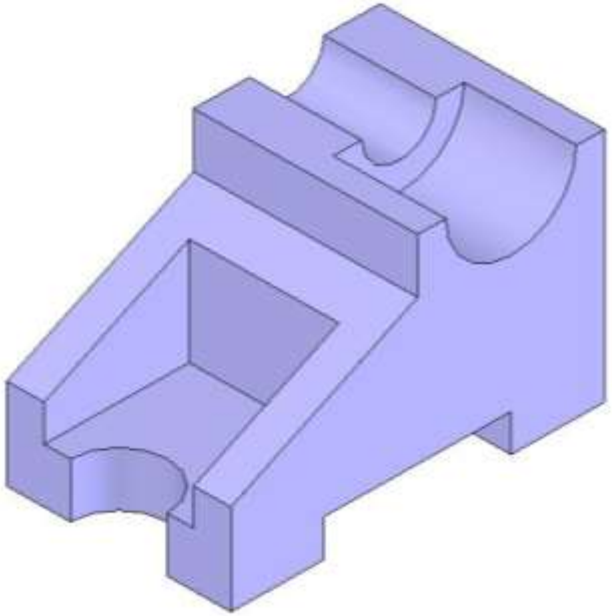


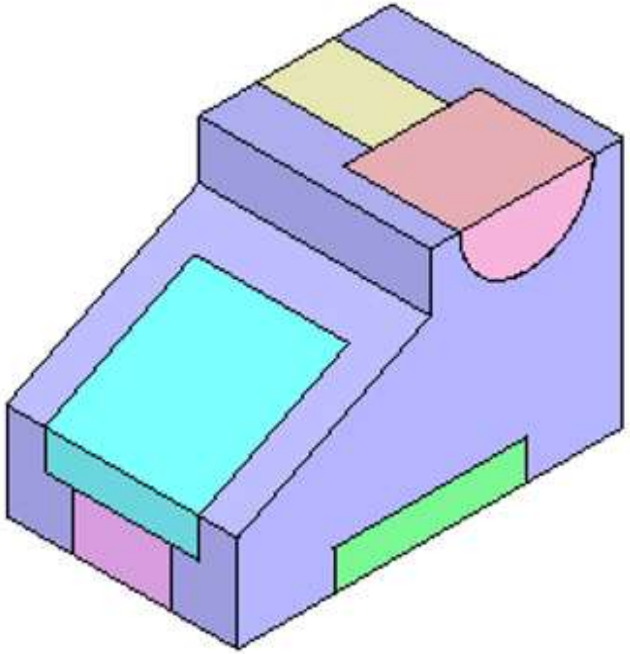


叠加

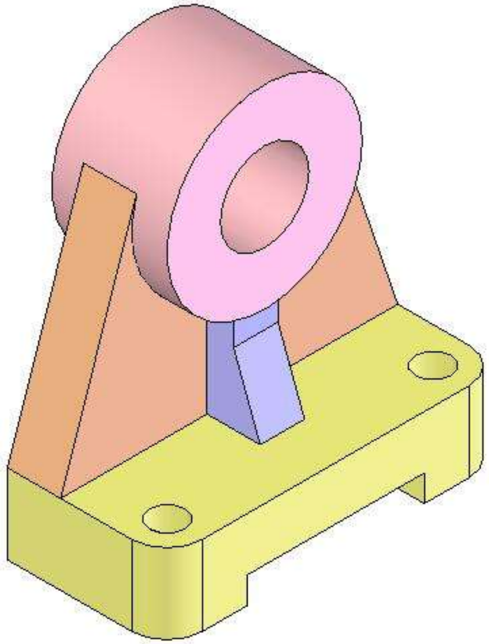


切割





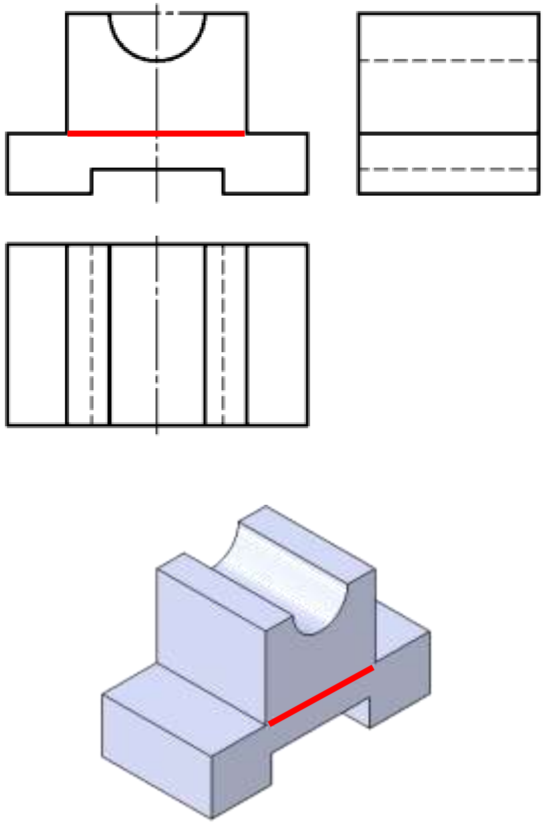
叠加和切割



组合体

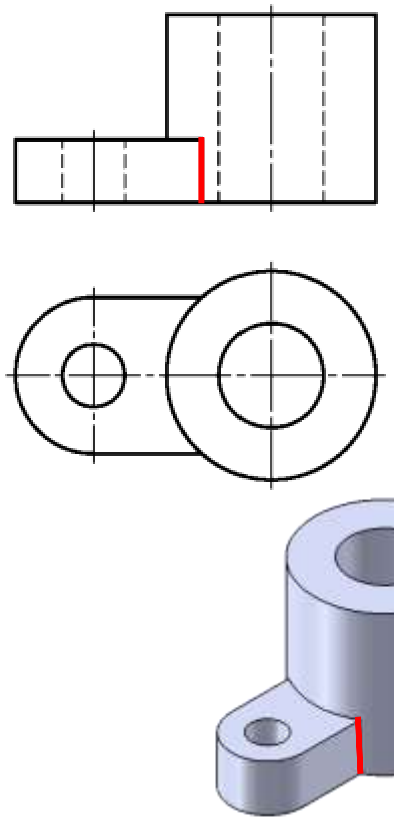
常见的表面结合关系

共面



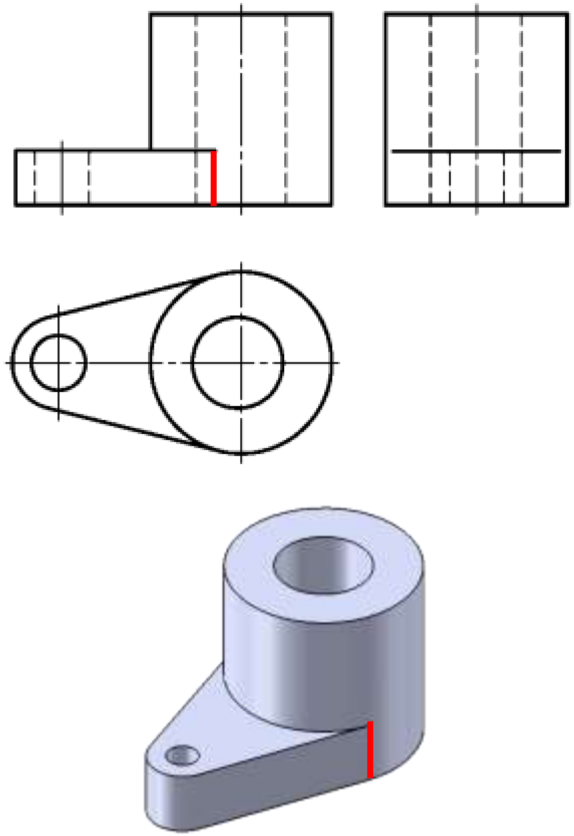
不画分界线

相交



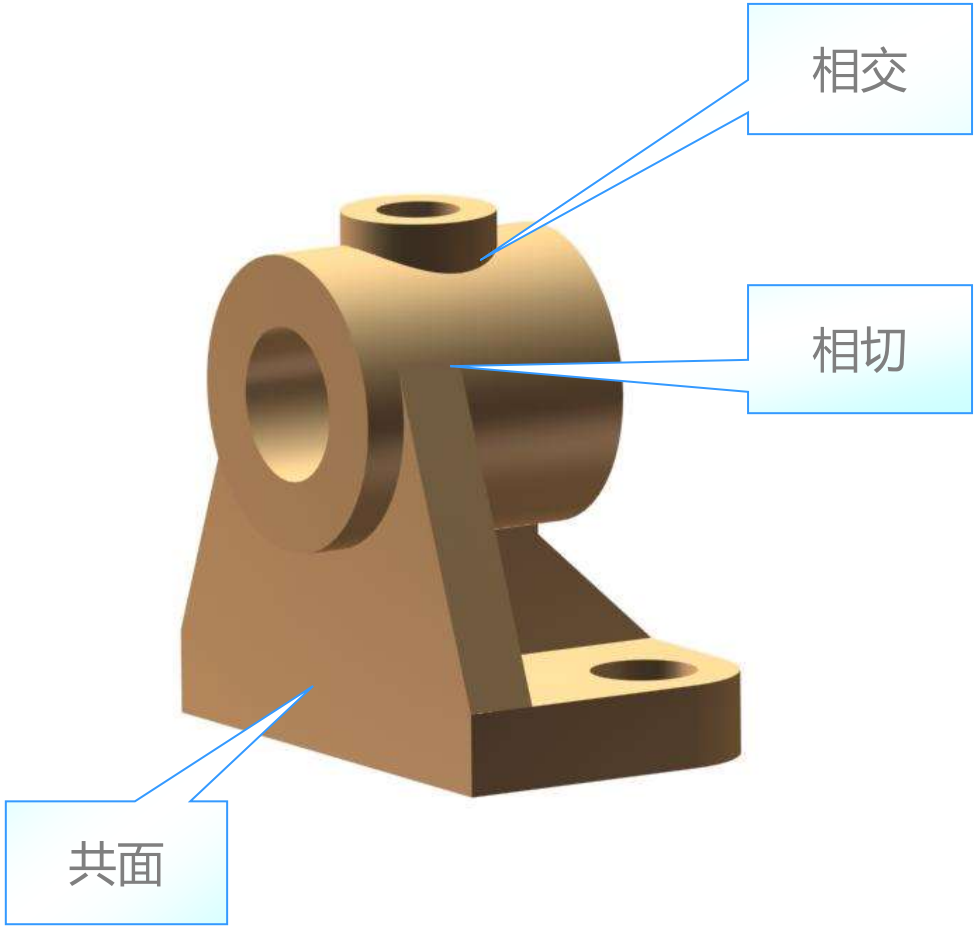
有分界线

相切

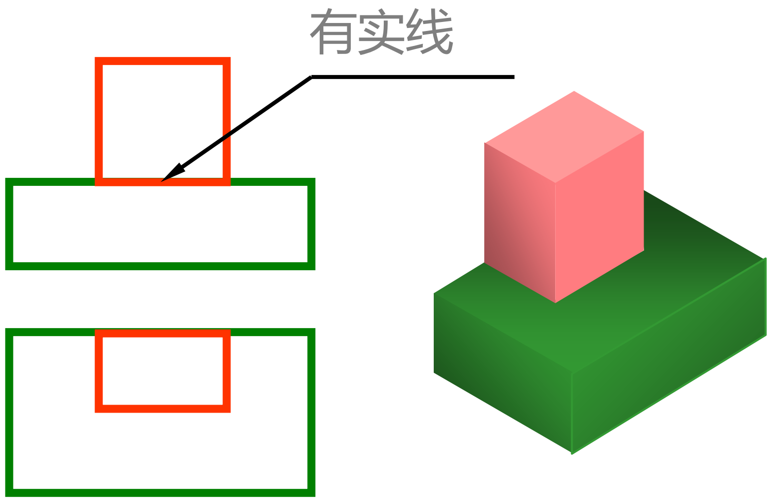


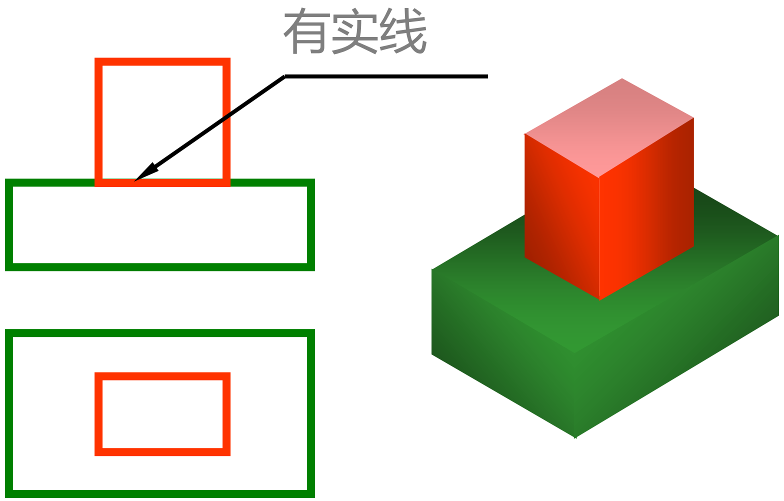
无分界线

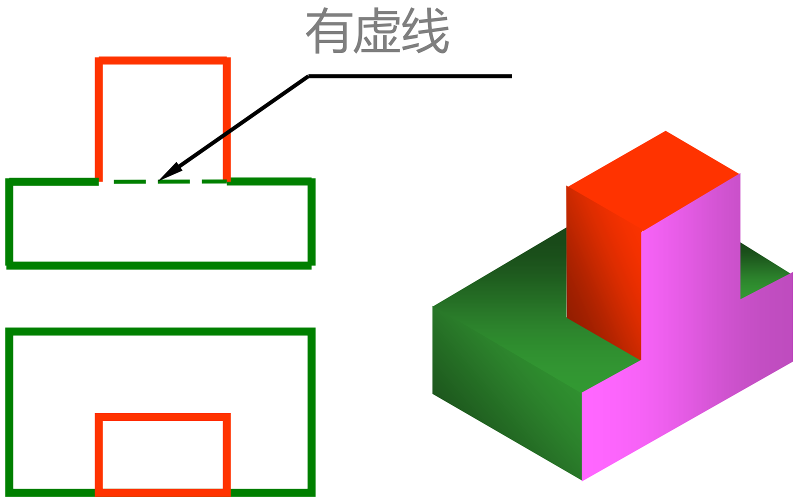
组合体

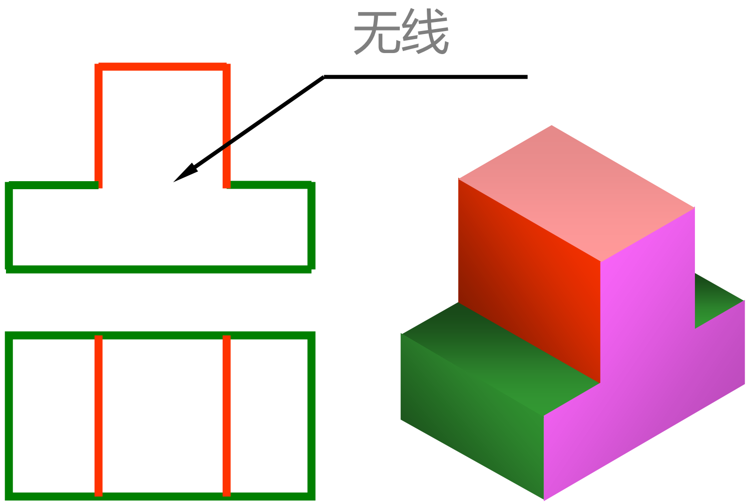


共面

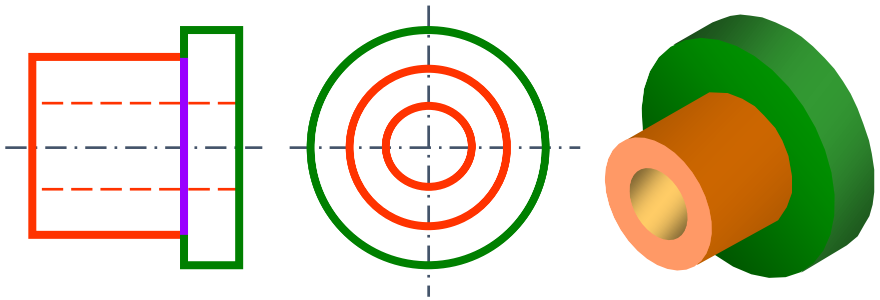


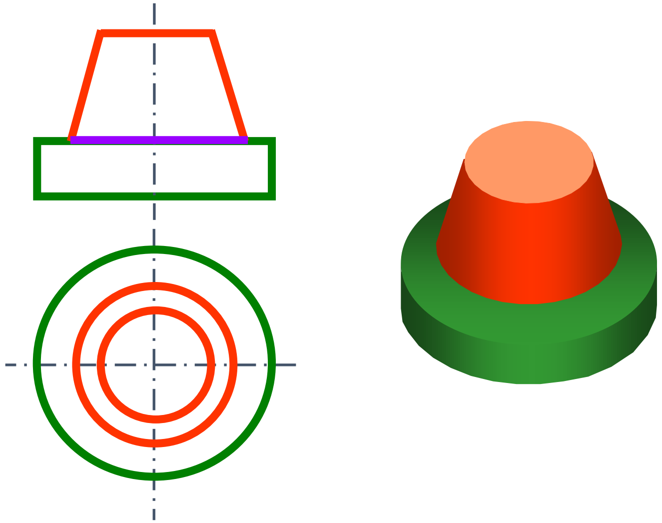


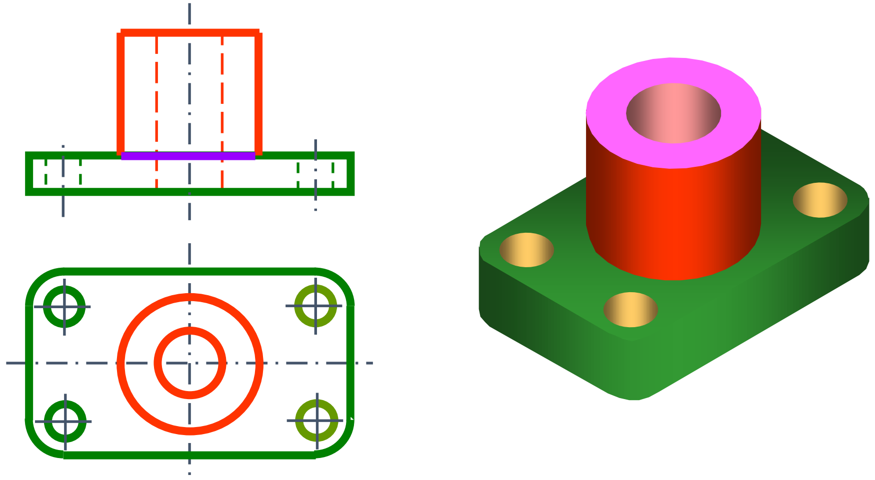


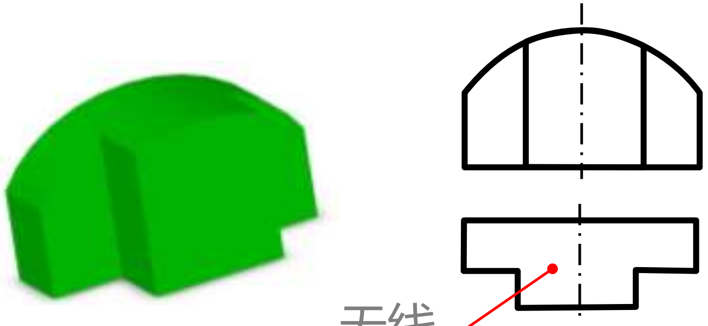


共面

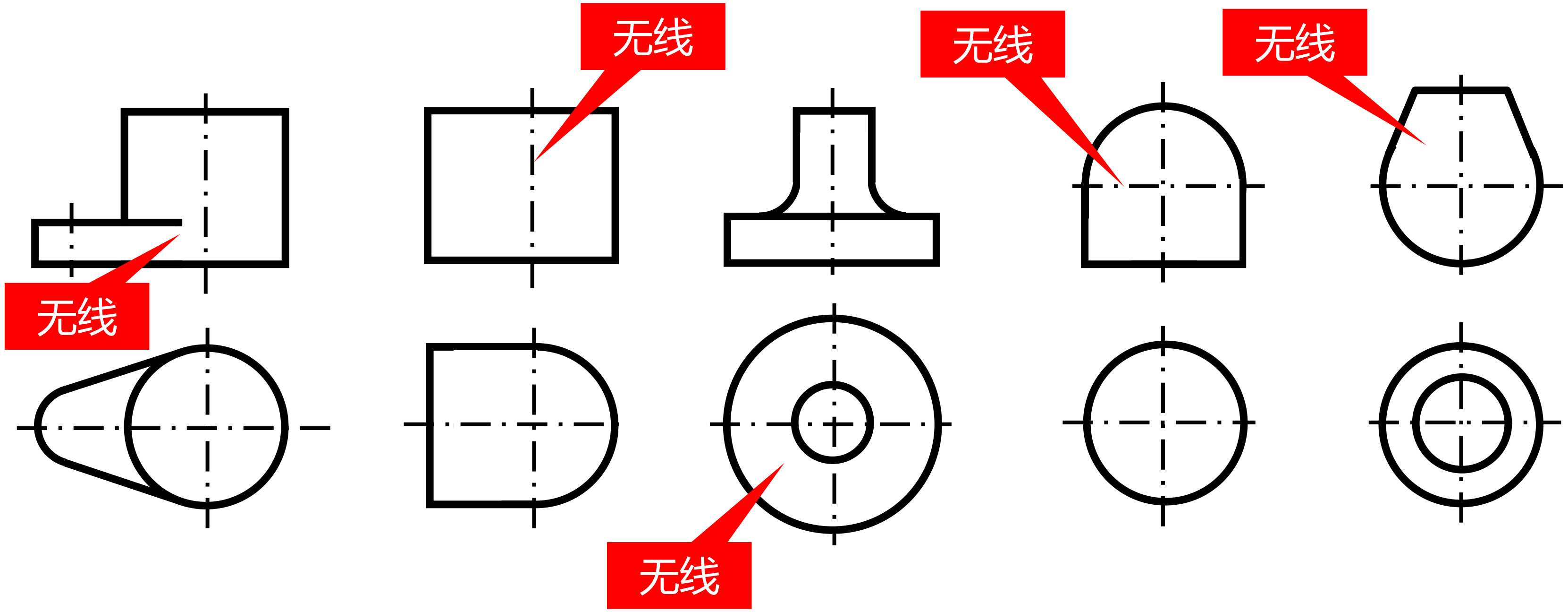




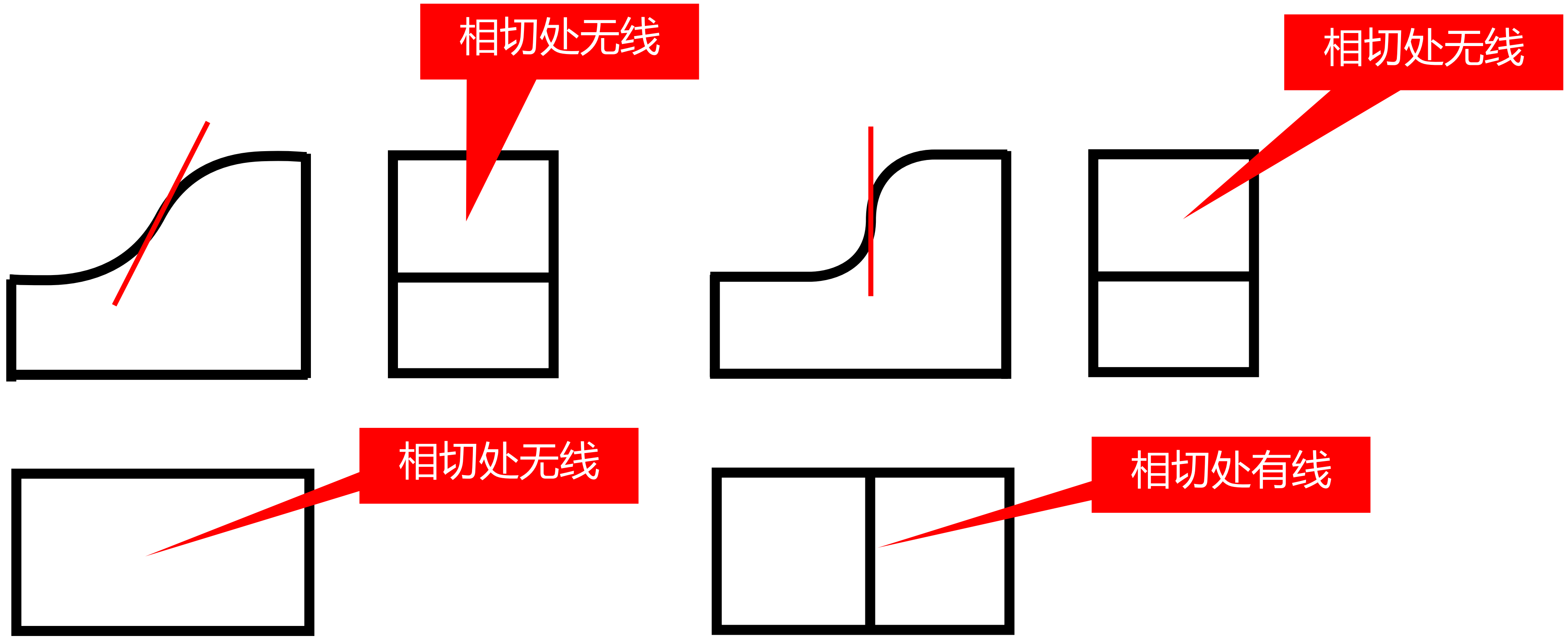




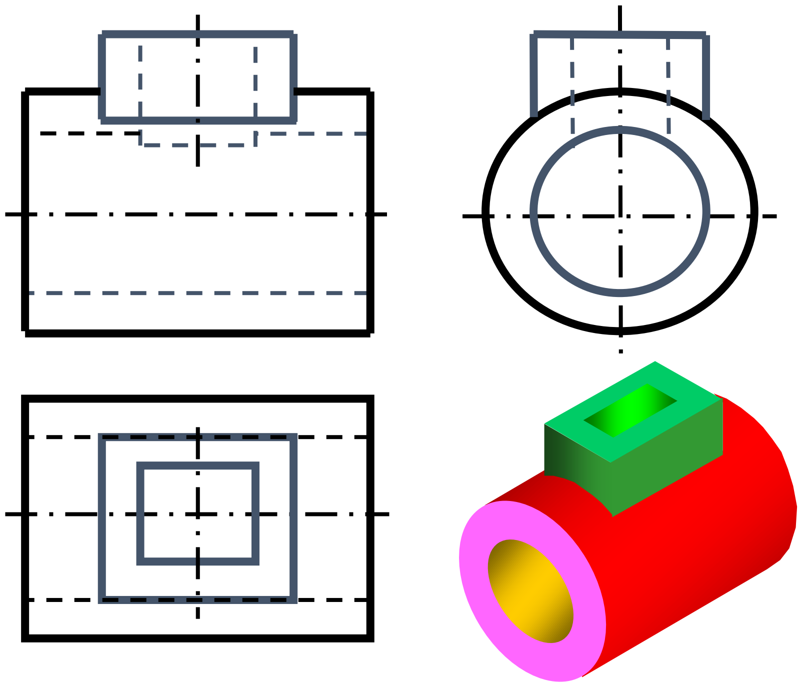
相切

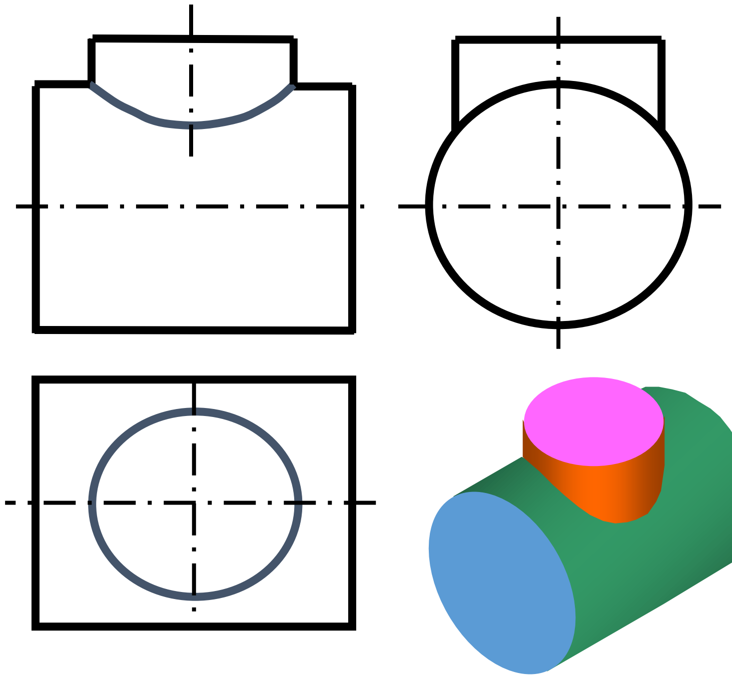


相切

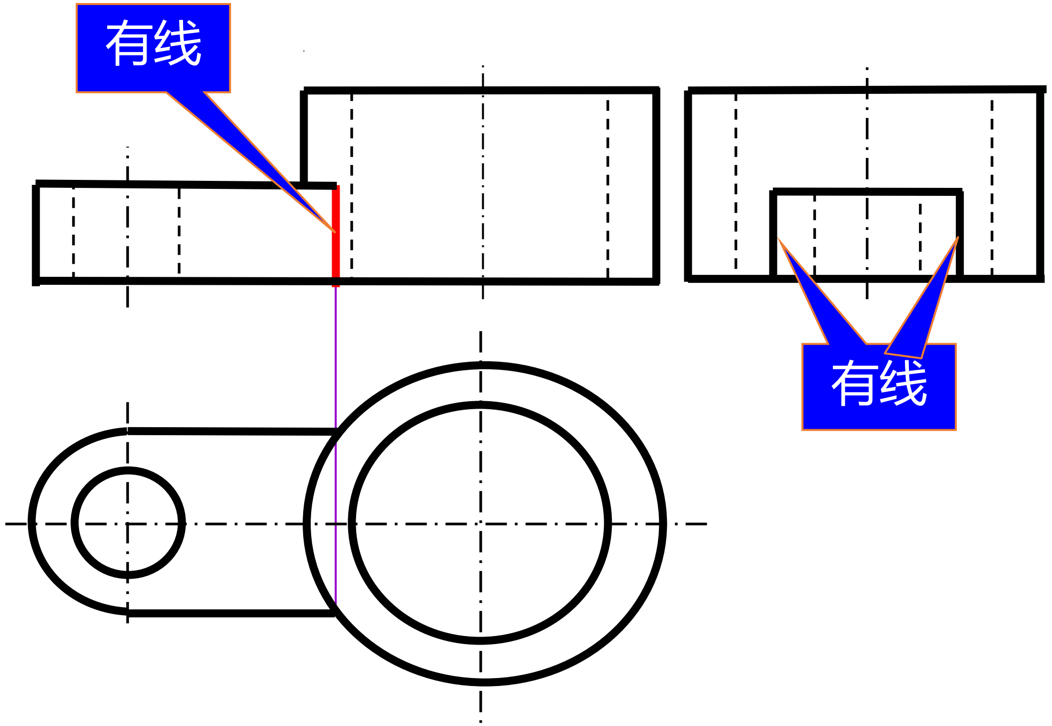


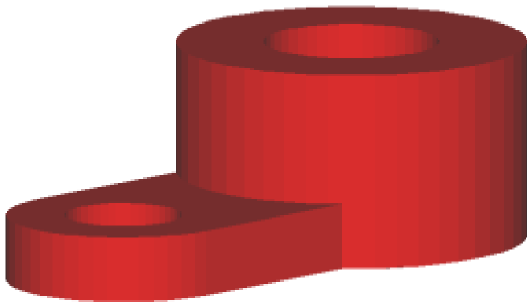
✓ 当两圆柱面相切时，它们的公共圆柱面的公切平面垂直于投影面时，应画出相切的素线在该投影面上的投影，也就是两个柱面的分界线。





■ 相贯线（截交线）

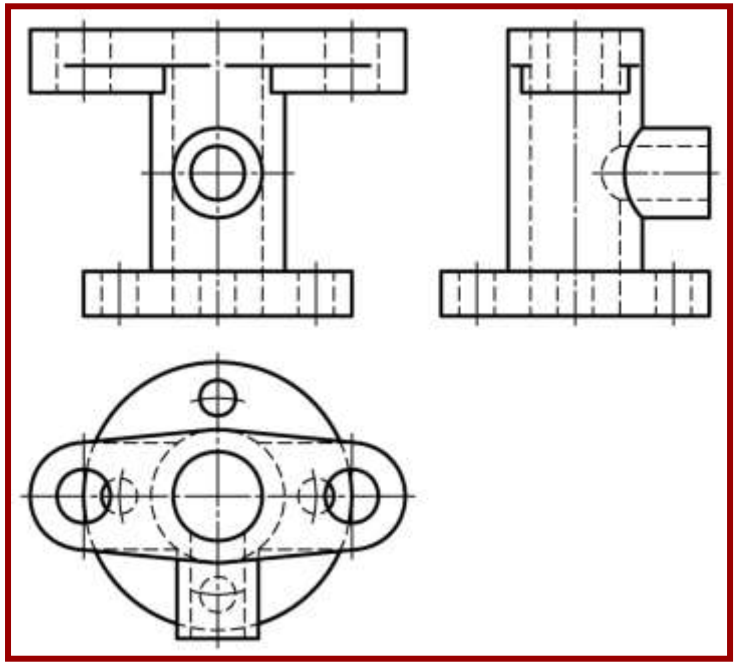




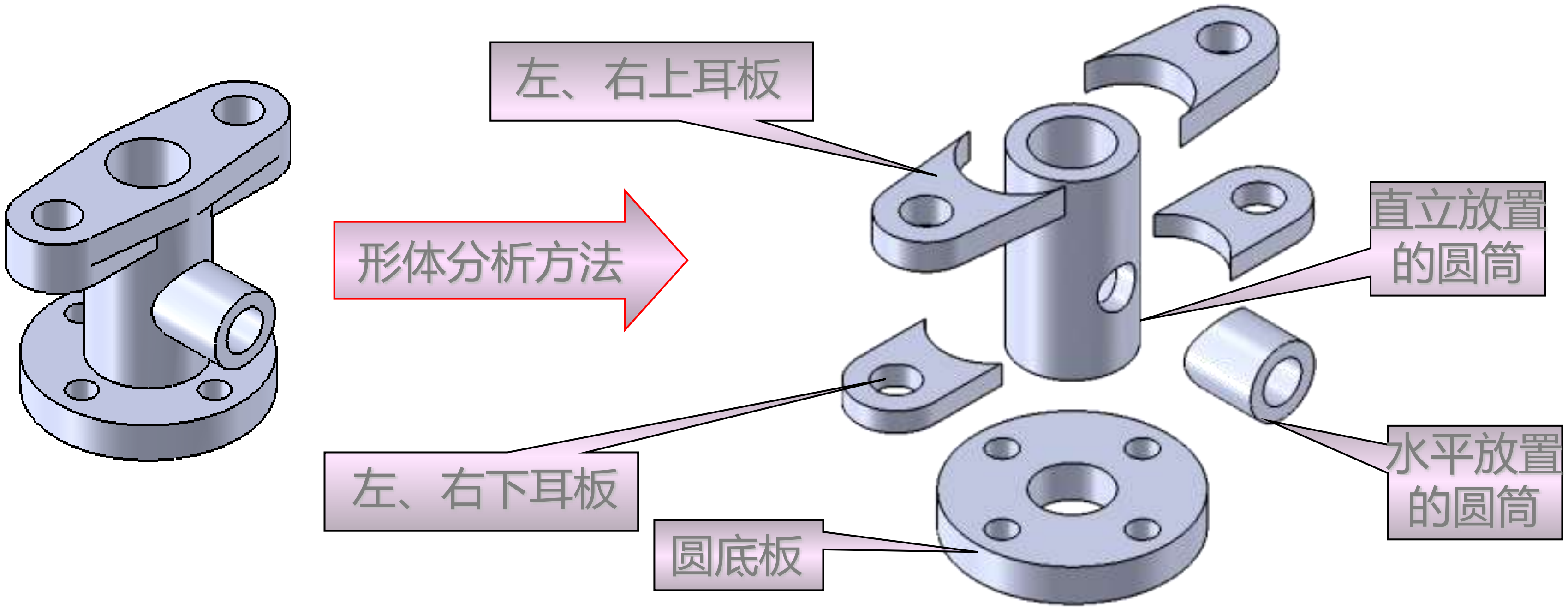
✓ 平面与其他表面的交线

形体分析

1. 进行形体分析
2. 确定主视图
3. 选比例、定图幅
4. 布图、画基准线
5. 逐个画出各形体三视图
    - 三个视图同时画
    - 每叠加一形体，考虑相邻两表面间关系
6. 检查、描深

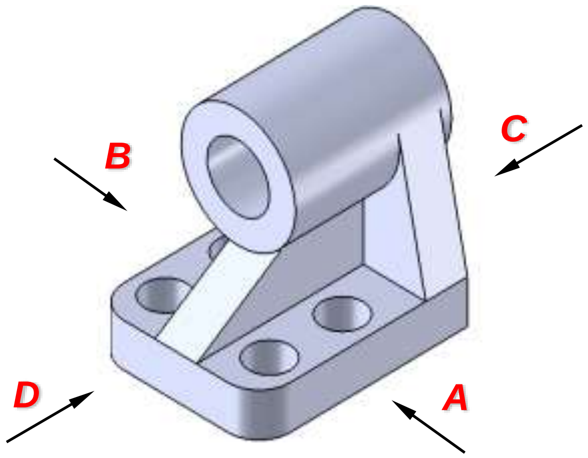


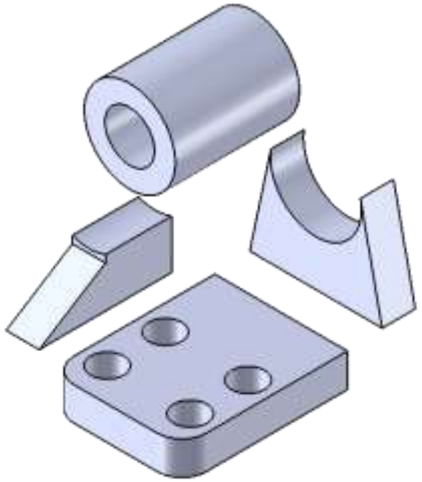
形体分析



- ✓ 根据组合体的形状，将其分解成若干部分，弄清各部分的形状和它们的相对位置及组合方式，分别画出各部分的投影

视图

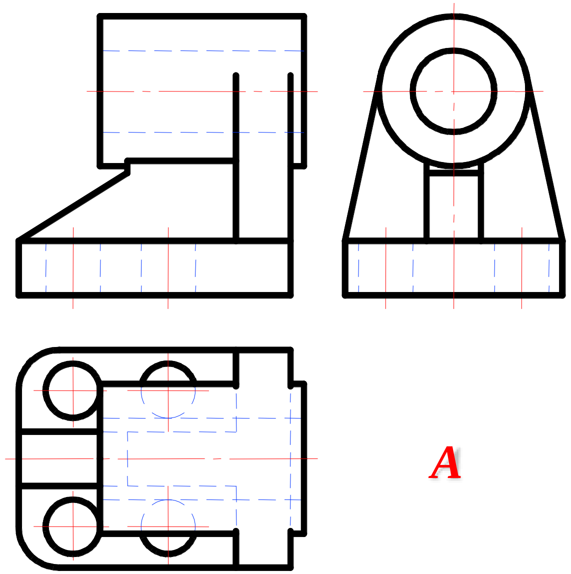




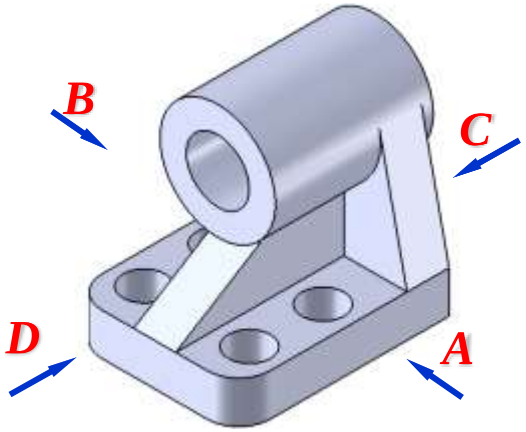
主视图：最重要的视图 → 先定放置方式 → 再定投射方向

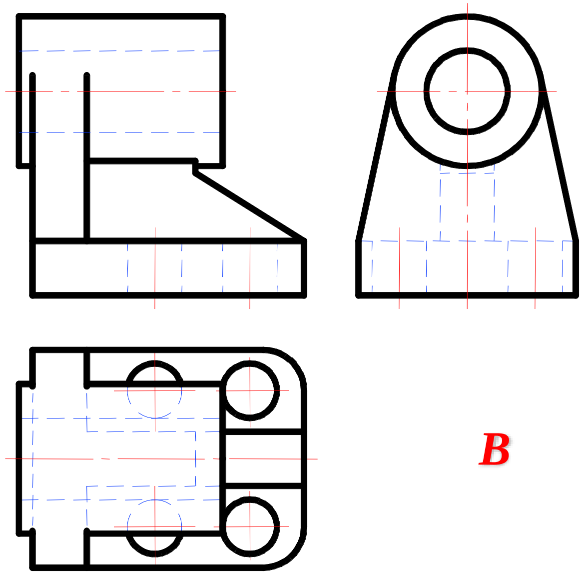
- **放置方式** 自然放置，一般将较大的平面作为底面
- **投射方向** 反映特征，可见性好（图中虚线最少）

视图

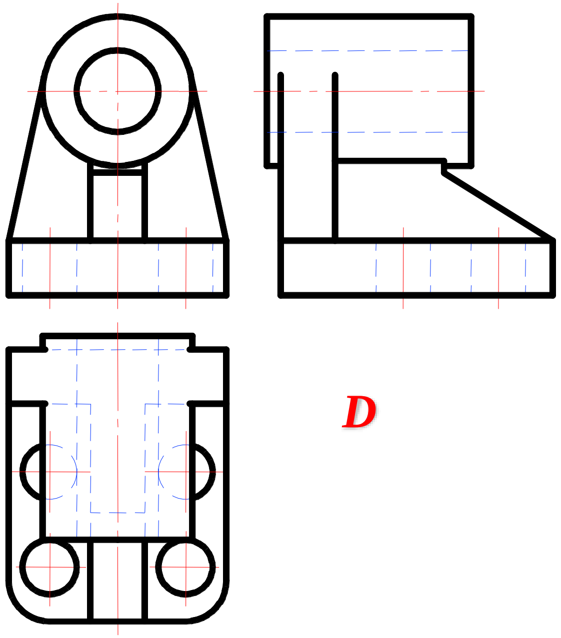


A

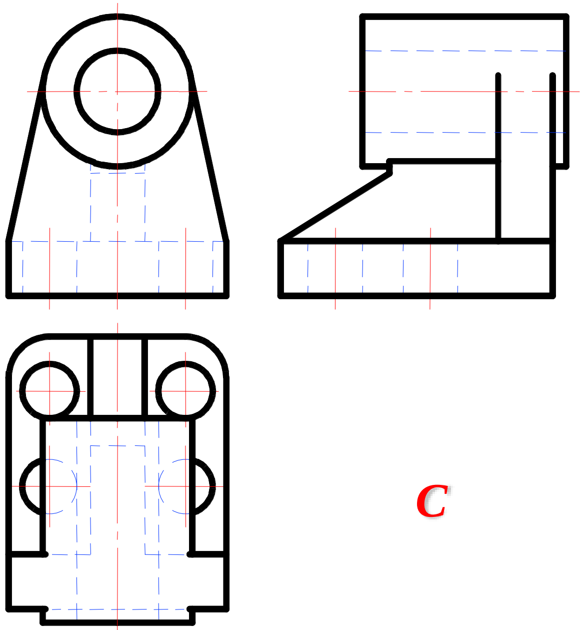




B

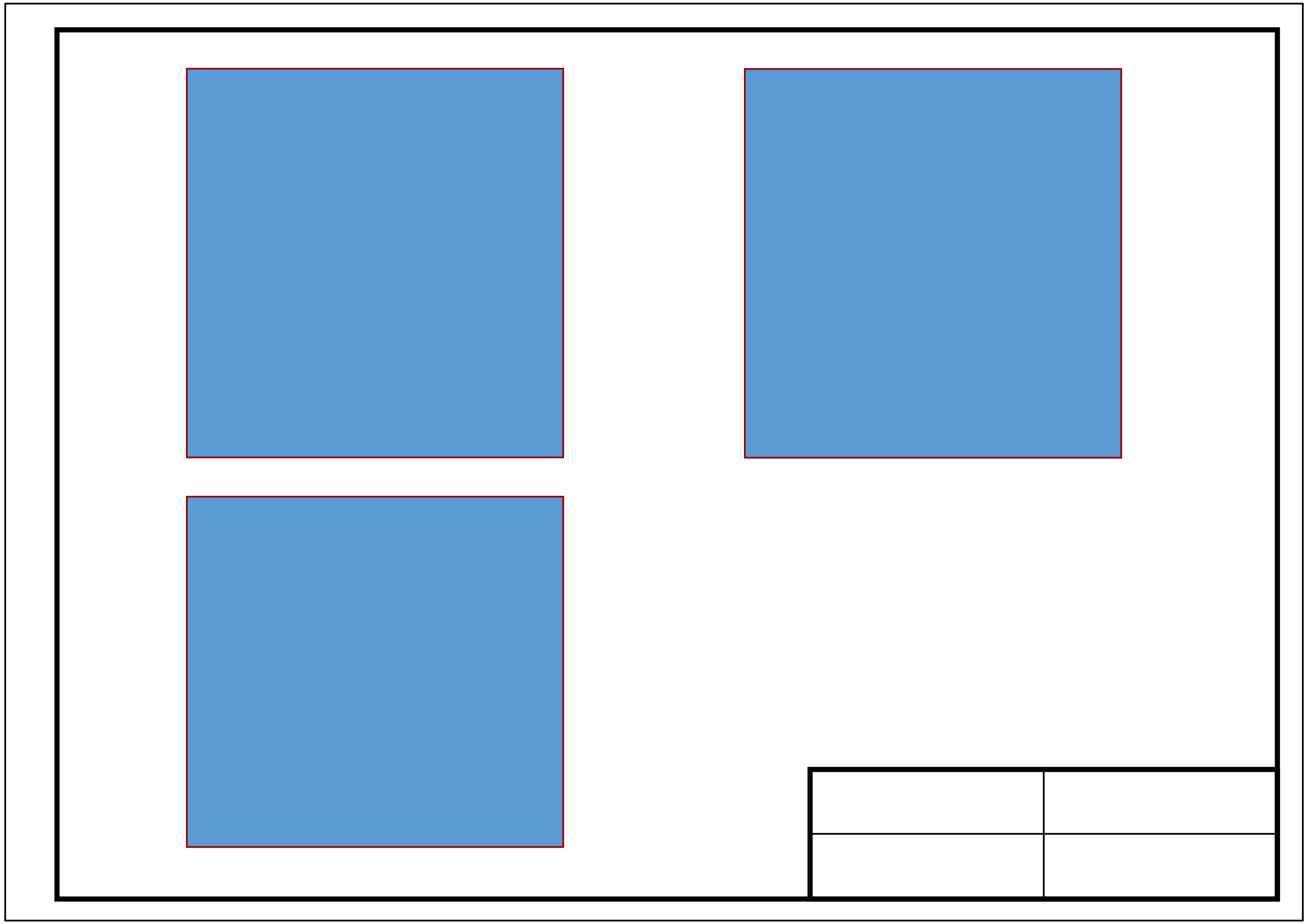


D

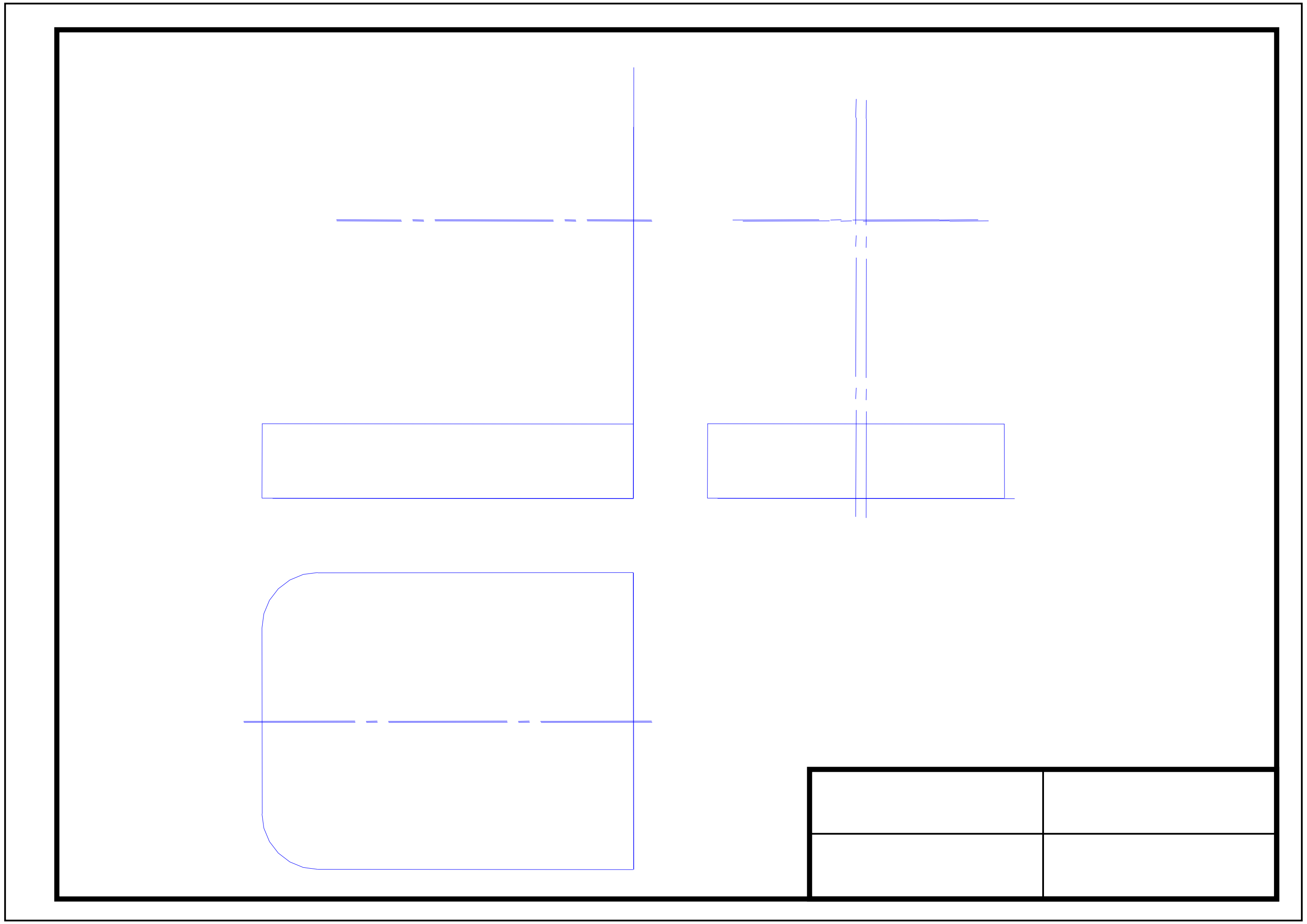


C

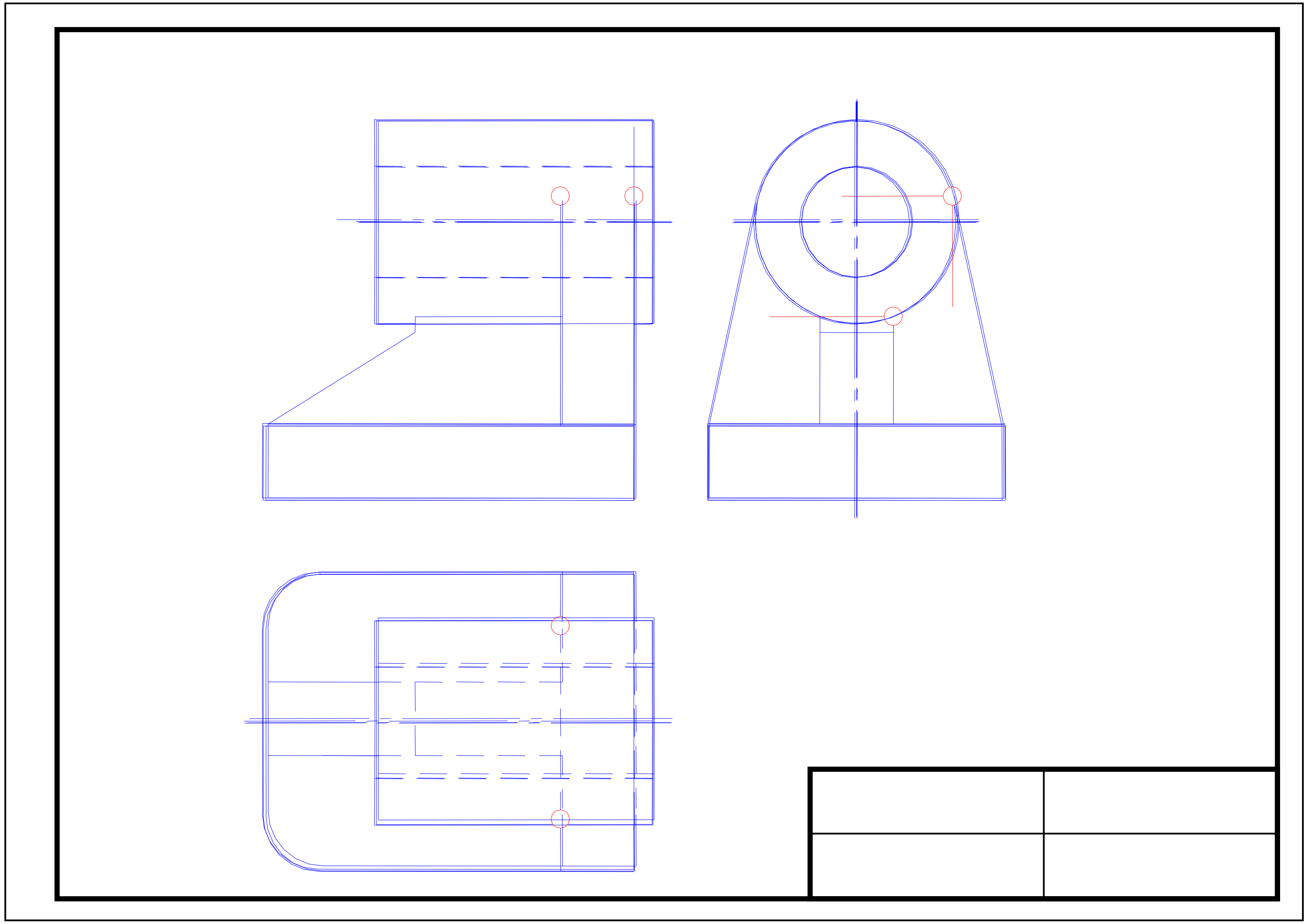
图纸



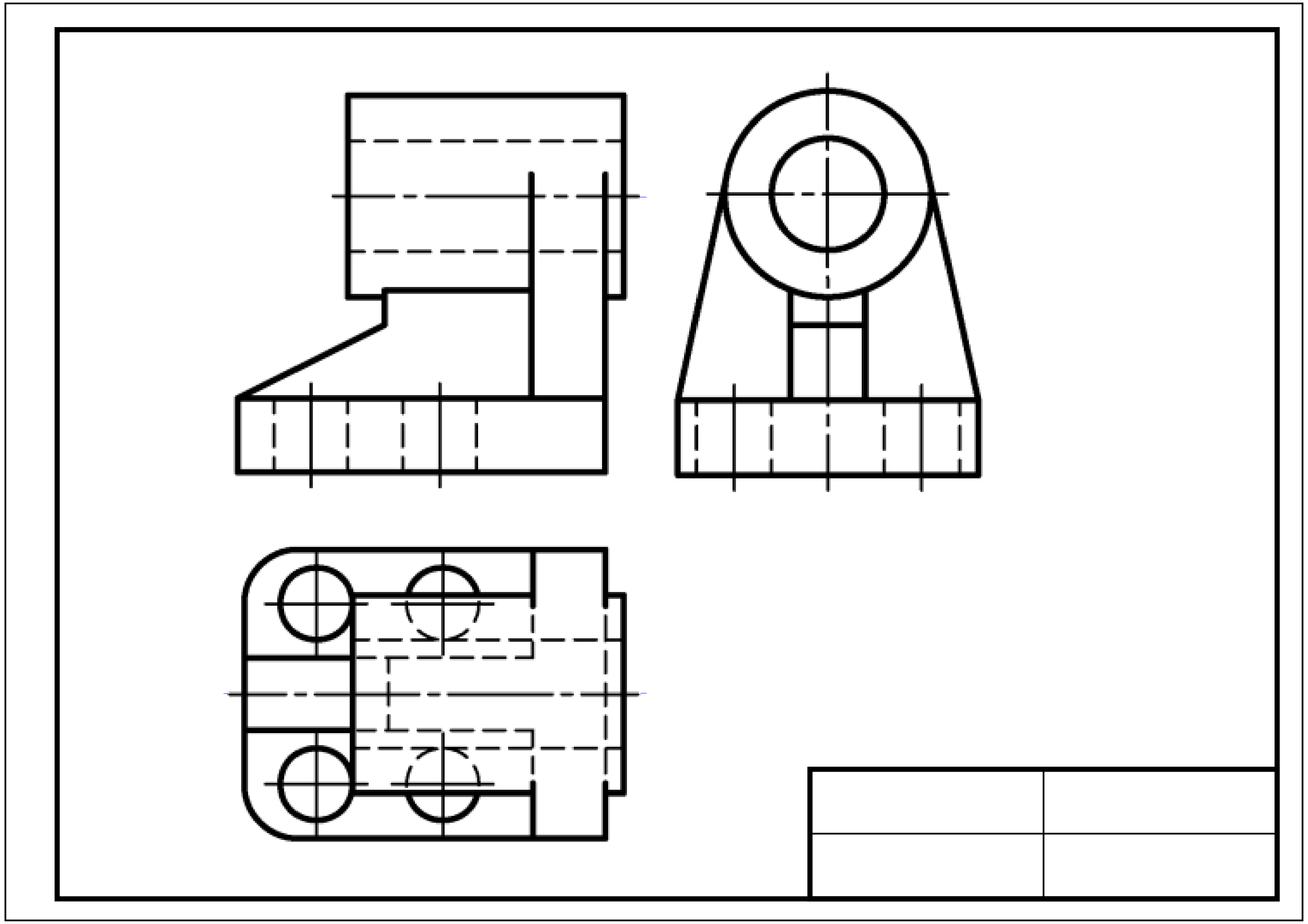
布置



绘图

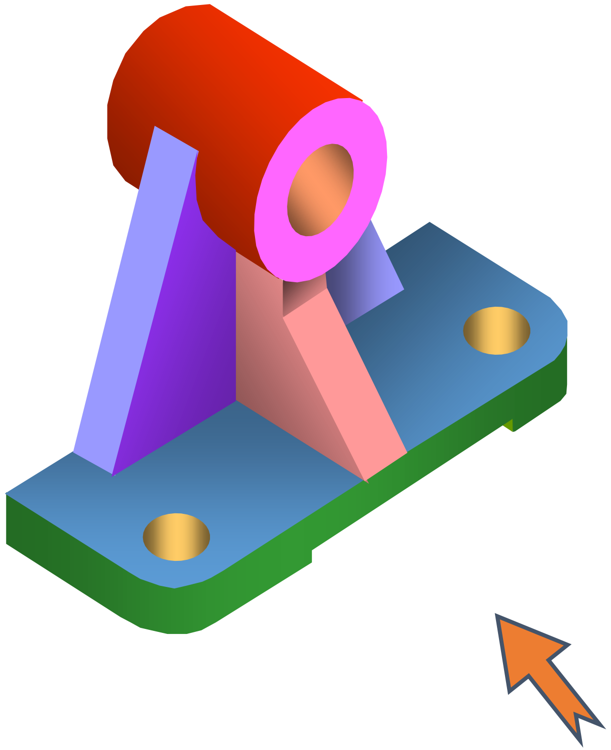


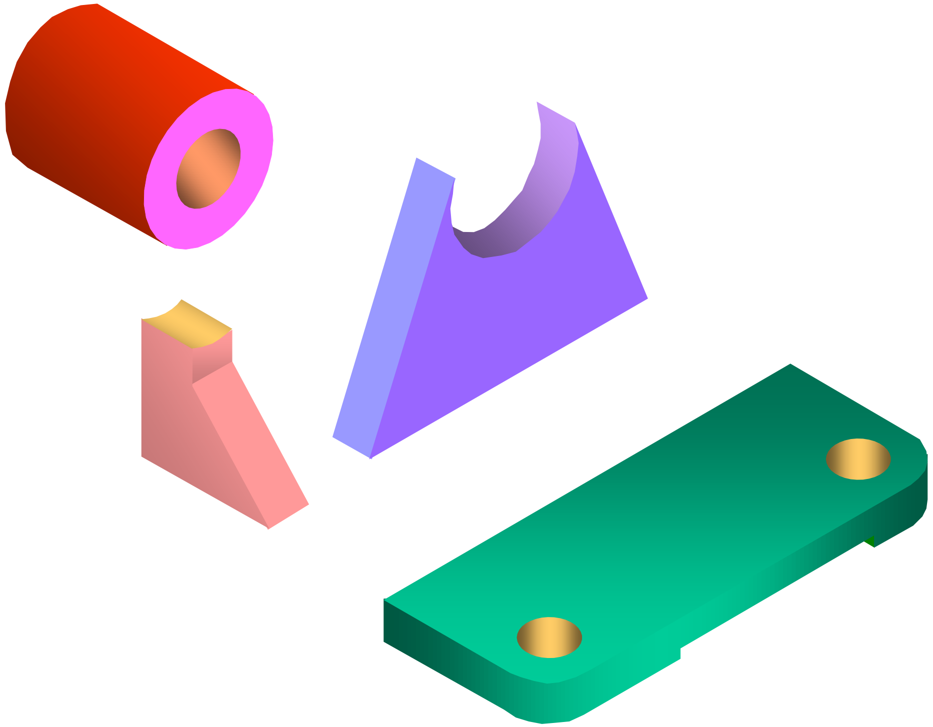
检查



【例】

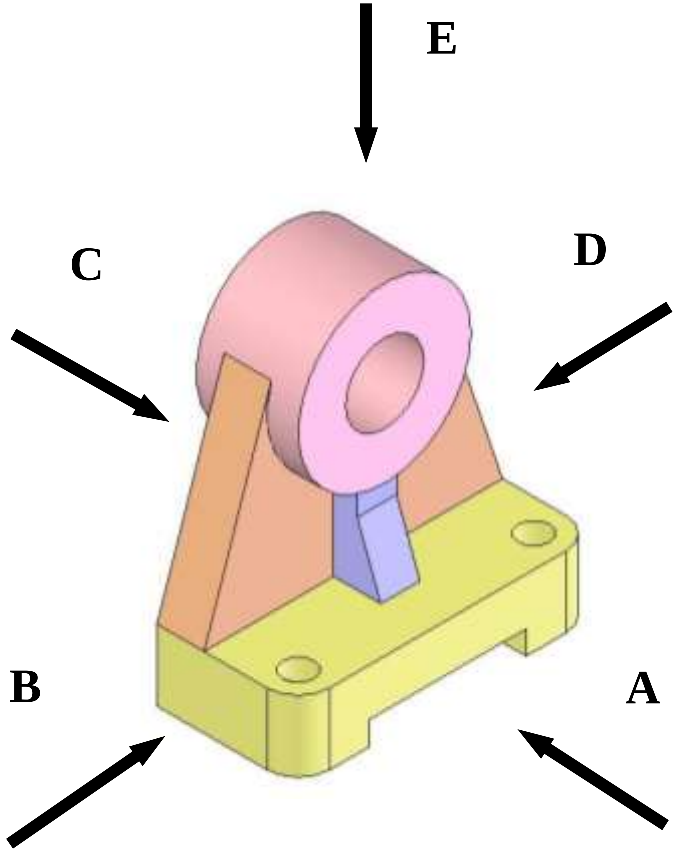
支架的三视图画法。

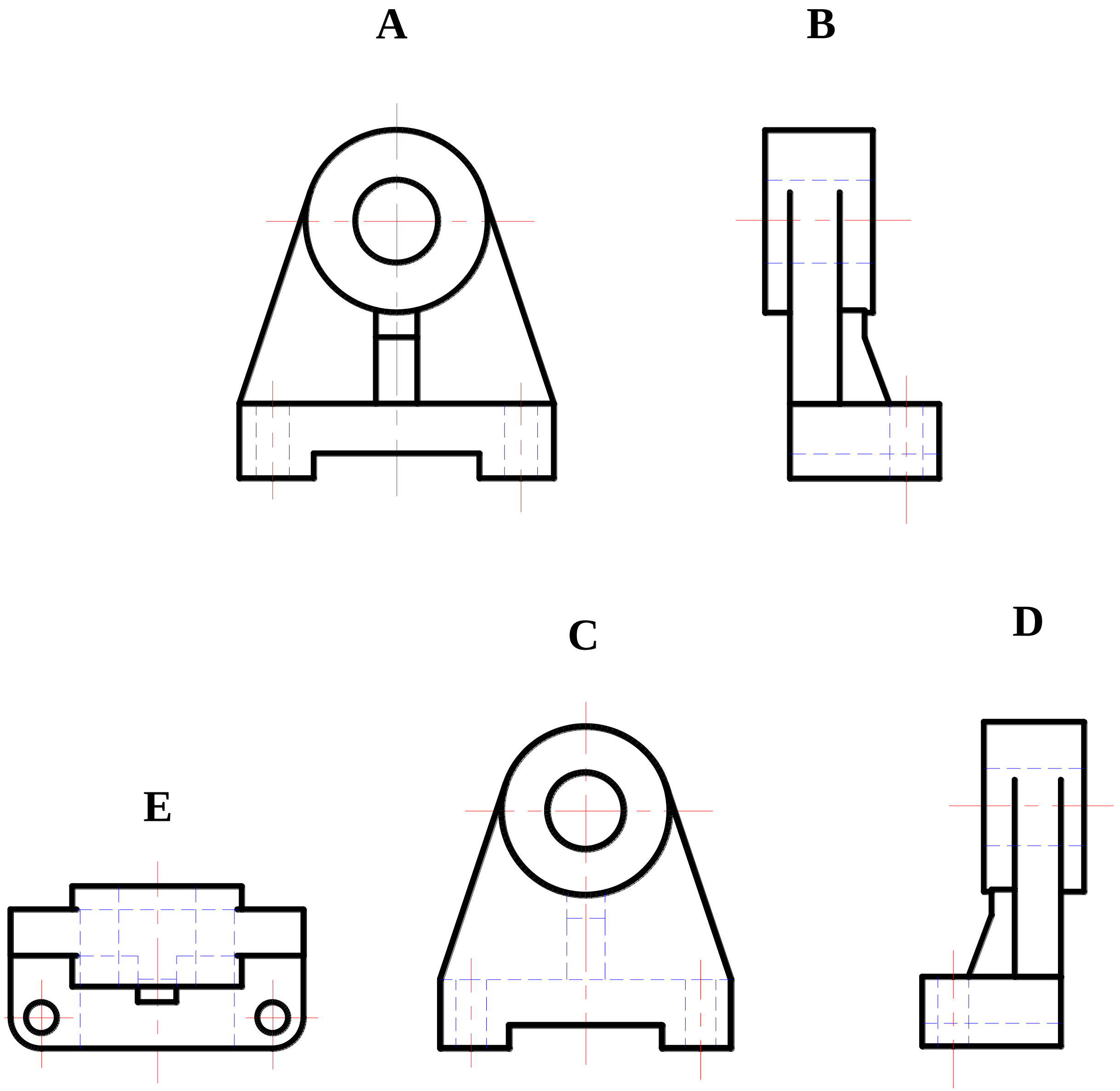




【例】

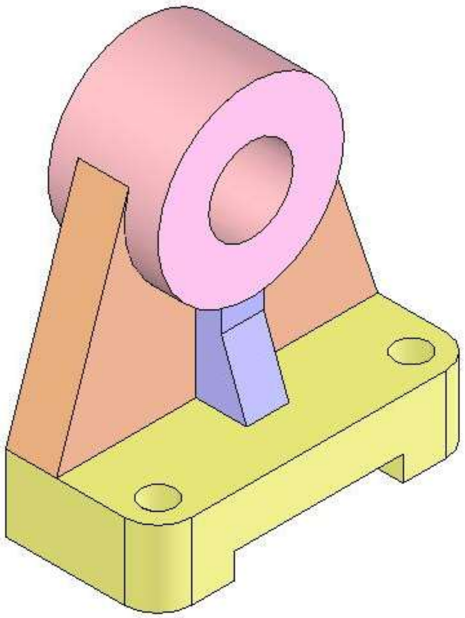
支架的三视图画法。

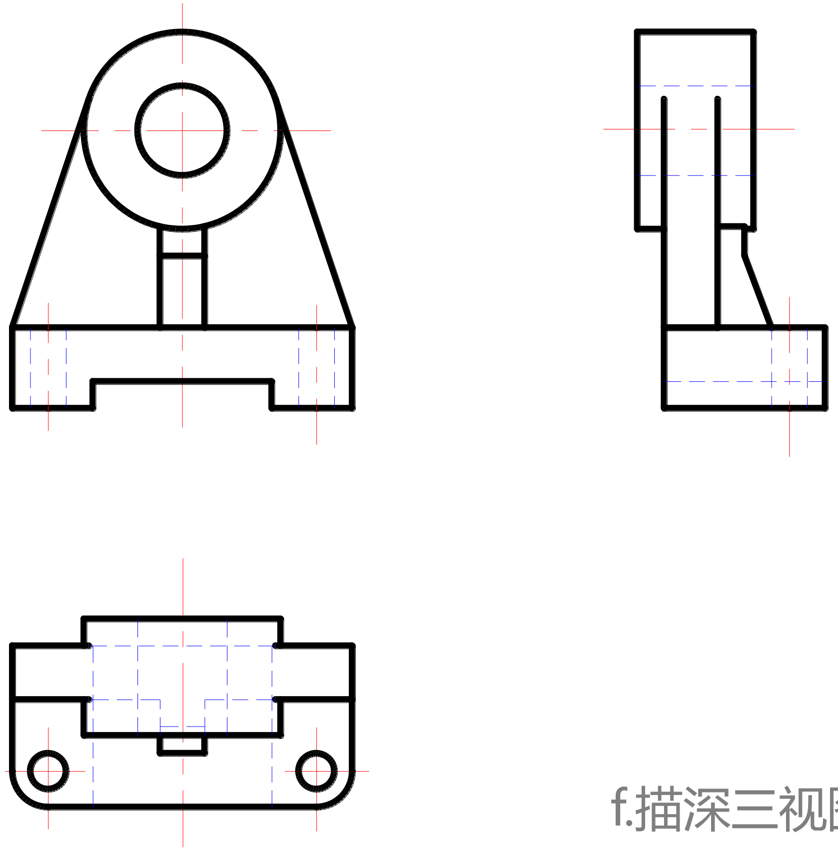




【例】

支架的三视图画法。

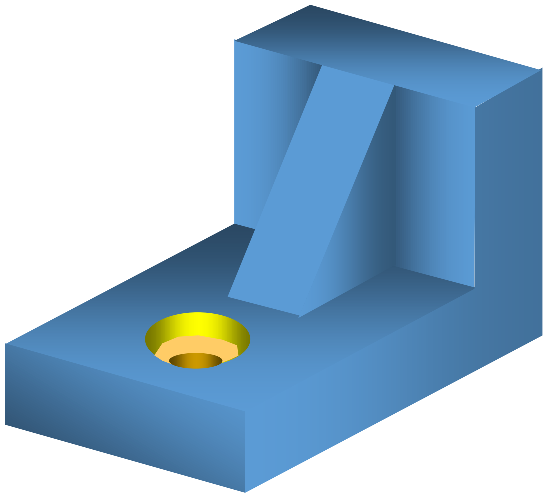


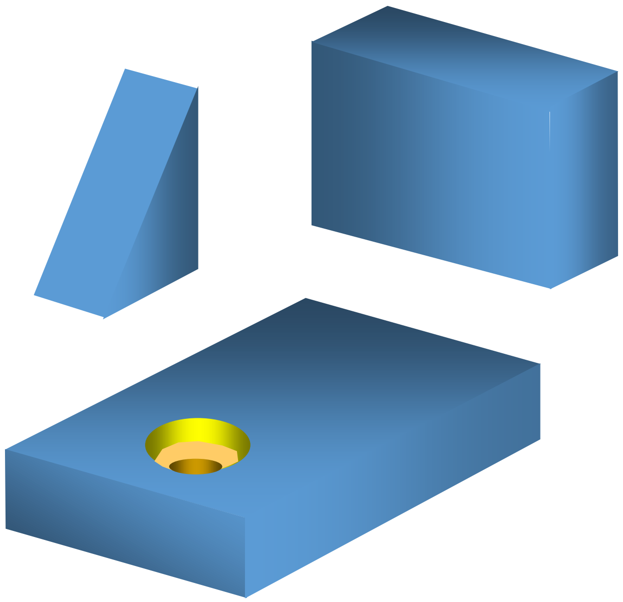


f.描深三视图

【例】

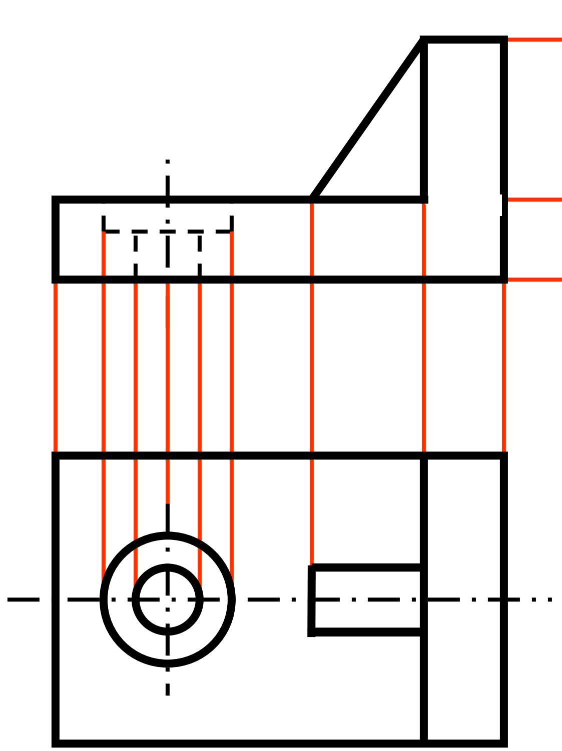
支架的三视图画法。

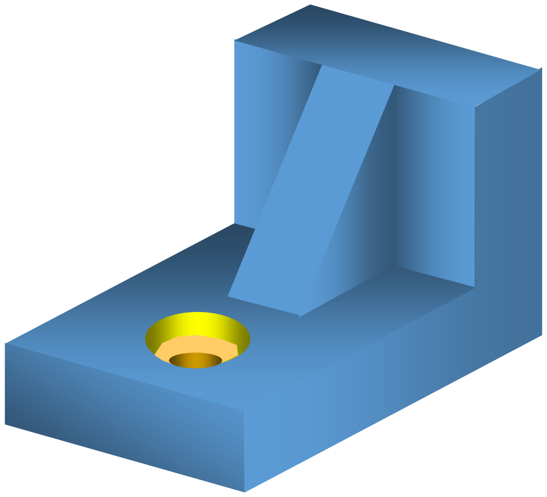




【例】

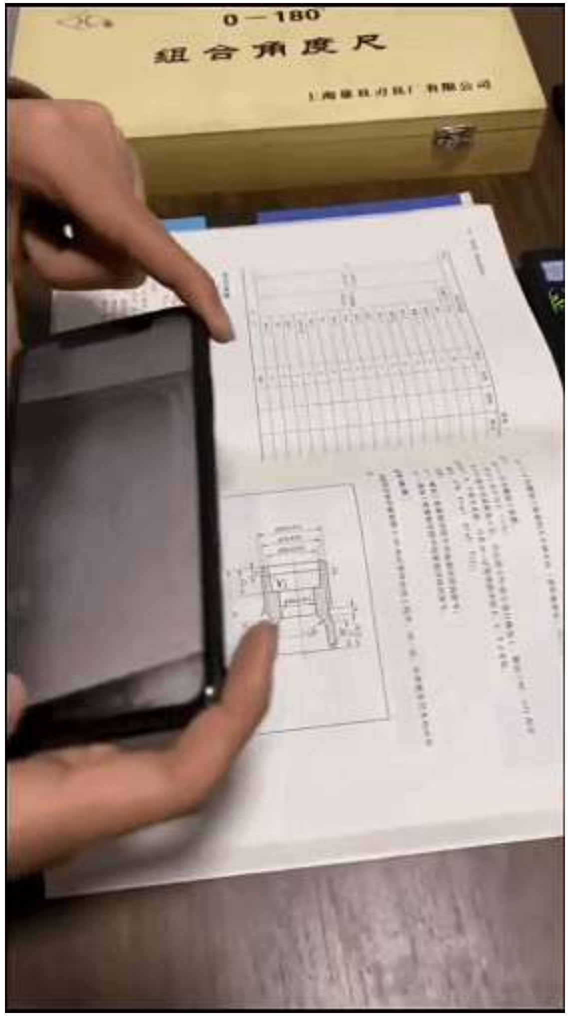
支架的三视图画法。



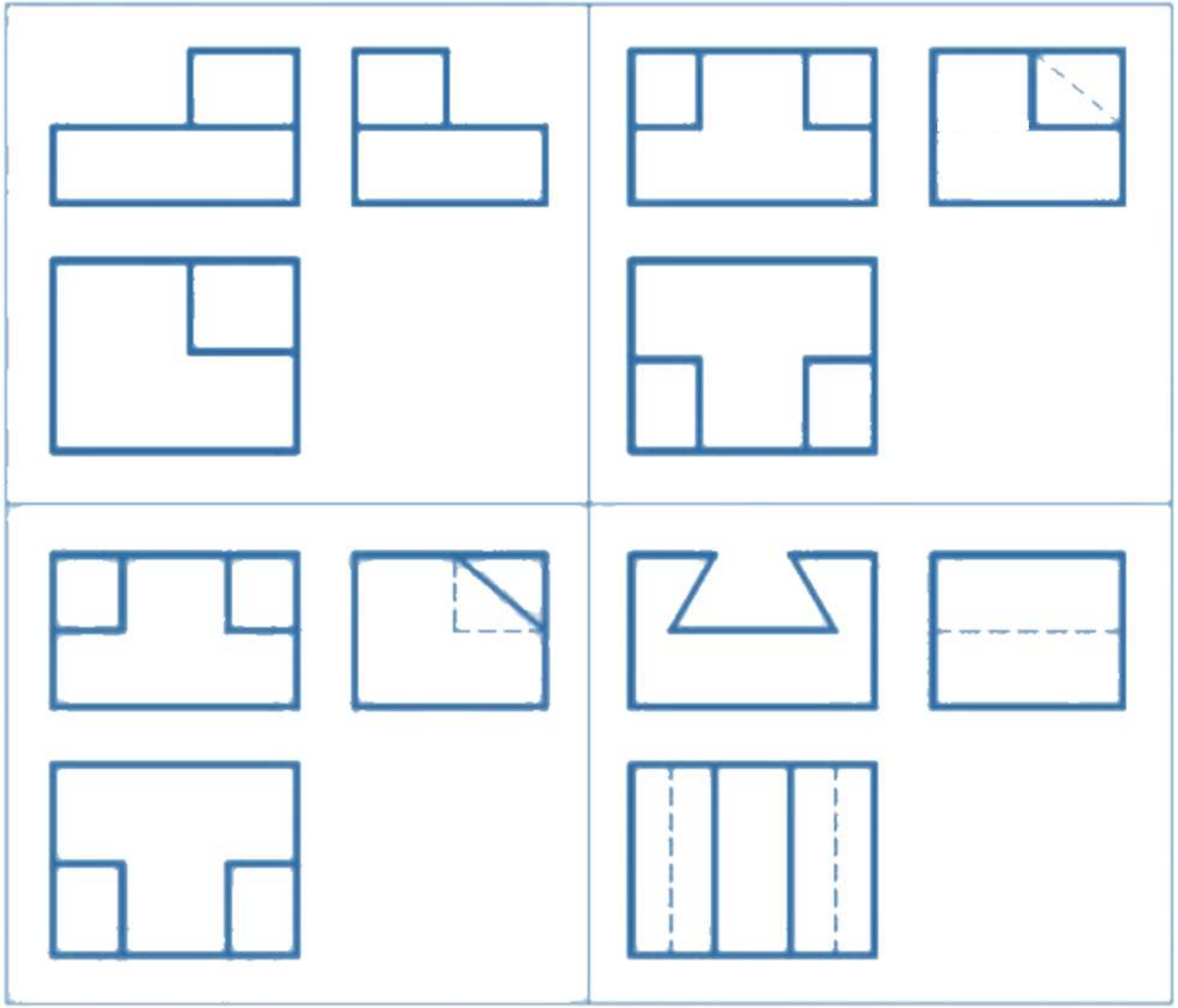


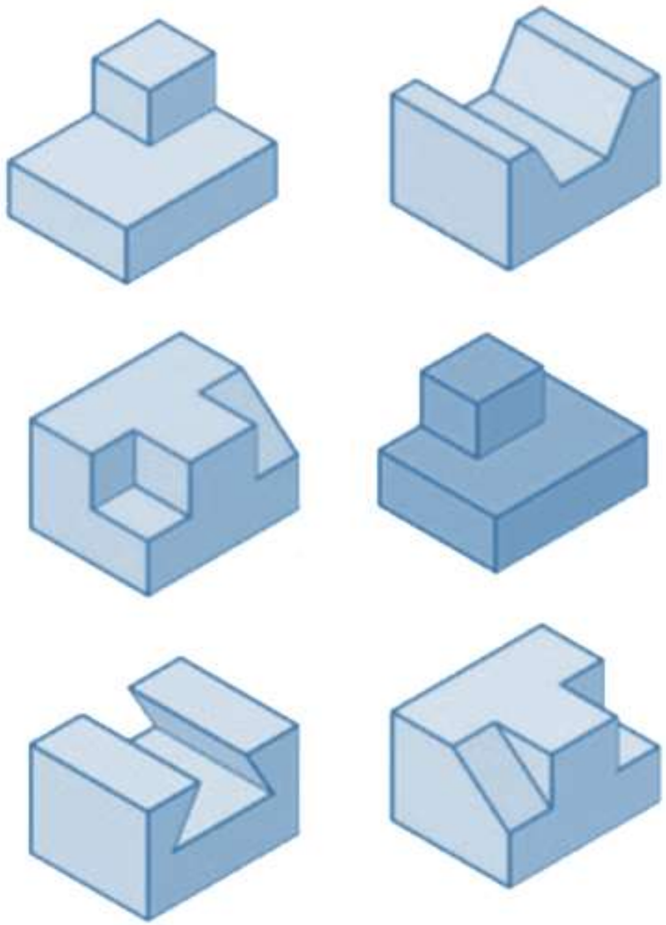
■ 底板 ■ 立板 ■ 肋板

读图



#【例】

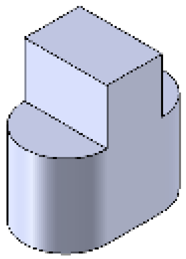


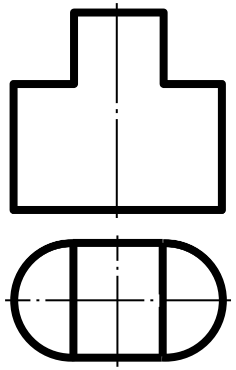


画图

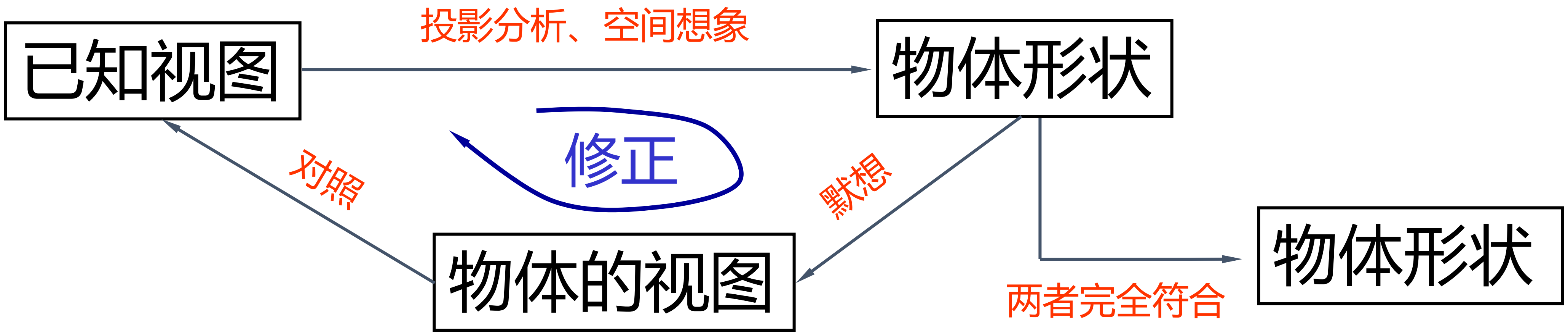
实体　　正投影原理　　视图

读图





读图



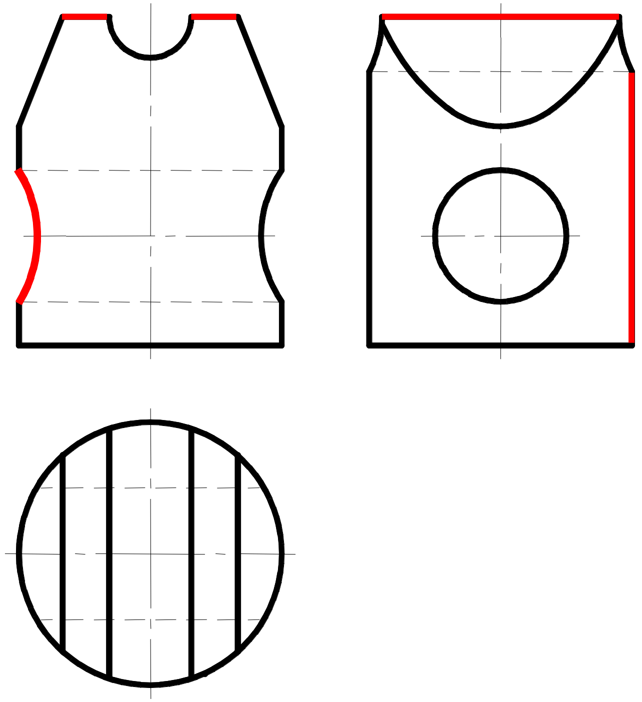
读图

- ✓ 从主视图入手
  - 最能反映物体的形状特征
  - 反映物体的工作位置
  - 反映物体的加工位置

- ✓ 结合其他视图
  - 一个视图不能完全表达物体的结构
  - 剖视、其他视图表达方法

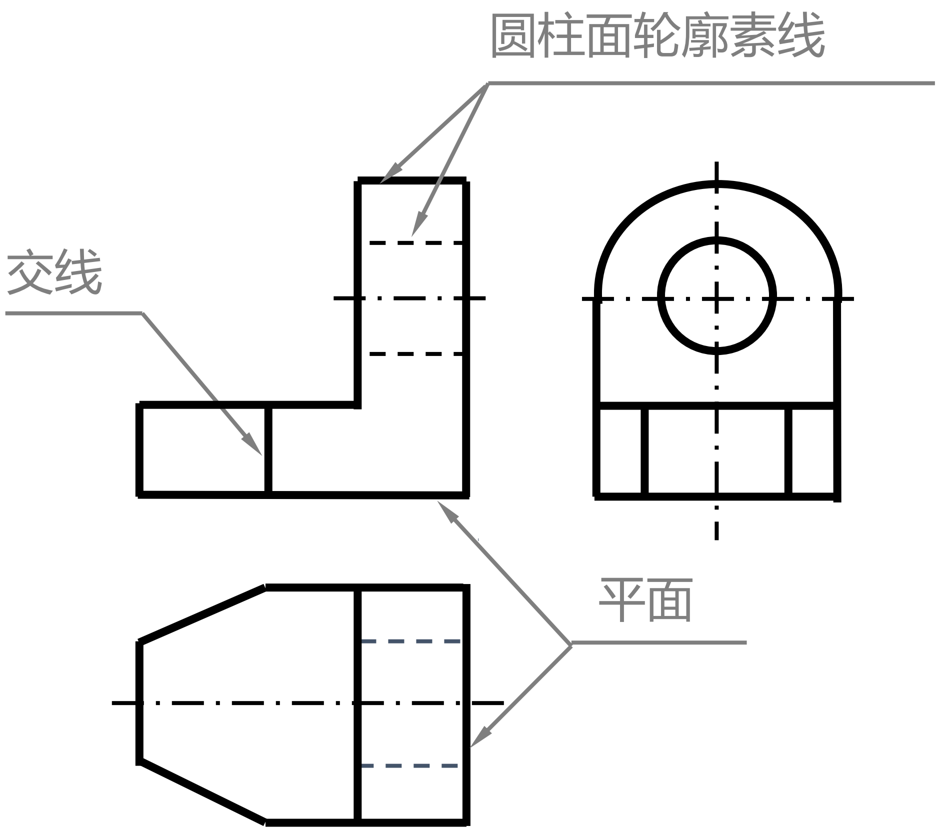
图线的含义

- ✓ 表面的积聚性投影
- ✓ 表面交线的投影
- ✓ 回转面轮廓素线的投影

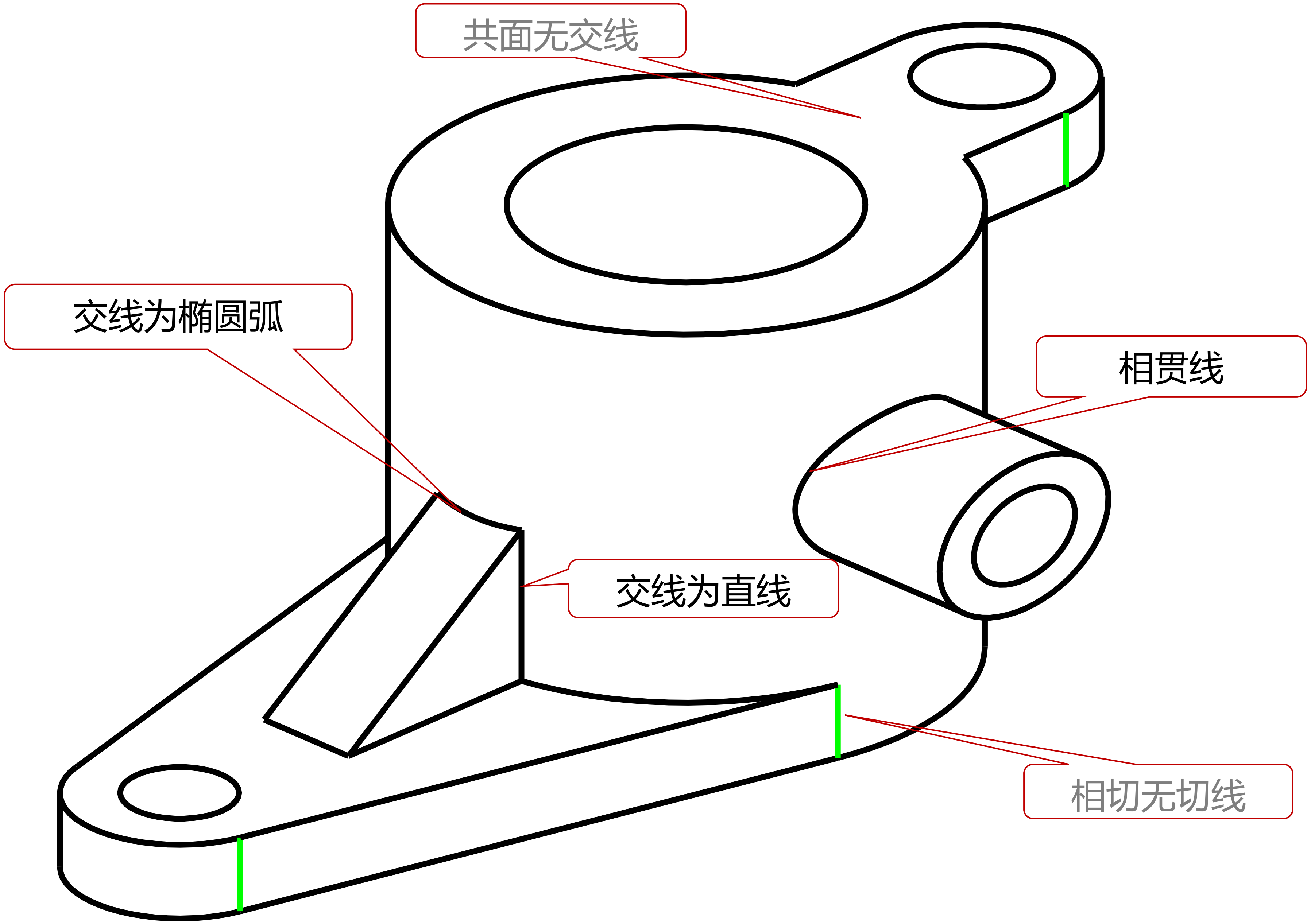


轮廓线

- ✓ 面的积聚投影
- ✓ 面与面的交线
- ✓ 回转面轮廓素线的投影

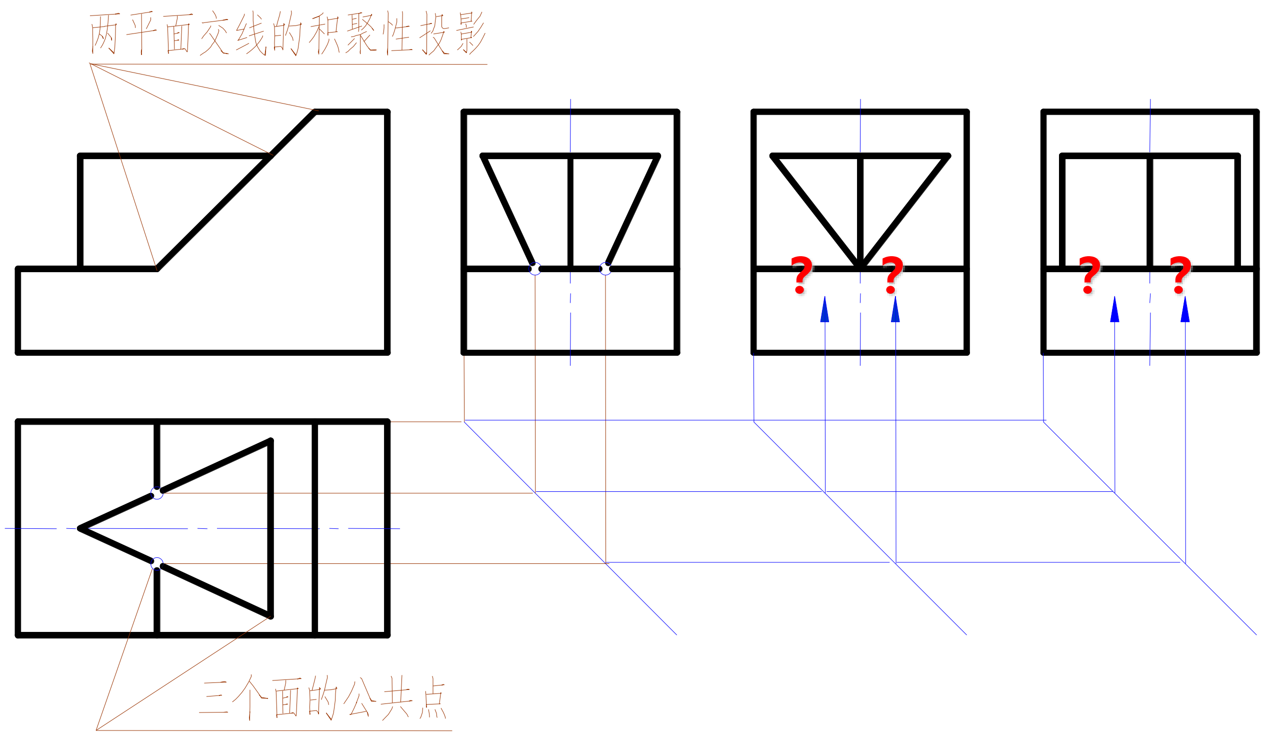


轮廓线



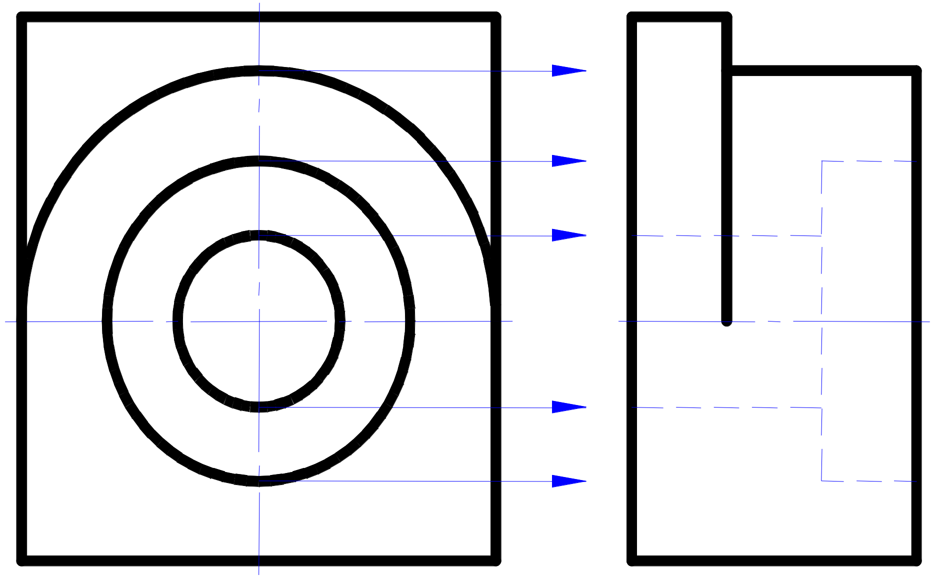
有效交点

✓ 已知视图上线段的有效交点一定可以在另外的视图上找到下落



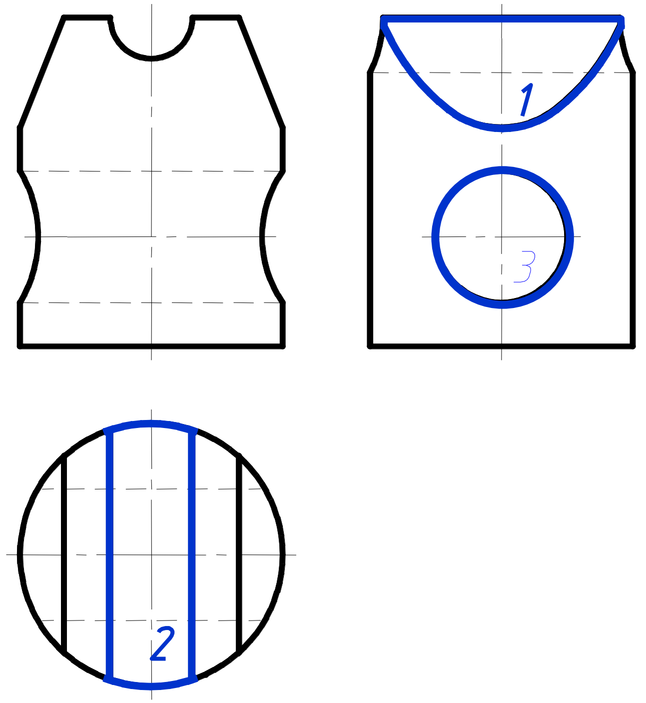
圆的象限点

- ✓ 视图上的圆往往是孔洞或圆柱的投影，因此圆的象限分界点所在的位置在另外的视图上一般应有界线素线的投影存在



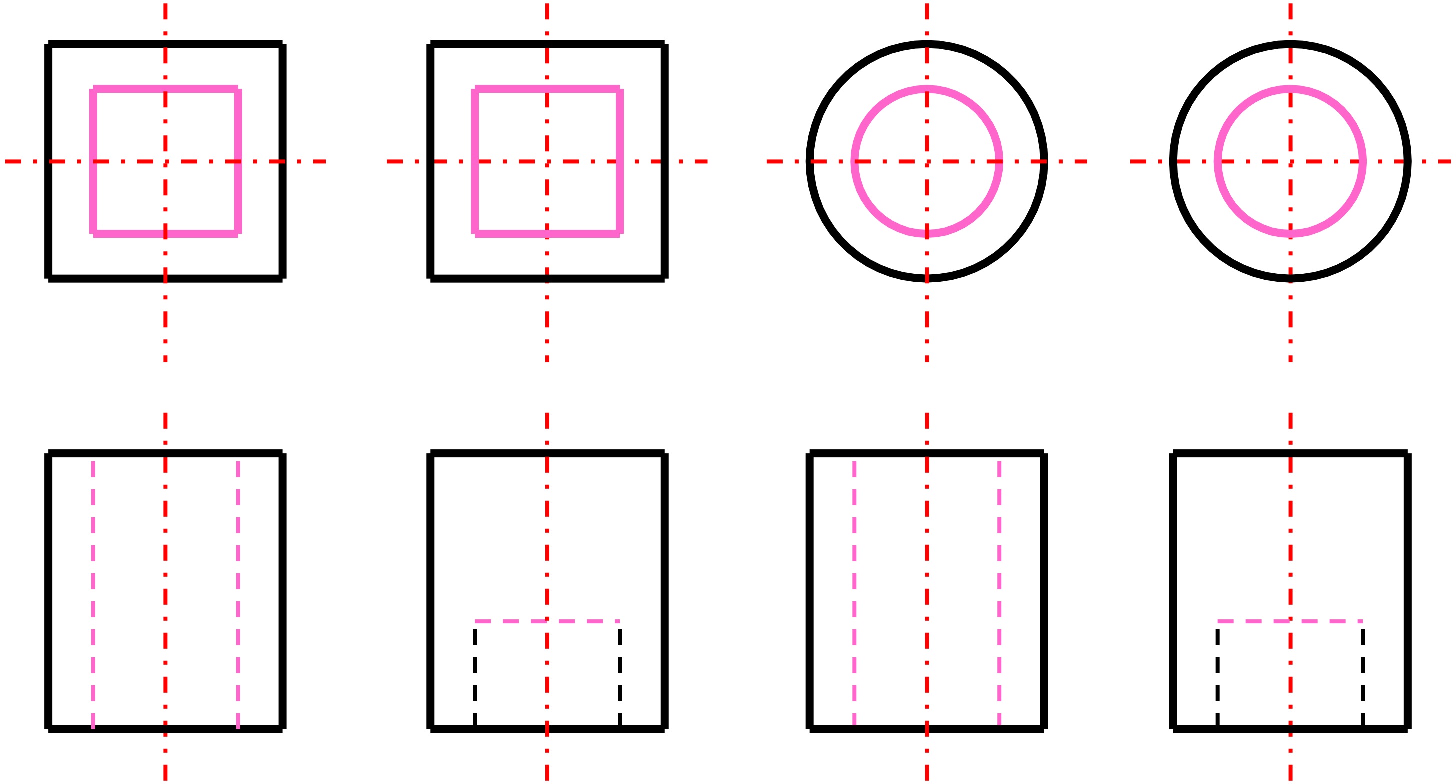
线框的含义

- ✓ 平面的投影
- ✓ 曲面的投影
- ✓ 孔的投影
- ✓ 光滑过渡表面的投影



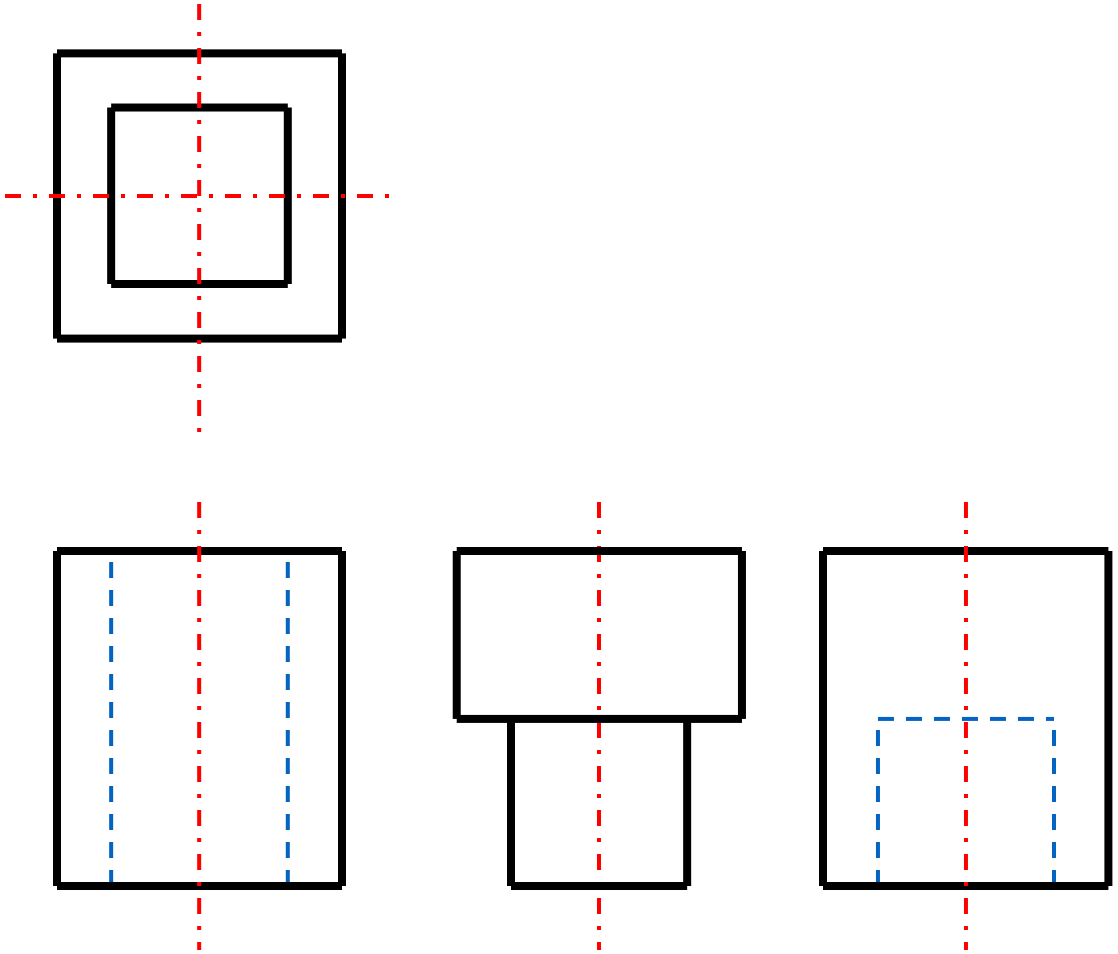
线框的含义

✓ 框中有框，不凸必凹

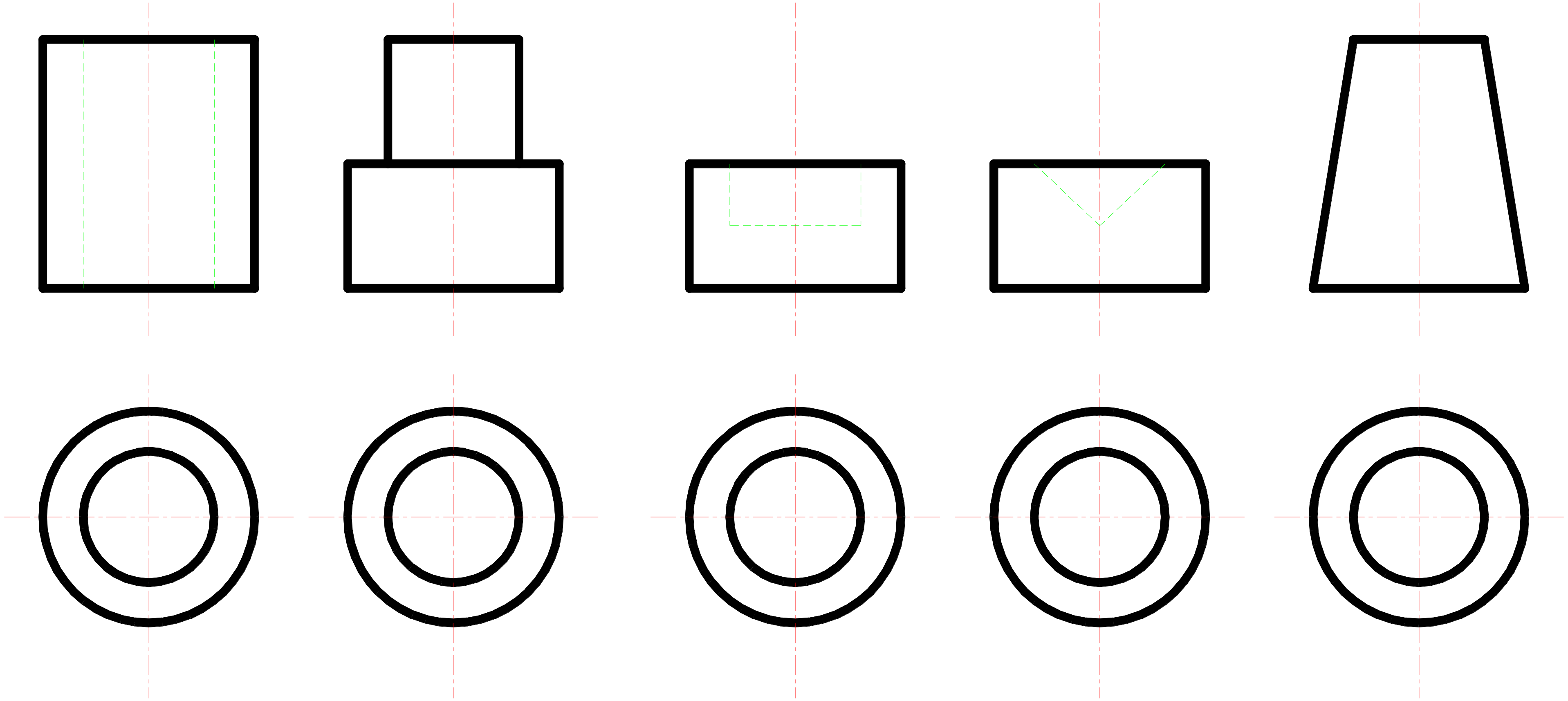


线框的含义

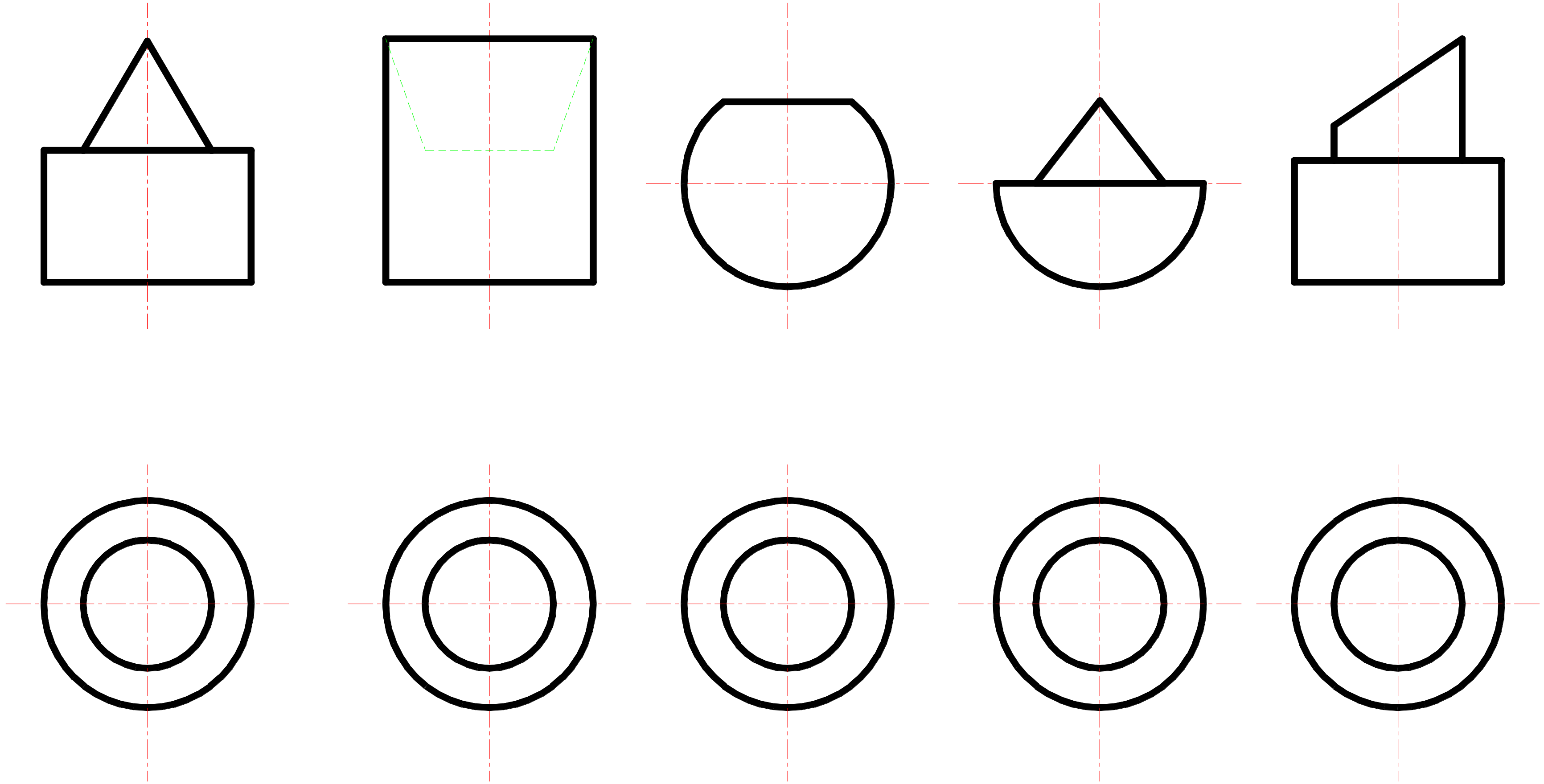
- ✓ 通孔
- ✓ 凸台
- ✓ 凹坑



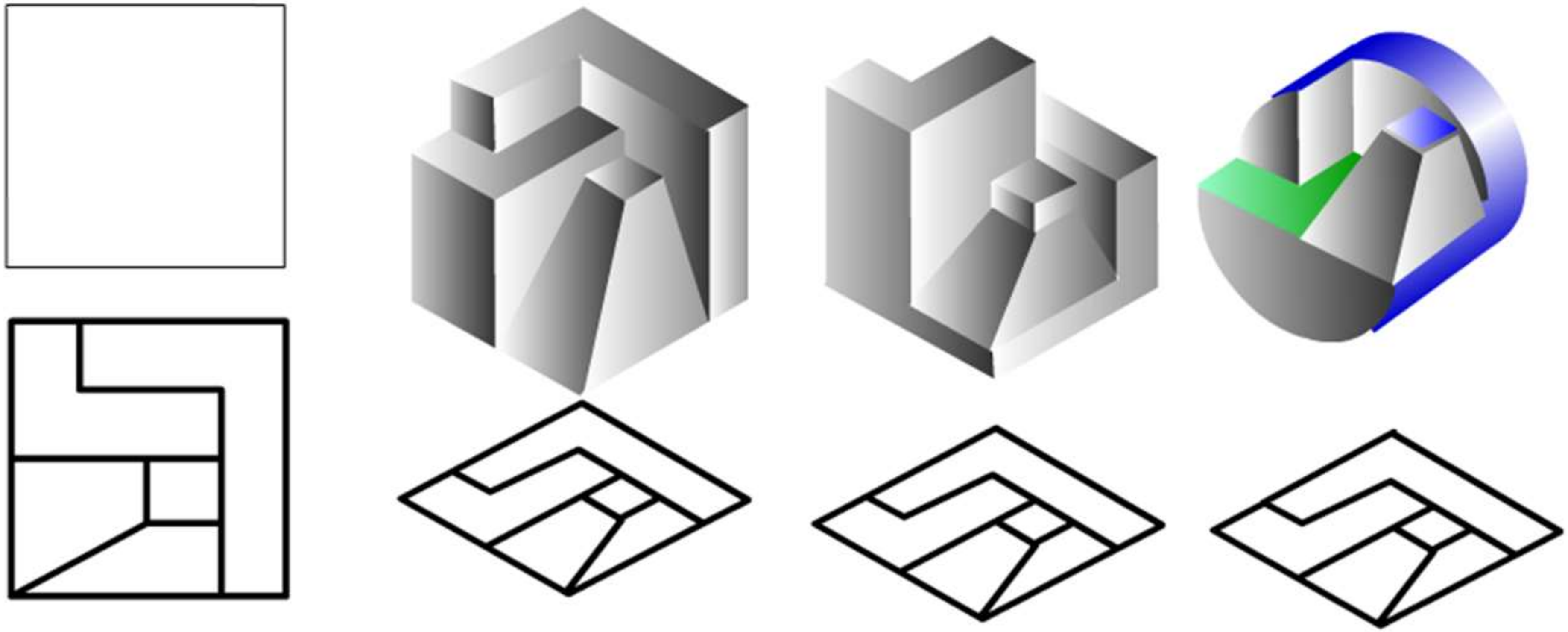
线框的含义



线框的含义

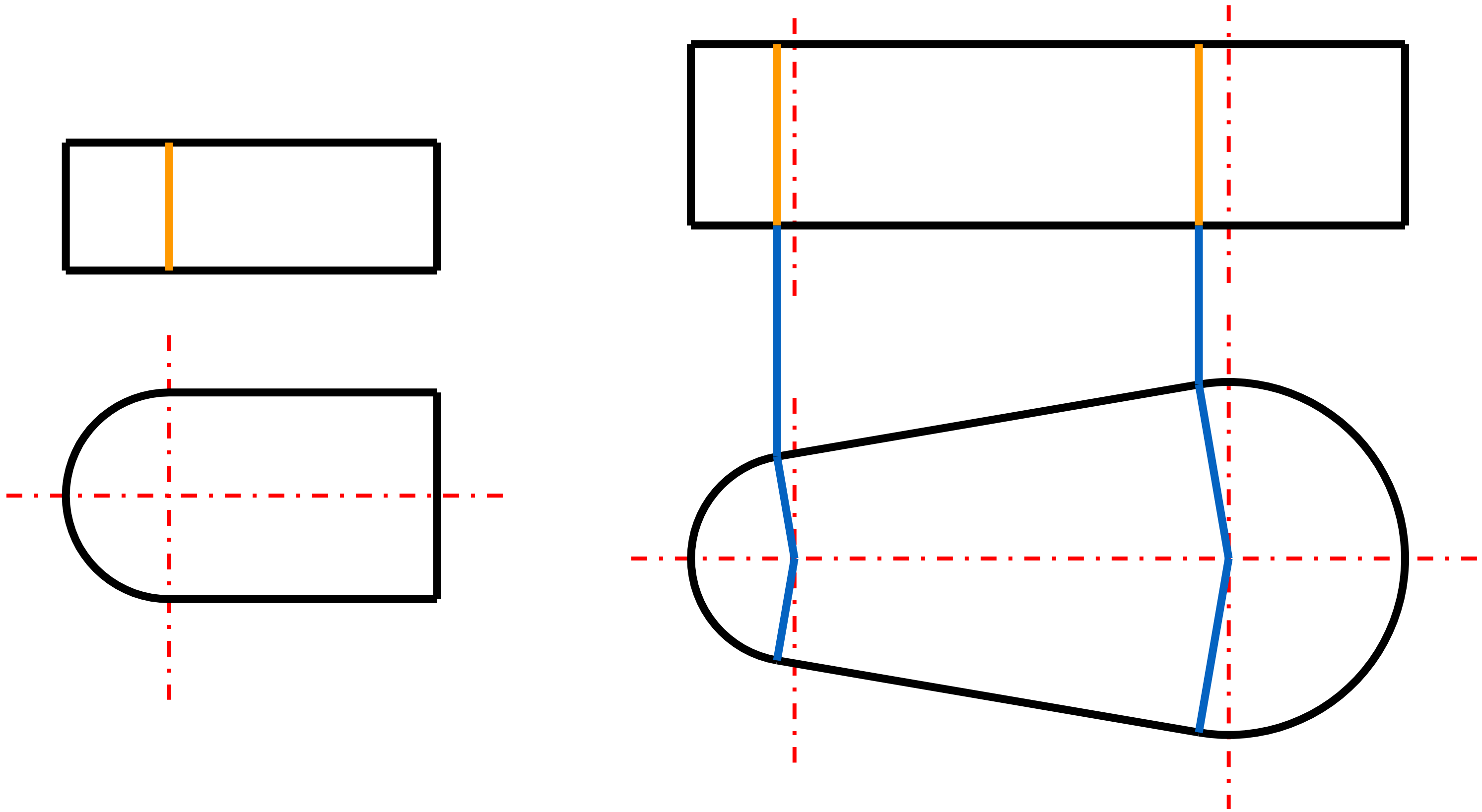


线框的含义



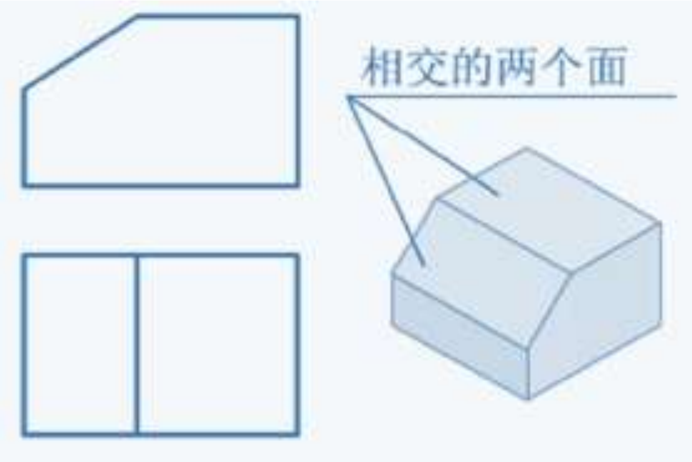
线框的含义

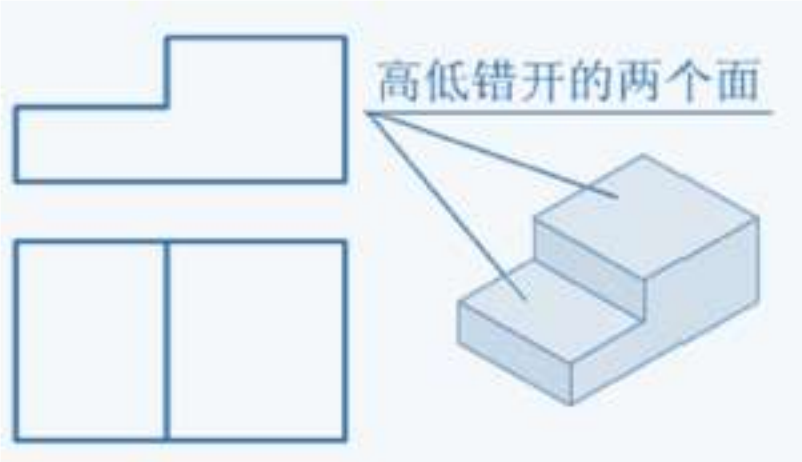
- ✓ 多个光滑过渡表面的投影



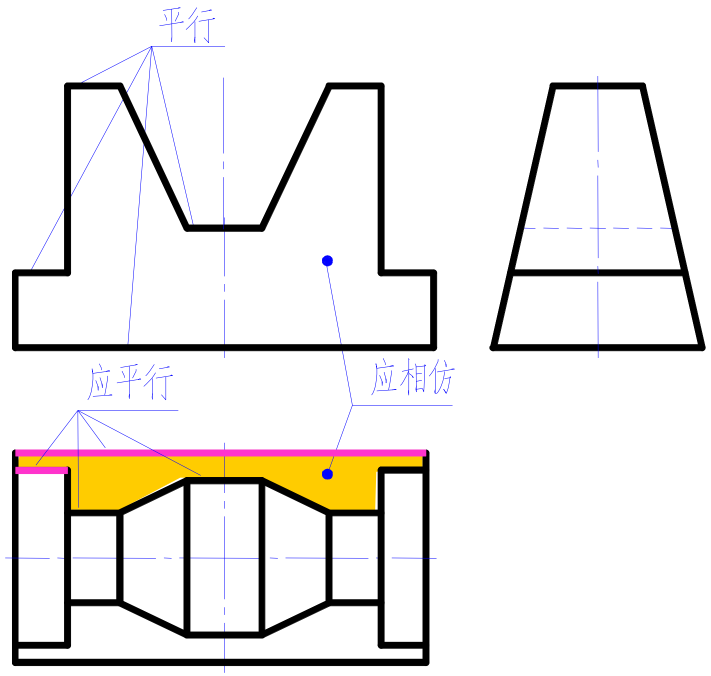
相邻两线框的含义

- ✓ 相交两个面
- ✓ 相隔一定距离的两个面

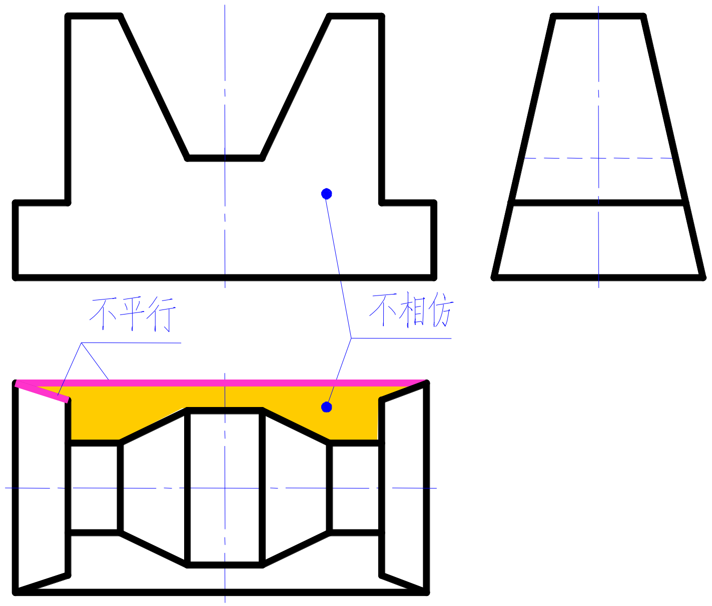




相仿性和平行关系

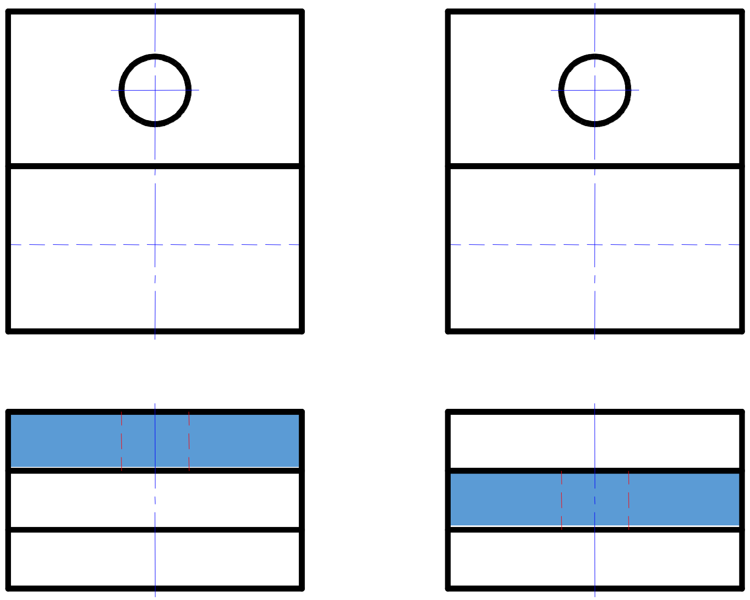


正确



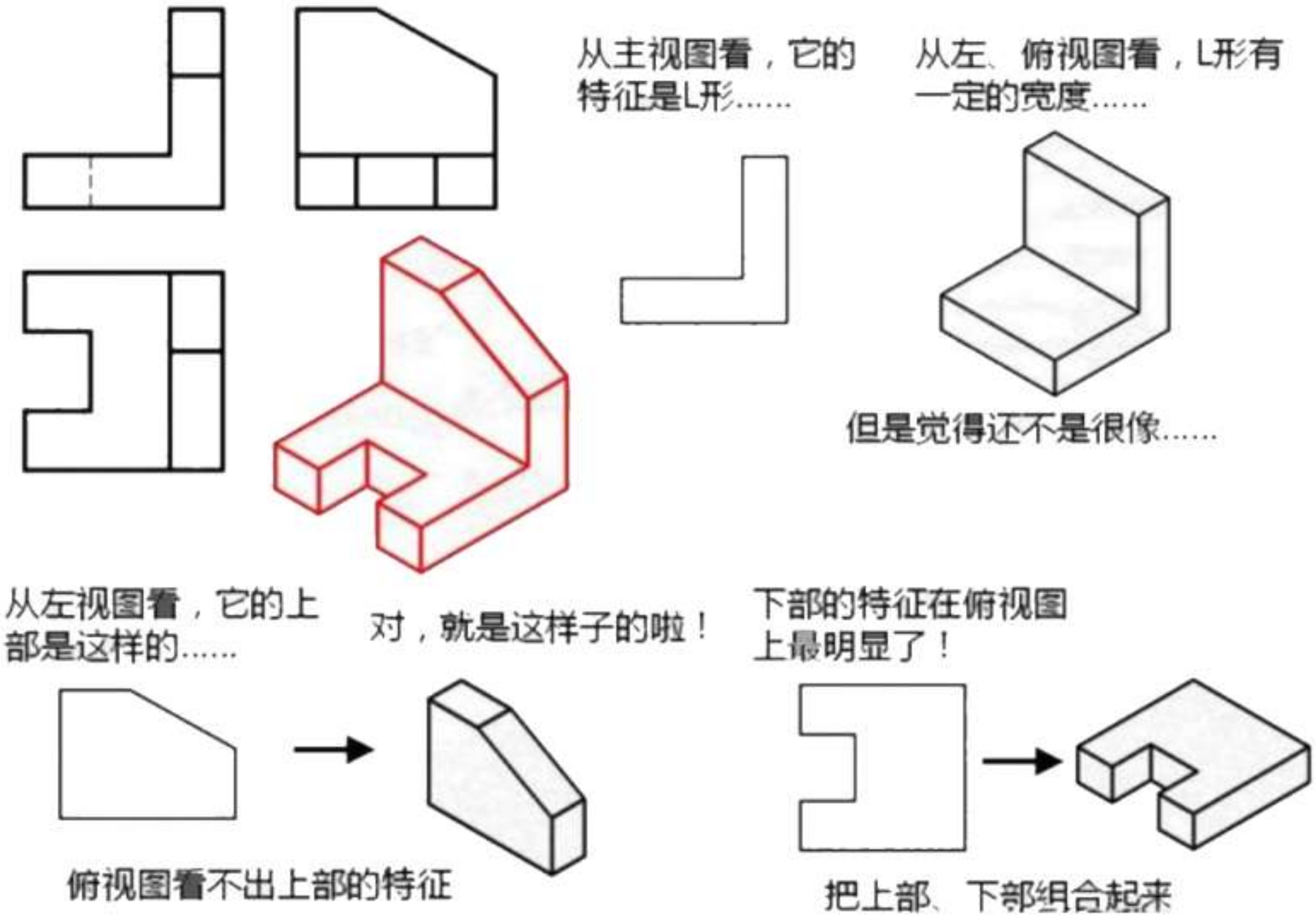
不正确

通孔与板厚



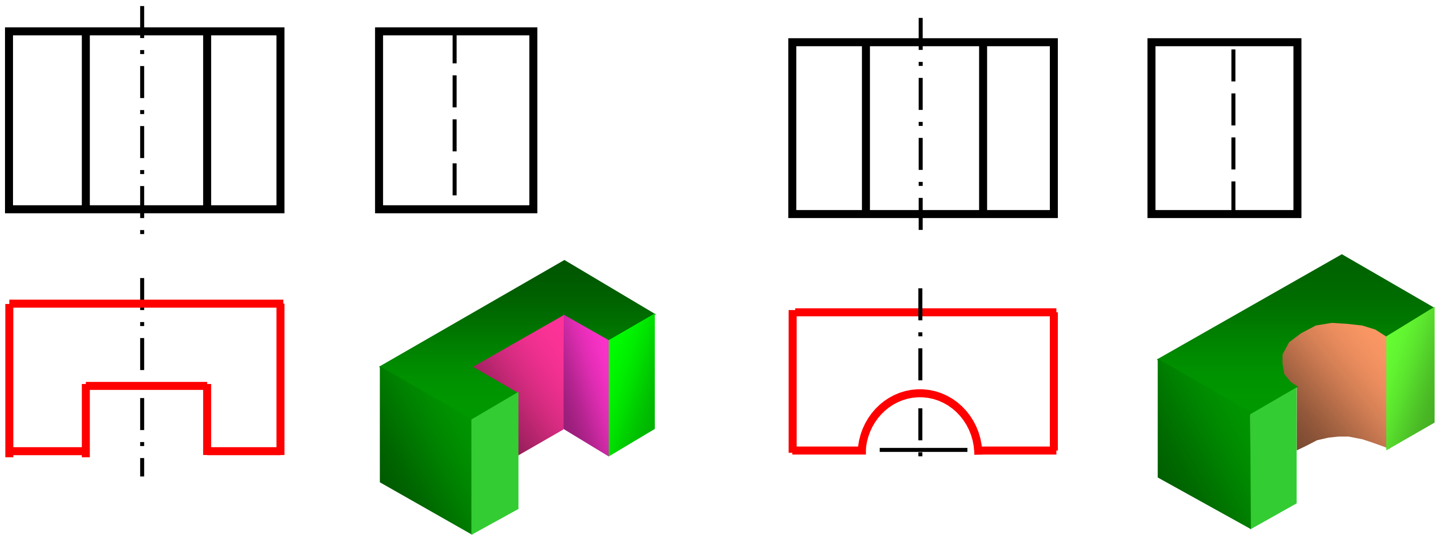
特征

- ✓ 最能反映物体特征的那个视图

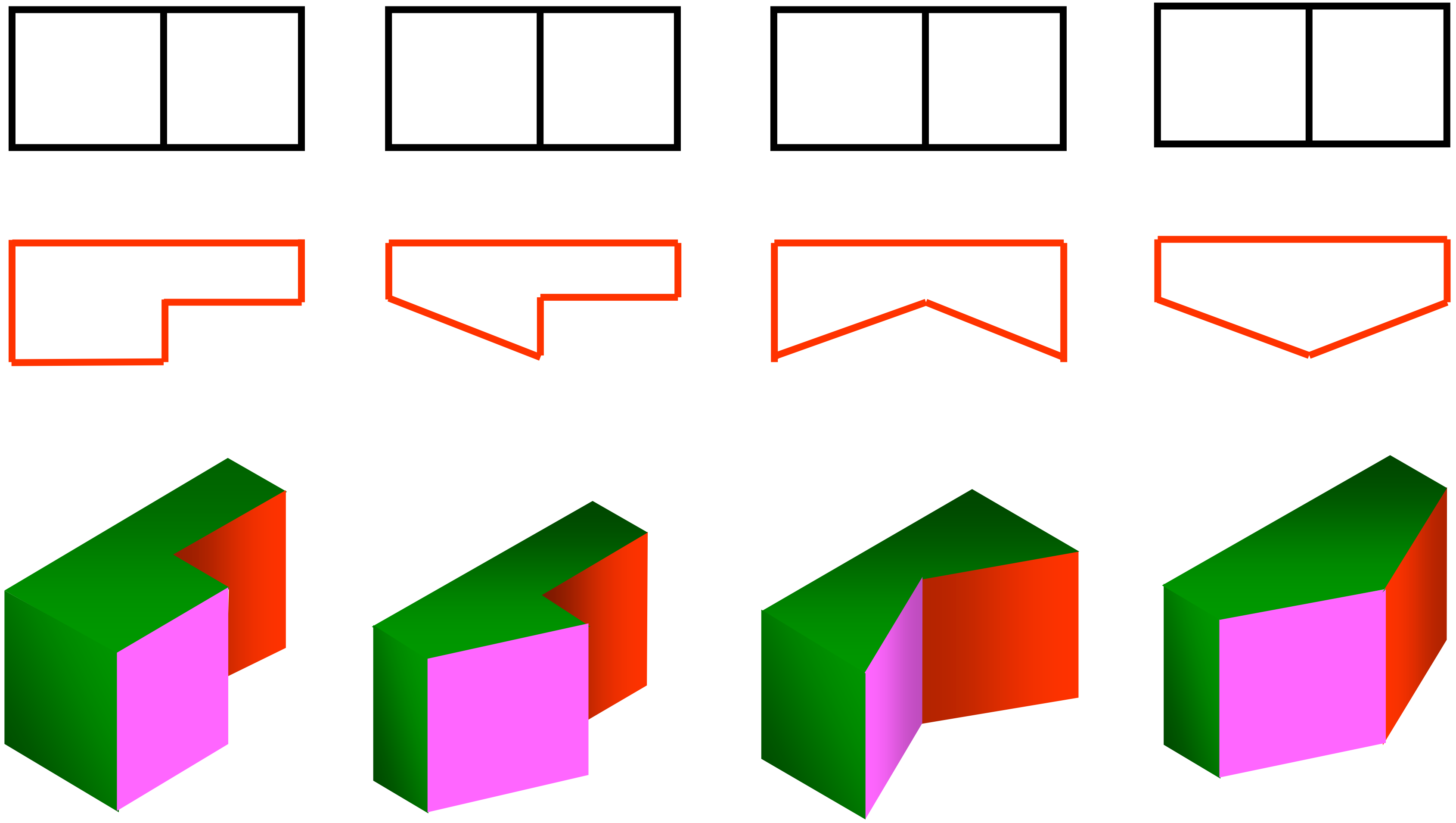


特征

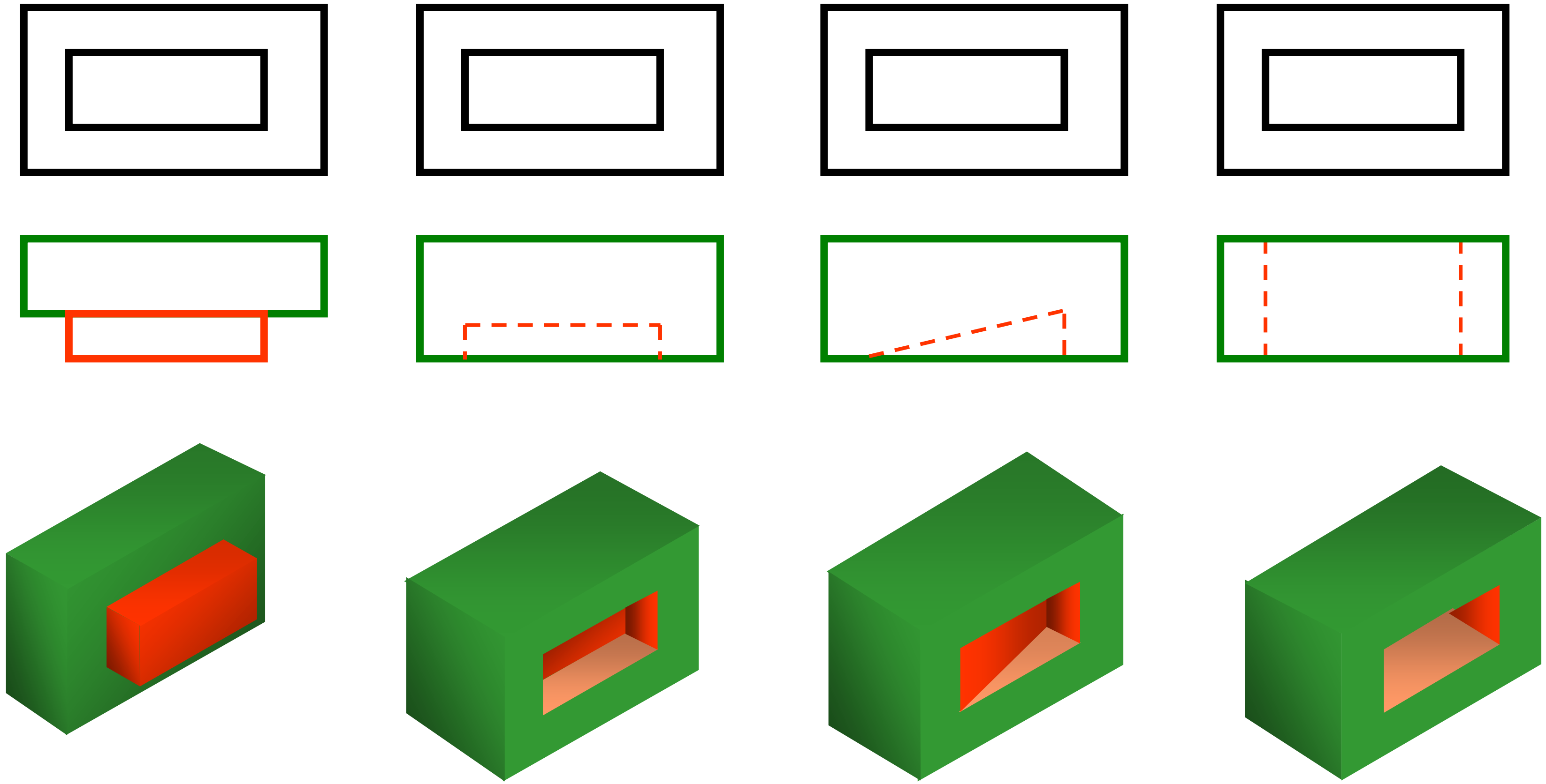
✓ 最能反映物体特征的那个视图



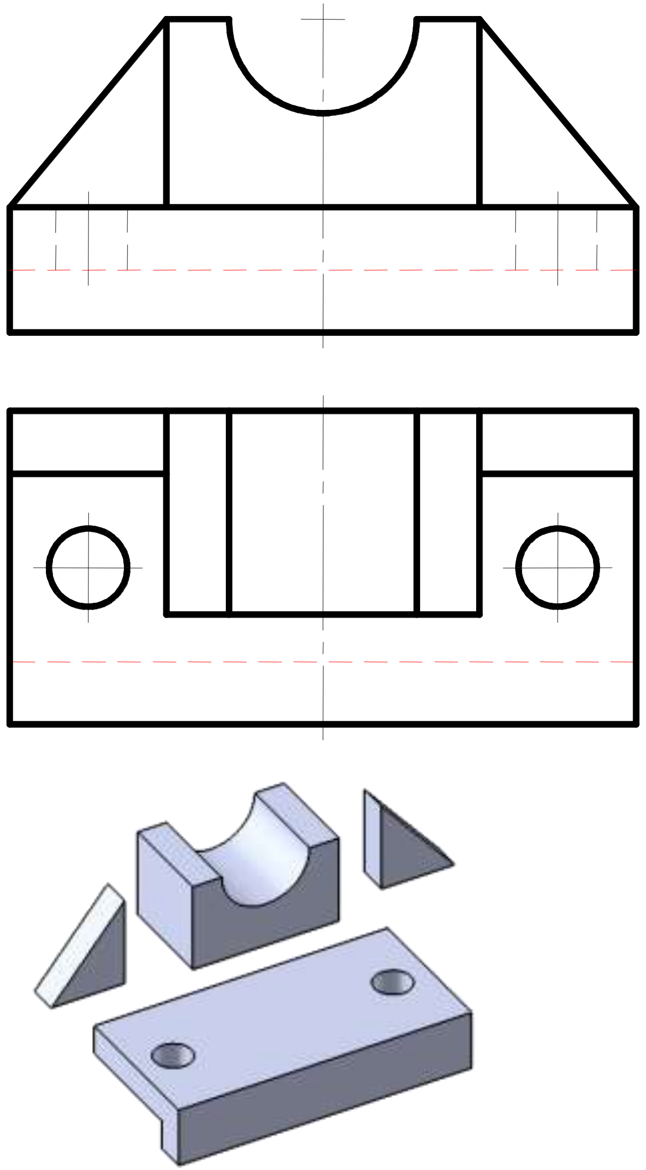
特征

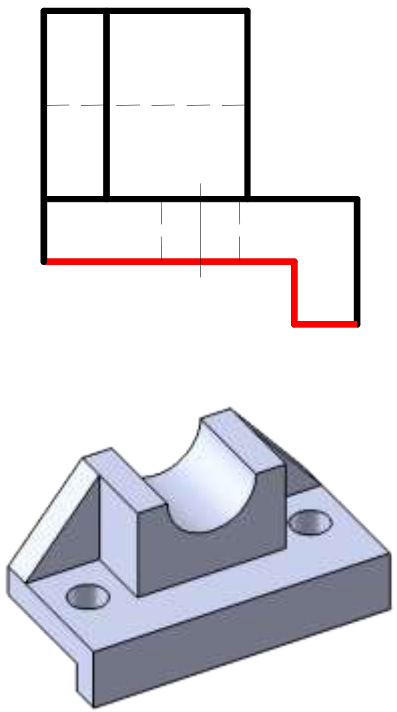


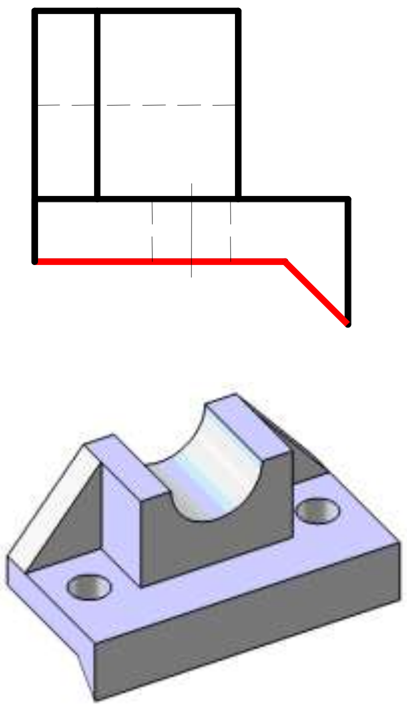
特征

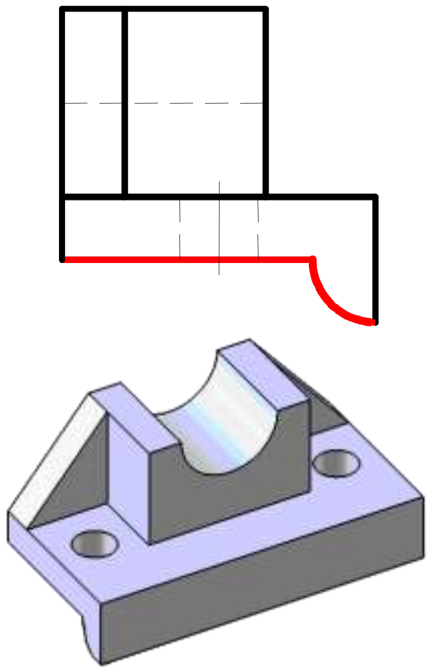


特征

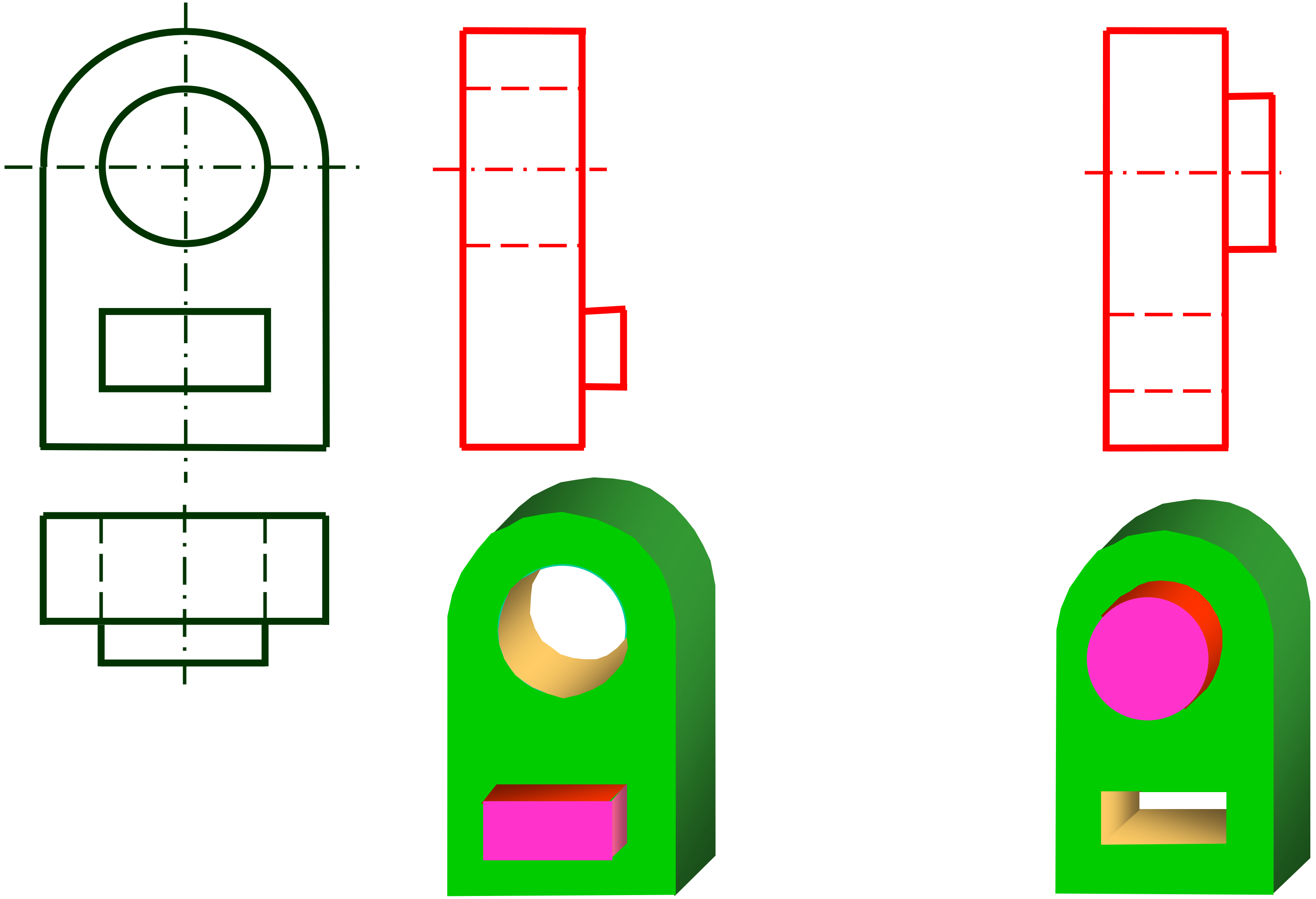




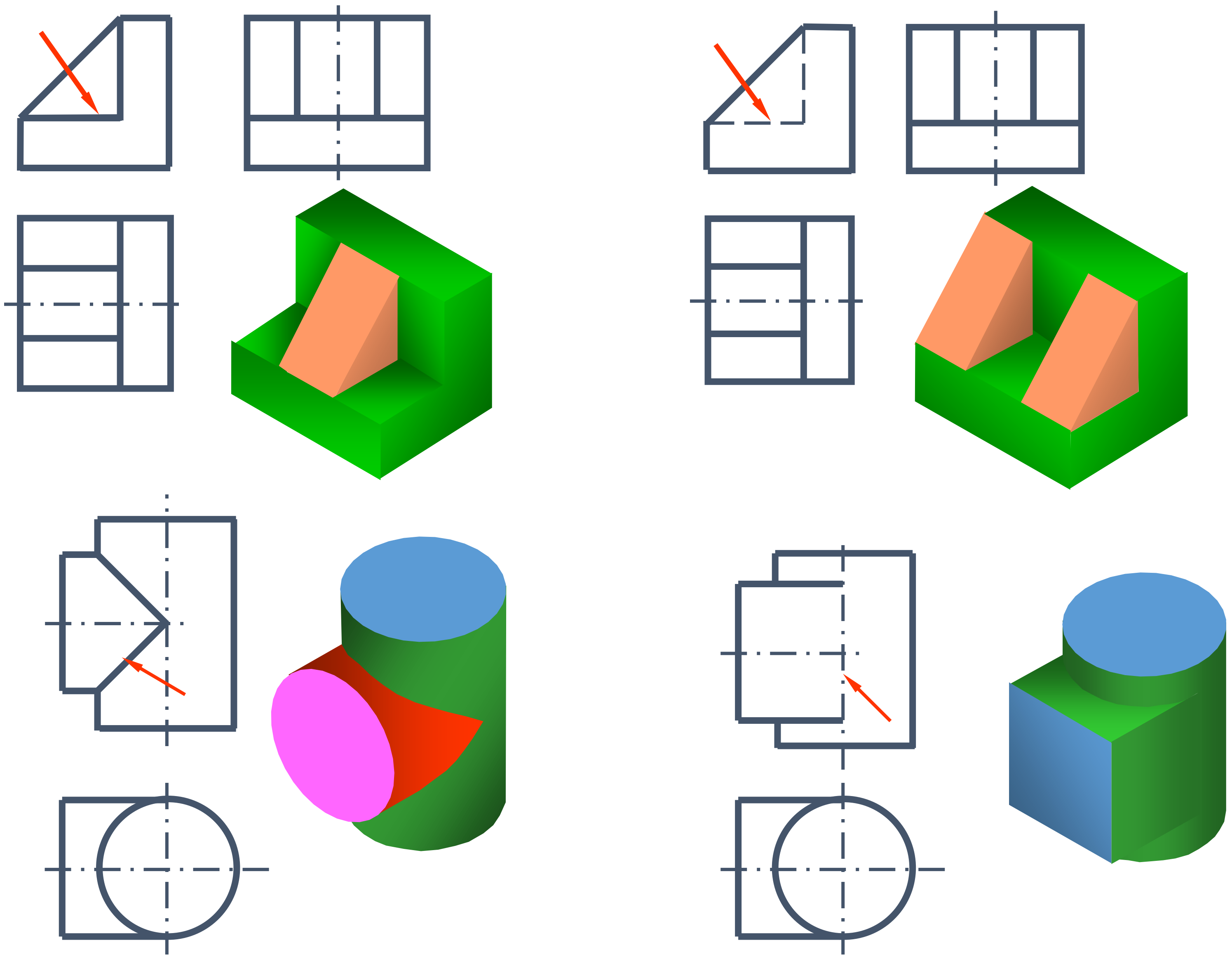




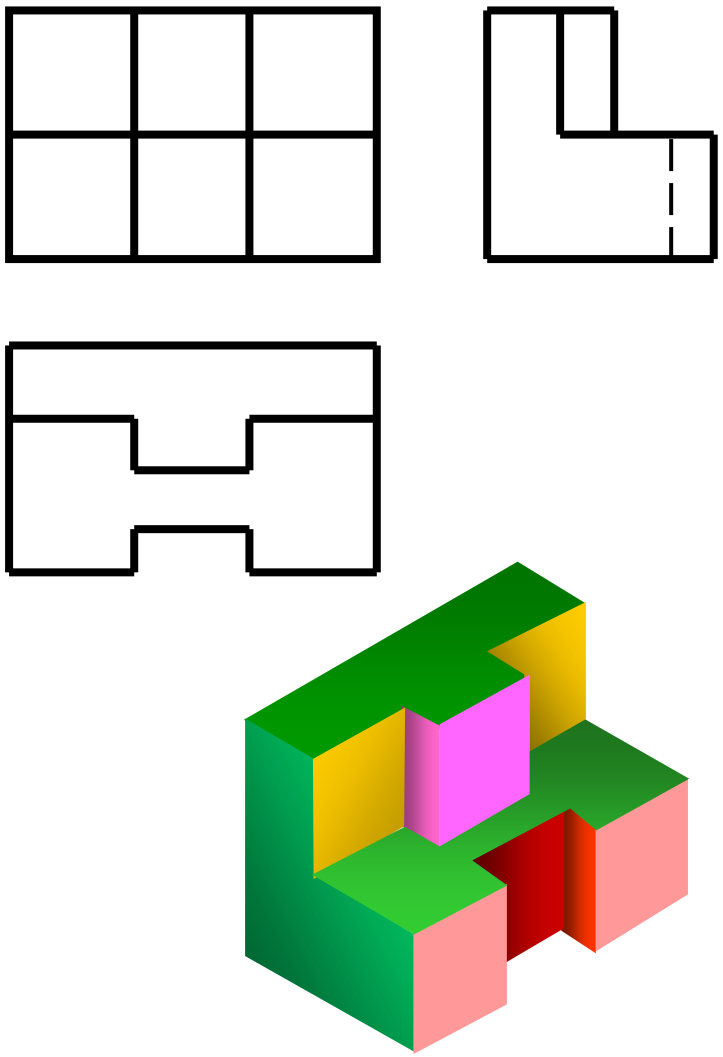
特征

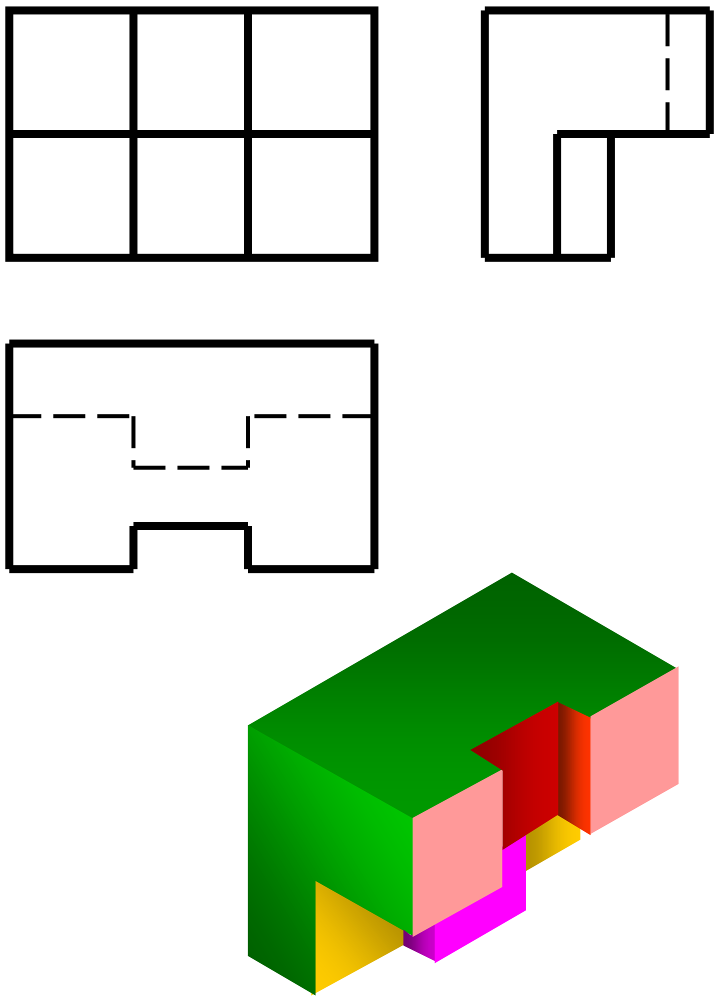


✓ 注意反映形体之间连接关系的图线

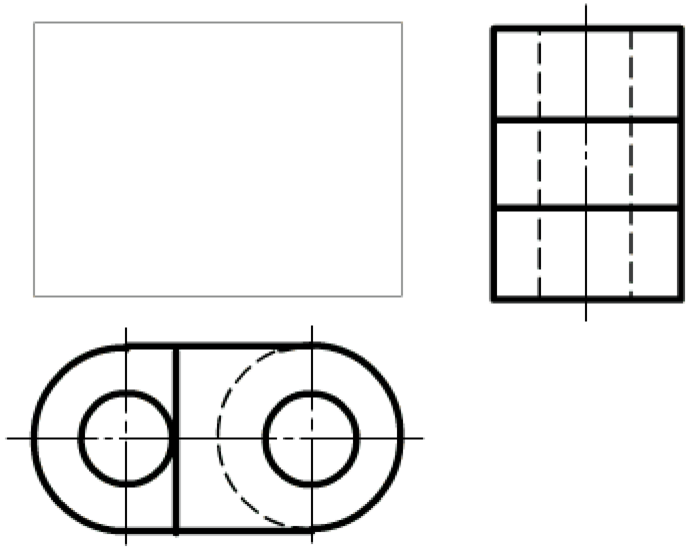


特征





想象



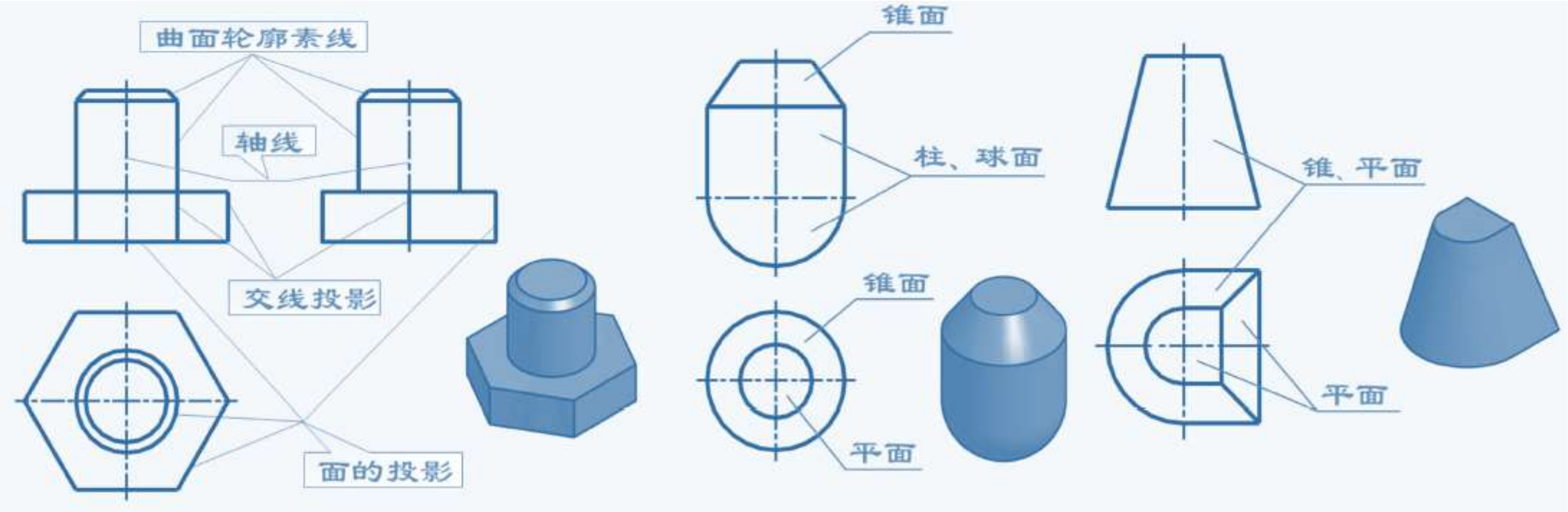
构思形体

形体1

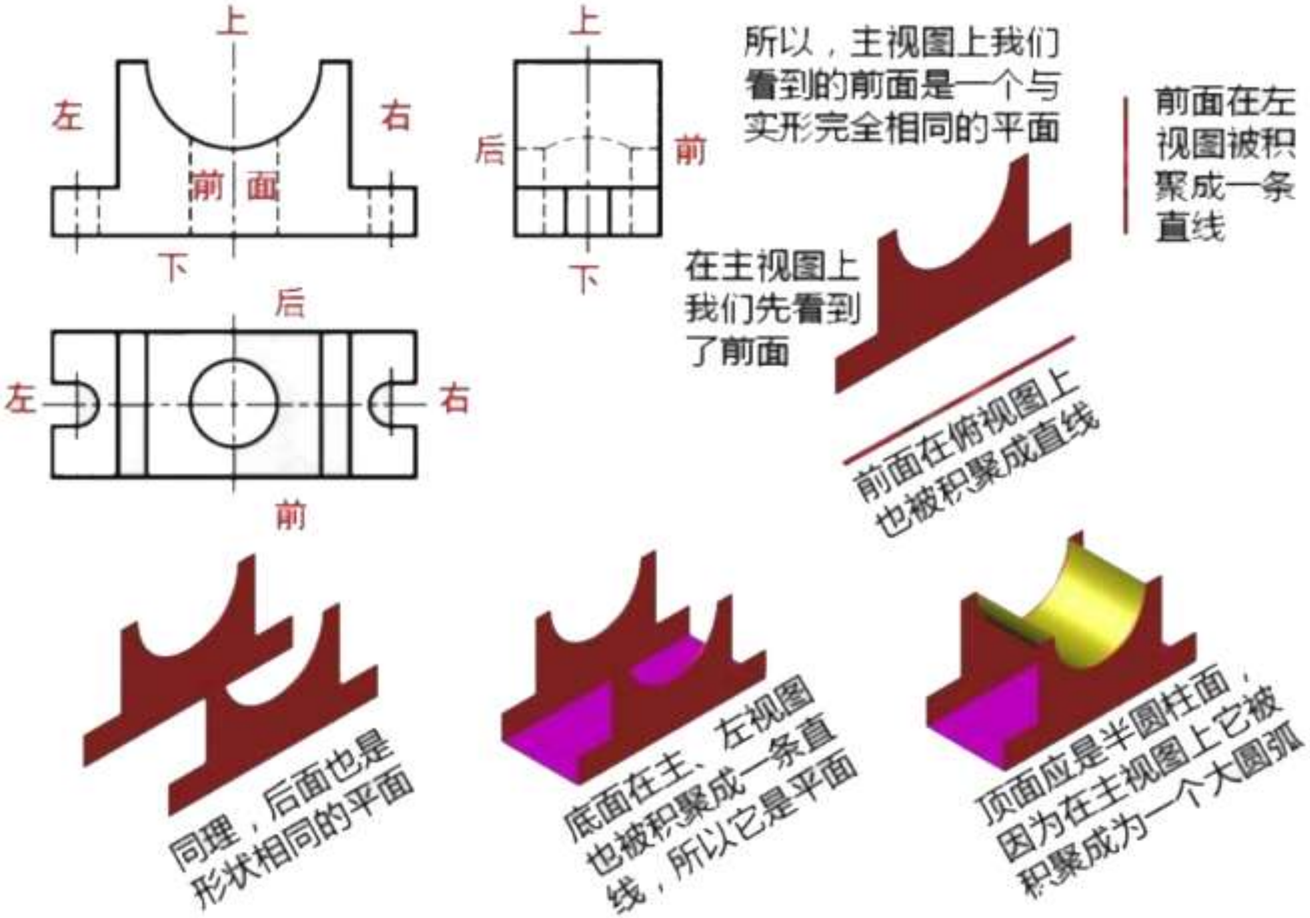
形体2

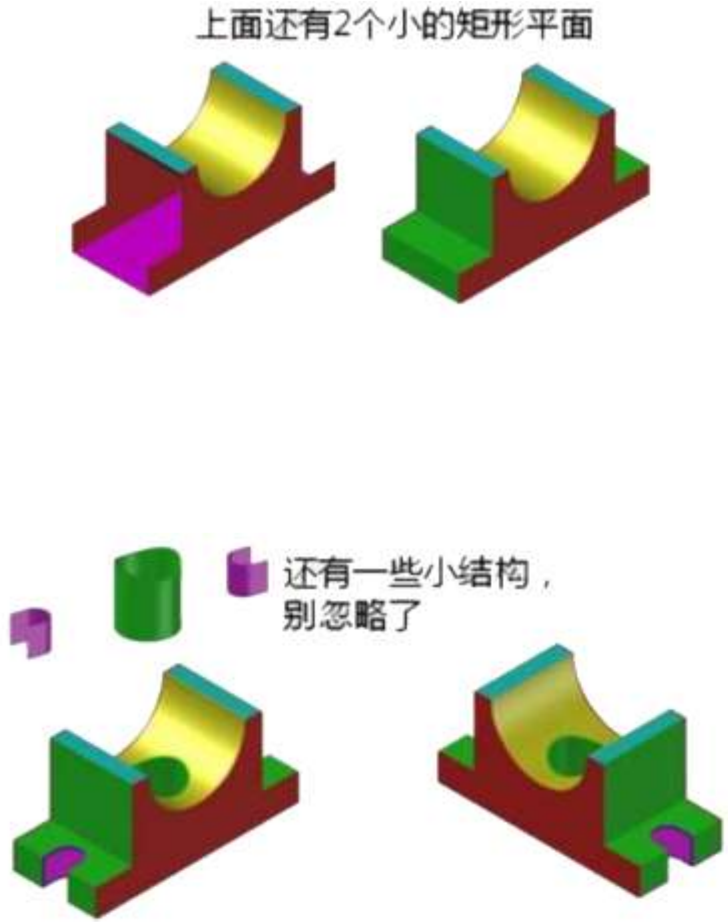
形体3

线面分析法

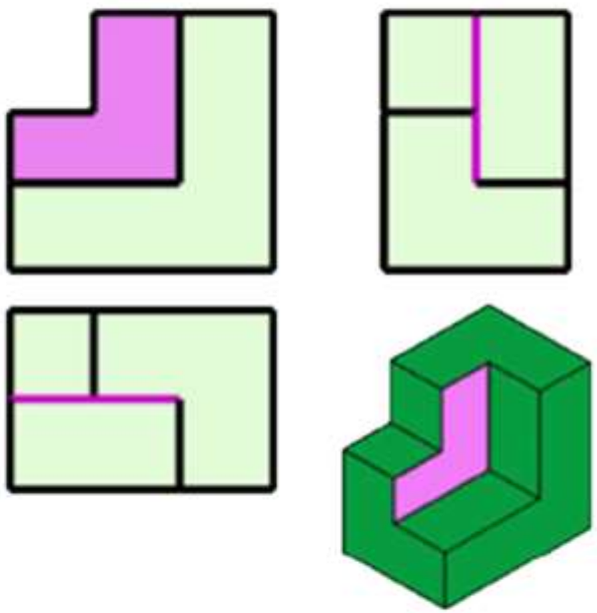


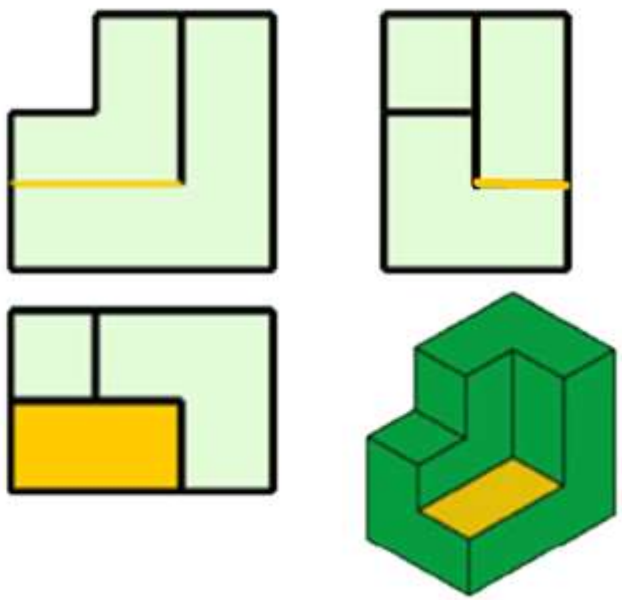
线面分析法

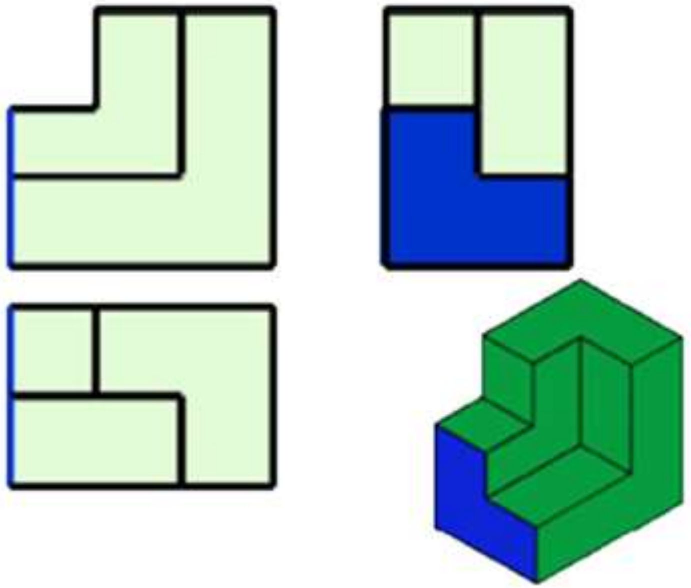

上
左
右
前 面
下
上
后
前
下
后
左
右
前
所以，主视图上我们
看到的前面是一个与
实形完全相同的平面
在主视图上
我们先看到
了前面
前面在左
视图被积
聚成一条
直线
前面在俯视图上
也被积聚成直线
同理，后面也是
形状相同的平面
底面在主、左视图
也被积聚成一条直
线，所以它是平面
顶面应是半圆柱面，
因为在主视图上它被
积聚成为一个大圆弧


上面还有2个小的矩形平面
还有一些小结构，
别忽略了

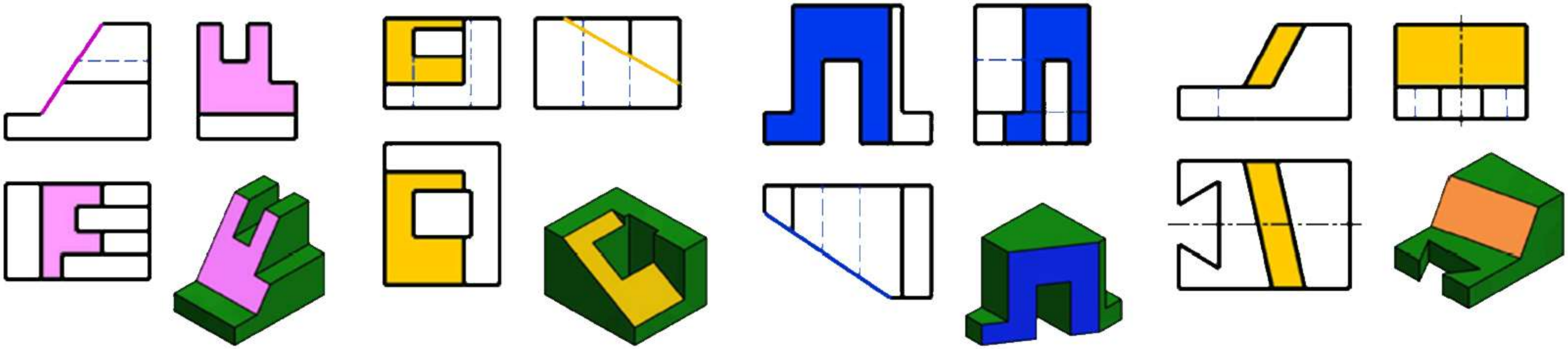
线面分析法



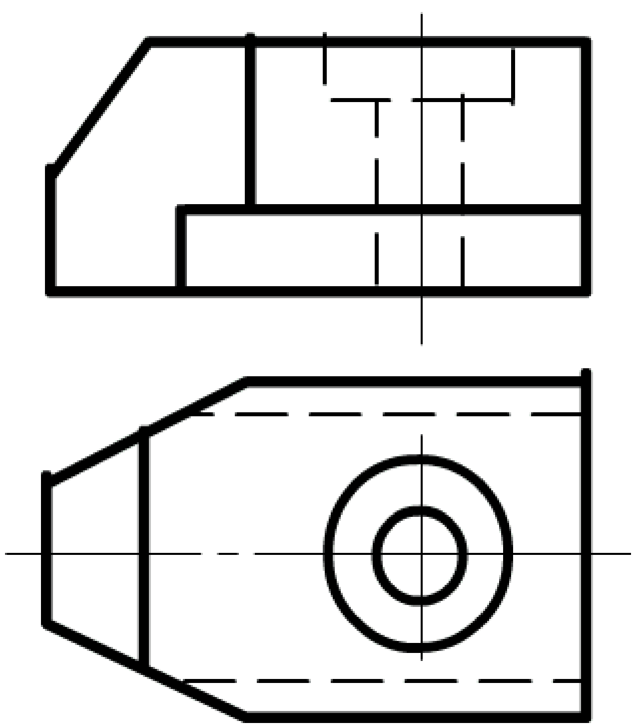




线面分析法



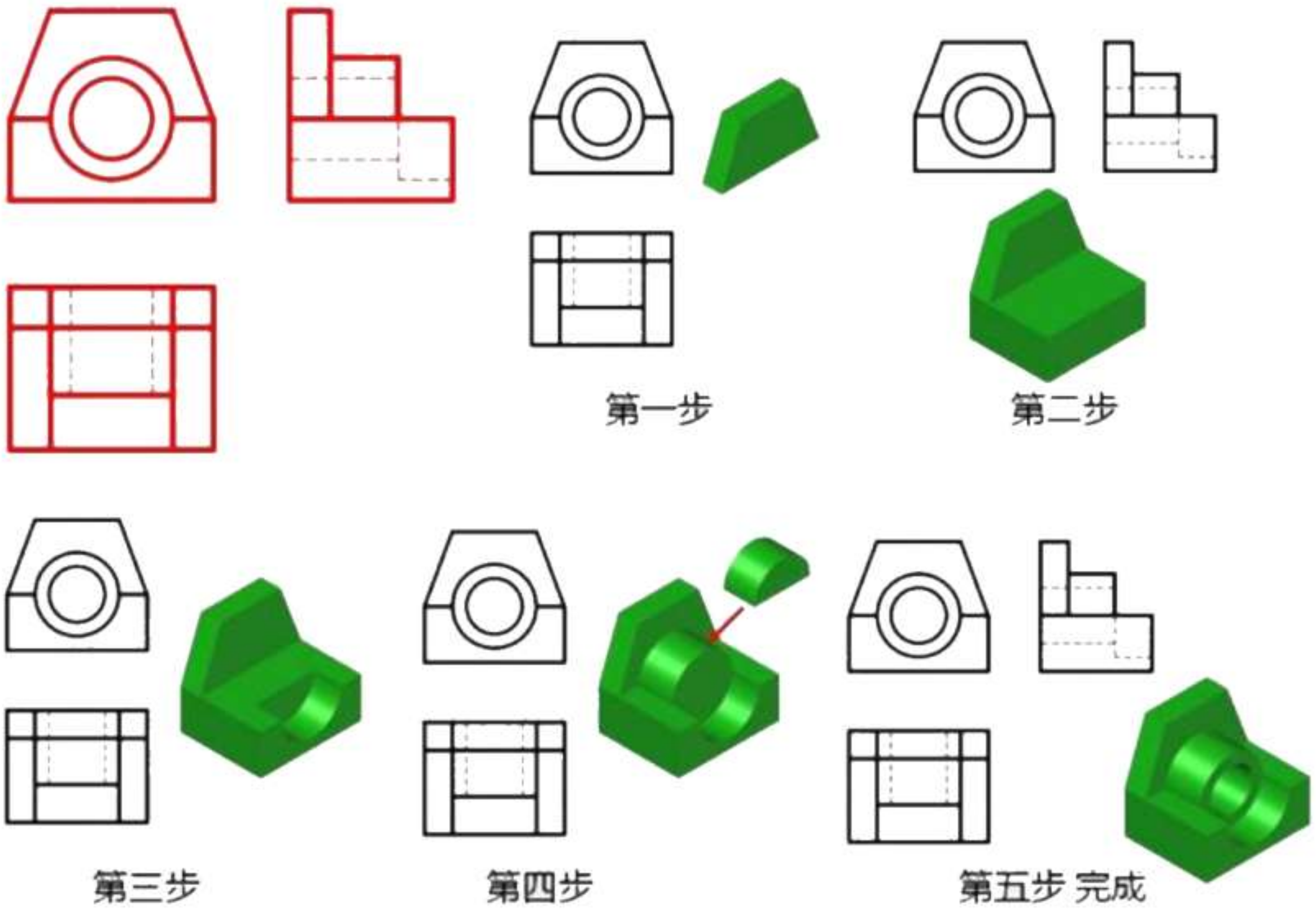
线面分析法



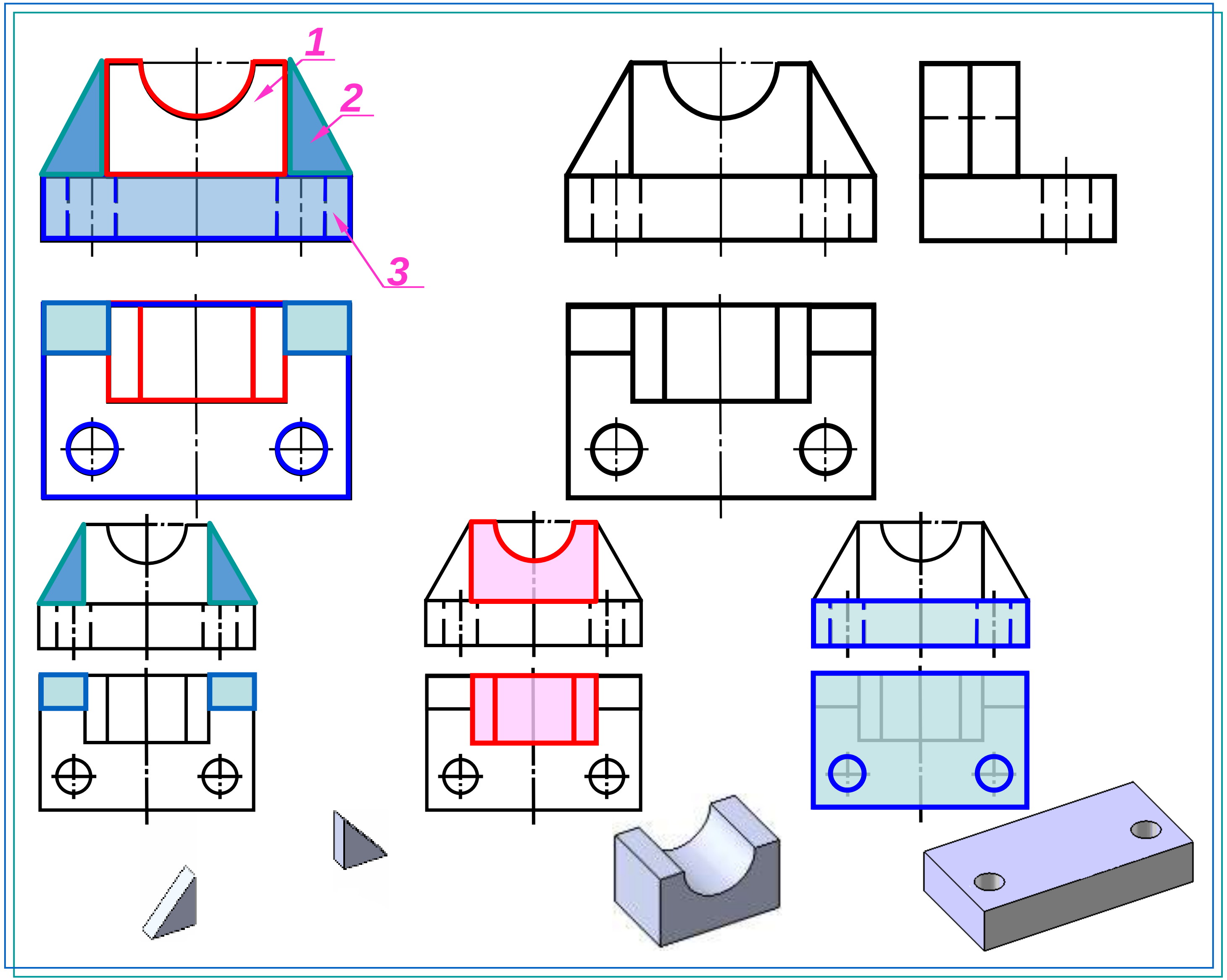
构思形体表面

- 表面1
- 表面2
- 表面3
- 表面4
- 表面5
- 其作表面

形体分析法



形体分析法



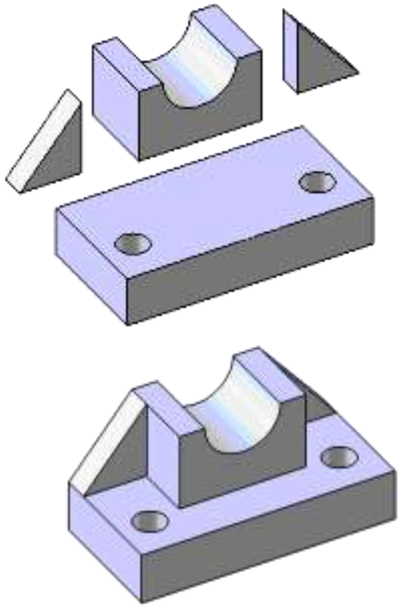
步骤：

1）图形分割

2）对出投影

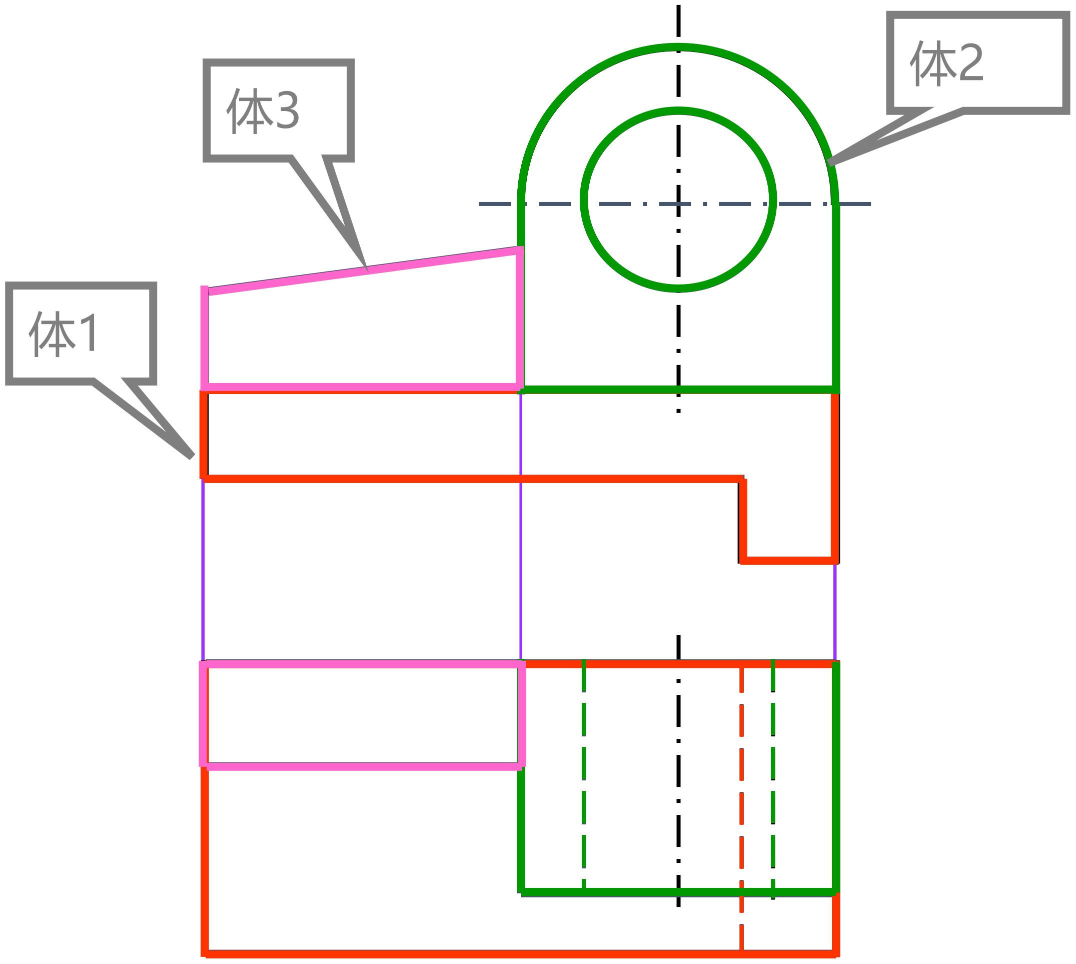
3）想像**堆积**

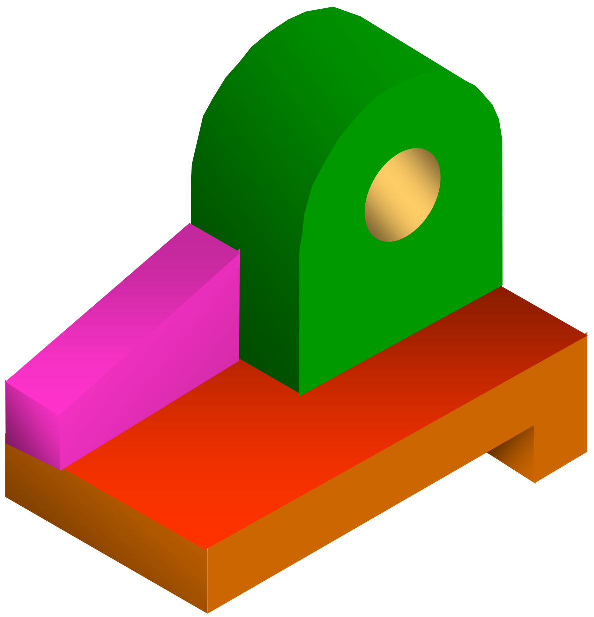
4）补画
第三视图



形体分析法

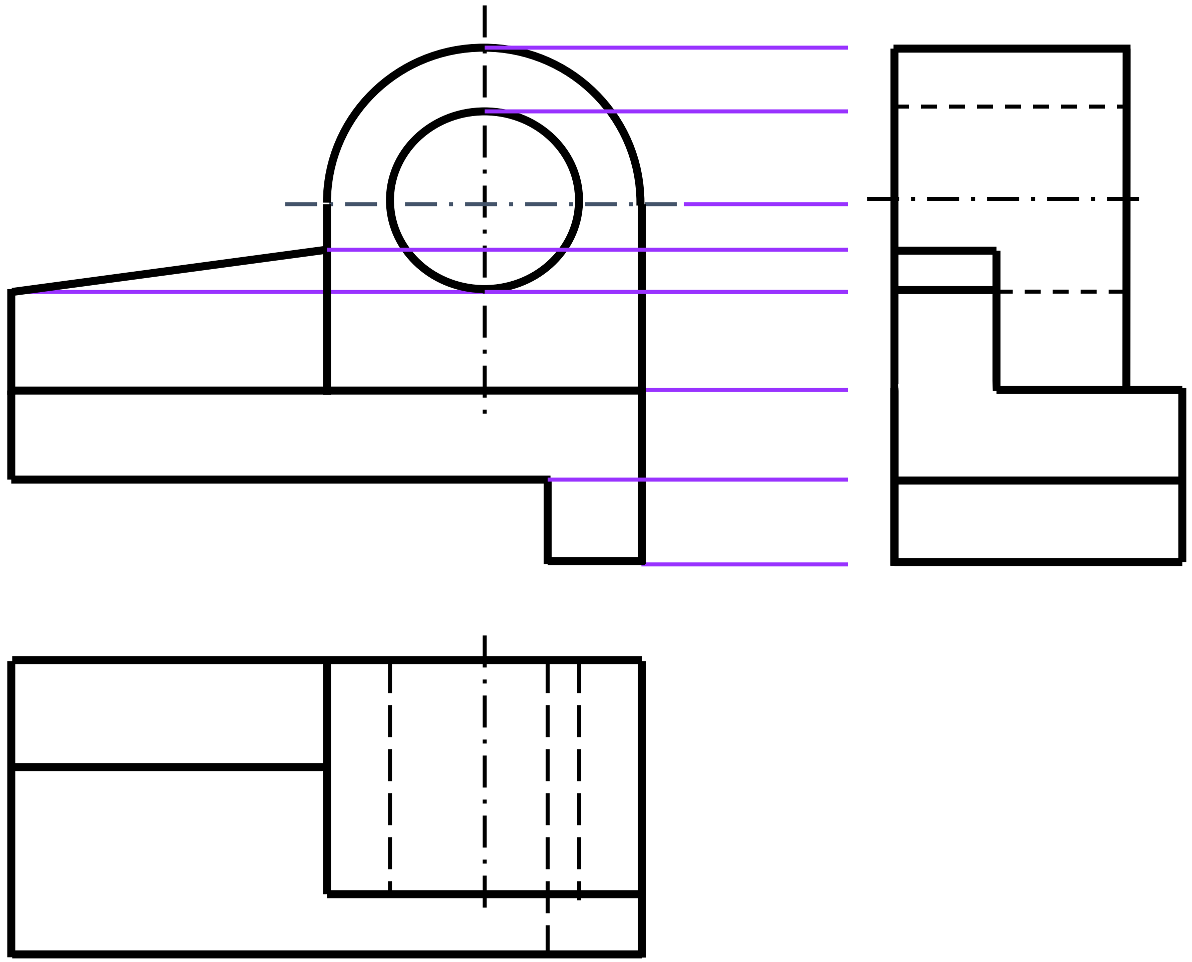
作左视图。

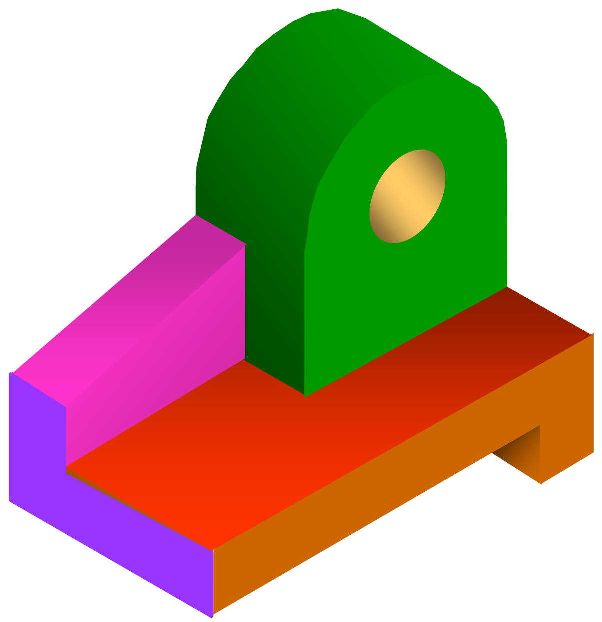




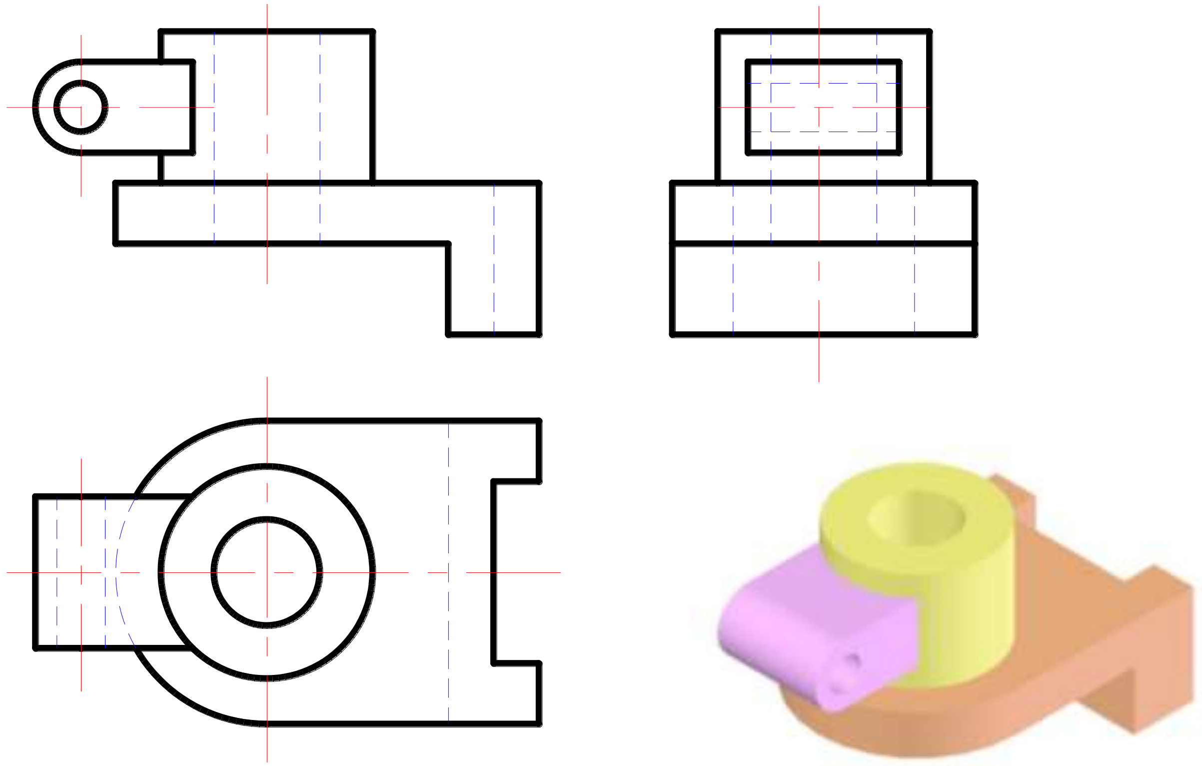
形体分析法

作左视图。



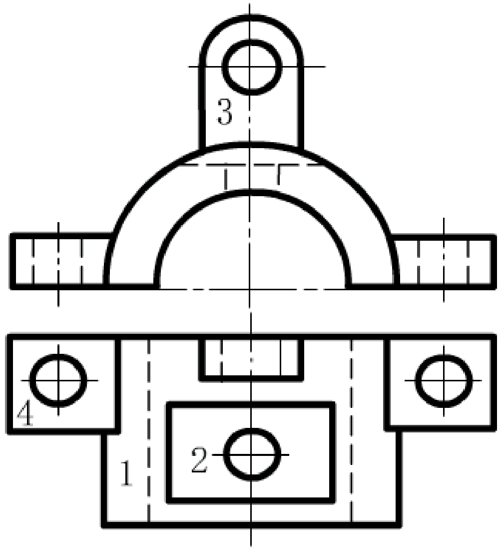


形体分析法



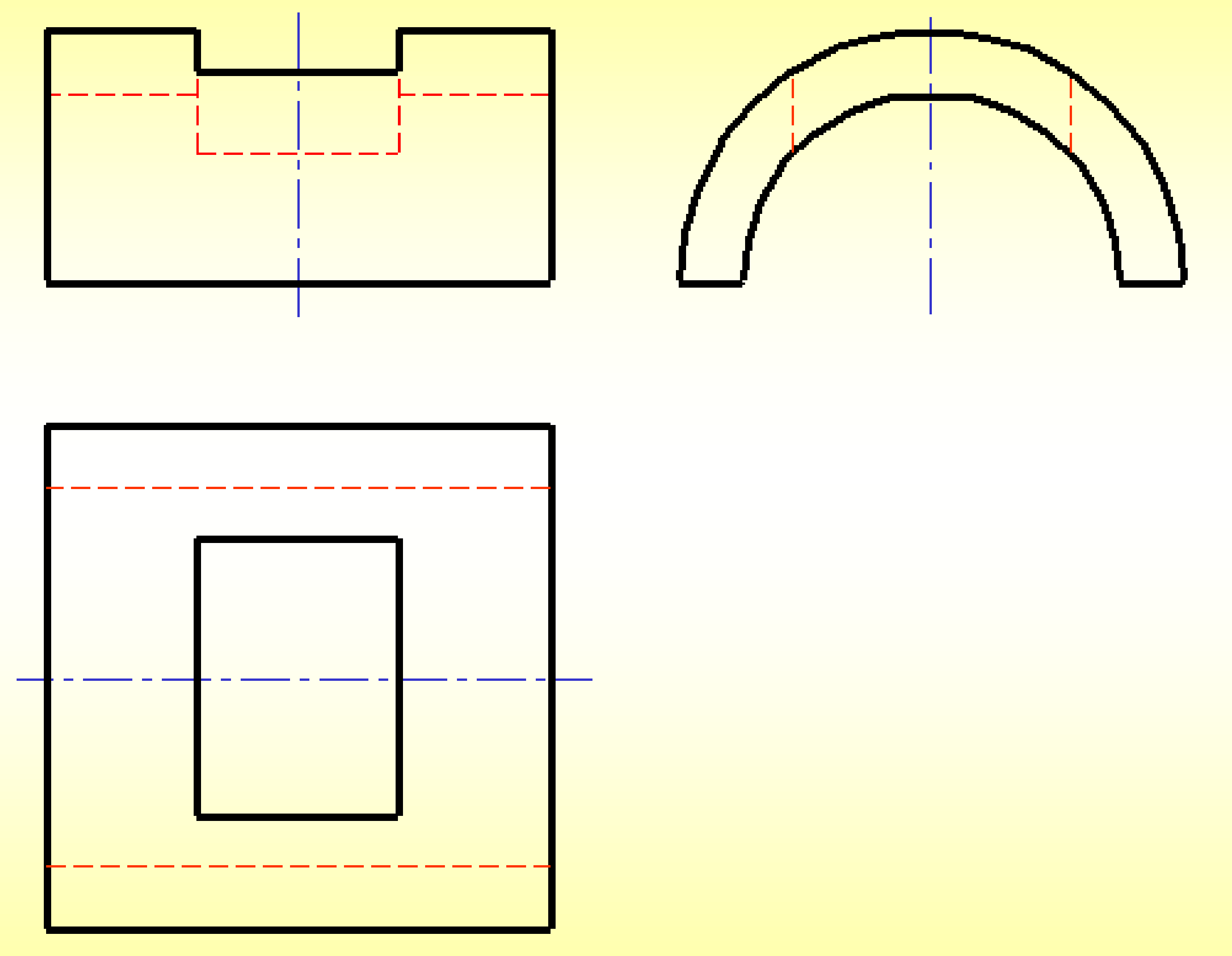
e.合起来,想整体

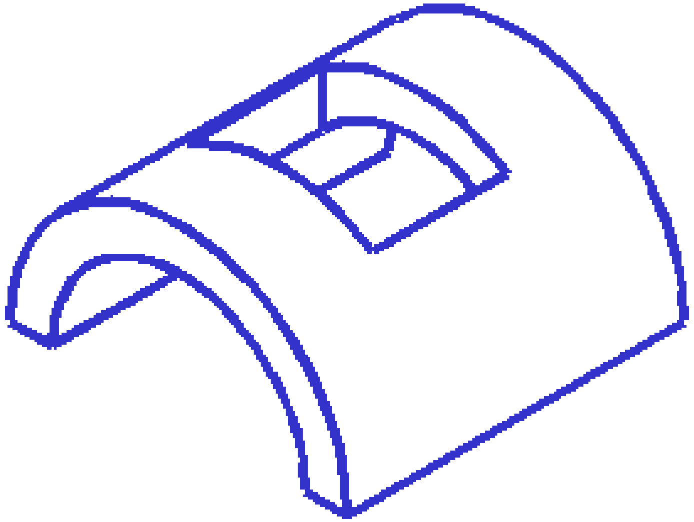
形体分析法



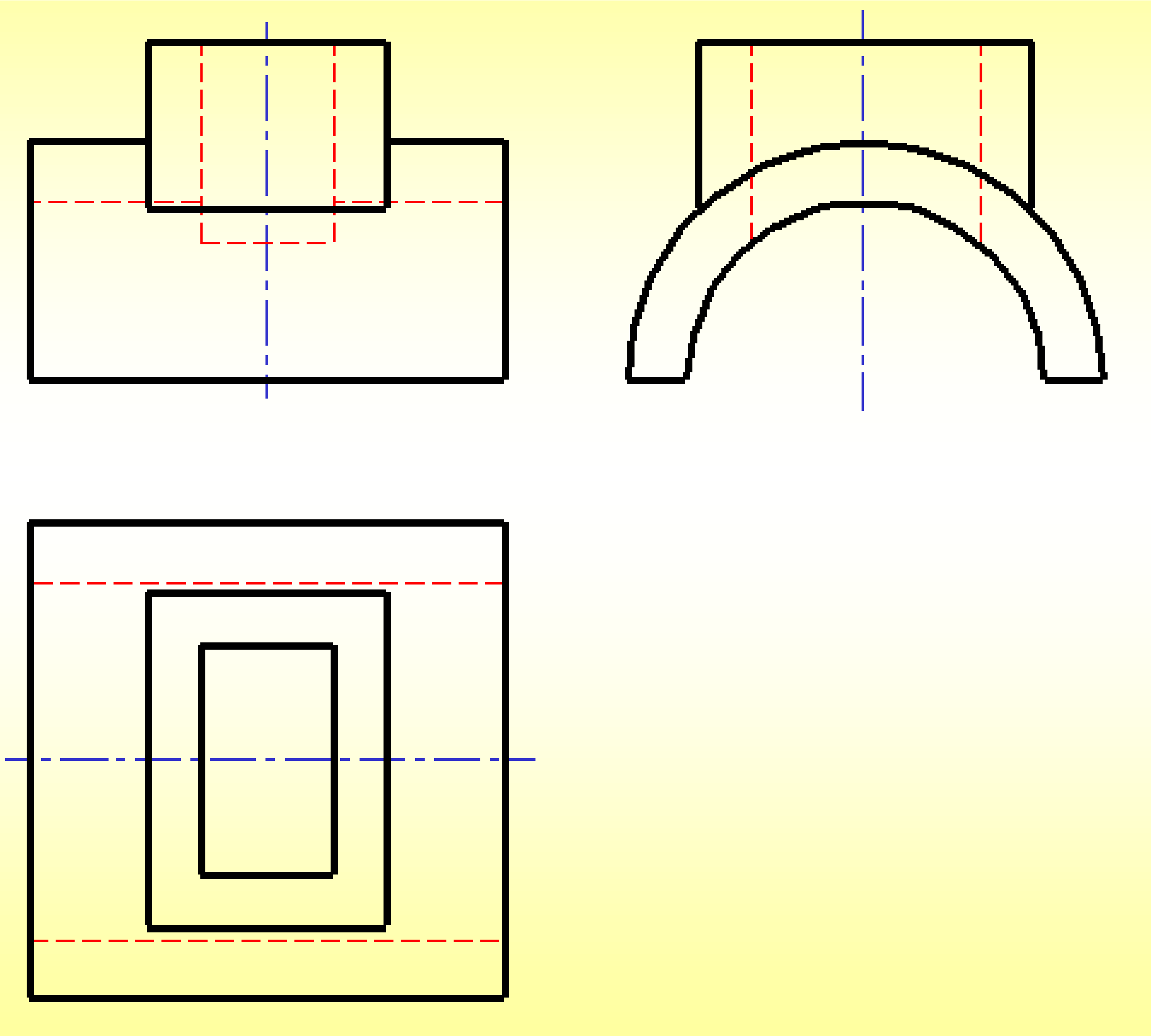
构思形体和线框

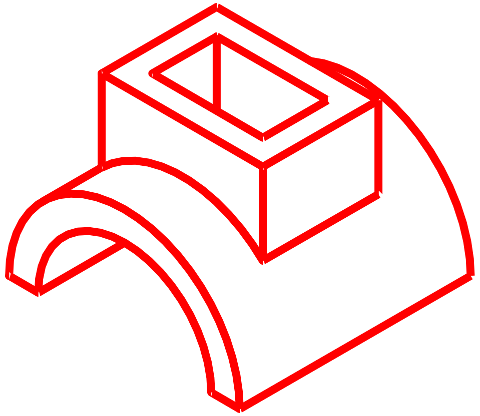
- 形体1
- 线框2
- 线框2中的圆
- 形体3
- 形体4





【例】





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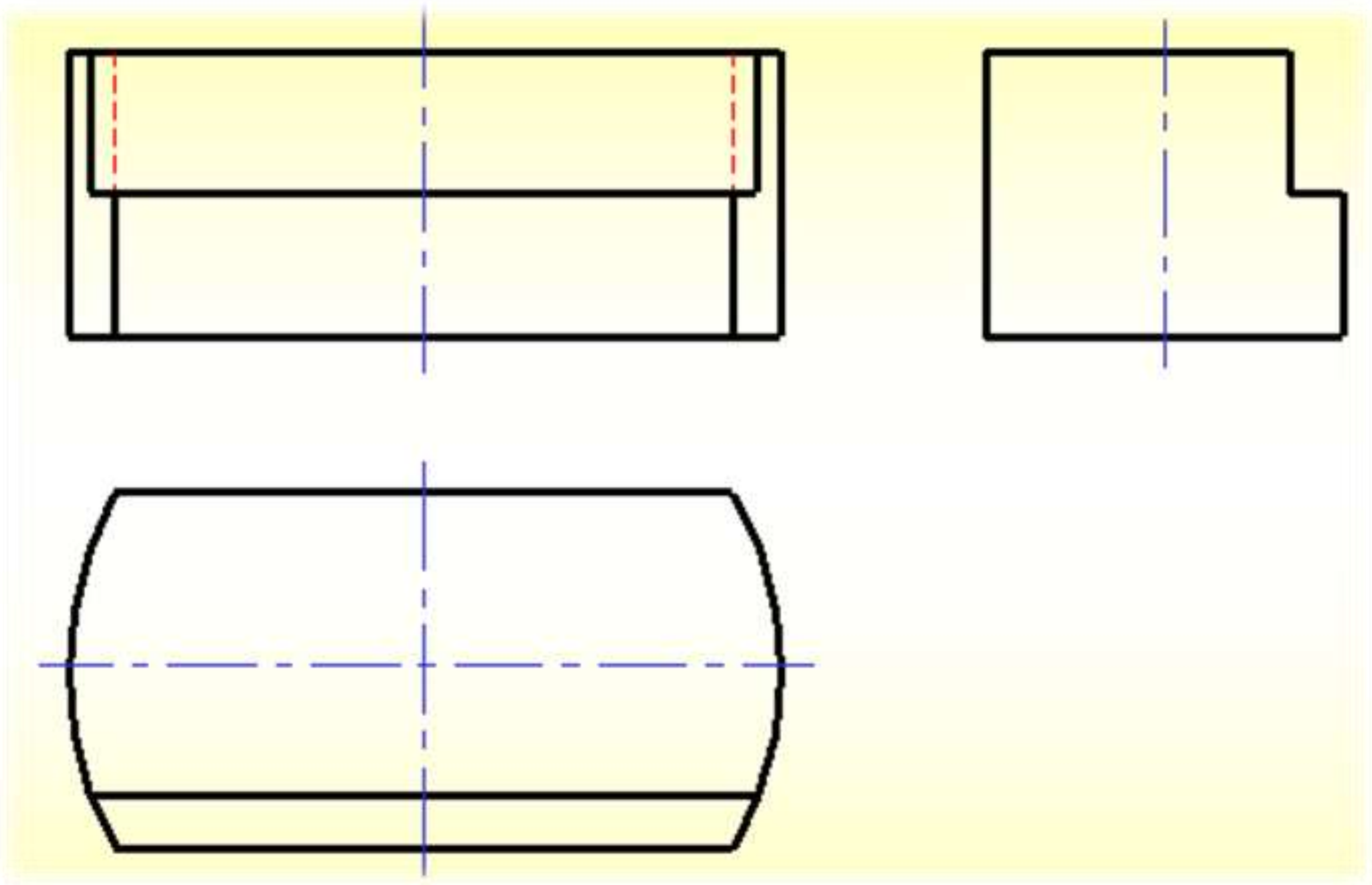

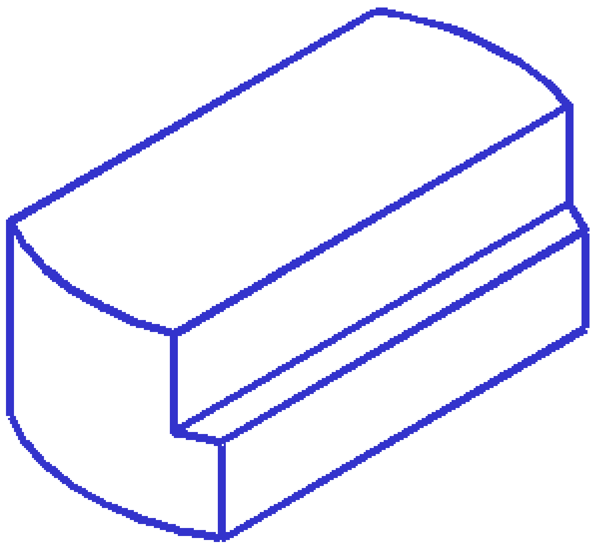

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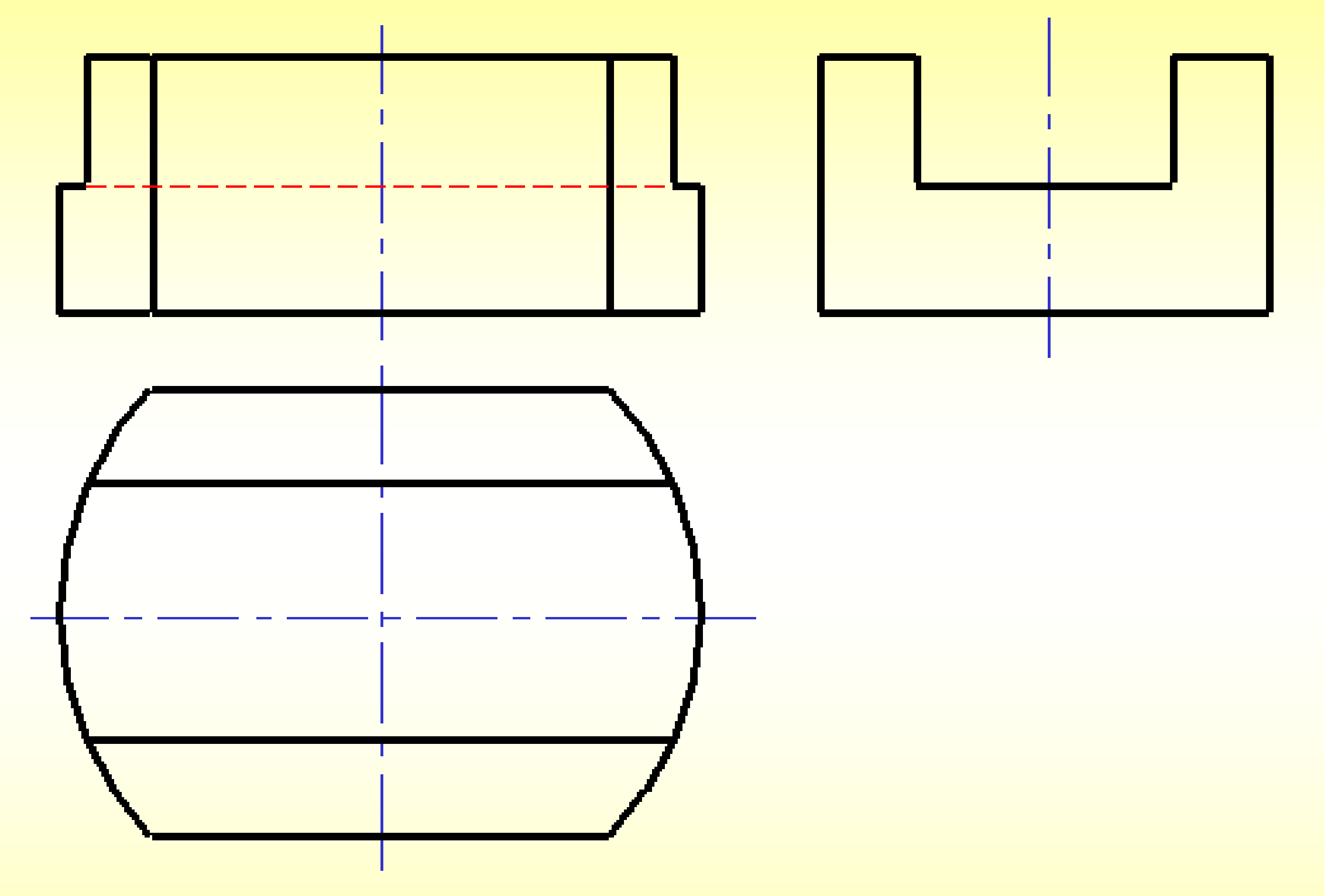
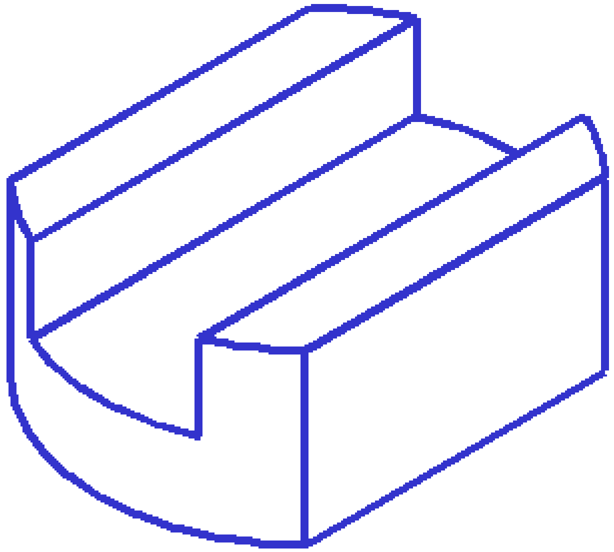
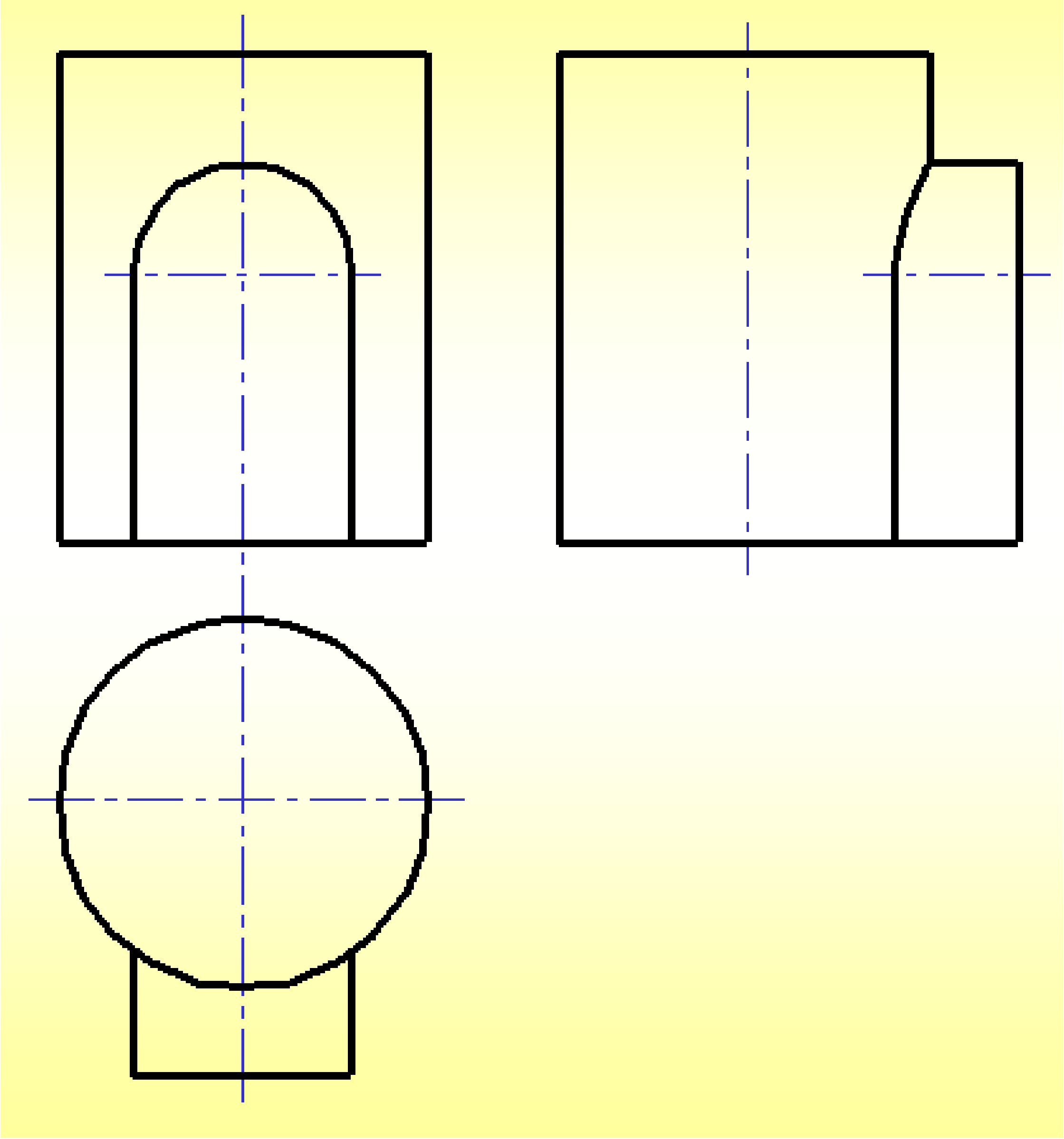
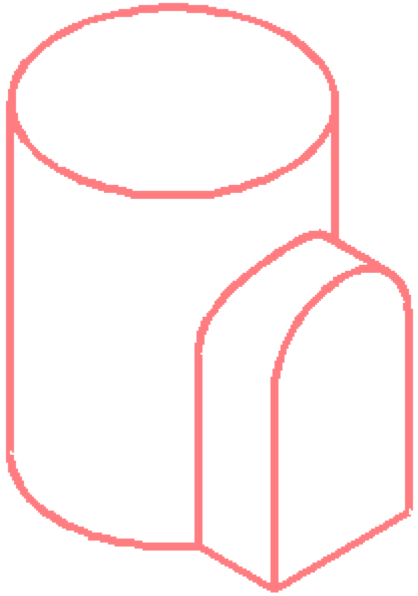




【例】





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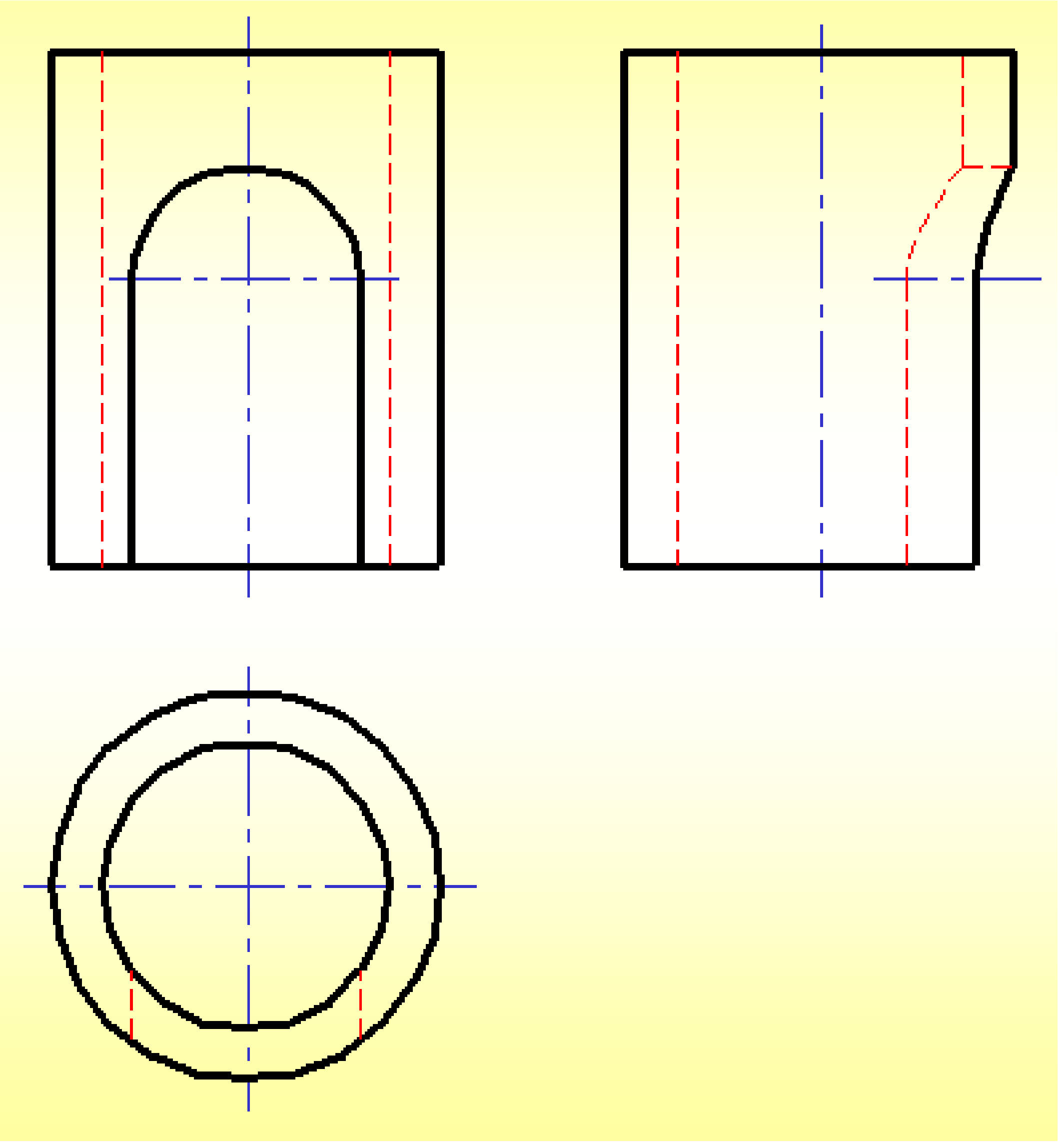

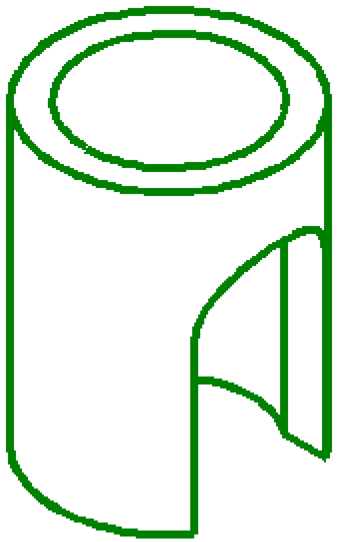

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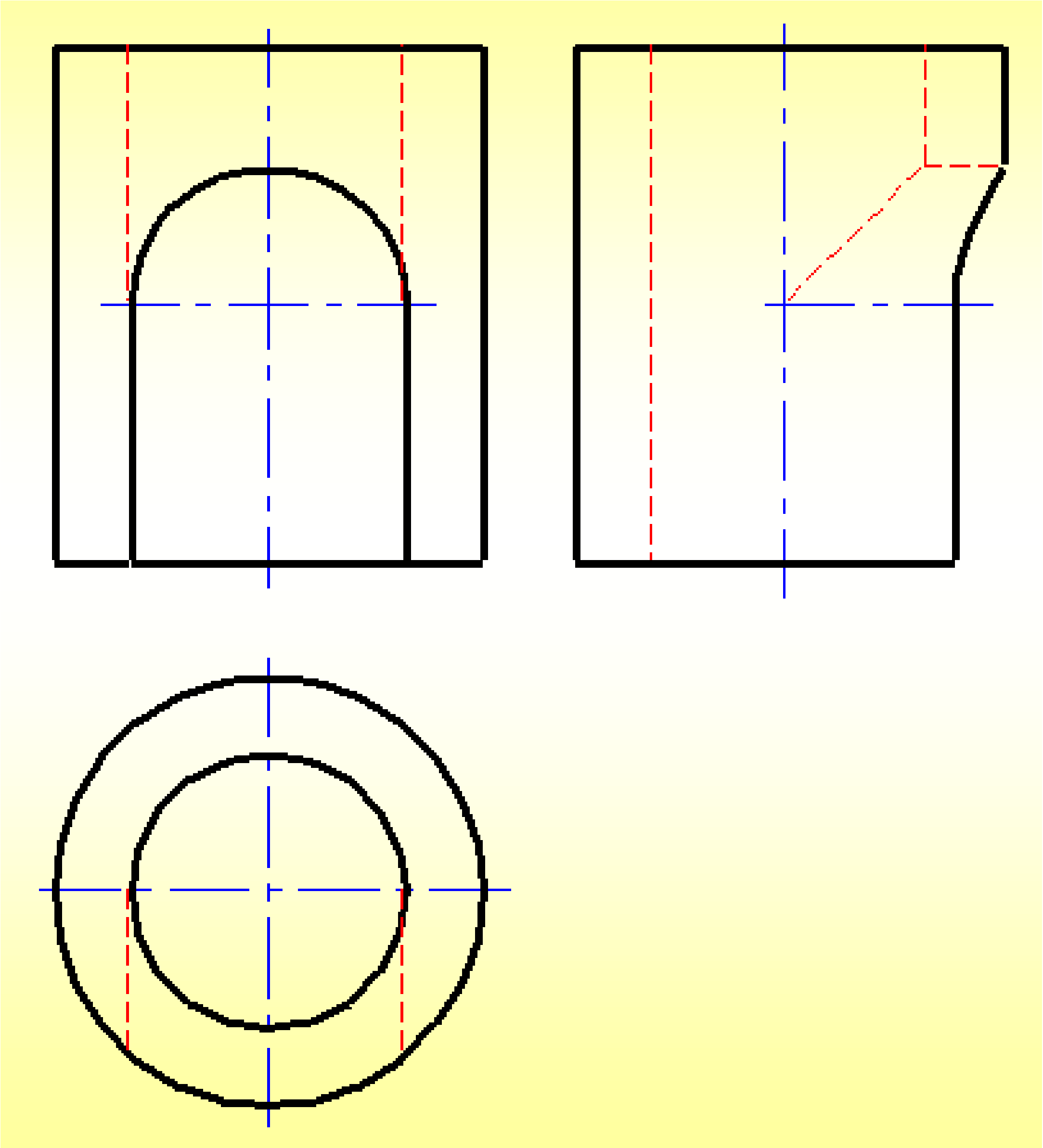

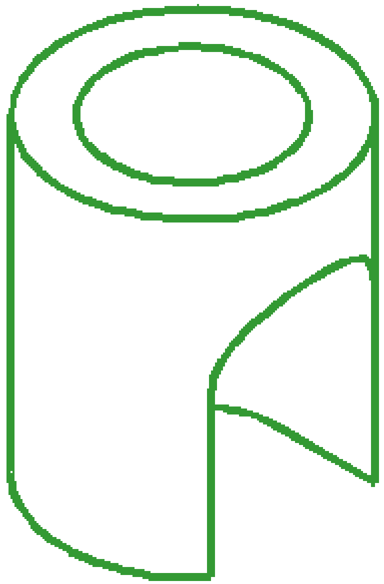

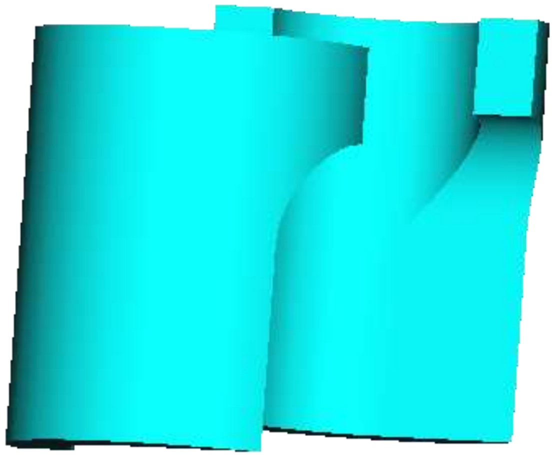

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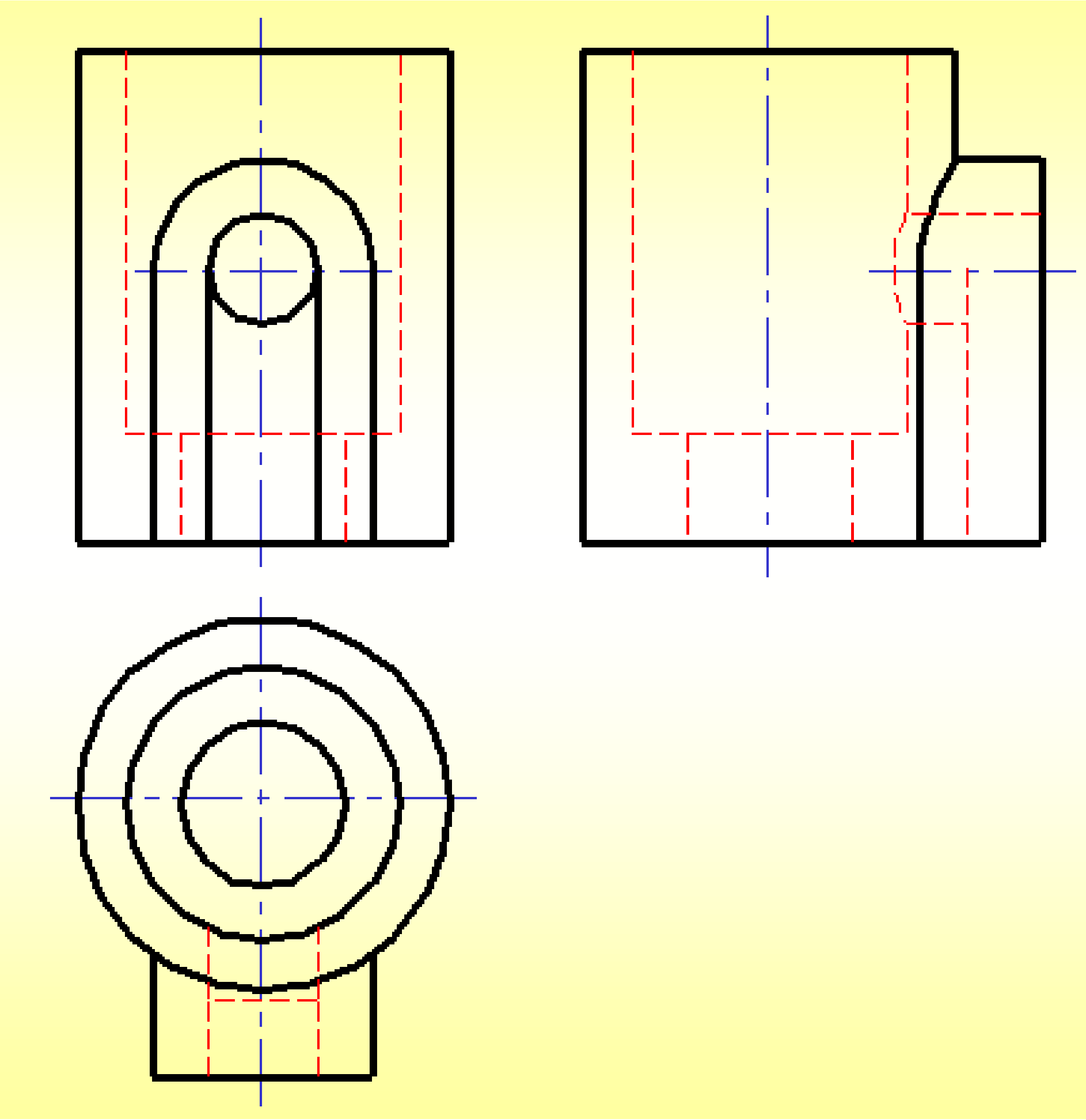

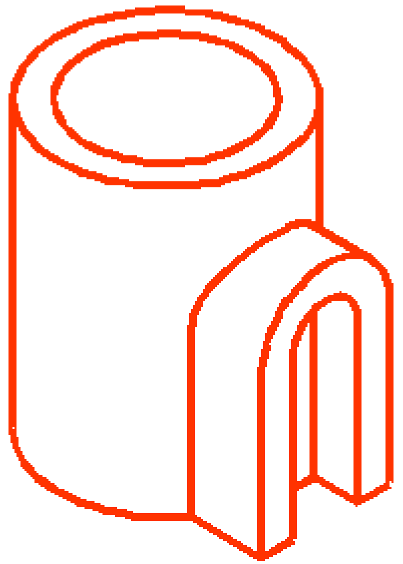

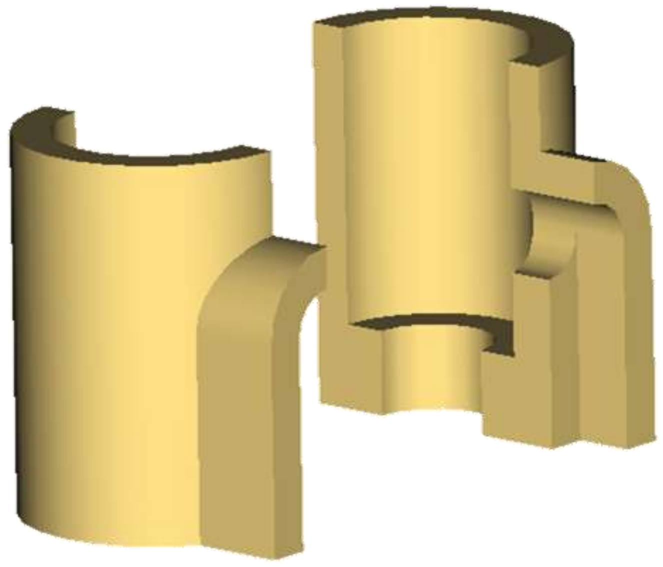

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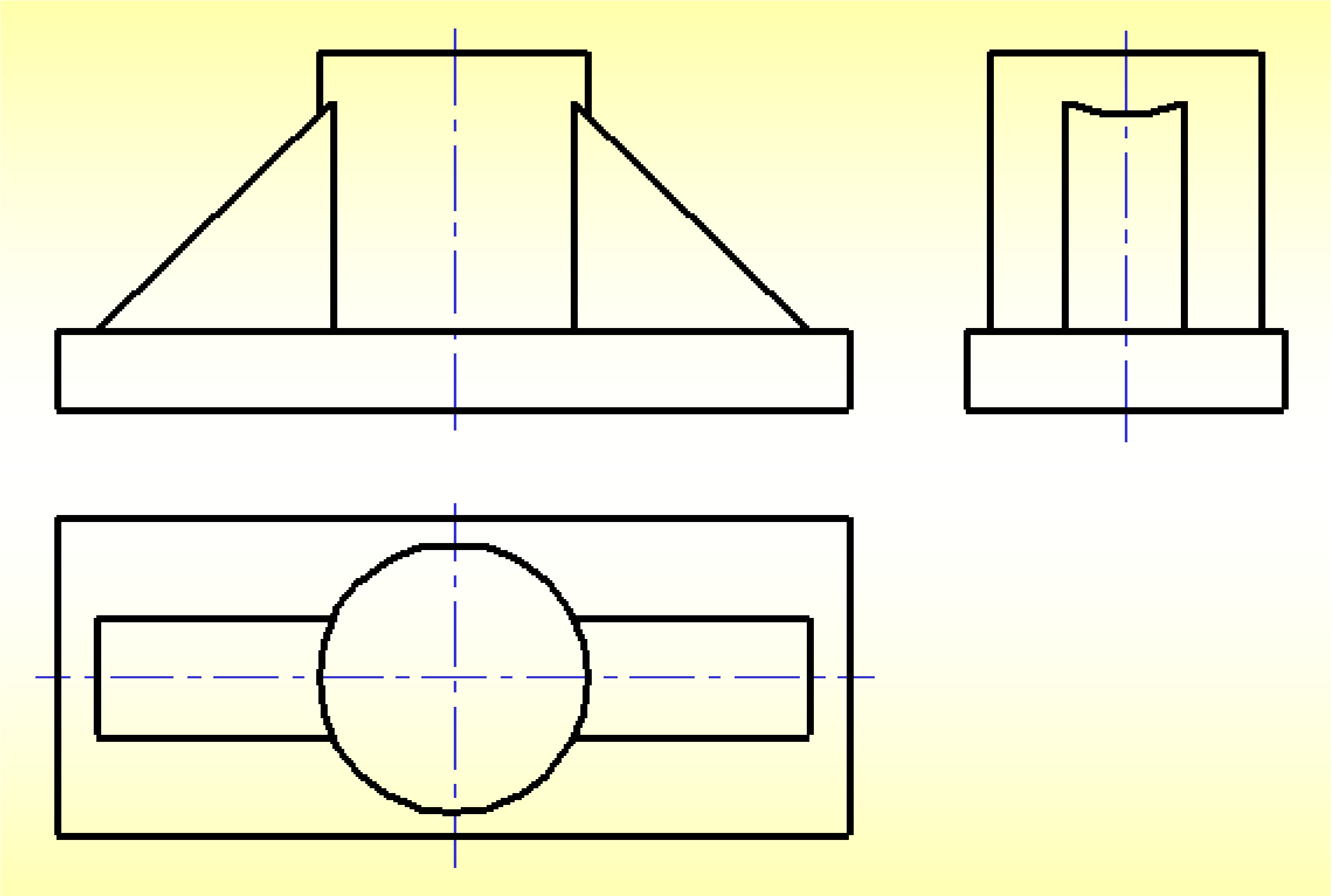
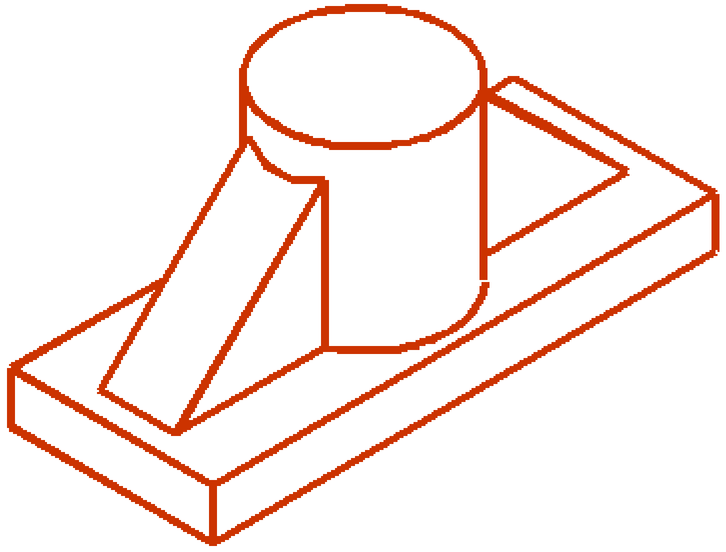
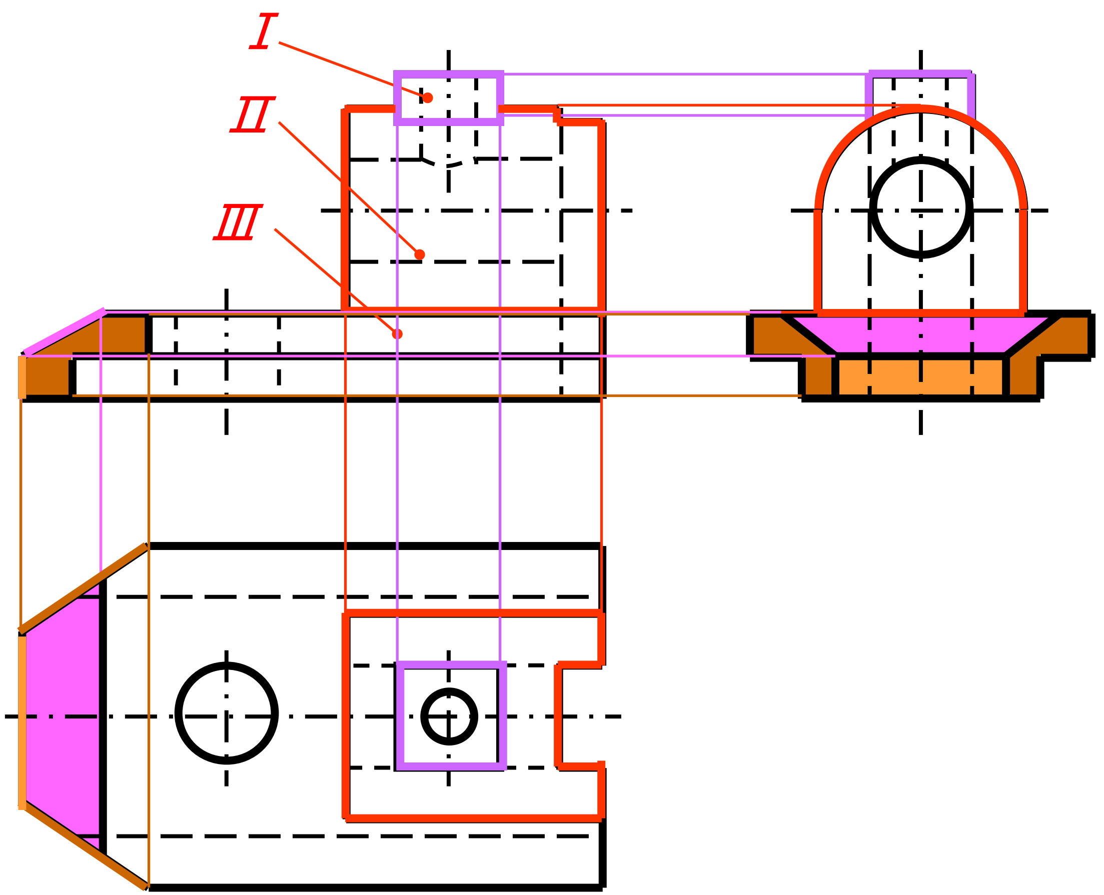
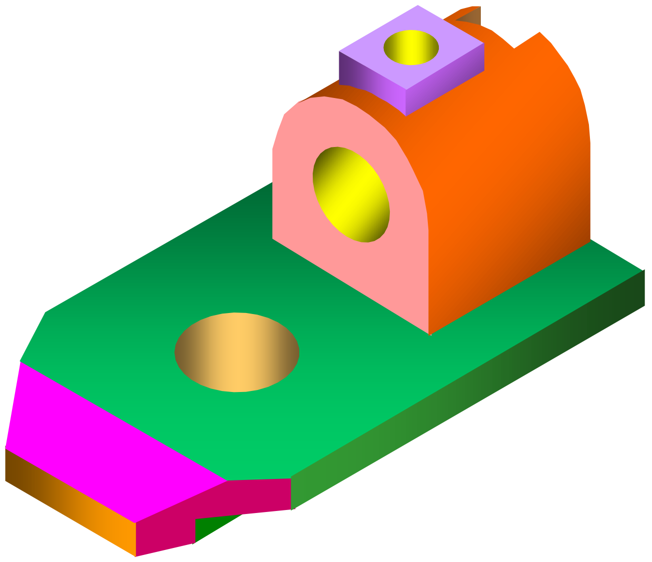




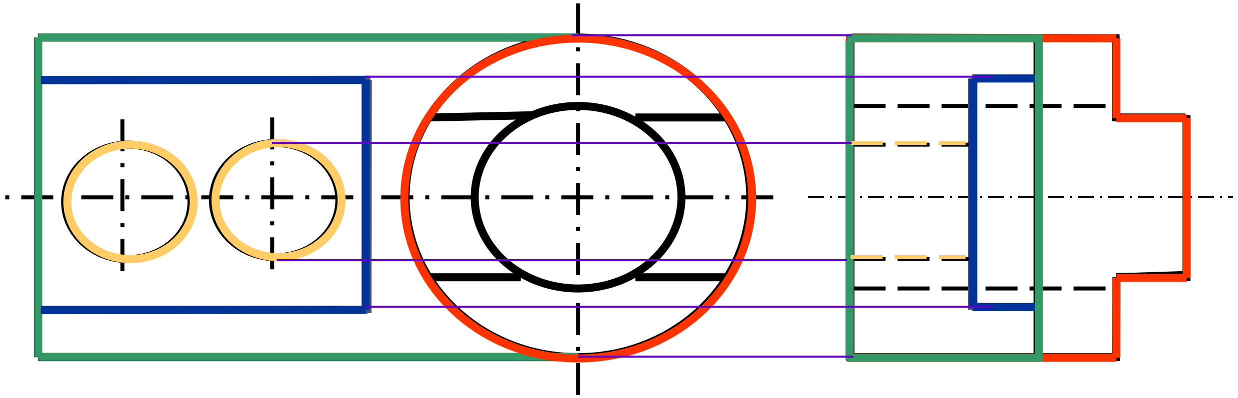
【例】

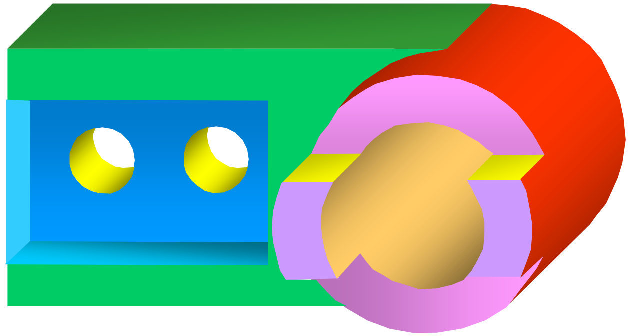
已知物体的三视图，想象出物体的形状。





【例】





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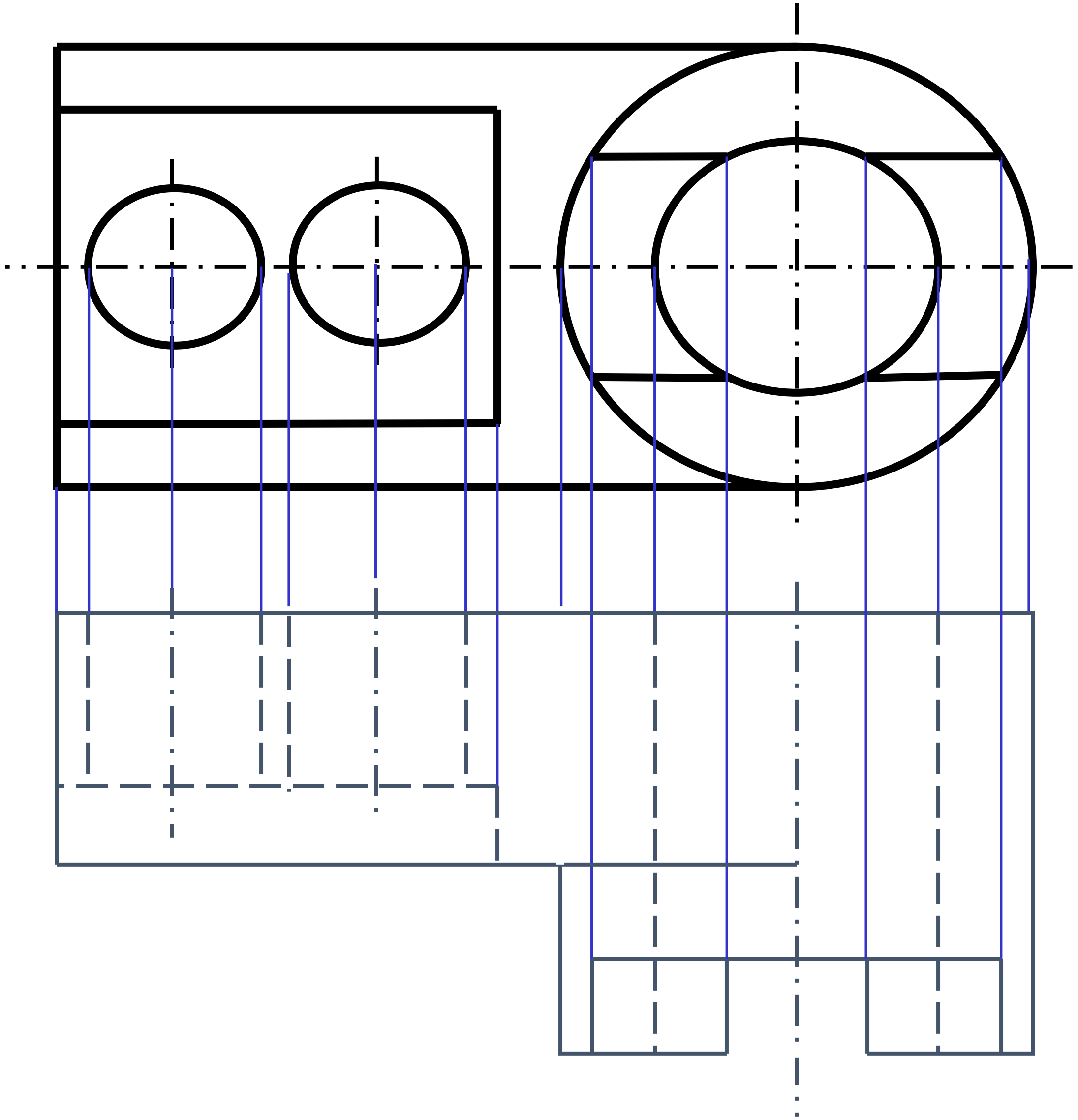

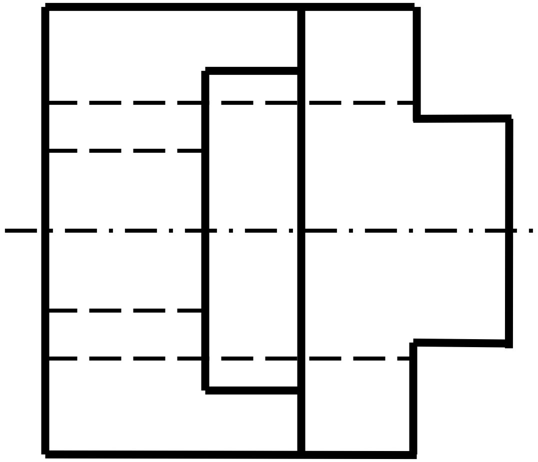

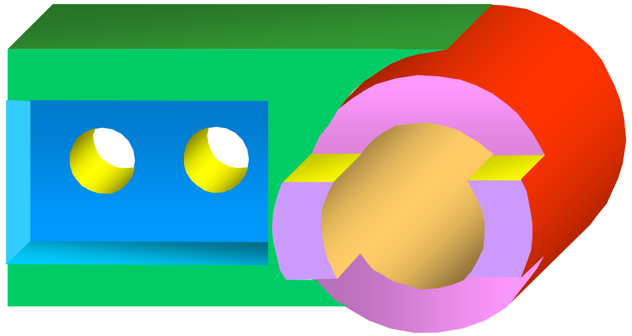

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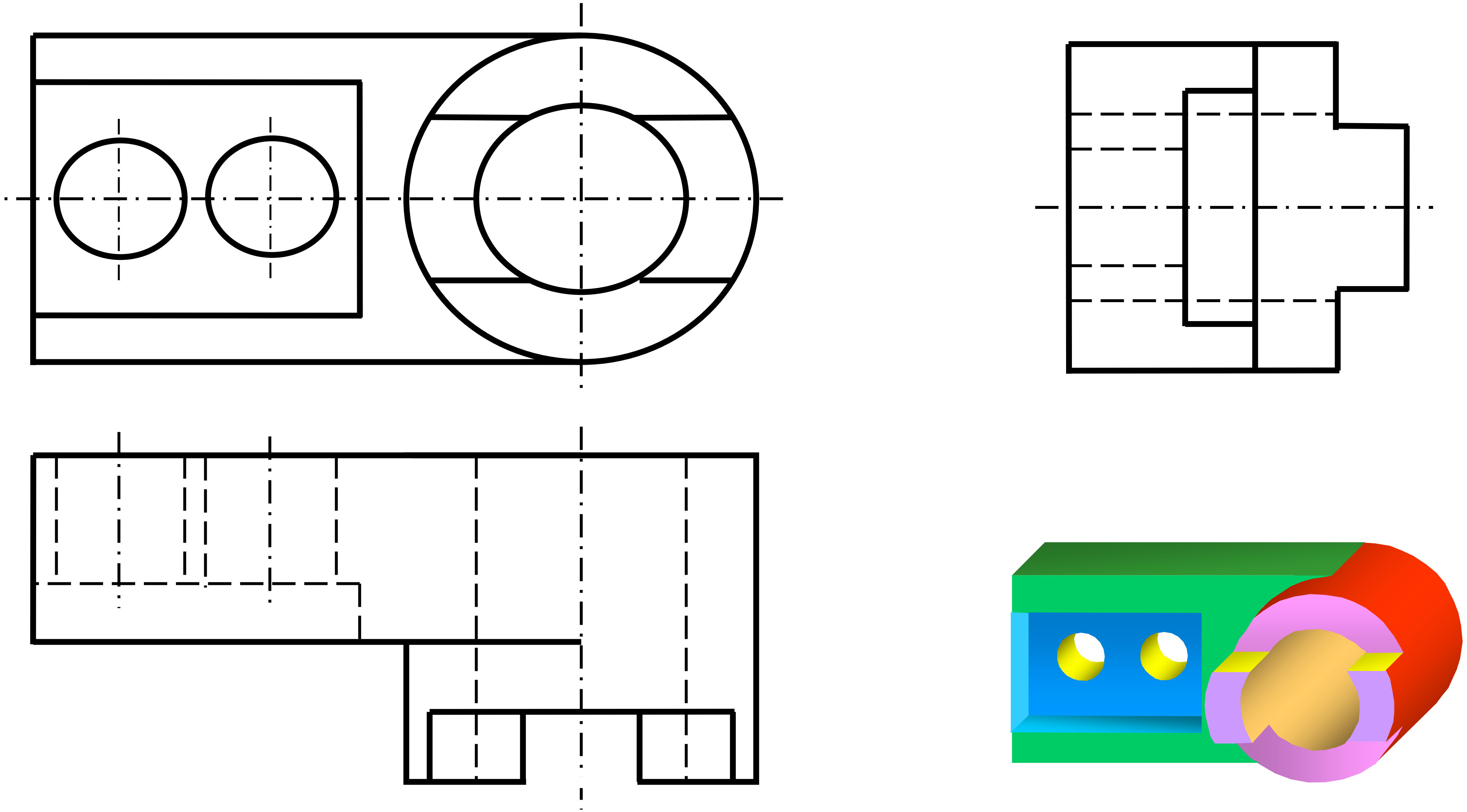

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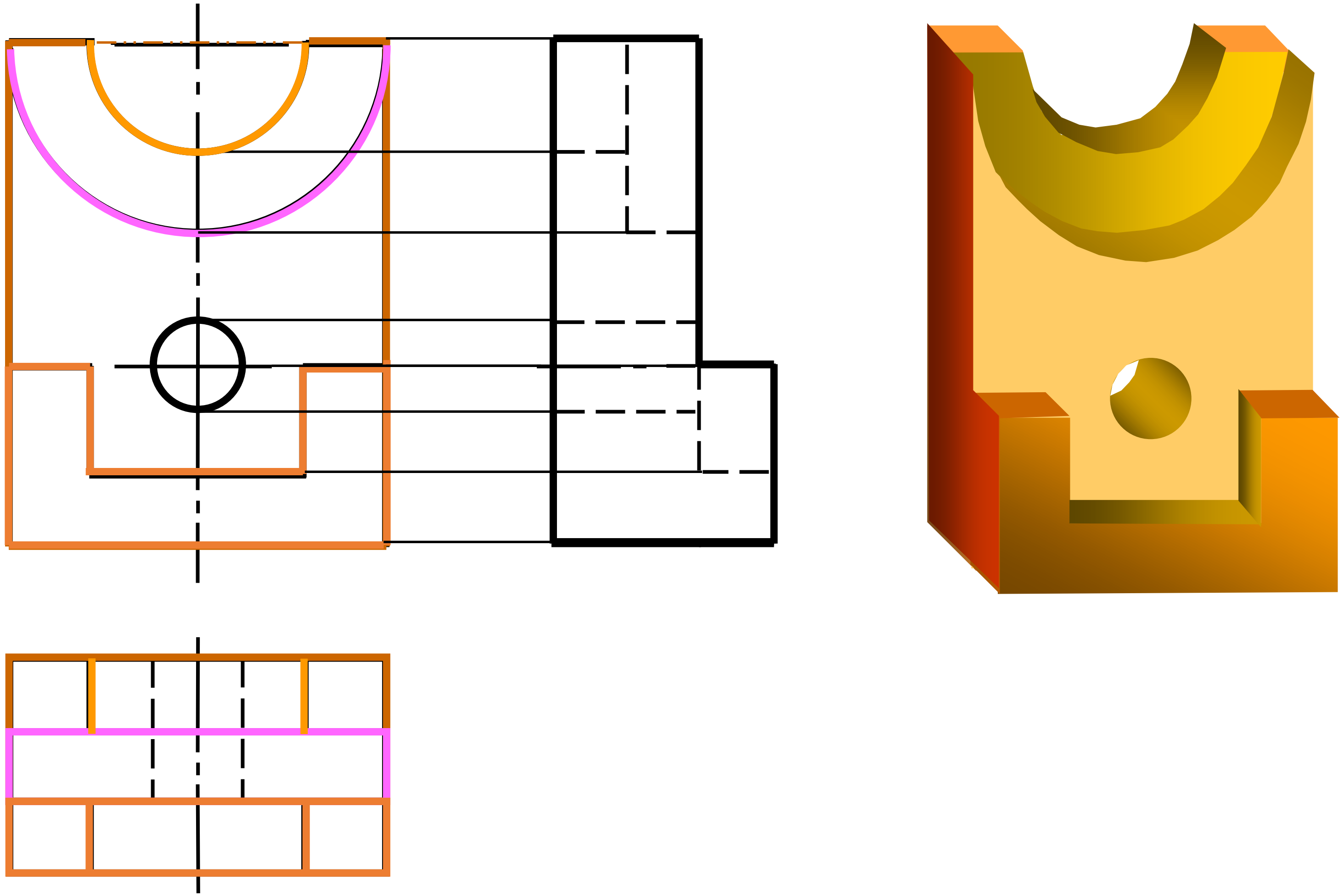

【例】

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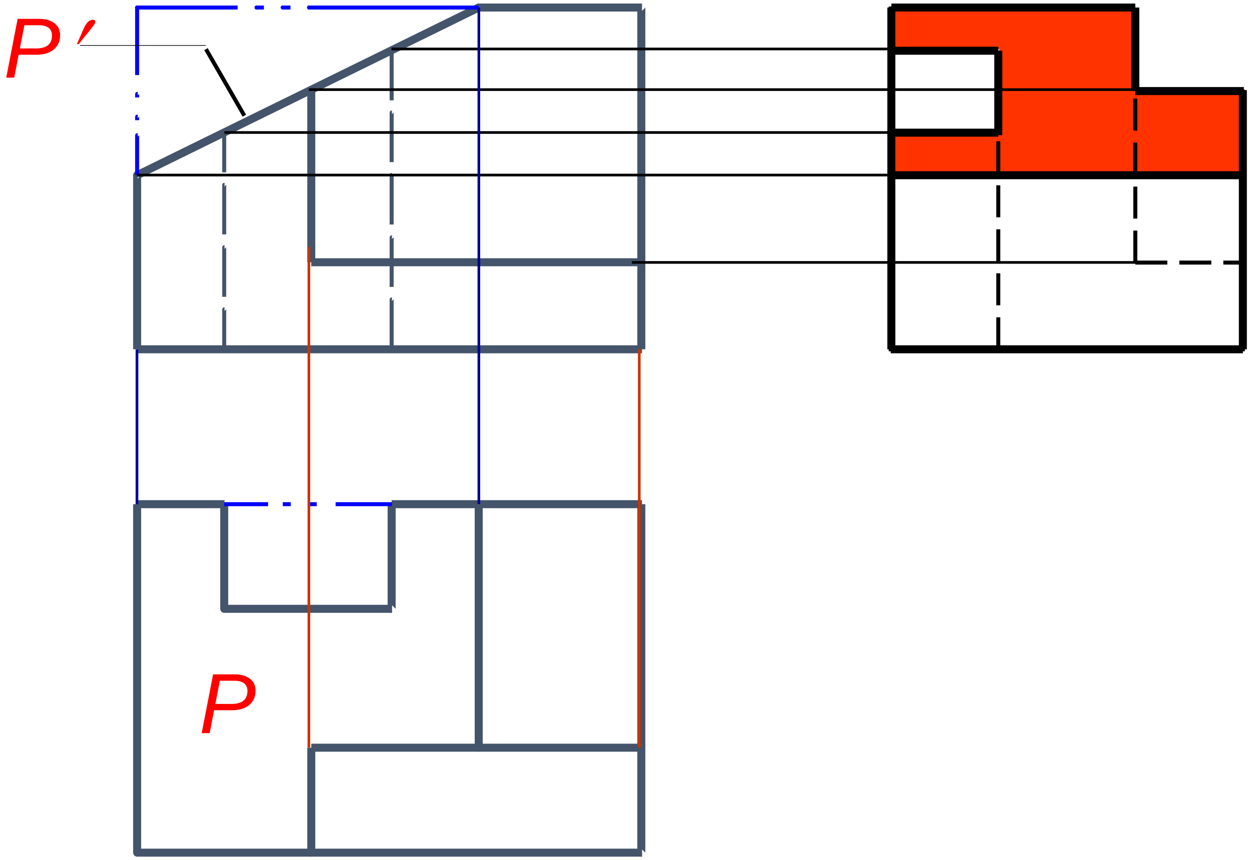

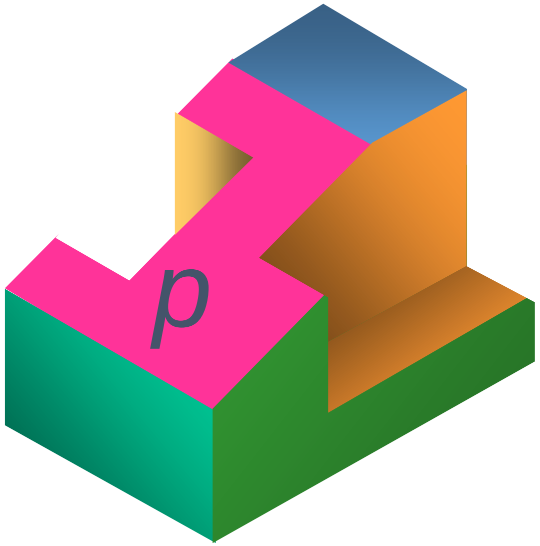

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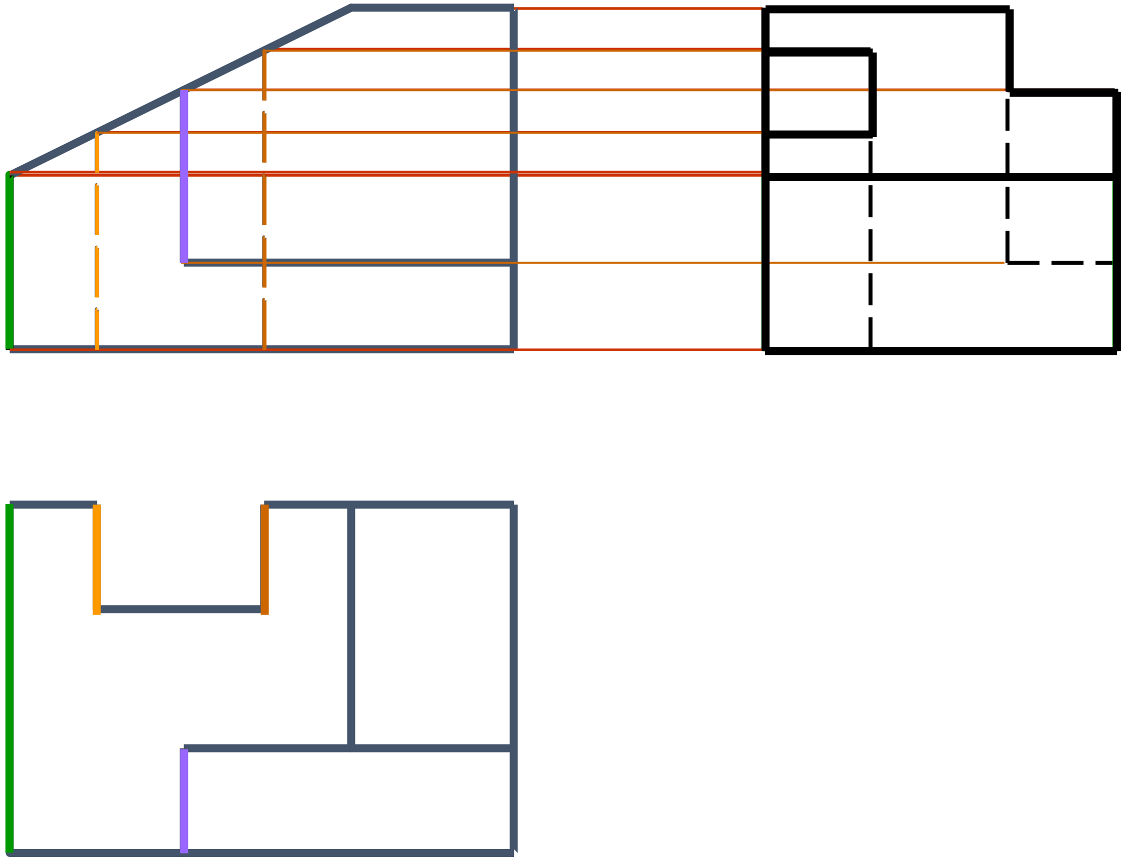

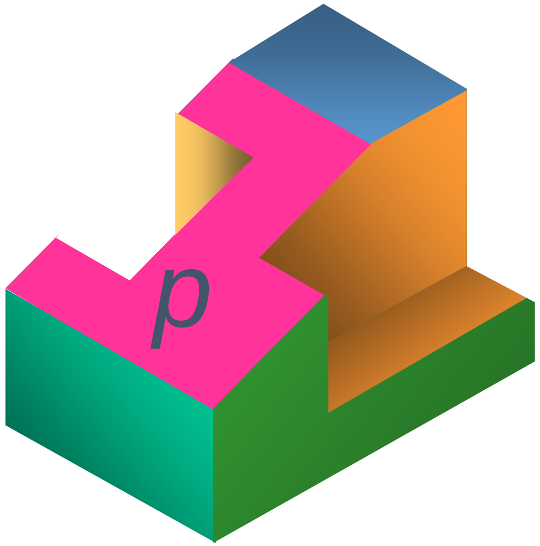

【例】

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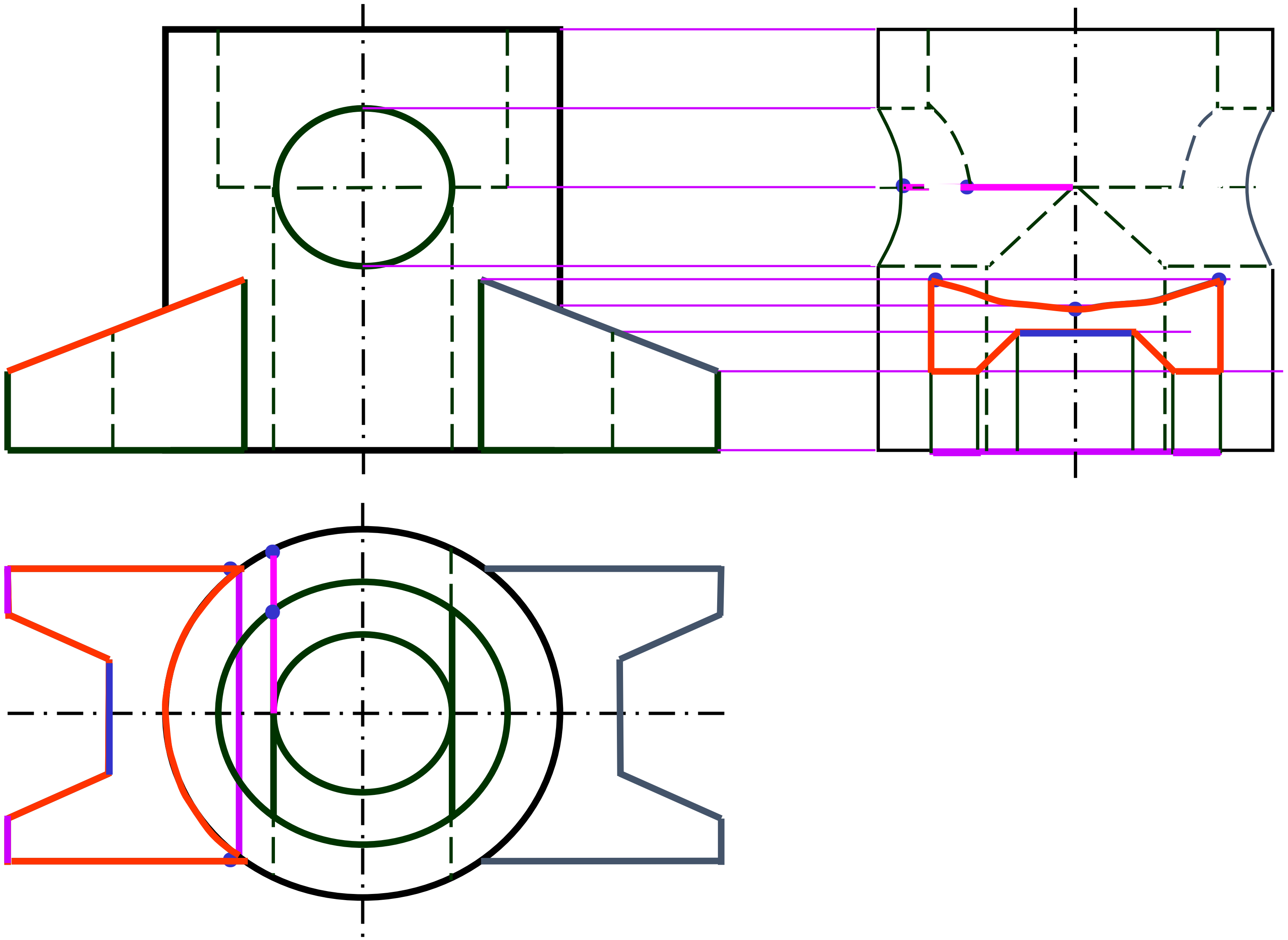

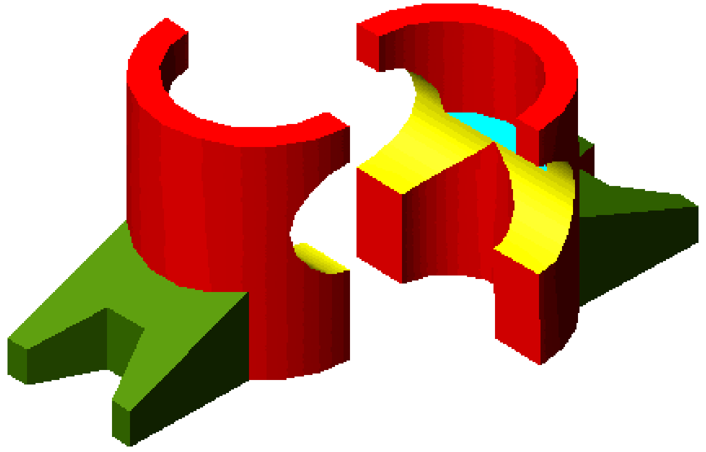

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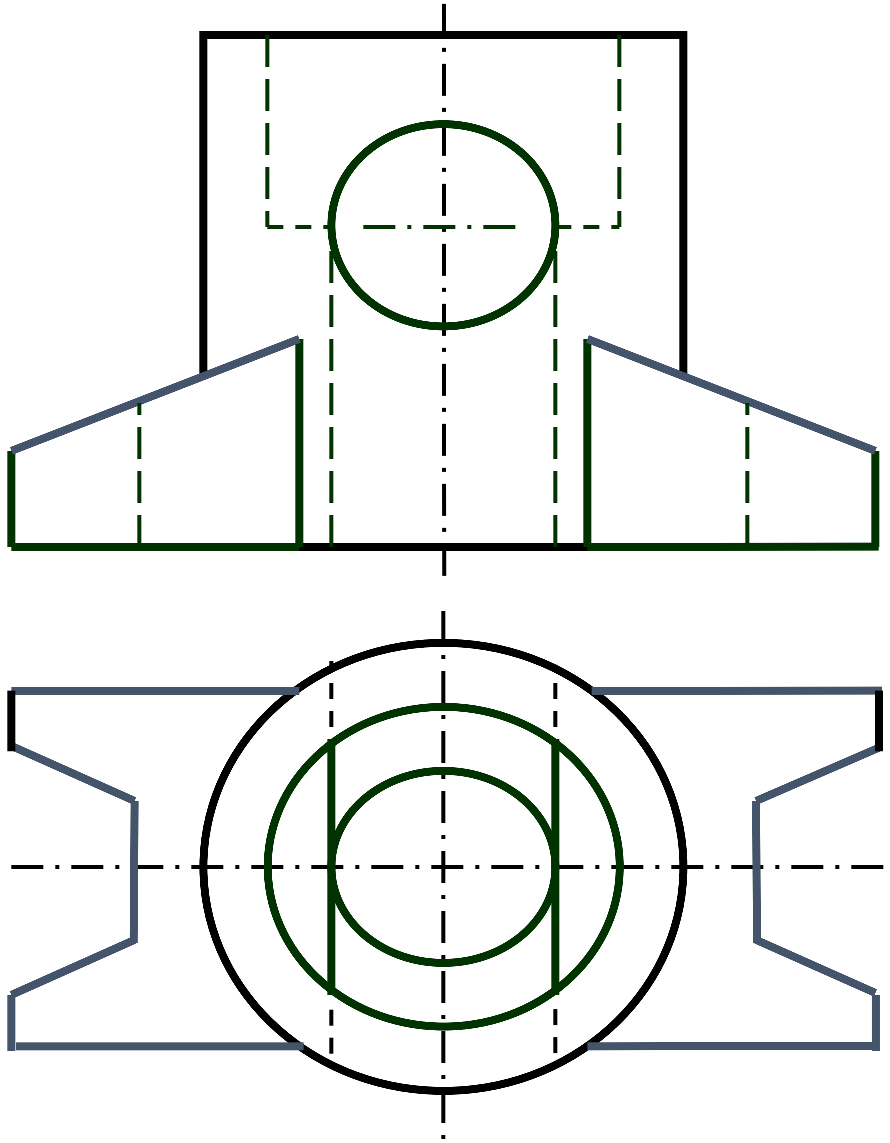

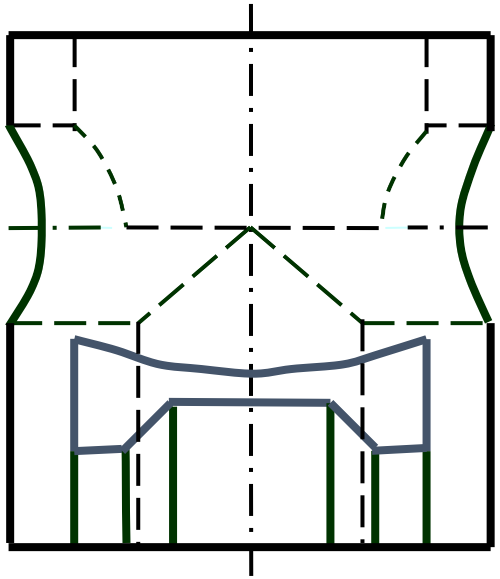

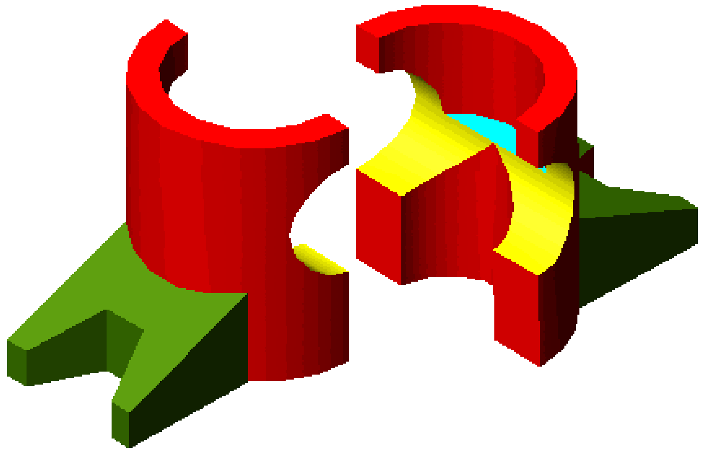

【例】